

Gat Andres Bonifacio High School

NEXORA 

Nexora Master Technical & Maintenance Service Manual

Repository-grounded engineering reference · 2026-07-13 · commit
3d0c93e

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Chapter 00 — Master Index and Field Guide

Snapshot authority. This chapter describes commit `3d0c93e5270d44b9912deeeae0218e95c9a311dd5` on branch `development`. Source paths named below are the authority if the implementation changes after 2026-07-13.

This is the front door to the appliance-grade engineering reference for Nexora LMS/LXP. It defines the evidence boundary, terminology, update discipline, and safe paths to every subsystem.

Source map

- `docs/CODEX_MASTER_MANUAL_PROMPT.md`
- `AGENTS.md`
- `backend/AGENTS.md`
- `next-frontend/AGENTS.md`
- `ai-service/AGENTS.md`
- `mobile/AGENTS.md`

Manual control record

Control	Recorded value	Meaning
Repository	capstone-nest-react-lms	Nexora LMS/LXP monorepo for web, mobile, backend, AI, and operations.
Branch	development	Branch inspected while the manual was generated.
Commit	3d0c93e5270d44b9912deeeae0218e95c9a311dd5	Immutable source snapshot for every catalog in this edition.
Inspection date	2026-07-13 Asia/Manila	Calendar date for staleness checks.
Schema scope	89 tables and 55 PostgreSQL enums	Derived from active files under backend/src/drizzle/schema.
Backend API scope	385 route handlers	Derived from Nest HTTP decorators in every controller source.
AI API scope	60 FastAPI route handlers	Derived from app and router decorators.
Web scope	103 App Router pages	Derived from every page.tsx in next-frontend/app.
Mobile scope	57 screen files	Derived from active mobile/src/screens, excluding tests.

Orientation map

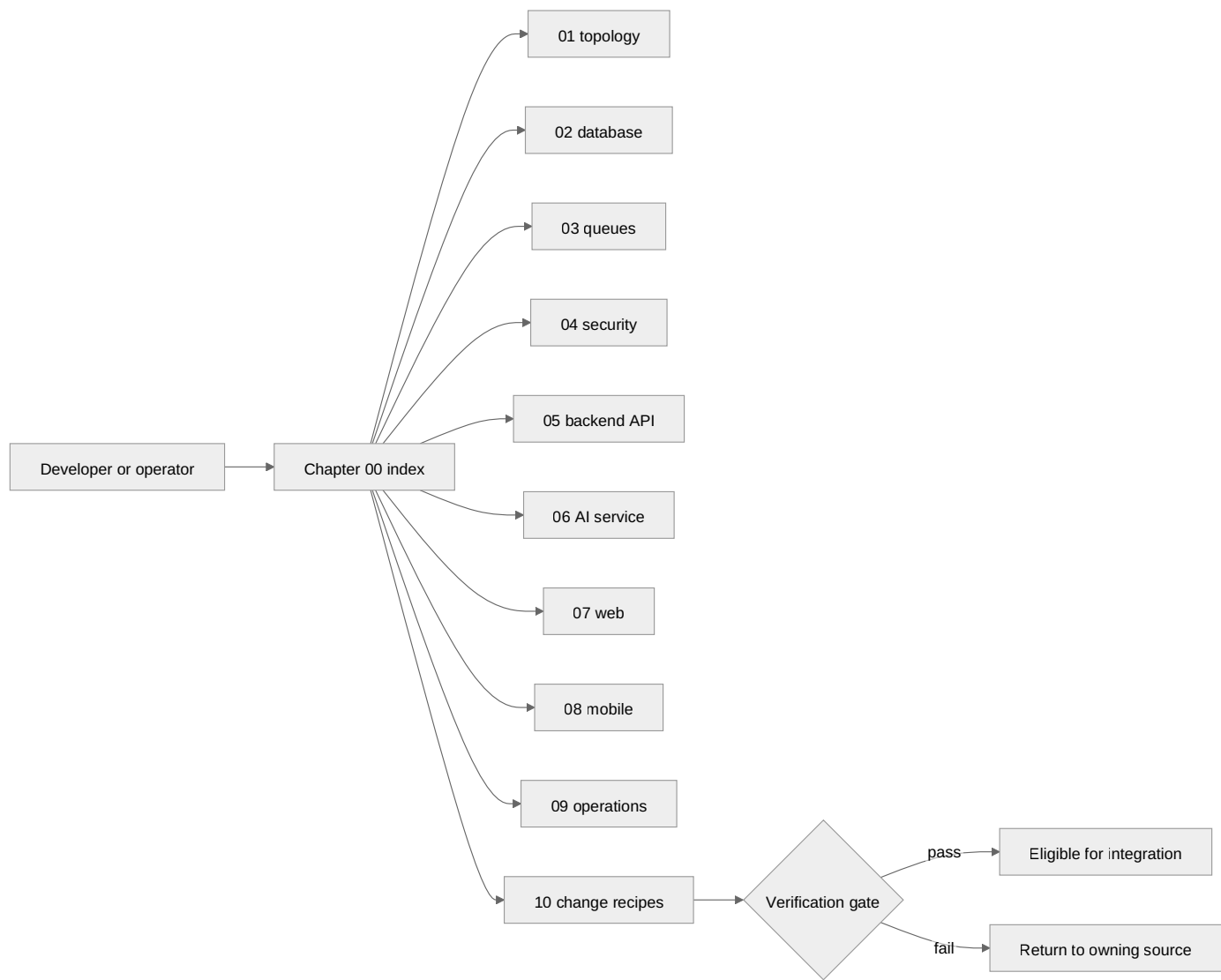


Diagram 1

Chapter inventory

Chapter	Title	Use it when
00	Master index and guide	How to navigate, validate, rebuild, and maintain this manual.
01	System topology and architecture	Processes, containers, networks, ports, volumes, trust boundaries, and ownership.
02	Drizzle database schema and entity dictionary	All 89 tables, 55 enums, columns, indexes, foreign keys, and deletion behavior.
03	BullMQ queue and job orchestration	All seven queues, producers, job payloads, processors, retries, retention, and recovery.
04	Auth, RBAC, security, and session lifecycle	Web and mobile login, refresh rotation, guards, role access, and internal service authentication.
05	NestJS backend module and API catalog	All 37 modules, 385 controller routes, handler inputs, DTOs, responses, and downstream delegates.
06	FastAPI AI service and vector engine	All 60 AI routes, Pydantic contracts, runtime models, retrieval, indexing, and guardrails.
07	Next.js web frontend and role workspaces	All 103 pages, service wrappers, providers, caching, role gates, components, and tokens.
08	Expo mobile architecture and navigation	

Chapter	Title	Use it when
		Boot composition, role navigation, screens, API clients, secure storage, and offline boundaries.
09	Observability, telemetry, and diagnostics	Metrics, logs, traces, dashboards, alerts, health endpoints, and incident procedures.
10	Developer modification cookbook	Exact maintenance recipes for schema, API, queue, AI, client, dependency, and verification work.

How to use the manual

1. Start with Chapter 01 when a request crosses process or trust boundaries.
2. Open Chapter 02 before changing persistence, because the database source is `backend/src/drizzle/schema/`, not a legacy `src/db/schema` path.
3. Open Chapter 03 before moving work into or out of a request cycle. BullMQ is the durability boundary for long-running AI and fan-out work.
4. Open Chapter 04 before changing cookies, bearer tokens, route decorators, forwarded headers, account state, or user-identifying logs.
5. Use the exhaustive catalogs in Chapters 05 through 08 to find the exact owner. Then inspect the named source file before editing.
6. Use Chapter 09 before claiming runtime health. A stopped stack and an old log stream are not live evidence.
7. Execute the recipe in Chapter 10 and its verification gate as one change unit.

Authority and conflict rules

Priority	Authority	Rule
1	Active source and migrations	Runtime code, active Drizzle schema, active migrations, and Compose files win over prose.
2	Repository kernel and folder guides	AGENTS.md and subsystem AGENTS.md define ownership and safety invariants.
3	This manual at its recorded commit	Use the catalog directly only while the source snapshot still matches.
4	Older reports and archived migrations	Historical evidence can explain intent but cannot override active files.

Architectural invariants

- The NestJS backend owns public authentication, role authorization, official academic state, audit history, API envelopes, and durable queue orchestration.
- Web and mobile call backend routes under `/api`. Neither client calls the Python AI service directly.
- AI output is assistive. It does not become an official grade, enrollment, class record, or intervention action without backend-owned policy and, where required, teacher approval.
- Core Compose remains usable without the `observability` profile and without `docker-compose.debug.yml`.
- Long-running extraction and generation work remains restart-safe through backend-owned BullMQ contracts and durable database job records.
- The default generic mobile target is `mobile/`. Archived mobile folders are not active architecture.

Naming and notation

Notation	Interpretation
<code>/api/<route></code>	Public NestJS route after the global prefix configured in <code>backend/src/main.ts</code> .

Notation	Interpretation
<code>/internal/<route></code>	AI-service route requiring the shared internal service token unless the source states otherwise.
<code>:id</code>	NestJS or manual route parameter notation.
<code>{id}</code>	FastAPI path-parameter notation.
JWT authenticated	Global JwtAuthGuard applies and the handler is not marked public.
JWT plus roles	The bearer token is validated and RolesGuard accepts at least one listed role.
Derived state	A recomputable projection that is not the sole official academic record.
Official state	Backend-owned durable academic record requiring explicit authorization and audit discipline.

Edition maintenance procedure

1. Record the new commit with `git rev-parse HEAD`.
2. Compare active source counts for tables, routes, queues, pages, and screens against the control record.
3. Regenerate or reconcile the affected chapter before changing the recorded commit.
4. Search the manual for removed paths and renamed symbols.
5. Run the compile script and inspect the PDF table of contents, diagrams, wide tables, code blocks, and page footer.
6. Commit the source chapters and the compiler together so a reader can reproduce the edition.

PDF build entrypoint

Run `bash docs/master-manual/compile_pdf.sh` from any directory. The script resolves the repository root, concatenates chapters 00 through 10, renders Mermaid in a local browser context, and writes `docs/master-manual/nexora-master-service-manual.pdf`. Its temporary build directory is removed after a successful compilation.

Field glossary

Term	Operational definition
BullMQ job ID	Queue-level deduplication and cancellation identifier, distinct from an AI database job UUID.
AI generation job	Durable row in <code>ai_generation_jobs</code> describing a teacher AI request and state.
Extraction	Durable <code>extracted_modules</code> workflow that converts an uploaded source into reviewable class content.
RAG	Retrieval-augmented generation over <code>content_chunks</code> and <code>content_chunk_embeddings</code> .
Class record	Official grade workbook rooted in <code>class_records</code> and its category, item, score, and final-grade children.
LXP intervention	Teacher-governed assistive remediation rooted in <code>intervention_cases</code> .
Ja	Student-facing J.A.K.I.P.I.R practice, review, ask, and tutoring capabilities.
Core topology	PostgreSQL, Redis, Ollama, backend, AI service, and frontend.
Observability topology	Opt-in Prometheus, blackbox exporter, Loki, Tempo, Grafana, Promtail, node-exporter, and cAdvisor services.

Chapter 01 — System Topology and Cross-Subsystem Architecture

Snapshot authority. This chapter describes commit `3d0c93e5270d44b9912deee0218e95c9a311dd5` on branch `developement`. Source paths named below are the authority if the implementation changes after 2026-07-13.

This chapter is the deployment and ownership schematic. It follows every public request from client to backend, every durable asynchronous handoff through Redis, every AI execution boundary, and every optional telemetry path.

Source map

- `docker-compose.yml`
- `docker-compose.debug.yml`
- `backend/src/main.ts`
- `backend/src/app.module.ts`
- `backend/Dockerfile`
- `backend/docker-entrypoint.sh`
- `next-frontend/Dockerfile`
- `next-frontend/next.config.ts`
- `ai-service/Dockerfile`
- `ai-service/app/main.py`
- `monitoring/`

Deployment topology

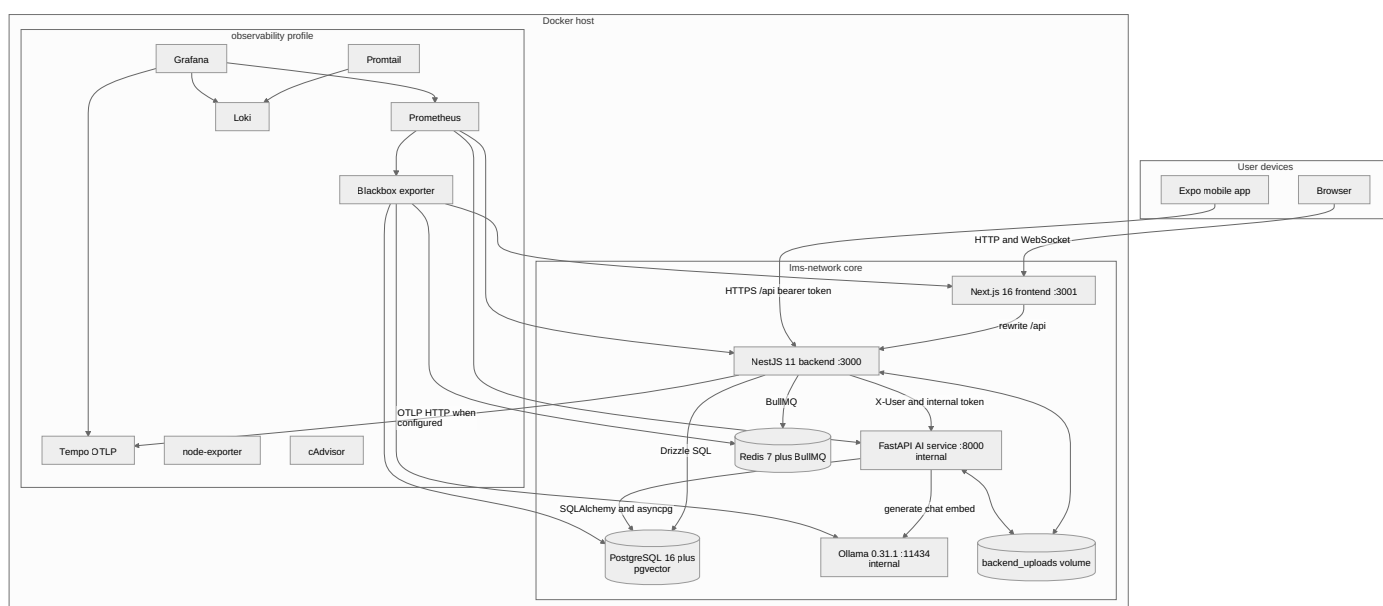


Diagram 2

Public request path

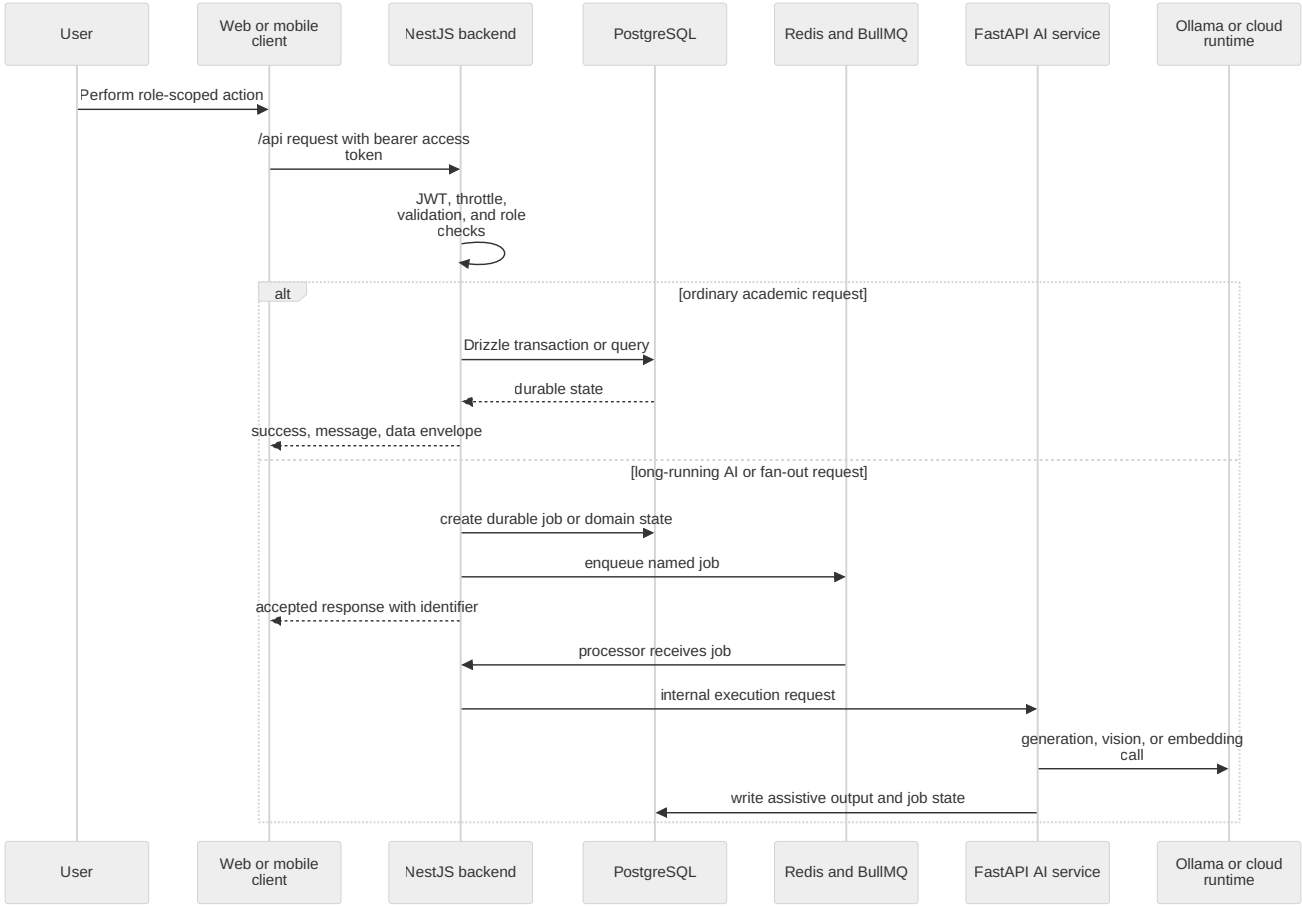


Diagram 3

Service responsibility matrix

Capability	Owning service	Durable authority	Prohibited shortcut
Authentication and RBAC	NestJS backend	users, roles, user_roles, re-fresh_tokens, OTP and audit data	Clients and AI service may not become independent auth authorities.
Official classes, enrollment, assessments, scores, and class records	NestJS backend	Drizzle tables and audited service transactions	AI output may not directly overwrite official records.
Web presentation and browser session orchestration	Next.js frontend	Backend API; browser holds only transient access state and refresh cookie	Web code may not query PostgreSQL or AI service directly.
Mobile presentation and secure token persistence	Expo mobile	Backend API; device secure storage holds session tokens	Mobile code may not assume browser cookie semantics.
AI tutoring, extraction, retrieval, and draft generation	FastAPI AI service behind backend	Assistive tables and backend-owned job rows	Shared-secret access does not grant public or academic authority.
Long-running work and retries	NestJS BullMQ workers	Redis queue state plus domain and job tables	FastAPI process-local background tasks may not replace durable queue orchestration.
Metrics, logs, and traces	Optional observability profile	Prometheus, Loki, Tempo, Grafana volumes	Core startup may not depend on observability services.

Compose service inventory

Exhaustive inventory rule. The 17 Compose service declarations below were extracted from `docker-compose.yml` and `docker-compose.debug.yml` at commit `3d0c93e`. A later source change requires regenerating or manually reconciling this chapter.

Compose file	Service	Image or build	Profile	Host ports	Runtime user	Volumes	Depends on
docker-compose.yml	postgres	pgvector/pgvector:pg16	core or override	not host-published	image default or Dockerfile user	postgres_data:/var/lib/postgresql/data	none
docker-compose.yml	redis	redis:7-alpine	core or override	not host-published	image default or Dockerfile user	none	none
docker-compose.yml	ollama	ollama/ollama:0.31.1	core or override	not host-published	image default or Dockerfile user	ollama_data:/root/.ollama	none
docker-compose.yml	backend	{"context":"./backend","dockerfile":"Dockerfile"}	core or override	3000:3000	image default or Dockerfile user	backend_uploads:/app/uploads	postgres, redis
docker-compose.yml	ai-service	{"context":"./ai-service","dockerfile":"Dockerfile"}	core or override	not host-published	image default or Dockerfile user	backend_uploads:/app/uploads	postgres, ollama
docker-compose.yml	frontend	{"context":"./next-frontend","dockerfile":"Dockerfile","args":{"NEXT_PUBLIC_API_URL":"\${NEXT_PUBLIC_API_URL:-http://backend:3000}","NEXT_PUBLIC_WS_URL":"\${NEXT_PUBLIC_WS_URL:-http://localhost:3000}"} }	core or override	\${FRONTEND_PORT:-3001}:3001	image default or Dockerfile user	none	backend
docker-compose.yml	prometheus	prom/prometheus:v2.54.1	observability	\${PROMETHEUS_PORT:-9090}:9090	image default or Dockerfile user	./monitoring/prometheus.yml:/etc/prometheus/prometheus.yml:ro, prometheus_data:/prometheus	none
docker-compose.yml	blackbox-exporter	prom/blackbox-exporter:v0.25.0	observability	not host-published	image default or Dockerfile user	./monitoring/blackbox.yml:/etc/blackbox_exporter/config.yml:ro	none
docker-compose.yml	loki	grafana/loki:3.2.1	observability	\${LOKI_PORT:-3100}:3100	image default or Dockerfile user	./monitoring/loki-config.yml:/etc/loki/local-config.yml:ro, loki_data:/loki	none
docker-compose.yml	tempo	grafana/tempo:2.6.1	observability	\${TEMPO_PORT:-3200}:3200, \${TEMPO_OTLP_HTTP_PORT:-4318}:4318	0	./monitoring/tempo-config.yml:/etc/tempo.yml:ro, tempo_data:/tmp/tempo	none
docker-compose.yml	grafana	grafana/grafana:11.6.16	observability	\${GRAFANA_PORT:-3002}:3000	image default or Dockerfile user	./monitoring/grafana/provisioning:/etc/grafana/provisioning:ro, ./monitoring/grafana/dashboards:/var/lib/grafana/dashboards:ro, grafana_data:/var/lib/grafana	prometheus, loki, tempo
docker-compose.yml	promtail	grafana/promtail:3.2.1	observability	not host-published	image default or Dockerfile user	./monitoring/promtail-con-	loki

Compose file	Service	Image or build	Profile	Host ports	Runtime user	Volumes	Depends on
						fig.yml:/etc/promtail/config.yml:ro, promtail_positions:/run/promtail, /var/lib/docker/containers:/var/lib/docker/containers:ro, /var/run/docker.sock:/var/run/docker.sock:ro	
docker-compose.yml	node-exporter	prom/node-exporter:v1.8.2	observability	not host-published	image default or Dockerfile user	/proc:/host/proc:ro, /sys:/host/sys:ro, /:/rootfs:ro, /run/udev/data:/run/udev/data:ro	none
docker-compose.yml	cadvisor	ghcr.io/google/cadvisor:v0.60.5	observability	not host-published	image default or Dockerfile user	:/rootfs:ro, /var/run:/var/run:ro, /sys:/sys:ro, /var/lib/docker:/var/lib/docker:ro	none
docker-compose.debug.yml	postgres	None	core or override	\${POSTGRES_HOST_PORT:-55432}:5432	image default or Dockerfile user	none	none
docker-compose.debug.yml	redis	None	core or override	\${REDIS_HOST_PORT:-6379}:6379	image default or Dockerfile user	none	none
docker-compose.debug.yml	ai-service	None	core or override	\${AI_SERVICE_PORT:-8000}:8000	image default or Dockerfile user	none	none

Network and port boundaries

Port	Owner	Core host exposure	Debug or observability exposure	Purpose
3000	backend	3000:3000	unchanged	Public Nest HTTP API, Swagger outside production, Web-Socket namespace, health, and metrics.
3001	frontend	FRONTEND_PORT default 3001	unchanged	Next.js web application.
5432	postgres	internal only	POSTGRES_HOST_PORT default 55432	PostgreSQL and pgvector.
6379	redis	internal only	REDIS_HOST_PORT default 6379	BullMQ transport and token rotation grace cache.
8000	ai-service	internal only	AI_SERVICE_PORT default 8000	FastAPI execution, health, readiness, and metrics.
11434	ollama	internal only	no supplied host mapping	Local text, vision, and embedding runtime.
9090	prometheus	absent unless profile enabled	PROMETHEUS_PORT default 9090	Metrics query and storage.
3100	loki	absent unless profile enabled	LOKI_PORT default 3100	Log aggregation API.
3200	tempo	absent unless profile enabled	TEMPO_PORT default 3200	Tempo query and readiness API.
4318	tempo	absent unless profile enabled	TEMPO_OTLP_HTTP_PORT default 4318	OTLP HTTP trace ingestion.
3000 container	grafana	absent unless profile enabled	GRAFANA_PORT default 3002	Dashboards and alerting UI.
9115	blackbox-exporter	internal only	profile-internal	HTTP and TCP probes.
9080	promtail	internal only	profile-internal	Promtail readiness.
9100	node-exporter	internal only	profile-internal	Host metrics.
8080	cadvisor	internal only	profile-internal	Container metrics.

All services join the bridge network `lms-network`. Core data stores and AI execution remain unexposed by default. The debug override adds only PostgreSQL, Redis, and AI service host mappings and must be named explicitly with a second `-f` argument.

Topology modes

Mode	Command shape	Services	Use
Core	<code>docker compose --env-file .env.compose up --build</code>	postgres, redis, ollama, backend, ai-service, frontend	Normal local or server topology.
Core plus observability	<code>docker compose --env-file .env.compose --profile observability up --build</code>	Core plus nine telemetry services	Operational dashboards, alerts, logs, probes, and traces.
Core plus debug ports	<code>docker compose -f docker-compose.yml -f docker-compose.debug.yml --env-file .env.compose up --build</code>	Core with host access to PostgreSQL, Redis, and FastAPI	Explicit local diagnostics and direct client tools.
Core plus both	<code>docker compose -f docker-compose.yml -f docker-compose.debug.yml --env-file .env.compose --profile observability up --build</code>	Core, debug mappings, and telemetry profile	Full local engineering laboratory.

Startup and readiness chain

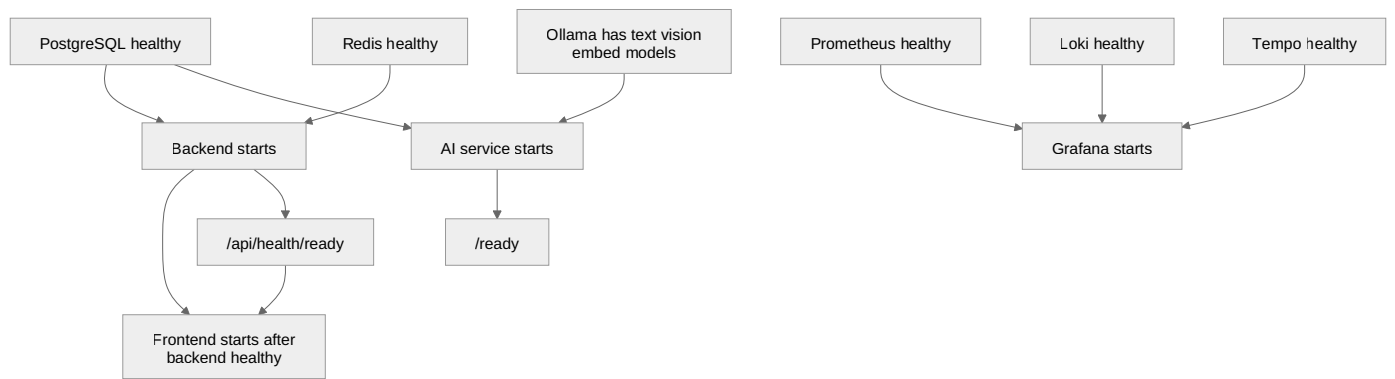


Diagram 4

- PostgreSQL uses `pg_isready` against database `capstone`.
- Redis uses `redis-cli ping`.
- Ollama readiness requires all configured text, vision, and embedding models to resolve with `ollama show`. Its entrypoint pulls each model before reporting ready.
- Backend readiness calls `/api/health/ready` after its entrypoint verifies the database, optionally runs migrations, optionally seeds, and drops privileges to the Node user.
- AI readiness calls `/ready` and is allowed to return a non-5xx degraded response only according to its configured readiness logic.
- Frontend waits for backend health but has no Compose healthcheck of its own.

Volumes and durability

Volume	Mounted by	Data class	Recovery implication
postgres_data	postgres	All PostgreSQL and pgvector records	Back up before destructive migration or credential-volume reset.
ollama_data	ollama	Downloaded model layers	Rebuildable from pinned model names but expensive to redownload.
backend_uploads	backend and ai-service	Uploaded PDFs, rosters, and library files	Shared path is required for extraction and protected download flows.

Volume	Mounted by	Data class	Recovery implication
prometheus_data	prometheus	Metrics time series	Optional operations history.
grafana_data	grafana	Grafana runtime state	Provisioned dashboards remain in source; local UI state may be lost.
loki_data	loki	Aggregated logs	Optional incident history.
tempo_data	tempo	Trace blocks	Optional distributed tracing history.
promtail_positions	promtail	Docker log read offsets	Prevents duplicate log ingestion after restart.

Container user policy

- Backend image ownership is baked for the official `node` account. The entrypoint may begin as root only to repair the mounted upload volume, then re-executes migrations, seeding, and the server through `su-exec node`.
- Frontend creates UID 1001 `nextjs`, copies the standalone build with that ownership, and declares `USER nextjs`.
- AI service creates the system account `app`, copies source with `app:app` ownership, and declares `USER app`.
- Tempo explicitly runs as UID 0 in Compose because the supplied volume path and image contract require it. Treat this as an observability-profile exception, not a pattern for application containers.
- PostgreSQL, Redis, Ollama, Grafana, Prometheus, Loki, exporters, and collectors use their image defaults unless Compose states otherwise.

Trust-boundary rules

- External users terminate at Next.js or the backend. The AI service is not host-published in core mode.
- The backend validates JWTs and roles before forwarding identity headers to FastAPI.
- FastAPI accepts forwarded identity only when the configured internal service token matches, if a secret is configured. Internal-only execution routes always require a non-empty matching secret.
- PostgreSQL and Redis do not have core host mappings. Add the debug override only for a controlled workstation.
- Observability mounts Docker and host paths read-only where specified. cAdvisor requires device and host filesystem visibility; Promtail reads Docker container logs and the Docker socket.

Failure domains

Failure	What remains available	Expected symptom	First evidence
Ollama unavailable	Core academic API and clients	AI readiness or AI requests degrade or fail; cloud fallback may serve configured tasks	ai-service /ready, Ollama health, AI metrics.
AI service unavailable	Non-AI backend routes and clients	Backend AI proxy returns 503 or 504; queued jobs retry	backend logs, BullMQ state, ai-service container state.
Redis unavailable	Read paths not requiring Redis may work	Queues and refresh grace cache fail; backend health may report dependency failure	Redis health and backend readiness.
PostgreSQL unavailable	Static frontend process may still answer	Backend and AI durable operations fail readiness	pg_isready, backend entrypoint output, /api/health/ready.
Observability profile unavailable	Entire core product	No dashboards, stored logs, or traces	docker compose --profile observability ps.
Shared upload volume unavailable or misowned	Routes not touching files	Upload, extraction, and protected download failures	backend entrypoint ownership message and file-service logs.

Chapter 02 — Drizzle ORM Database Schema and Entity Dictionary

Snapshot authority. This chapter describes commit `3d0c93e5270d44b9912deeeae0218e95c9a311dd5` on branch `developement`. Source paths named below are the authority if the implementation changes after 2026-07-13.

This chapter is the persistent-data service sheet. It records every active PostgreSQL table, enum, column builder, default, index, outgoing foreign key, vector dimension, and declared deletion action.

Source map

- `backend/src/drizzle/schema/`
- `backend/drizzle/0000_baseline_nexora.sql`
- `backend/drizzle/0001_mixed_morgan_stark.sql`
- `backend/drizzle/0002_small_photon.sql`
- `backend/drizzle/0003_enable_pgvector.sql`
- `backend/drizzle/meta/`
- `backend/drizzle.config.ts`

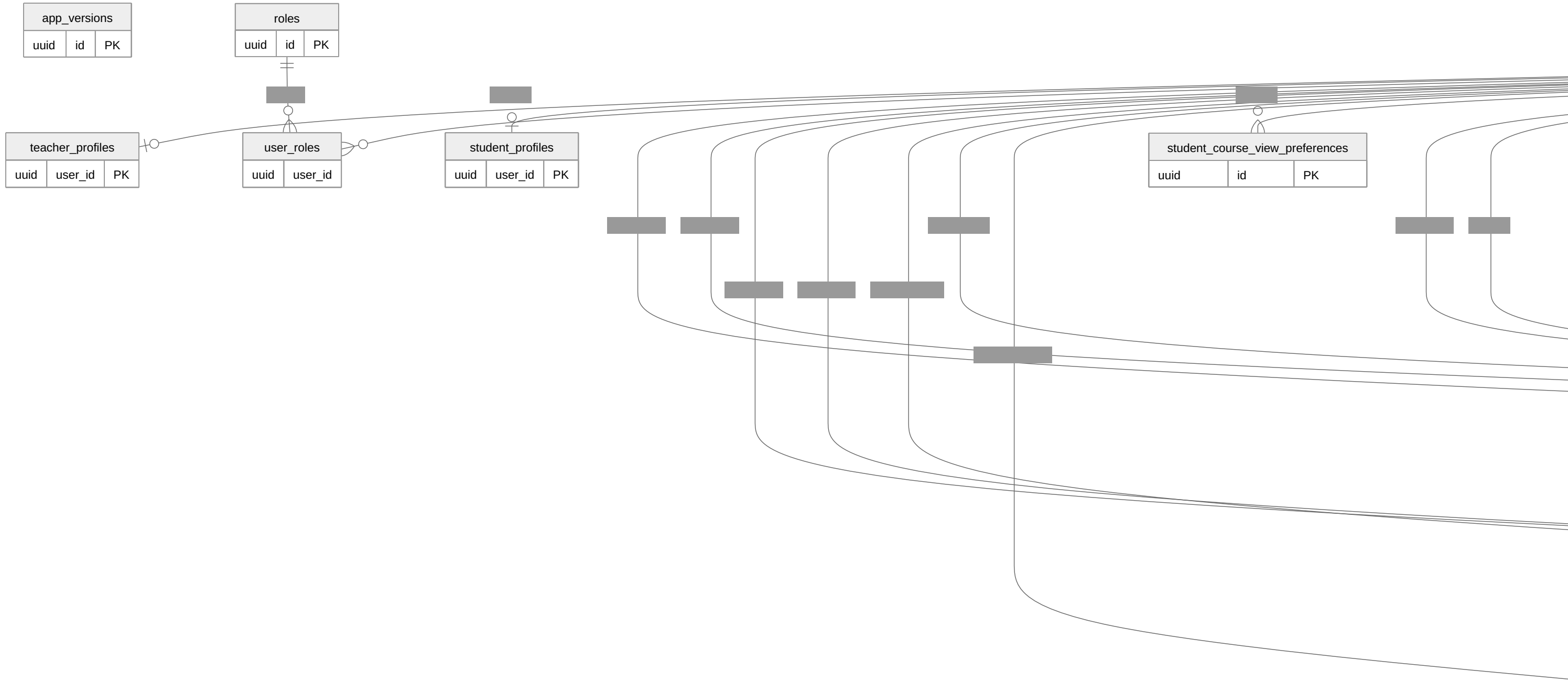
Data authority and migration chain

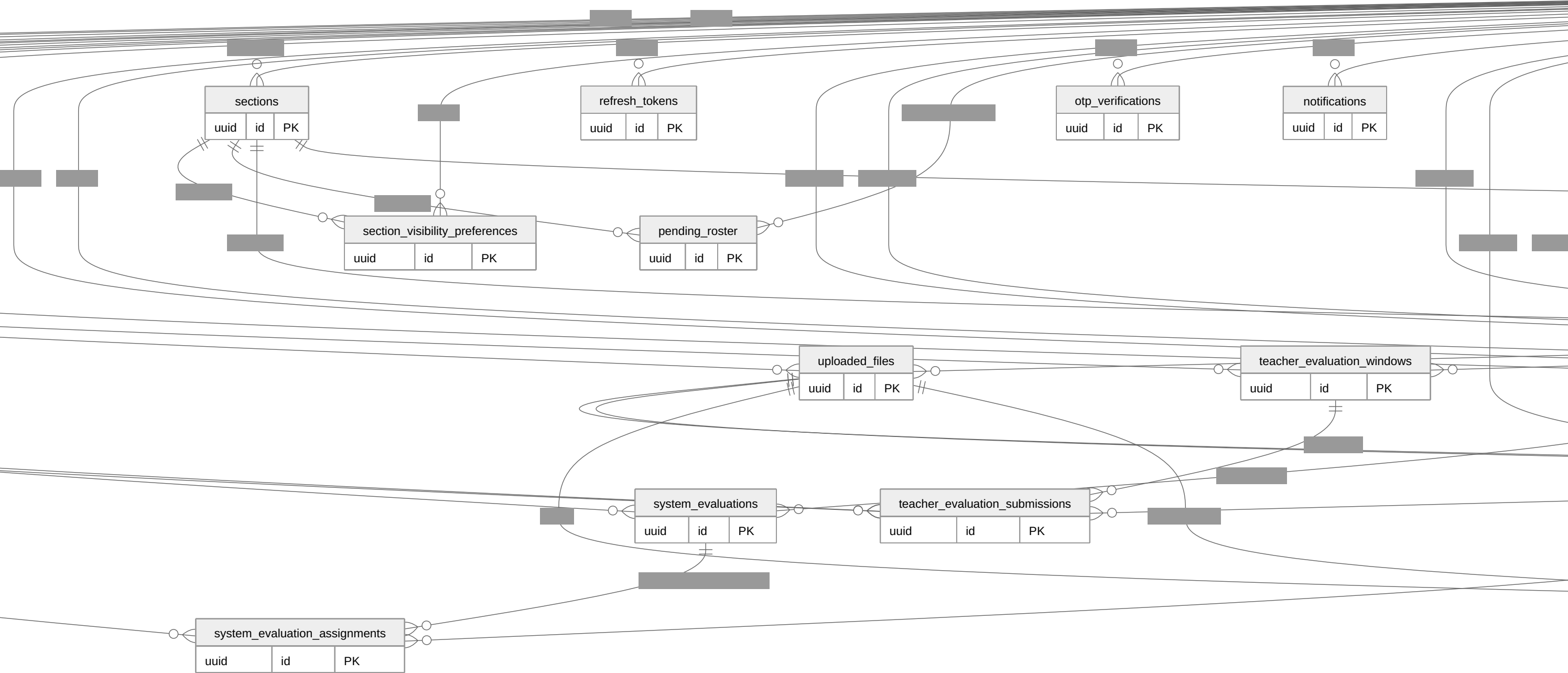
- Active TypeScript schema authority is `backend/src/drizzle/schema/`.
- Active SQL migrations are ordered `0000_baseline_nexora.sql`, `0001_mixed_morgan_stark.sql`, `0002_small_photon.sql`, and `0003_enable_pgvector.sql`.
- Migration 0003 executes `CREATE EXTENSION IF NOT EXISTS vector;`. This must run before a fresh database creates or uses the 768-dimensional vector column from the baseline.
- The baseline declares the complete current table set. Later migrations adjust it. Drizzle metadata snapshots support generation but do not replace the TypeScript source.
- Legacy or archived migration folders are historical evidence only and are not part of the active sequence.

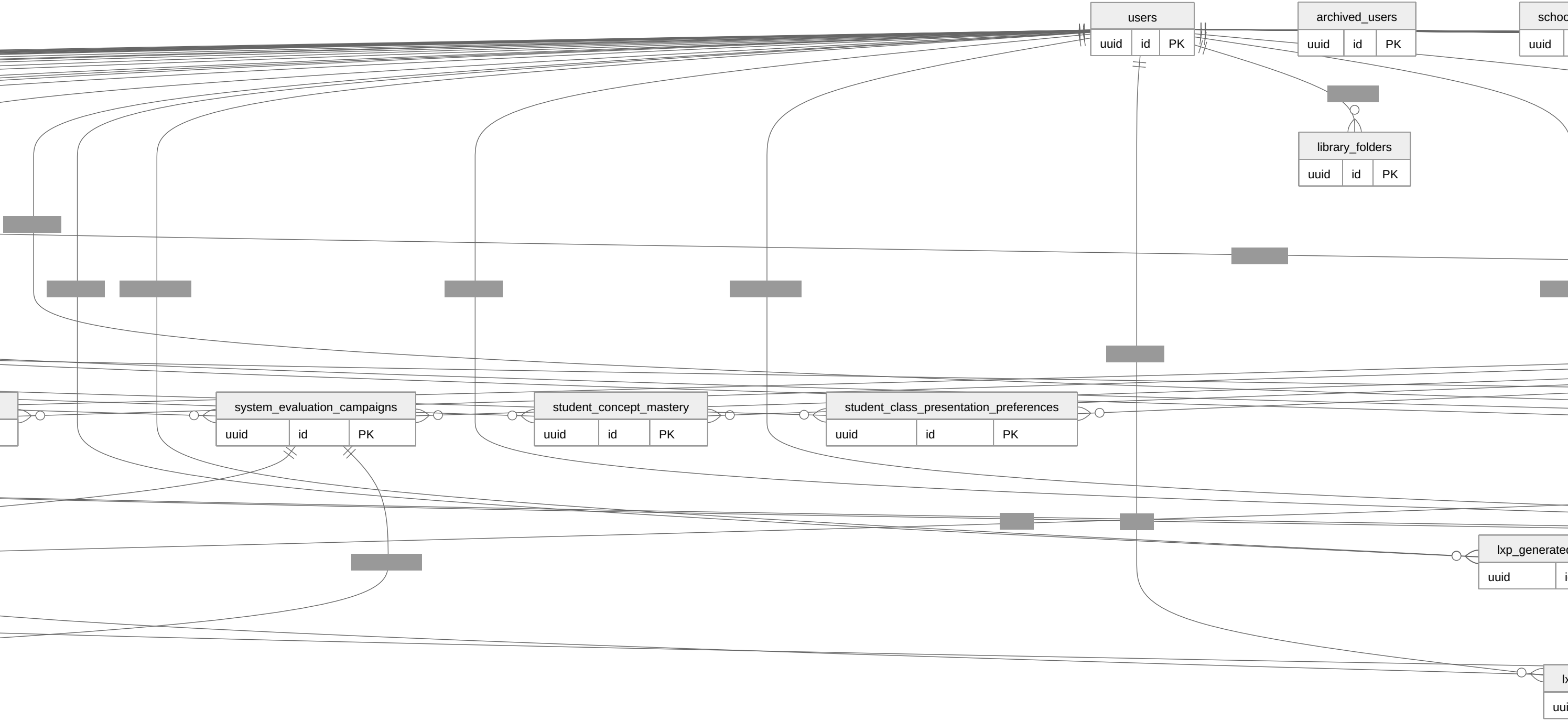
Master entity-relationship diagram

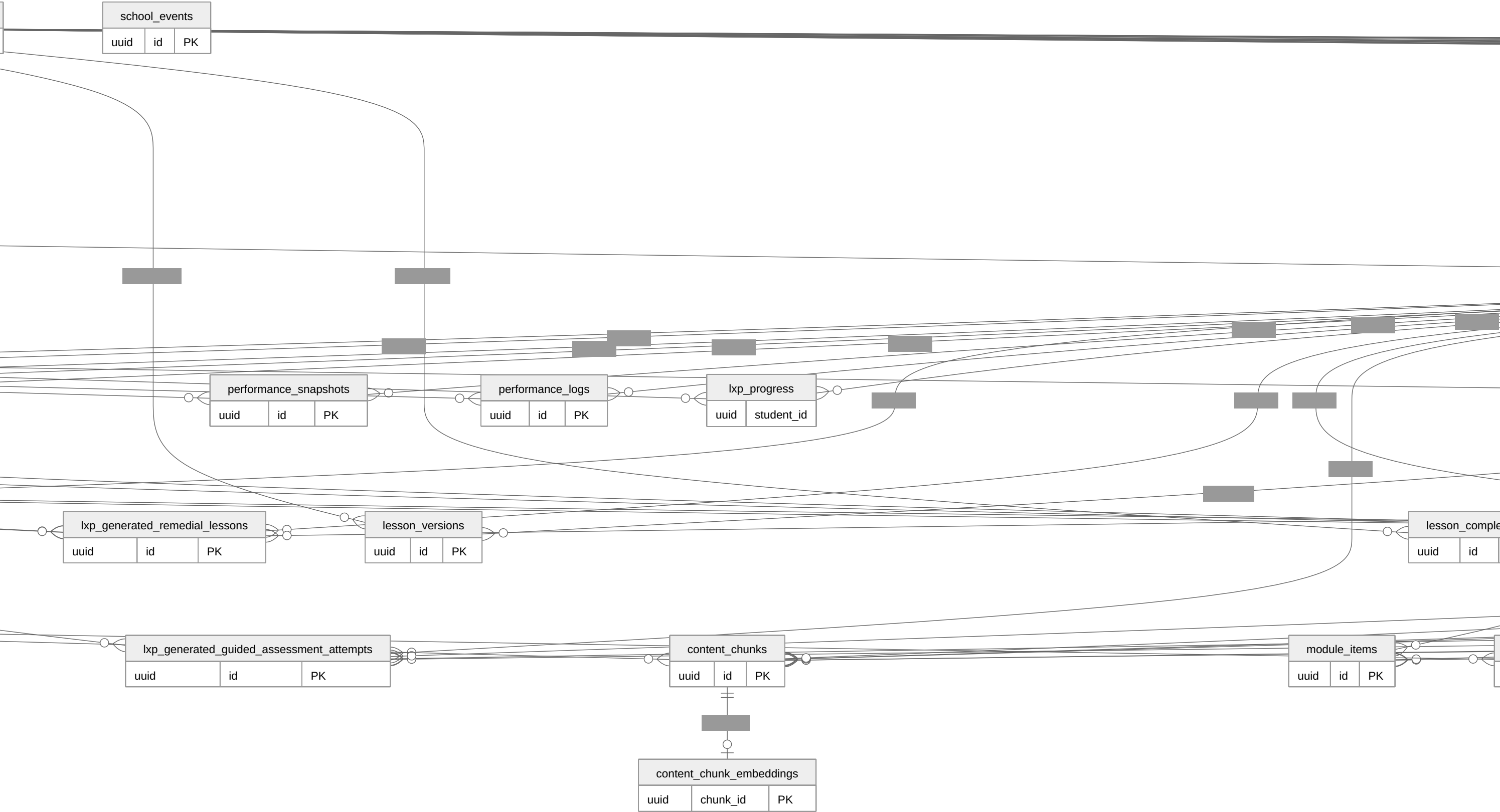
Exhaustive inventory rule. The 89 active tables below were extracted from `backend/src/drizzle/schema/*.ts` at commit `3d0c93e`. A later source change requires regenerating or manually reconciling this chapter.

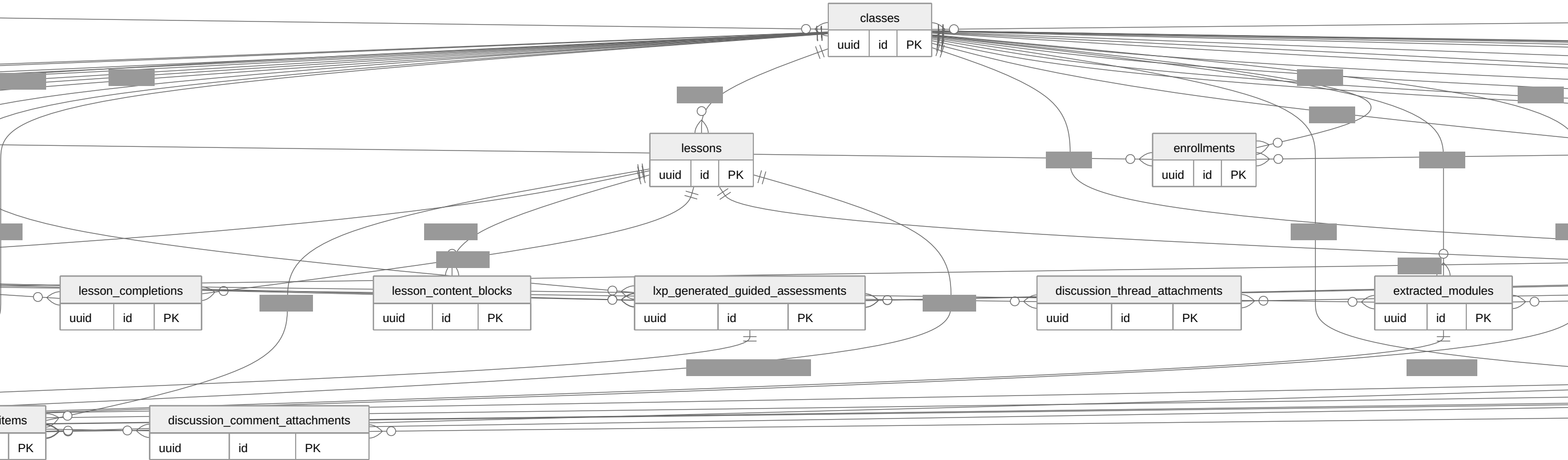
The master ERD includes every active table. For PDF legibility it shows one representative key field per entity; the complete columns follow in the encyclopedia.

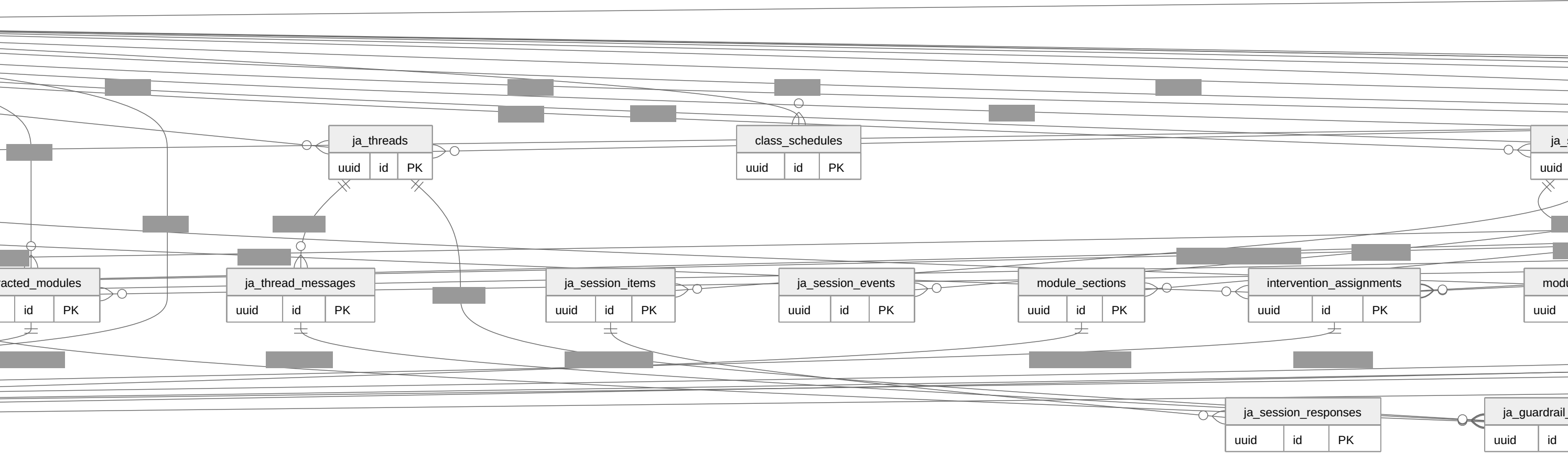


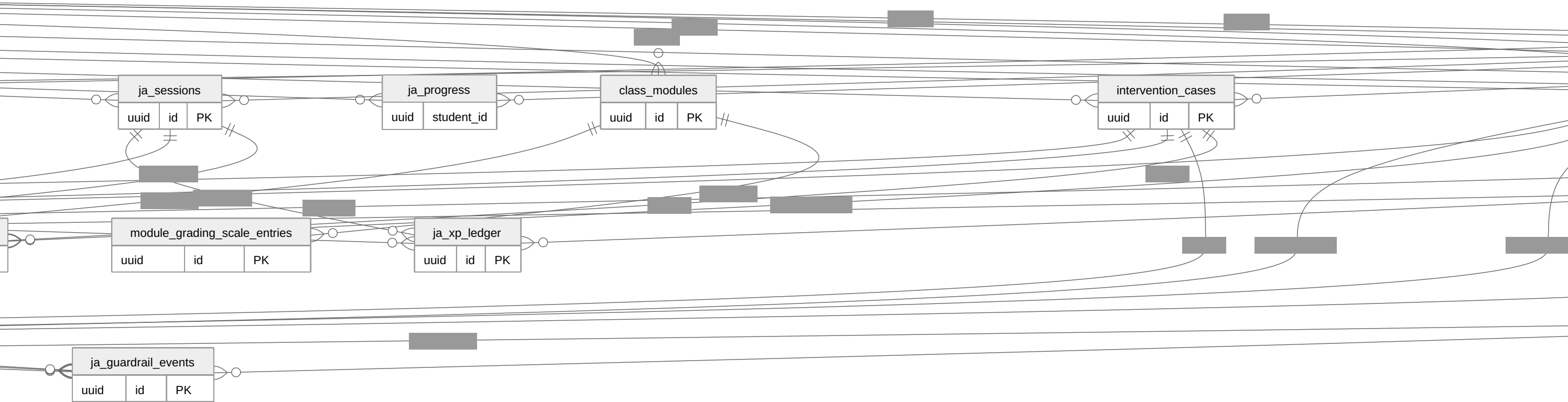


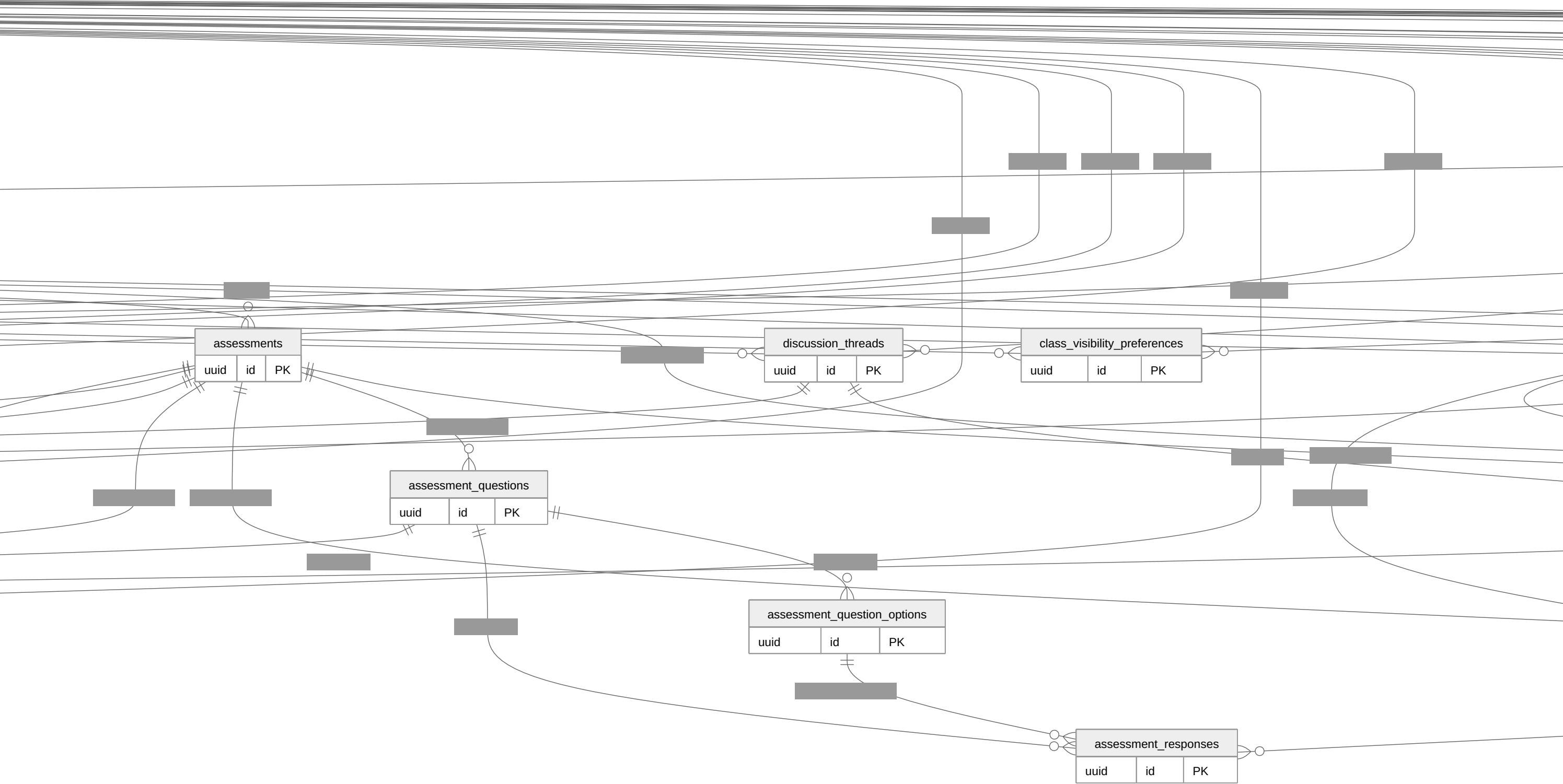


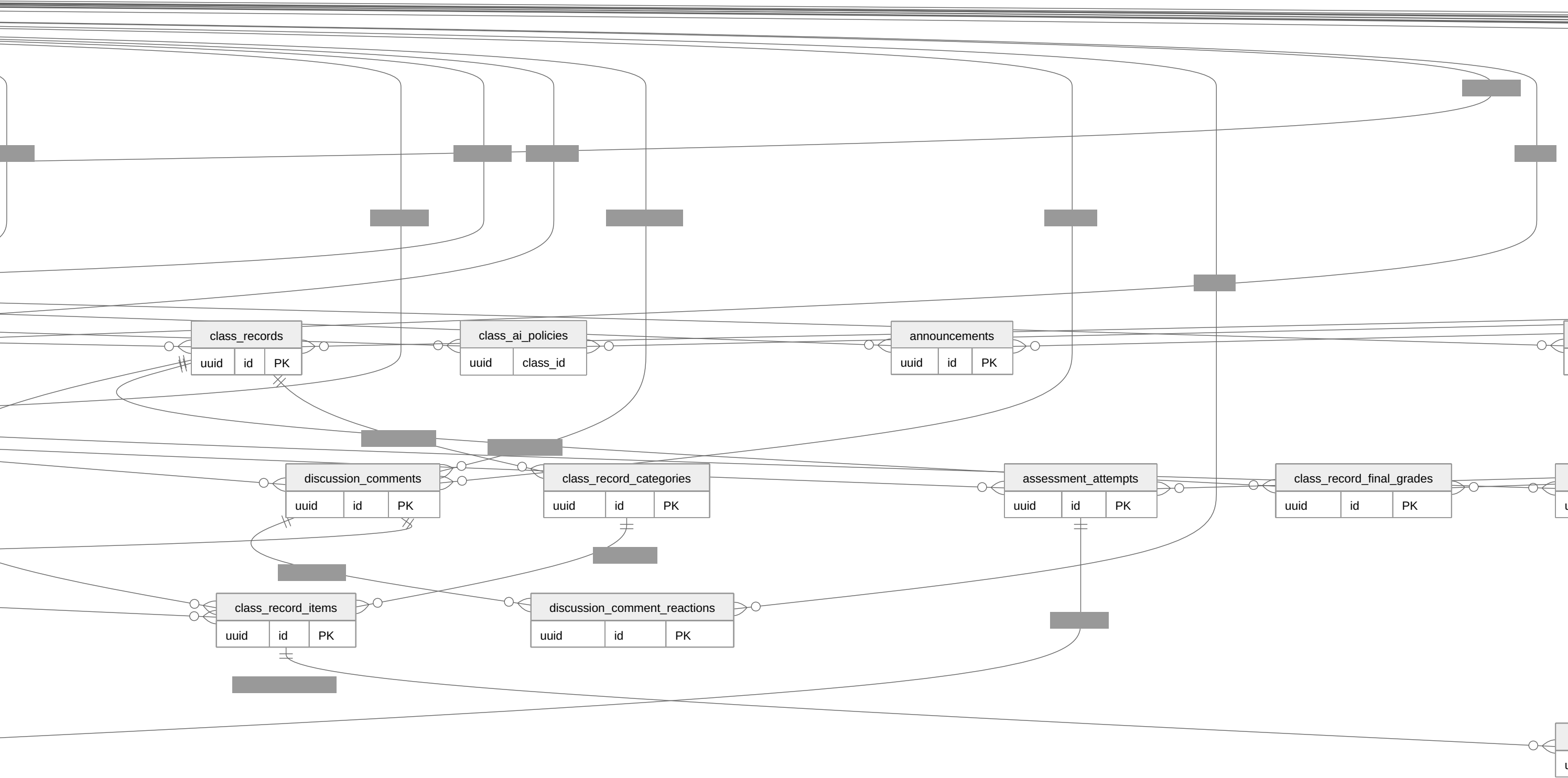


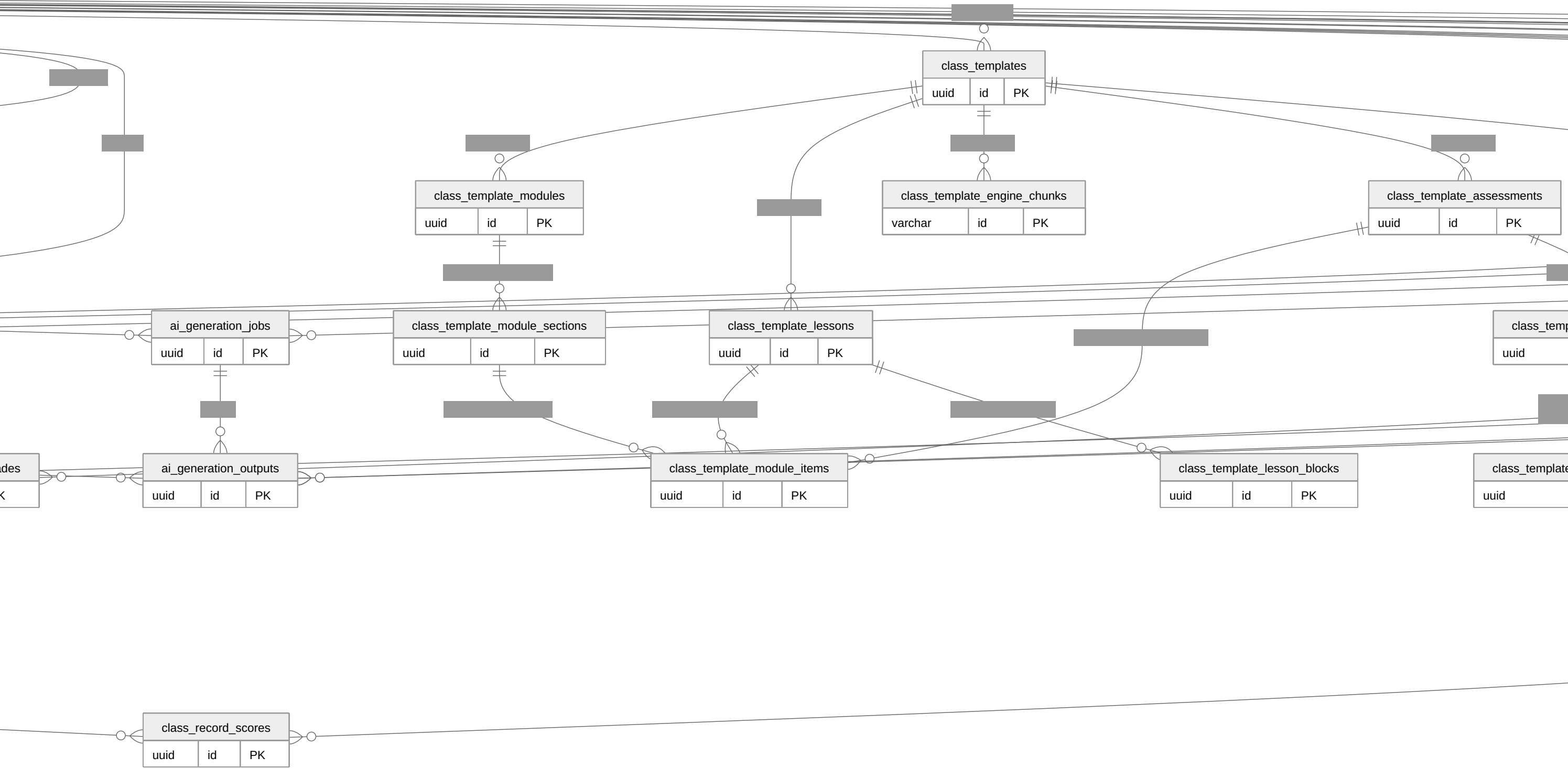


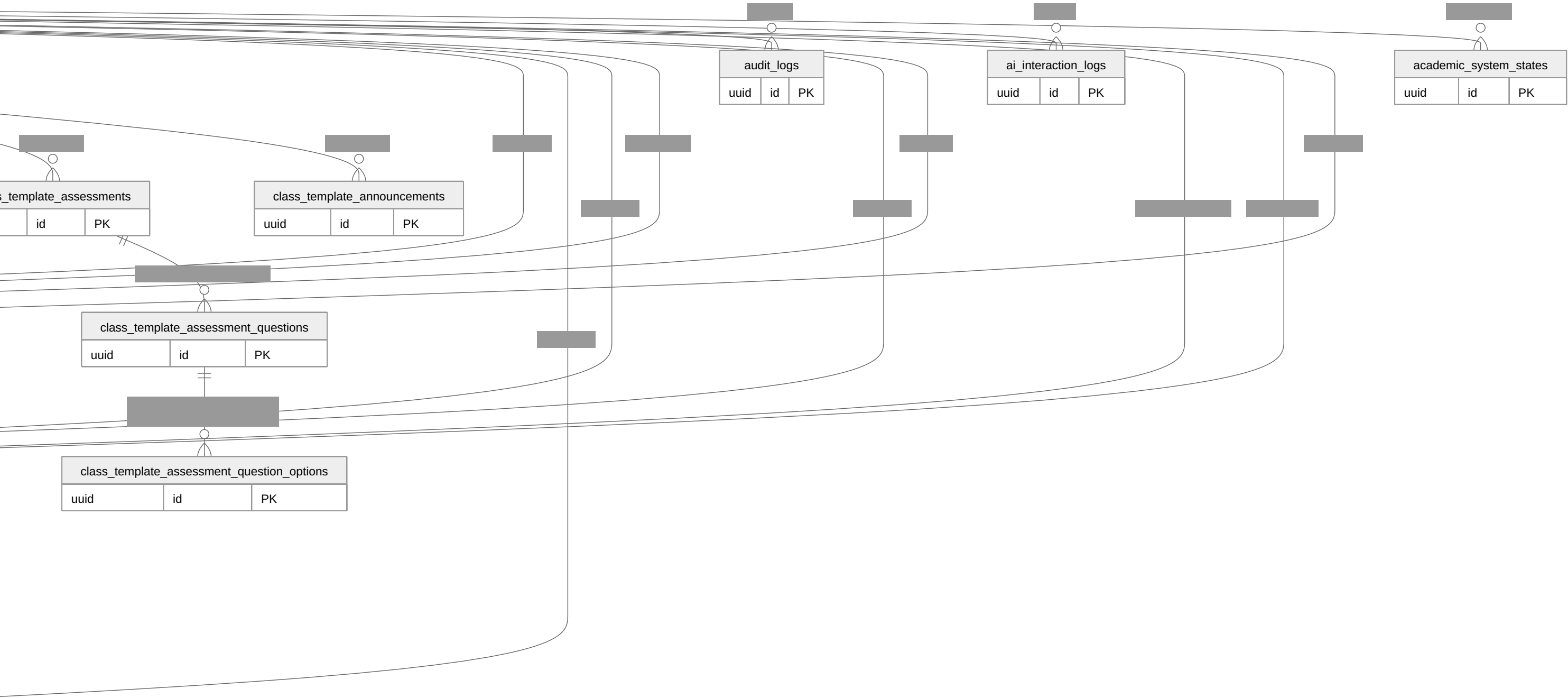












Domain ownership map

Schema source	Tables	Primary ownership
backend/src/drizzle/schema/academic-state.schema.ts	academic_system_states	Academic-State persistence
backend/src/drizzle/schema/ai-mentor.schema.ts	ai_interaction_logs, extracted_modules	Ai-Mentor persistence
backend/src/drizzle/schema/announcements-notifications.schema.ts	announcements, notifications	Announcements-Notifications persistence
backend/src/drizzle/schema/app-version.schema.ts	app_versions	App-Version persistence
backend/src/drizzle/schema/base.schema.ts	roles, users, user_roles, audit_logs, sections, classes, class_schedules, student_profiles, teacher_profiles, enrollments, lessons, lesson_content_blocks, lesson_completions, lesson_versions, assessments, assessment_questions, assessment_question_options, assessment_attempts, assessment_responses, archived_users, class_modules, module_sections, module_items, module_grading_scale_entries, uploaded_files, class_visibility_preferences, student_class_presentation_preferences, student_course_view_preferences, section_visibility_preferences, library_folders, pending_roster	Base persistence
backend/src/drizzle/schema/class-record.schema.ts	class_records, class_record_categories, class_record_items, class_record_scores, class_record_final_grades	Class-Record persistence
backend/src/drizzle/schema/class-templates.schema.ts	class_templates, class_template_modules, class_template_module_sections, class_template_assessments, class_template_lessons, class_template_lesson_blocks, class_template_assessment_questions, class_template_assessment_question_options, class_template_engine_chunks, class_template_module_items, class_template_announcements	Class-Templates persistence
backend/src/drizzle/schema/discussion-board.schema.ts	discussion_threads, discussion_thread_attachments, discussion_comments, discussion_comment_attachments, discussion_comment_reactions	Discussion-Board persistence
backend/src/drizzle/schema/ja.schema.ts	ja_sessions, ja_session_items, ja_session_responses, ja_session_events, ja_progress, ja_xp_ledger, ja_threads, ja_thread_messages, ja_guardrail_events	Ja persistence
backend/src/drizzle/schema/lxp.schema.ts	intervention_cases, intervention_assignments, lxp_generated_remedial_lessons, lxp_generated_guided_assessments, lxp_generated_guided_assessment_attempts, lxp_progress, system_evaluations, system_evaluation_campaigns, system_evaluation_assignments, class_ai_policies, teacher_evaluation_windows, teacher_evaluation_submissions	Lxp persistence
backend/src/drizzle/schema/otp.schema.ts	otp_verifications	Otp persistence
backend/src/drizzle/schema/performance.schema.ts	performance_snapshots, performance_logs	Performance persistence
backend/src/drizzle/schema/rag.schema.ts	content_chunks, content_chunk_embeddings, student_concept_mastery, ai_generation_jobs, ai_generation_outputs	Rag persistence
backend/src/drizzle/schema/refresh-tokens.schema.ts	refresh_tokens	Refresh-Tokens persistence
backend/src/drizzle/schema/school-events.schema.ts	school_events	School-Events persistence

PostgreSQL enum dictionary

Exhaustive inventory rule. The 55 active enum declarations below were extracted from `backend/src/drizzle/schema/*.ts` at commit `3d0c93e`. A later source change requires regenerating or manually reconciling this chapter.

Database enum	TypeScript symbol	Allowed values	Source
ai_session_type	aiSessionTypeEnum	module_extraction, mentor_chat, admin_analytics_chat, mistake_explanation	backend/src/drizzle/schema/ai-mentor.schema.ts
extraction_status	extractionStatusEnum	pending, processing, completed, failed, applied	backend/src/drizzle/schema/ai-mentor.schema.ts
notification_type	notificationTypeEnum	announcement_posted, discussion_thread_posted, discussion_comment_posted, assessment_assigned, grade_updated, assessment_due, assessment_graded	backend/src/drizzle/schema/announcements-notifications.schema.ts
account_status	accountStatusEnum	ACTIVE, PENDING, SUSPENDED, DELETED	backend/src/drizzle/schema/base.schema.ts
content_type	contentTypeEnum	video, document, quiz, link	backend/src/drizzle/schema/base.schema.ts
assessment_type	assessmentTypeEnum	quiz, exam, assignment, file_upload	backend/src/drizzle/schema/base.schema.ts
rubric_parse_status	rubricParseStatusEnum	pending, parsed, reviewed, failed	backend/src/drizzle/schema/base.schema.ts
class_record_category	classRecordCategoryEnum	written_work, performance_task, quarterly_assessment	backend/src/drizzle/schema/base.schema.ts
grading_period	gradingPeriodEnum	Q1, Q2, Q3, Q4	backend/src/drizzle/schema/base.schema.ts
enrollment_status	enrollmentStatusEnum	enrolled, dropped, completed	backend/src/drizzle/schema/base.schema.ts
grade_level	gradeLevelEnum	7, 8, 9, 10	backend/src/drizzle/schema/base.schema.ts
lesson_content_type	lessonContentTypeEnum	text, image, video, question, file, divider	backend/src/drizzle/schema/base.schema.ts
lesson_version_type	lessonVersionTypeEnum	auto, manual, restore	backend/src/drizzle/schema/base.schema.ts
question_type	questionTypeEnum	multiple_choice, multiple_select, true_false, short_answer, fill_blank, dropdown	backend/src/drizzle/schema/base.schema.ts
feedback_level	feedbackLevelEnum	immediate, standard, detailed	backend/src/drizzle/schema/base.schema.ts
file_scope	fileScopeEnum	private, general	backend/src/drizzle/schema/base.schema.ts
library_subject_key	librarySubjectKeyEnum	math, science, english, filipino, apte, mapeh, esp	backend/src/drizzle/schema/base.schema.ts
library_index_status	libraryIndexStatusEnum	not_indexed, pending, processing, completed, failed	backend/src/drizzle/schema/base.schema.ts
library_file_kind	libraryFileKindEnum	pdf, txt, pptx, image	backend/src/drizzle/schema/base.schema.ts
module_item_type	moduleItemTypeEnum	lesson, assessment, file	backend/src/drizzle/schema/base.schema.ts
student_presentation_mode	studentPresentationModeEnum	solid, gradient, preset	backend/src/drizzle/schema/base.schema.ts
student_course_view_mode	studentCourseViewModeEnum	card, wide	backend/src/drizzle/schema/base.schema.ts
class_record_status	classRecordStatusEnum	draft, finalized, locked	backend/src/drizzle/schema/class-record.schema.ts
class_record_remarks	classRecordRemarksEnum	Passed, For Intervention	backend/src/drizzle/schema/class-record.schema.ts
class_template_status	classTemplateStatusEnum	draft, published	

Database enum	TypeScript symbol	Allowed values	Source
			backend/src/drizzle/schema/class-templates.schema.ts
class_template_item_type	classTemplateItemTypeEnum	assessment, lesson, file	backend/src/drizzle/schema/class-templates.schema.ts
discussion_thread_status	discussionThreadStatusEnum	draft, published, closed, archived	backend/src/drizzle/schema/discussion-board.schema.ts
discussion_reaction_type	discussionReactionTypeEnum	like, heart, wow	backend/src/drizzle/schema/discussion-board.schema.ts
discussion_attachment_type	discussionAttachmentTypeEnum	image, pdf, link	backend/src/drizzle/schema/discussion-board.schema.ts
ja_session_mode	jaSessionModeEnum	practice, review	backend/src/drizzle/schema/ja.schema.ts
ja_session_status	jaSessionStatusEnum	active, completed, deleted	backend/src/drizzle/schema/ja.schema.ts
ja_reward_state	jaRewardStateEnum	pending, awarded	backend/src/drizzle/schema/ja.schema.ts
ja_session_event_type	jaSessionEventTypeEnum	focus_lost, focus_restored, focus_strike, resumed, completed, deleted	backend/src/drizzle/schema/ja.schema.ts
ja_xp_event_type	jaXpEventTypeEnum	session_completion	backend/src/drizzle/schema/ja.schema.ts
ja_thread_status	jaThreadStatusEnum	active, archived	backend/src/drizzle/schema/ja.schema.ts
ja_thread_message_role	jaThreadMessageRoleEnum	student, assistant, system	backend/src/drizzle/schema/ja.schema.ts
ja_guardrail_event_type	jaGuardrailEventTypeEnum	blocked_prompt	backend/src/drizzle/schema/ja.schema.ts
intervention_case_status	interventionCaseStatusEnum	pending, active, completed, dismissed	backend/src/drizzle/schema/lxp.schema.ts
lxp_assignment_type	lxpAssignmentTypeEnum	lesson_review, assessment_retry, generated_lesson_review, guided_assessment	backend/src/drizzle/schema/lxp.schema.ts
lxp_generated_artifact_status	lxpGeneratedArtifactStatusEnum	draft, approved, rejected	backend/src/drizzle/schema/lxp.schema.ts
lxp_guided_attempt_status	lxpGuidedAttemptStatusEnum	in_progress, submitted	backend/src/drizzle/schema/lxp.schema.ts
system_evaluation_target	systemEvaluationTargetEnum	lms, lxp, ai_mentor, intervention, overall	backend/src/drizzle/schema/lxp.schema.ts
system_evaluation_form_type	systemEvaluationFormTypeEnum	system, ja_hub	backend/src/drizzle/schema/lxp.schema.ts
system_evaluation_audience_role	systemEvaluationAudienceRoleEnum	student, teacher	backend/src/drizzle/schema/lxp.schema.ts
system_evaluation_campaign_status	systemEvaluationCampaignStatusEnum	draft, active, closed	backend/src/drizzle/schema/lxp.schema.ts
system_evaluation_assignment_status	systemEvaluationAssignmentStatusEnum	pending, submitted, expired	backend/src/drizzle/schema/lxp.schema.ts
teacher_evaluation_type	teacherEvaluationTypeEnum	teacher_class, ja_hub, learners_path	backend/src/drizzle/schema/lxp.schema.ts
teacher_evaluation_window_status	teacherEvaluationWindowStatusEnum	active, closed	backend/src/drizzle/schema/lxp.schema.ts
ai_policy_source_scope	aiPolicySourceScopeEnum	recommended_only, class_materials	backend/src/drizzle/schema/lxp.schema.ts
otp_purpose	otpPurposeEnum	email_verification, password_reset, login_2fa	backend/src/drizzle/schema/otp.schema.ts
content_source_type	contentSourceTypeEnum	lesson_block, extracted_module, assessment_question, library_file	backend/src/drizzle/schema/rag.schema.ts
ai_generation_job_type	aiGenerationJobTypeEnum	quiz_generation, remedial_plan_generation, performance_diagnostics, class_lesson_plan_generation, reindexing, backfill	backend/src/drizzle/schema/rag.schema.ts
ai_generation_output_type	aiGenerationOutputTypeEnum	assessment_draft, intervention_recommendation, performance_diagnostic, class_lesson_plan	backend/src/drizzle/schema/rag.schema.ts

Database enum	TypeScript symbol	Allowed values	Source
ai_generation_status	aiGenerationStatusEnum	pending, processing, completed, ap-proved, cancelled, rejected, failed	backend/src/drizzle/schema/rag.schema.ts
school_event_type	schoolEventTypeEnum	school_event, holiday_break	backend/src/drizzle/schema/school-events.schema.ts

Vector and retrieval storage

Concern	Current implementation	Maintenance consequence
Extension	Migration 0003 runs CREATE EXTENSION IF NOT EXISTS vector	Do not remove the migration while the baseline contains vector(768).
Vector table	content_chunk_embeddings	Embedding lifecycle is separated from chunk text and cascades with chunk deletion.
Vector column	embedding vector(768) NOT NULL	Embedding providers must return exactly 768 numeric values.
Model identity	embedding_model text NOT NULL	A model change must update stored vectors or filter by model during mixed migration.
Timestamp	embedded_at defaults to the current time	Use it to identify stale or partial reindex runs.
ANN index	No HNSW or IVFFlat index is declared in active schema or migrations	Current similarity search has no declared approx-imate-nearest-neighbor index; add one only with measured query evidence and a migration.
B-tree index	content_chunk_embeddings_model_idx on embed-ding_model	Supports model filtering but does not accelerate vector distance ordering.

Relationship and cascade conventions

- Child content that has no independent meaning usually declares `onDelete: 'cascade'`. Examples include user-role assignments, lesson blocks, question options, and chunk embeddings.
- Historical or accountability references may declare `onDelete: 'set null'` so the event survives identity deletion.
- A nullable column is not proof of SET NULL behavior. The foreign-key callback must declare it.
- Junction uniqueness is normally implemented by a unique index or composite primary key. Read the exact table-level index list before creating duplicate-cleanup logic.
- Derived totals and risk flags must remain recomputable. Do not introduce a second authoritative copy in a new column.

Complete entity encyclopedia

Source file: backend/src/drizzle/schema/academic-state.schema.ts

This source file declares 1 active table.

academic_system_states

Purpose. Current school-year and quarter state used for controlled academic transitions.

Drizzle declaration. `academicSystemStates` in `backend/src/drizzle/schema/academic-state.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKe-y().defaultRandom()

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
schoolYear	school_year	text	NOT NULL	No default	No PK, FK, or unique marker	text('school_year').notNull()
quarter	quarter	gradingPeriodEnum	NOT NULL	default('Q1')	No PK, FK, or unique marker	gradingPeriodEnum('quarter').notNull().default('Q1')
updatedAt	updated_by	uuid	NULL allowed	No default	FK → users.id (onDelete: 'set null',)	uuid('updated_by').references(() => users.id, { onDelete: 'set null', })
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `schoolYearIdx: index('academic_system_states_school_year_idx').on( table.schoolYear, )`
- `quarterIdx: index('academic_system_states_quarter_idx').on(table.quarter)`
- `updatedAtIdx: index('academic_system_states_updated_at_idx').on( table.updatedAt, )`

Relationship map.

- `updated_by: FK → users.id (onDelete: 'set null',)`

Delete and lifecycle behavior.

- No outgoing foreign key declares cascading deletion.
- Parent deletes set these references to NULL: `updated_by: FK → users.id (onDelete: 'set null',)`.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

Source file: backend/src/drizzle/schema/ai-mentor.schema.ts

This source file declares 2 active tables.

ai_interaction_logs

Purpose. AI conversation and generation interaction logs separated from official academic records.

Drizzle declaration. `aiInteractionLogs` in `backend/src/drizzle/schema/ai-mentor.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
userId	user_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('user_id').notNull().references(() => users.id, { onDelete: 'cascade' })
sessionType	session_type	aiSessionTypeEnum	NOT NULL	No default	No PK, FK, or unique marker	aiSessionTypeEnum('session_type').notNull()
inputText	input_text	text	NOT NULL	No default	No PK, FK, or unique marker	text('input_text').notNull()
outputText	output_text	text	NOT NULL	No default		

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
					No PK, FK, or unique marker	text('output_text').notNull()
modelUsed	model_used	text	NOT NULL	No default	No PK, FK, or unique marker	text('model_used').notNull()
contextMetadata	context_metadata	json	NULL allowed	No default	No PK, FK, or unique marker	json('context_metadata')
responseTimeMs	response_time_ms	integer	NULL allowed	No default	No PK, FK, or unique marker	integer('response_time_ms')
sessionId	session_id	uuid	NULL allowed	No default	No PK, FK, or unique marker	uuid('session_id')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()

Indexes and table constraints.

- `userIdIdx: index('ai_interaction_logs_user_id_idx').on(table.userId)`
- `sessionTypeIdx: index('ai_interaction_logs_session_type_idx').on( table.sessionType, )`
- `createdAtIdx: index('ai_interaction_logs_created_at_idx').on( table.createdAt, )`
- `sessionIdIdx: index('ai_interaction_logs_session_id_idx').on( table.sessionId, )`

Relationship map.

- `user_id`: FK → `users.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `user_id`: FK → `users.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

extracted_modules

Purpose. Durable PDF or source extraction jobs, review payloads, progress, error, and apply state.

Drizzle declaration. `extractedModules` in `backend/src/drizzle/schema/ai-mentor.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
fileId	file_id	uuid	NOT NULL	No default	FK → uploaded-Files.id (onDelete: 'cascade')	uuid('file_id').notNull().references(() => uploaded-Files.id, { onDelete: 'cascade' })
classId	class_id	uuid	NOT NULL	No default	FK → classes.id (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
teacherId	teacher_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('teacher_id').notNull().references(() => users.id, { onDelete: 'cascade' })
rawText	raw_text	text	NOT NULL	No default	No PK, FK, or unique marker	text('raw_text').notNull()
structuredContent	structured_content	json	NULL allowed	No default	No PK, FK, or unique marker	json('structured_content')

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
extractionStatus	extraction_status	extraction-StatusEnum	NOT NULL	default('pending')	No PK, FK, or unique marker	extraction-StatusEnum('extraction_status').notNull().default('pending')
errorMessage	error_message	text	NULL allowed	No default	No PK, FK, or unique marker	text('error_message')
modelUsed	model_used	text	NULL allowed	No default	No PK, FK, or unique marker	text('model_used')
isApplied	is_applied	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('is_applied').notNull().default(false)
progressPercent	progress_percent	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('progress_percent').notNull().default(0)
totalChunks	total_chunks	integer	NULL allowed	No default	No PK, FK, or unique marker	integer('total_chunks')
processedChunks	processed_chunks	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('processed_chunks').notNull().default(0)
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `fileIdx: index('extracted_modules_file_id_idx').on(table.fileId)`
- `classIdx: index('extracted_modules_class_id_idx').on(table.classId)`
- `teacherIdx: index('extracted_modules_teacher_id_idx').on(table.teacherId)`
- `statusIdx: index('extracted_modules_status_idx').on(table.extractionStatus)`

Relationship map.

- `file_id`: FK → `uploadedFiles.id` (onDelete: 'cascade')
- `class_id`: FK → `classes.id` (onDelete: 'cascade')
- `teacher_id`: FK → `users.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `file_id`: FK → `uploadedFiles.id` (onDelete: 'cascade'); `class_id`: FK → `classes.id` (onDelete: 'cascade'); `teacher_id`: FK → `users.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

Source file: backend/src/drizzle/schema/announcements-notifications.schema.ts

This source file declares 2 active tables.

announcements

Purpose. Scheduled or published class announcements and author ownership.

Drizzle declaration. `announcements` in `backend/src/drizzle/schema/announcements-notifications.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
classId	class_id	uuid	NOT NULL	No default	FK → classes.id (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
authorId	author_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('author_id').notNull().references(() => users.id, { onDelete: 'cascade' })
title	title	varchar	NOT NULL	No default	No PK, FK, or unique marker	varchar('title', { length: 255 }).notNull()
content	content	text	NOT NULL	No default	No PK, FK, or unique marker	text('content').notNull()
isPinned	is_pinned	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('is_pinned').notNull().default(false)
isVisible	is_visible	boolean	NOT NULL	default(true)	No PK, FK, or unique marker	boolean('is_visible').notNull().default(true)
isCoreTemplateAsset	is_core_template_asset	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('is_core_template_asset').notNull().default(false)
templateId	template_id	uuid	NULL allowed	No default	No PK, FK, or unique marker	uuid('template_id')
templateSourceId	template_source_id	uuid	NULL allowed	No default	No PK, FK, or unique marker	uuid('template_source_id')
scheduledAt	scheduled_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('scheduled_at')
publishedAt	published_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('published_at')
archivedAt	archived_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('archived_at')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `classIdx: index('announcements_class_id_idx').on(table.classId)`
- `authorIdx: index('announcements_author_id_idx').on(table.authorId)`
- `classPublishedIdx: index('announcements_class_published_idx').on( table.classId, table.publishedAt, )`

Relationship map.

- `class_id`: FK → `classes.id` (onDelete: 'cascade')
- `author_id`: FK → `users.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `class_id`: FK → `classes.id` (onDelete: 'cascade'); `author_id`: FK → `users.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

notifications

Purpose. Deduplicated per-user notification inbox records.

Drizzle declaration. `notifications` in `backend/src/drizzle/schema/announcements-notifications.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
userId	user_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('user_id').notNull().references(() => users.id, { onDelete: 'cascade' })
type	type	notification-TypeEnum	NOT NULL	No default	No PK, FK, or unique marker	notificationTypeEnum('type').notNull()
referenceId	reference_id	uuid	NULL allowed	No default	No PK, FK, or unique marker	uuid('reference_id')
title	title	varchar	NOT NULL	No default	No PK, FK, or unique marker	varchar('title', { length: 255 }).notNull()
body	body	text	NOT NULL	No default	No PK, FK, or unique marker	text('body').notNull()
isRead	is_read	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('is_read').notNull().default(false)
readAt	read_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('read_at')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()

Indexes and table constraints.

- `userUnreadIdx: index('notifications_user_unread_idx').on( table.userId, table.isRead, )`
- `userCreatedIdx: index('notifications_user_created_idx').on( table.userId, table.createdAt, )`
- `userTypeReferenceUniqueIdx: uniqueIndex( 'notifications_user_type_reference_unique_idx', ) .on(table.userId, table.type, table.referenceId) .where(sqlreference_id IS NOT NULL)`

Relationship map.

- `user_id`: FK → `users.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `user_id`: FK → `users.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

Source file: backend/src/drizzle/schema/app-version.schema.ts

This source file declares 1 active table.

app_versions

Purpose. Mobile platform version policy, minimum version, store links, and rollout messaging.

Drizzle declaration. `appVersions` in `backend/src/drizzle/schema/app-version.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
platform	platform	text	NOT NULL	default('android')	No PK, FK, or unique marker	text('platform').notNull().default('android')
versionCode	version_code	integer	NOT NULL	No default	No PK, FK, or unique marker	integer('version_code').notNull()
minSupportedVersionCode	min_supported_version_code	integer	NOT NULL	No default	No PK, FK, or unique marker	integer('min_supported_version_code').notNull()
nativeVersion	native_version	text	NOT NULL	No default	No PK, FK, or unique marker	text('native_version').notNull()
otaRuntimeVersion	ota_runtime_version	text	NOT NULL	No default	No PK, FK, or unique marker	text('ota_runtime_version').notNull()
apkDownloadUrl	apk_download_url	text	NOT NULL	No default	No PK, FK, or unique marker	text('apk_download_url').notNull()
apkSha256	apk_sha256	text	NULL allowed	No default	No PK, FK, or unique marker	text('apk_sha256')
apkSizeBytes	apk_size_bytes	integer	NULL allowed	No default	No PK, FK, or unique marker	integer('apk_size_bytes')
isForceUpdate	is_force_update	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('is_force_update').notNull().default(false)
requiresFullApk	requires_full_apk	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('requires_full_apk').notNull().default(false)
releaseNotes	release_notes	text	NULL allowed	No default	No PK, FK, or unique marker	text('release_notes')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `platformIdx: index('app_versions_platform_idx').on(table.platform)`
- `versionCodeIdx: index('app_versions_version_code_idx').on( table.versionCode, )`

Relationship map.

- No outgoing foreign key is declared on this table.

Delete and lifecycle behavior.

- No outgoing foreign key declares cascading deletion.
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

Source file: `backend/src/drizzle/schema/base.schema.ts`

This source file declares 31 active tables.

roles

Purpose. Canonical role records used by the user-to-role junction and RBAC checks.

Drizzle declaration. `roles` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
name	name	text	NOT NULL	No default	UNIQUE	text('name').notNull().unique()
description	description	text	NULL allowed	No default	No PK, FK, or unique marker	text('description')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()

Indexes and table constraints.

- No table-level index callback is declared in the active Drizzle source.

Relationship map.

- No outgoing foreign key is declared on this table.

Delete and lifecycle behavior.

- No outgoing foreign key declares cascading deletion.
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

users

Purpose. Application identities, credentials, account status, profile-completion state, and account lifecycle timestamps.

Drizzle declaration. `users` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
email	email	text	NOT NULL	No default	UNIQUE	text('email').notNull().unique()
password	password	text	NOT NULL	No default	No PK, FK, or unique marker	text('password').notNull()
firstName	first_name	text	NOT NULL	No default	No PK, FK, or unique marker	text('first_name').notNull()
middleName	middle_name	text	NULL allowed	No default	No PK, FK, or unique marker	text('middle_name')
lastName	last_name	text	NOT NULL	No default	No PK, FK, or unique marker	text('last_name').notNull()
status	account_status	accountStatusEnum	NOT NULL	default('ACTIVE')	No PK, FK, or unique marker	account-StatusEnum('account_status').notNull().default('ACTIVE')
isEmailVerified	is_email_verified	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('is_email_verified').notNull().default(false)
lastLoginAt	last_login_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('last_login_at')
createdAt	created_at	timestamp	NOT NULL	defaultNow()		

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
					No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `emailIdx: index('users_email_idx').on(table.email)`
- `statusIdx: index('users_status_idx').on(table.status)`

Relationship map.

- No outgoing foreign key is declared on this table.

Delete and lifecycle behavior.

- No outgoing foreign key declares cascading deletion.
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

user_roles

Purpose. Many-to-many assignments connecting users to canonical roles.

Drizzle declaration. `userRoles` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
userId	user_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('user_id').notNull().references(() => users.id, { onDelete: 'cascade' })
roleId	role_id	uuid	NOT NULL	No default	FK → roles.id (onDelete: 'cascade')	uuid('role_id').notNull().references(() => roles.id, { onDelete: 'cascade' })
assignedAt	assigned_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('assigned_at').notNull().defaultNow()
assignedBy	assigned_by	text	NOT NULL	No default	No PK, FK, or unique marker	text('assigned_by').notNull()

Indexes and table constraints.

- `pk: primaryKey({ columns: [table.userId, table.roleId] })`
- `userIdIdx: index('user_roles_user_id_idx').on(table.userId)`
- `roleIdIdx: index('user_roles_role_id_idx').on(table.roleId)`

Relationship map.

- `user_id`: FK → users.id (onDelete: 'cascade')
- `role_id`: FK → roles.id (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `user_id`: FK → users.id (onDelete: 'cascade'); `role_id`: FK → roles.id (onDelete: 'cascade').

- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

audit_logs

Purpose. Append-only security and academic mutation evidence with actor, action, target, metadata, and time.

Drizzle declaration. `auditLogs` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
actorId	actor_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('actor_id').notNull().references(() => users.id, { onDelete: 'cascade' })
action	action	text	NOT NULL	No default	No PK, FK, or unique marker	text('action').notNull()
targetType	target_type	text	NOT NULL	No default	No PK, FK, or unique marker	text('target_type').notNull()
targetId	target_id	uuid	NOT NULL	No default	No PK, FK, or unique marker	uuid('target_id').notNull()
metadata	metadata	json	NULL allowed	No default	No PK, FK, or unique marker	json('metadata')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()

Indexes and table constraints.

- `actorIdIdx: index('audit_logs_actor_id_idx').on(table.actorId)`
- `actionIdx: index('audit_logs_action_idx').on(table.action)`
- `createdAtIdx: index('audit_logs_created_at_idx').on(table.createdAt)`

Relationship map.

- actor_id: FK → users.id (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: actor_id: FK → users.id (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

sections

Purpose. School section master data, adviser ownership, grade level, school year, schedule presentation, and lifecycle state.

Drizzle declaration. `sections` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
name	name	text	NOT NULL	No default	No PK, FK, or unique marker	text('name').notNull()

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
gradeLevel	grade_level	text	NOT NULL	No default	No PK, FK, or unique marker	text('grade_level').notNull()
schoolYear	school_year	text	NOT NULL	No default	No PK, FK, or unique marker	text('school_year').notNull()
capacity	capacity	integer	NOT NULL	default(40)	No PK, FK, or unique marker	integer('capacity').notNull().default(40)
roomNumber	room_number	text	NULL allowed	No default	No PK, FK, or unique marker	text('room_number')
cardBannerUrl	card_banner_url	text	NULL allowed	No default	No PK, FK, or unique marker	text('card_banner_url')
adviserId	adviser_id	uuid	NULL allowed	No default	FK → users.id (onDelete: 'set null',)	uuid('adviser_id').references(() => users.id, { onDelete: 'set null', })
isActive	is_active	boolean	NOT NULL	default(true)	No PK, FK, or unique marker	boolean('is_active').notNull().default(true)
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `adviserIdx: index('sections_adviser_idx').on(table.adviserId)`
- `gradeLevelIdx: index('sections_grade_level_idx').on(table.gradeLevel)`
- `schoolYearIdx: index('sections_school_year_idx').on(table.schoolYear)`
- `uniqueSection: unique().on(table.name, table.gradeLevel, table.schoolYear)`

Relationship map.

- `adviser_id`: FK → `users.id` (onDelete: 'set null',)

Delete and lifecycle behavior.

- No outgoing foreign key declares cascading deletion.
- Parent deletes set these references to NULL: `adviser_id`: FK → `users.id` (onDelete: 'set null',).
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

classes

Purpose. Subject-class offerings connected to teachers and sections, including presentation and school-year settings.

Drizzle declaration. `classes` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
subjectName	subject_name	text	NOT NULL	No default	No PK, FK, or unique marker	text('subject_name').notNull()
subjectCode	subject_code	text	NOT NULL	No default	No PK, FK, or unique marker	text('subject_code').notNull()
subjectGradeLevel	subject_grade_level	gradeLevelEnum	NULL allowed	No default	No PK, FK, or unique marker	

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
						gradeLevelEnum('subject_grade_level')
sectionId	section_id	uuid	NOT NULL	No default	FK → sections.id (onDelete: 'cascade')	uuid('section_id').notNull().references(() => sections.id, { onDelete: 'cascade' })
teacherId	teacher_id	uuid	NULL allowed	No default	FK → users.id (onDelete: 'set null',)	uuid('teacher_id').references(() => users.id, { onDelete: 'set null', })
room	room	text	NULL allowed	No default	No PK, FK, or unique marker	text('room')
cardPreset	card_preset	text	NOT NULL	default('aurora')	No PK, FK, or unique marker	text('card_preset').notNull().default('aurora')
cardBannerUrl	card_banner_url	text	NULL allowed	No default	No PK, FK, or unique marker	text('card_banner_url')
schoolYear	school_year	text	NOT NULL	No default	No PK, FK, or unique marker	text('school_year').notNull()
writtenWorkGradingWeight	written_work_grading_weight	integer	NOT NULL	default(30)	No PK, FK, or unique marker	integer('written_work_grading_weight').notNull().default(30)
performanceTaskGradingWeight	performance_task_grading_weight	integer	NOT NULL	default(50)	No PK, FK, or unique marker	integer('performance_task_grading_weight').notNull().default(50)
quarterlyAssessmentGradingWeight	quarterly_assessment_grading_weight	integer	NOT NULL	default(20)	No PK, FK, or unique marker	integer('quarterly_assessment_grading_weight').notNull().default(20)
isActive	is_active	boolean	NOT NULL	default(true)	No PK, FK, or unique marker	boolean('is_active').notNull().default(true)
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `teacherIdx: index('classes_teacher_idx').on(table.teacherId)`
- `sectionIdx: index('classes_section_idx').on(table.sectionId)`
- `subjectCodeIdx: index('classes_subject_code_idx').on(table.subjectCode)`
- `subjectNameIdx: index('classes_subject_name_idx').on(table.subjectName)`
- `schoolYearIdx: index('classes_school_year_idx').on(table.schoolYear)`
- `uniqueClass: unique().on( table.subjectCode, table.sectionId, table.schoolYear, )`

Relationship map.

- `section_id`: FK → `sections.id` (onDelete: 'cascade')
- `teacher_id`: FK → `users.id` (onDelete: 'set null',)

Delete and lifecycle behavior.

- Cascading parent deletes affect: `section_id`: FK → `sections.id` (onDelete: 'cascade').

- Parent deletes set these references to NULL: teacher_id: FK → users.id (onDelete: 'set null',).
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

class_schedules

Purpose. Normalized day, start-time, end-time, and room slots for a class.

Drizzle declaration. `classSchedules` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
classId	class_id	uuid	NOT NULL	No default	FK → classes.id (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
days	days	text	NOT NULL	No default	No PK, FK, or unique marker	text('days').array().notNull()
startTime	start_time	text	NOT NULL	No default	No PK, FK, or unique marker	text('start_time').notNull()
endTime	end_time	text	NOT NULL	No default	No PK, FK, or unique marker	text('end_time').notNull()
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `classIdIdx: index('class_schedules_class_id_idx').on(table.classId)`

Relationship map.

- class_id: FK → classes.id (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: class_id: FK → classes.id (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

student_profiles

Purpose. Role-specific profile data for student profiles.

Drizzle declaration. `studentProfiles` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
userId	user_id	uuid	NOT NULL	No default	PRIMARY KEY; FK → users.id (onDelete: 'cascade')	uuid('user_id').primaryKey().references(() => users.id, { onDelete: 'cascade' })
dateOfBirth	date_of_birth	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('date_of_birth')
profilePicture	profile_picture	text	NULL allowed	No default	No PK, FK, or unique marker	text('profile_picture')

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
gender	gender	text	NULL allowed	No default	No PK, FK, or unique marker	text('gender')
phone	phone	text	NULL allowed	No default	No PK, FK, or unique marker	text('phone')
address	address	text	NULL allowed	No default	No PK, FK, or unique marker	text('address')
familyName	family_name	text	NULL allowed	No default	No PK, FK, or unique marker	text('family_name')
familyRelationship	family_relationship	text	NULL allowed	No default	No PK, FK, or unique marker	text('family_relationship')
familyContact	family_contact	text	NULL allowed	No default	No PK, FK, or unique marker	text('family_contact')
gradeLevel	grade_level	gradeLevelEnum	NULL allowed	No default	No PK, FK, or unique marker	gradeLevelEnum('grade_level')
lrn	lrn	varchar	NULL allowed	No default	No PK, FK, or unique marker	varchar('lrn', { length: 12 })
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `userIdx: index('student_profiles_user_id_idx').on(table.userId)`
- `gradeLevelIdx: index('student_profiles_grade_level_idx').on( table.gradeLevel, )`
- `lrnIdx: unique('student_profiles_lrn_unique').on(table.lrn)`

Relationship map.

- `user_id`: FK → `users.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `user_id`: FK → `users.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

teacher_profiles

Purpose. Role-specific profile data for teacher profiles.

Drizzle declaration. `teacherProfiles` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
userId	user_id	uuid	NOT NULL	No default	PRIMARY KEY; FK → <code>users.id</code> (onDelete: 'cascade')	<code>uuid('user_id').primaryKey().references(() => users.id, { onDelete: 'cascade' })</code>
department	department	text	NULL allowed	No default	No PK, FK, or unique marker	text('department')
specialization	specialization	text	NULL allowed	No default	No PK, FK, or unique marker	text('specialization')
profilePicture	profile_picture	text	NULL allowed	No default	No PK, FK, or unique marker	text('profile_picture')
contactNumber	contact_number	text	NULL allowed	No default	No PK, FK, or unique marker	text('contact_number')

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
dateOfBirth	date_of_birth	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('date_of_birth')
gender	gender	text	NULL allowed	No default	No PK, FK, or unique marker	text('gender')
address	address	text	NULL allowed	No default	No PK, FK, or unique marker	text('address')
employeeId	employee_id	varchar	NULL allowed	No default	No PK, FK, or unique marker	varchar('employee_id', { length: 20 })
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `userIdx: index('teacher_profiles_user_id_idx').on(table.userId)`
- `departmentIdx: index('teacher_profiles_department_idx').on( table.department, )`
- `employeeIdIdx: index('teacher_profiles_employee_id_idx').on( table.employeeId, )`

Relationship map.

- `user_id`: FK → `users.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `user_id`: FK → `users.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

enrollments

Purpose. Student membership in classes and sections with enrollment lifecycle status.

Drizzle declaration. `enrollments` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
studentId	student_id	uuid	NOT NULL	No default	FK → <code>users.id</code> (onDelete: 'cascade')	uuid('student_id').notNull().references(() => users.id, { onDelete: 'cascade' })
classId	class_id	uuid	NULL allowed	No default	FK → <code>classes.id</code> (onDelete: 'cascade',)	uuid('class_id').references(() => classes.id, { onDelete: 'cascade', })
sectionId	section_id	uuid	NOT NULL	No default	FK → <code>sections.id</code> (onDelete: 'cascade')	uuid('section_id').notNull().references(() => sections.id, { onDelete: 'cascade' })
status	status	enrollment-StatusEnum	NOT NULL	default('enrolled')	No PK, FK, or unique marker	enrollment-StatusEnum('status').notNull().default('enrolled')
enrolledAt	enrolled_at	timestamp	NOT NULL	defaultNow()		

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
					No PK, FK, or unique marker	timestamp('enrolled_at').notNull().defaultNow()
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()

Indexes and table constraints.

- `studentIdx: index('enrollments_student_idx').on(table.studentId)`
- `classIdx: index('enrollments_class_idx').on(table.classId)`
- `sectionIdx: index('enrollments_section_idx').on(table.sectionId)`
- `statusIdx: index('enrollments_status_idx').on(table.status)`
- `uniqueEnrollment: unique().on(table.studentId, table.classId)`

Relationship map.

- `student_id: FK → users.id (onDelete: 'cascade')`
- `class_id: FK → classes.id (onDelete: 'cascade',)`
- `section_id: FK → sections.id (onDelete: 'cascade')`

Delete and lifecycle behavior.

- Cascading parent deletes affect: `student_id: FK → users.id (onDelete: 'cascade')`; `class_id: FK → classes.id (onDelete: 'cascade',)`; `section_id: FK → sections.id (onDelete: 'cascade')`.
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

lessons

Purpose. Teacher-authored or generated lessons, ordering, publication state, template lineage, and extraction lineage.

Drizzle declaration. `lessons` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
title	title	text	NOT NULL	No default	No PK, FK, or unique marker	text('title').notNull()
description	description	text	NULL allowed	No default	No PK, FK, or unique marker	text('description')
classId	class_id	uuid	NOT NULL	No default	FK → classes.id (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
order	order	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('order').notNull().default(0)
isDraft	is_draft	boolean	NOT NULL	default(true)	No PK, FK, or unique marker	boolean('is_draft').notNull().default(true)
sourceExtractionId	source_extraction_id	uuid	NULL allowed	No default	No PK, FK, or unique marker	uuid('source_extraction_id')
isCoreTemplateAsset	is_core_template_asset	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('is_core_template_asset').notNull().default(false)
templateId	template_id	uuid	NULL allowed	No default		uuid('template_id')

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
					No PK, FK, or unique marker	
templateSourceId	template_source_id	uuid	NULL allowed	No default	No PK, FK, or unique marker	uuid('template_source_id')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `classIdIdx: index('lessons_class_id_idx').on(table.classId)`
- `classOrderIdx: index('lessons_class_order_idx').on( table.classId, table.order, )`
- `sourceExtractionIdx: index('lessons_source_extraction_idx').on( table.sourceExtractionId, )`
- `templateIdIdx: index('lessons_template_id_idx').on(table.templateId)`
- `templateSourceIdIdx: index('lessons_template_source_id_idx').on( table.templateSourceId, )`

Relationship map.

- `class_id`: FK → `classes.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `class_id`: FK → `classes.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

lesson_content_blocks

Purpose. Ordered typed content blocks that compose a lesson body.

Drizzle declaration. `lessonContentBlocks` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
lessonId	lesson_id	uuid	NOT NULL	No default	FK → <code>lessons.id</code> (onDelete: 'cascade')	uuid('lesson_id').notNull().references(() => lessons.id, { onDelete: 'cascade' })
type	type	lessonContent-TypeEnum	NOT NULL	No default	No PK, FK, or unique marker	lessonContent-TypeEnum('type').notNull()
order	order	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('order').notNull().default(0)
content	content	json	NOT NULL	No default	No PK, FK, or unique marker	json('content').notNull()
metadata	metadata	json	NULL allowed	No default	No PK, FK, or unique marker	json('metadata')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `lessonIdIdx: index('lesson_content_blocks_lesson_id_idx').on( table.lessonId, )`
- `lessonOrderIdx: index('lesson_content_blocks_lesson_order_idx').on( table.lessonId, table.order, )`

Relationship map.

- `lesson_id: FK → lessons.id (onDelete: 'cascade')`

Delete and lifecycle behavior.

- Cascading parent deletes affect: `lesson_id: FK → lessons.id (onDelete: 'cascade')`.
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

lesson_completions

Purpose. Per-student lesson completion evidence.

Drizzle declaration. `lessonCompletions` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
<code>id</code>	<code>id</code>	<code>uuid</code>	NOT NULL	<code>defaultRandom()</code>	PRIMARY KEY	<code>uuid('id').primaryKey().defaultRandom()</code>
<code>studentId</code>	<code>student_id</code>	<code>uuid</code>	NOT NULL	No default	FK → <code>users.id</code> (onDelete: 'cascade')	<code>uuid('student_id').notNull().references(() => users.id, { onDelete: 'cascade' })</code>
<code>lessonId</code>	<code>lesson_id</code>	<code>uuid</code>	NOT NULL	No default	FK → <code>lessons.id</code> (onDelete: 'cascade')	<code>uuid('lesson_id').notNull().references(() => lessons.id, { onDelete: 'cascade' })</code>
<code>completedAt</code>	<code>completed_at</code>	<code>timestamp</code>	NOT NULL	<code>defaultNow()</code>	No PK, FK, or unique marker	<code>timestamp('completed_at').notNull().defaultNow()</code>
<code>progressPercentage</code>	<code>progress_percentage</code>	<code>integer</code>	NOT NULL	<code>default(0)</code>	No PK, FK, or unique marker	<code>integer('progress_percentage').notNull().default(0)</code>

Indexes and table constraints.

- `studentIdIdx: index('lesson_completions_student_id_idx').on( table.studentId, )`
- `lessonIdIdx: index('lesson_completions_lesson_id_idx').on(table.lessonId)`
- `uniqueCompletion: unique('lesson_completions_student_lesson_unique').on( table.studentId, table.lessonId, )`

Relationship map.

- `student_id: FK → users.id (onDelete: 'cascade')`
- `lesson_id: FK → lessons.id (onDelete: 'cascade')`

Delete and lifecycle behavior.

- Cascading parent deletes affect: `student_id: FK → users.id (onDelete: 'cascade')`; `lesson_id: FK → lessons.id (onDelete: 'cascade')`.
- No outgoing foreign key declares SET NULL behavior.

- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

lesson_versions

Purpose. Saved lesson revision payloads used for version history.

Drizzle declaration. `lessonVersions` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
lessonId	lesson_id	uuid	NOT NULL	No default	FK → lessons.id (onDelete: 'cascade')	uuid('lesson_id').notNull().references(() => lessons.id, { onDelete: 'cascade' })
versionNumber	version_number	integer	NOT NULL	No default	No PK, FK, or unique marker	integer('version_number').notNull()
type	type	lessonVersion-TypeEnum	NOT NULL	default('auto')	No PK, FK, or unique marker	lessonVersion-TypeEnum('type').notNull().default('auto')
label	label	text	NULL allowed	No default	No PK, FK, or unique marker	text('label')
snapshot	snapshot	json	NOT NULL	No default	No PK, FK, or unique marker	json('snapshot').notNull()
createdBy	created_by	uuid	NULL allowed	No default	FK → users.id (onDelete: 'set null',)	uuid('created_by').references(() => users.id, { onDelete: 'set null', })
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()

Indexes and table constraints.

- `lessonIdIdx: index('lesson_versions_lesson_id_idx').on(table.lessonId)`
- `lessonVersionUnique: unique('lesson_versions_lesson_version_unique').on( table.lessonId, table.versionNumber, )`

Relationship map.

- lesson_id: FK → lessons.id (onDelete: 'cascade')
- created_by: FK → users.id (onDelete: 'set null',)

Delete and lifecycle behavior.

- Cascading parent deletes affect: lesson_id: FK → lessons.id (onDelete: 'cascade').
- Parent deletes set these references to NULL: created_by: FK → users.id (onDelete: 'set null').
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

assessments

Purpose. Assessment configuration, scheduling, attempt limits, publication, scoring, feedback, and class-record integration.

Drizzle declaration. `assessments` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
title	title	text	NOT NULL	No default	No PK, FK, or unique marker	text('title').notNull()
description	description	text	NULL allowed	No default	No PK, FK, or unique marker	text('description')
classId	class_id	uuid	NOT NULL	No default	FK → classes.id (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
type	type	assessment-TypeEnum	NOT NULL	default('quiz')	No PK, FK, or unique marker	assessment-TypeEnum('type').notNull().default('quiz')
dueDate	due_date	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('due_date')
closeWhenDue	close_when_due	boolean	NOT NULL	default(true)	No PK, FK, or unique marker	boolean('close_when_due').notNull().default(true)
randomizeQuestions	randomize_questions	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('randomize_questions').notNull().default(false)
timedQuestionsEnabled	timed_questions_enabled	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('timed_questions_enabled').notNull().default(false)
questionTimeLimitSeconds	question_time_limit_seconds	integer	NULL allowed	No default	No PK, FK, or unique marker	integer('question_time_limit_seconds')
strictMode	strict_mode	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('strict_mode').notNull().default(false)
fileUploadInstructions	file_upload_instructions	text	NULL allowed	No default	No PK, FK, or unique marker	text('file_upload_instructions')
teacherAttachmentFileId	teacher_attachment_file_id	uuid	NULL allowed	No default	No PK, FK, or unique marker	uuid('teacher_attachment_file_id')
rubricSourceFileId	rubric_source_file_id	uuid	NULL allowed	No default	No PK, FK, or unique marker	uuid('rubric_source_file_id')
rubricParseStatus	rubric_parse_status	rubricParseStatusEnum	NOT NULL	default('pending')	No PK, FK, or unique marker	rubricParseStatusEnum('rubric_parse_status').notNull().default('pending')
rubricParsedAt	rubric_parsed_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('rubric_parsed_at')
rubricRawText	rubric_raw_text	text	NULL allowed	No default	No PK, FK, or unique marker	text('rubric_raw_text')
rubricParseError	rubric_parse_error	text	NULL allowed	No default	No PK, FK, or unique marker	text('rubric_parse_error')
rubricCriteria	rubric_criteria	json	NULL allowed	No default	No PK, FK, or unique marker	json('rubric_criteria')
allowedUploadMimeTypes	allowed_upload_mime_types	text	NULL allowed	No default	No PK, FK, or unique marker	text('allowed_upload_mime_types').array()
allowedUploadExtensions	allowed_upload_extensions	text	NULL allowed	No default	No PK, FK, or unique marker	text('allowed_upload_extensions').array()
maxUploadSizeBytes	max_upload_size_bytes	integer	NULL allowed	default(104857600)	No PK, FK, or unique marker	integer('max_upload_size_bytes').default(104857600)
totalPoints	total_points	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('total_points').notNull().default(0)
passingScore	passing_score	integer	NULL allowed	default(60)		

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
					No PK, FK, or unique marker	integer('passing_score').default(60)
maxAttempts	max_attempts	integer	NOT NULL	default(1)	No PK, FK, or unique marker	integer('max_attempts').notNull().default(1)
timeLimitMinutes	time_limit_minutes	integer	NULL allowed	No default	No PK, FK, or unique marker	integer('time_limit_minutes')
isPublished	is_published	boolean	NULL allowed	default(false)	No PK, FK, or unique marker	boolean('is_published').default(false)
feedbackLevel	feedback_level	feedbackLevelEnum	NULL allowed	default('standard')	No PK, FK, or unique marker	feedbackLevelEnum('feedback_level').default('standard')
feedbackDelayHours	feedback_delay_hours	integer	NULL allowed	default(24)	No PK, FK, or unique marker	integer('feedback_delay_hours').default(24)
isCoreTemplateAsset	is_core_template_asset	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('is_core_template_asset').notNull().default(false)
templateId	template_id	uuid	NULL allowed	No default	No PK, FK, or unique marker	uuid('template_id')
templateSourceId	template_source_id	uuid	NULL allowed	No default	No PK, FK, or unique marker	uuid('template_source_id')
classRecordCategory	class_record_category	classRecordCategoryEnum	NULL allowed	No default	No PK, FK, or unique marker	classRecordCategoryEnum('class_record_category')
quarter	quarter	gradingPeriodEnum	NULL allowed	No default	No PK, FK, or unique marker	gradingPeriodEnum('quarter')
aiOrigin	ai_origin	text	NULL allowed	No default	No PK, FK, or unique marker	text('ai_origin')
aiGenerationOutputId	ai_generation_output_id	uuid	NULL allowed	No default	No PK, FK, or unique marker	uuid('ai_generation_output_id')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `index('assessments_class_id_idx').on(table.classId)`

Relationship map.

- class_id: FK → classes.id (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: class_id: FK → classes.id (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

assessment_questions

Purpose. Ordered scored questions belonging to assessments.

Drizzle declaration. `assessmentQuestions` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
assessmentId	assessment_id	uuid	NOT NULL	No default	FK → assessments.id (onDelete: 'cascade')	uuid('assessment_id').notNull().references(() => assessments.id, { onDelete: 'cascade' })
type	type	questionTypeEnum	NOT NULL	default('multiple_choice')	No PK, FK, or unique marker	questionTypeEnum('type').notNull().default('multiple_choice')
content	content	text	NOT NULL	No default	No PK, FK, or unique marker	text('content').notNull()
points	points	integer	NOT NULL	default(1)	No PK, FK, or unique marker	integer('points').notNull().default(1)
order	order	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('order').notNull().default(0)
isRequired	is_required	boolean	NULL allowed	default(true)	No PK, FK, or unique marker	boolean('is_required').default(true)
explanation	explanation	text	NULL allowed	No default	No PK, FK, or unique marker	text('explanation')
imageUrl	image_url	text	NULL allowed	No default	No PK, FK, or unique marker	text('image_url')
metadata	metadata	json	NULL allowed	No default	No PK, FK, or unique marker	json('metadata')
conceptTags	concept_tags	json	NULL allowed	No default	No PK, FK, or unique marker	json('concept_tags')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `assessmentIdIdx: index('assessment_questions_assessment_id_idx').on( table.assessmentId, )`
- `orderIdIdx: index('assessment_questions_order_idx').on(table.order)`

Relationship map.

- `assessment_id: FK → assessments.id (onDelete: 'cascade')`

Delete and lifecycle behavior.

- Cascading parent deletes affect: `assessment_id: FK → assessments.id (onDelete: 'cascade')`.
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

assessment_question_options

Purpose. Selectable options and correctness metadata for objective questions.

Drizzle declaration. `assessmentQuestionOptions` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
questionId	question_id	uuid	NOT NULL	No default	FK → assessmentQuestions.id (onDelete: 'cascade')	uuid('question_id').notNull().references(() => assessmentQuestions.id, { onDelete: 'cascade' })
text	text	text	NOT NULL	No default	No PK, FK, or unique marker	text('text').notNull()
imageUrl	image_url	text	NULL allowed	No default	No PK, FK, or unique marker	text('image_url')
isCorrect	is_correct	boolean	NULL allowed	default(false)	No PK, FK, or unique marker	boolean('is_correct').default(false)
order	order	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('order').notNull().default(0)
metadata	metadata	json	NULL allowed	No default	No PK, FK, or unique marker	json('metadata')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()

Indexes and table constraints.

- `questionIdIdx: index('assessment_question_options_question_id_idx').on( table.questionId, )`

Relationship map.

- `question_id: FK → assessmentQuestions.id (onDelete: 'cascade')`

Delete and lifecycle behavior.

- Cascading parent deletes affect: `question_id: FK → assessmentQuestions.id (onDelete: 'cascade')`.
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

assessment_attempts

Purpose. Per-student assessment execution, timer, submission, review, score, and return state.

Drizzle declaration. `assessmentAttempts` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
studentId	student_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('student_id').notNull().references(() => users.id, { onDelete: 'cascade' })
assessmentId	assessment_id	uuid	NOT NULL	No default	FK → assessments.id (onDelete: 'cascade')	uuid('assessment_id').notNull().references(() => assessments.id, { onDelete: 'cascade' })
attemptNumber	attempt_number	integer	NOT NULL	default(1)	No PK, FK, or unique marker	integer('attempt_number').notNull().default(1)
startedAt	started_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('started_at').notNull().defaultNow()

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
expiresAt	expires_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('expires_at')
lastQuestionIndex	last_question_index	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('last_question_index').not-Null().default(0)
currentQuestion-StartedAt	current_question_started_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('current_question_started_at')
currentQuestion-DeadlineAt	current_question_deadline_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('current_question_deadline_at')
violationCount	violation_count	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('violation_count').not-Null().default(0)
questionOrder	question_order	text	NULL allowed	No default	No PK, FK, or unique marker	text('question_order').array()
draftResponses	draft_responses	json	NULL allowed	No default	No PK, FK, or unique marker	json('draft_responses')
submittedAt	submitted_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('submitted_at')
score	score	integer	NULL allowed	No default	No PK, FK, or unique marker	integer('score')
passed	passed	boolean	NULL allowed	No default	No PK, FK, or unique marker	boolean('passed')
isSubmitted	is_submitted	boolean	NULL allowed	default(false)	No PK, FK, or unique marker	boolean('is_submitted').default(false)
timeSpentSeconds	time_spent_seconds	integer	NULL allowed	default(0)	No PK, FK, or unique marker	integer('time_spent_seconds').default(0)
isReturned	is_returned	boolean	NULL allowed	default(false)	No PK, FK, or unique marker	boolean('is_returned').default(false)
returnedAt	returned_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('returned_at')
teacherFeedback	teacher_feedback	text	NULL allowed	No default	No PK, FK, or unique marker	text('teacher_feedback')
rubricScores	rubric_scores	json	NULL allowed	No default	No PK, FK, or unique marker	json('rubric_scores')
directScore	direct_score	integer	NULL allowed	No default	No PK, FK, or unique marker	integer('direct_score')
submittedFiles	submitted_files	json	NULL allowed	No default	No PK, FK, or unique marker	json('submitted_files')
submittedFileId	submitted_file_id	uuid	NULL allowed	No default	No PK, FK, or unique marker	uuid('submitted_file_id')
submittedFileOriginalName	submitted_file_original_name	text	NULL allowed	No default	No PK, FK, or unique marker	text('submitted_file_original_name')
submittedFileMimeType	submitted_file_mime_type	varchar	NULL allowed	No default	No PK, FK, or unique marker	varchar('submitted_file_mime_type', { length: 100, })
submittedFileSize-Bytes	submitted_file_size_bytes	bigint	NULL allowed	No default	No PK, FK, or unique marker	bigint('submitted_file_size_bytes', { mode: 'number', })
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').not-Null().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').not-Null().defaultNow()

Indexes and table constraints.

- `studentIdIdx: index('assessment_attempts_student_id_idx').on( table.studentId, )`
- `assessmentIdIdx: index('assessment_attempts_assessment_id_idx').on( table.assessmentId, )`
- `expiresAtIdx: index('assessment_attempts_expires_at_idx').on( table.expiresAt, )`
- `submittedIdx: index('assessment_attempts_submitted_idx').on( table.isSubmitted, )`
- `uniqueAttemptNumber: unique('assessment_attempts_student_assessment_attempt_unique',
) .on(table.studentId, table.assessmentId, table.attemptNumber)`

Relationship map.

- `student_id`: FK → `users.id` (onDelete: 'cascade')
- `assessment_id`: FK → `assessments.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `student_id`: FK → `users.id` (onDelete: 'cascade'); `assessment_id`: FK → `assessments.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

assessment_responses

Purpose. Per-attempt answers and awarded points for individual questions.

Drizzle declaration. `assessmentResponses` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
<code>id</code>	<code>id</code>	<code>uuid</code>	NOT NULL	<code>defaultRandom()</code>	PRIMARY KEY	<code>uuid('id').primaryKey().defaultRandom()</code>
<code>attemptId</code>	<code>attempt_id</code>	<code>uuid</code>	NOT NULL	No default	FK → <code>assessmentAttempts.id</code> (onDelete: 'cascade')	<code>uuid('attempt_id').notNull().references(() => assessmentAttempts.id, { onDelete: 'cascade' })</code>
<code>questionId</code>	<code>question_id</code>	<code>uuid</code>	NOT NULL	No default	FK → <code>assessmentQuestions.id</code> (onDelete: 'cascade')	<code>uuid('question_id').notNull().references(() => assessmentQuestions.id, { onDelete: 'cascade' })</code>
<code>studentAnswer</code>	<code>student_answer</code>	<code>text</code>	NULL allowed	No default	No PK, FK, or unique marker	<code>text('student_answer')</code>
<code>selectedOptionId</code>	<code>selected_option_id</code>	<code>uuid</code>	NULL allowed	No default	FK → <code>assessmentQuestionOptions.id</code> (onDelete: 'set null')	<code>uuid('selected_option_id').references(() => assessmentQuestionOptions.id, { onDelete: 'set null' },)</code>
<code>selectedOptionIds</code>	<code>selected_option_ids</code>	<code>text</code>	NULL allowed	No default	No PK, FK, or unique marker	<code>text('selected_option_ids').array()</code>
<code>isCorrect</code>	<code>is_correct</code>	<code>boolean</code>	NULL allowed	No default	No PK, FK, or unique marker	<code>boolean('is_correct')</code>
<code>pointsEarned</code>	<code>points_earned</code>	<code>integer</code>	NULL allowed	<code>default(0)</code>	No PK, FK, or unique marker	<code>integer('points_earned').default(0)</code>
<code>createdAt</code>	<code>created_at</code>	<code>timestamp</code>	NOT NULL	<code>defaultNow()</code>	No PK, FK, or unique marker	<code>timestamp('created_at').notNull().defaultNow()</code>

Indexes and table constraints.

- `attemptIdIdx: index('assessment_responses_attempt_id_idx').on( table.attemptId, )`
- `questionIdIdx: index('assessment_responses_question_id_idx').on( table.questionId, )`

Relationship map.

- `attempt_id: FK → assessmentAttempts.id (onDelete: 'cascade')`
- `question_id: FK → assessmentQuestions.id (onDelete: 'cascade')`
- `selected_option_id: FK → assessmentQuestionOptions.id (onDelete: 'set null')`

Delete and lifecycle behavior.

- Cascading parent deletes affect: `attempt_id: FK → assessmentAttempts.id (onDelete: 'cascade')`; `question_id: FK → assessmentQuestions.id (onDelete: 'cascade')`.
- Parent deletes set these references to NULL: `selected_option_id: FK → assessmentQuestionOptions.id (onDelete: 'set null')`.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

archived_users

Purpose. Retained identity snapshot for accounts removed from the active user table.

Drizzle declaration. `archivedUsers` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
<code>id</code>	<code>id</code>	<code>uuid</code>	NOT NULL	<code>defaultRandom()</code>	PRIMARY KEY	<code>uuid('id').primaryKey().defaultRandom()</code>
<code>originalUserId</code>	<code>original_user_id</code>	<code>uuid</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>uuid('original_user_id').notNull()</code>
<code>email</code>	<code>email</code>	<code>text</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>text('email').notNull()</code>
<code>fullName</code>	<code>full_name</code>	<code>text</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>text('full_name').notNull()</code>
<code>role</code>	<code>role</code>	<code>text</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>text('role').notNull()</code>
<code>archivedData</code>	<code>archived_data</code>	<code>json</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>json('archived_data').notNull()</code>
<code>archivedBy</code>	<code>archived_by</code>	<code>uuid</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>uuid('archived_by').notNull()</code>
<code>archivedAt</code>	<code>archived_at</code>	<code>timestamp</code>	NOT NULL	<code>defaultNow()</code>	No PK, FK, or unique marker	<code>timestamp('archived_at').notNull().defaultNow()</code>
<code>purgedAt</code>	<code>purged_at</code>	<code>timestamp</code>	NULL allowed	No default	No PK, FK, or unique marker	<code>timestamp('purged_at')</code>

Indexes and table constraints.

- `originalUserIdIdx: index('archived_users_original_user_id_idx').on( table.originalUserId, )`
- `emailIdx: index('archived_users_email_idx').on(table.email)`
- `archivedAtIdx: index('archived_users_archived_at_idx').on(table.archivedAt)`

Relationship map.

- No outgoing foreign key is declared on this table.

Delete and lifecycle behavior.

- No outgoing foreign key declares cascading deletion.

- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

class_modules

Purpose. Durable records for class modules, with the columns and relationships listed in this dictionary.

Drizzle declaration. `classModules` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
classId	class_id	uuid	NOT NULL	No default	FK → classes.id (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
title	title	text	NOT NULL	No default	No PK, FK, or unique marker	text('title').notNull()
description	description	text	NULL allowed	No default	No PK, FK, or unique marker	text('description')
order	order	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('order').notNull().default(0)
isVisible	is_visible	boolean	NOT NULL	default(true)	No PK, FK, or unique marker	boolean('is_visible').notNull().default(true)
isLocked	is_locked	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('is_locked').notNull().default(false)
teacherNotes	teacher_notes	text	NULL allowed	No default	No PK, FK, or unique marker	text('teacher_notes')
themeKind	theme_kind	text	NOT NULL	default('gradient')	No PK, FK, or unique marker	text('theme_kind').notNull().default('gradient')
gradientId	gradient_id	text	NOT NULL	default('oceanic-blue')	No PK, FK, or unique marker	text('gradient_id').notNull().default('oceanic-blue')
coverImageUrl	cover_image_url	text	NULL allowed	No default	No PK, FK, or unique marker	text('cover_image_url')
imagePositionX	image_position_x	integer	NOT NULL	default(50)	No PK, FK, or unique marker	integer('image_position_x').notNull().default(50)
imagePositionY	image_position_y	integer	NOT NULL	default(50)	No PK, FK, or unique marker	integer('image_position_y').notNull().default(50)
imageScale	image_scale	integer	NOT NULL	default(120)	No PK, FK, or unique marker	integer('image_scale').notNull().default(120)
isCoreTemplateAsset	is_core_template_asset	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('is_core_template_asset').notNull().default(false)
templateId	template_id	uuid	NULL allowed	No default	No PK, FK, or unique marker	uuid('template_id')
templateSourceId	template_source_id	uuid	NULL allowed	No default	No PK, FK, or unique marker	uuid('template_source_id')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
						timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `classIdIdx: index('class_modules_class_id_idx').on(table.classId)`
- `classOrderIdx: index('class_modules_class_order_idx').on( table.classId, table.order, )`
- `classTitleUnique: unique('class_modules_class_title_unique').on( table.classId, table.title, )`

Relationship map.

- `class_id`: FK → `classes.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `class_id`: FK → `classes.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

module_sections

Purpose. Durable records for module sections, with the columns and relationships listed in this dictionary.

Drizzle declaration. `moduleSections` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
moduleId	module_id	uuid	NOT NULL	No default	FK → <code>classModules.id</code> (onDelete: 'cascade')	uuid('module_id').notNull().references(() => classModules.id, { onDelete: 'cascade' })
title	title	text	NOT NULL	No default	No PK, FK, or unique marker	text('title').notNull()
description	description	text	NULL allowed	No default	No PK, FK, or unique marker	text('description')
order	order	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('order').notNull().default(0)
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `moduleIdIdx: index('module_sections_module_id_idx').on(table.moduleId)`
- `moduleOrderIdx: index('module_sections_module_order_idx').on( table.moduleId, table.order, )`

Relationship map.

- `module_id`: FK → `classModules.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `module_id`: FK → `classModules.id` (onDelete: 'cascade').

- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

module_items

Purpose. Durable records for module items, with the columns and relationships listed in this dictionary.

Drizzle declaration. `moduleItems` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
moduleSectionId	module_section_id	uuid	NOT NULL	No default	FK → moduleSections.id (onDelete: 'cascade')	uuid('module_section_id').notNull().references(() => moduleSections.id, { onDelete: 'cascade' })
itemType	item_type	moduleItem-TypeEnum	NOT NULL	No default	No PK, FK, or unique marker	moduleItem-TypeEnum('item_type').notNull()
lessonId	lesson_id	uuid	NULL allowed	No default	FK → lessons.id (onDelete: 'cascade')	uuid('lesson_id').references(() => lessons.id, { onDelete: 'cascade', })
assessmentId	assessment_id	uuid	NULL allowed	No default	FK → assessments.id (onDelete: 'cascade')	uuid('assessment_id').references(() => assessments.id, { onDelete: 'cascade', })
fileId	file_id	uuid	NULL allowed	No default	No PK, FK, or unique marker	uuid('file_id')
order	order	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('order').notNull().default(0)
isVisible	is_visible	boolean	NOT NULL	default(true)	No PK, FK, or unique marker	boolean('is_visible').notNull().default(true)
isRequired	is_required	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('is_required').notNull().default(false)
isGiven	is_given	boolean	NOT NULL	default(true)	No PK, FK, or unique marker	boolean('is_given').notNull().default(true)
isCoreTemplateAsset	is_core_template_asset	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('is_core_template_asset').notNull().default(false)
templateId	template_id	uuid	NULL allowed	No default	No PK, FK, or unique marker	uuid('template_id')
templateSourceId	template_source_id	uuid	NULL allowed	No default	No PK, FK, or unique marker	uuid('template_source_id')
metadata	metadata	json	NULL allowed	No default	No PK, FK, or unique marker	json('metadata')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `sectionIdIdx: index('module_items_section_id_idx').on( table.moduleSectionId, )`

- `sectionOrderIdx: index('module_items_section_order_idx').on( table.moduleSectionId, table.order, )`
- `lessonIdIdx: index('module_items_lesson_id_idx').on(table.lessonId)`
- `assessmentIdIdx: index('module_items_assessment_id_idx').on( table.assessmentId, )`
- `fileIdIdx: index('module_items_file_id_idx').on(table.fileId)`
- `uniqueLessonItem: unique('module_items_lesson_id_unique').on( table.lessonId, )`
- `uniqueAssessmentItem: unique('module_items_assessment_id_unique').on( table.assessmentId, )`

Relationship map.

- `module_section_id`: FK → `moduleSections.id` (onDelete: 'cascade')
- `lesson_id`: FK → `lessons.id` (onDelete: 'cascade',)
- `assessment_id`: FK → `assessments.id` (onDelete: 'cascade',)

Delete and lifecycle behavior.

- Cascading parent deletes affect: `module_section_id`: FK → `moduleSections.id` (onDelete: 'cascade'); `lesson_id`: FK → `lessons.id` (onDelete: 'cascade',); `assessment_id`: FK → `assessments.id` (onDelete: 'cascade',).
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

module_grading_scale_entries

Purpose. Durable records for module grading scale entries, with the columns and relationships listed in this dictionary.

Drizzle declaration. `moduleGradingScaleEntries` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
<code>id</code>	<code>id</code>	<code>uuid</code>	NOT NULL	<code>defaultRandom()</code>	PRIMARY KEY	<code>uuid('id').primaryKey().defaultRandom()</code>
<code>moduleId</code>	<code>module_id</code>	<code>uuid</code>	NOT NULL	No default	FK → <code>classModules.id</code> (onDelete: 'cascade')	<code>uuid('module_id').notNull().references(() => classModules.id, { onDelete: 'cascade' })</code>
<code>letter</code>	<code>letter</code>	<code>varchar</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>varchar('letter', { length: 8 }).notNull()</code>
<code>label</code>	<code>label</code>	<code>text</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>text('label').notNull()</code>
<code>minScore</code>	<code>min_score</code>	<code>integer</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>integer('min_score').notNull()</code>
<code>maxScore</code>	<code>max_score</code>	<code>integer</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>integer('max_score').notNull()</code>
<code>description</code>	<code>description</code>	<code>text</code>	NULL allowed	No default	No PK, FK, or unique marker	<code>text('description')</code>
<code>order</code>	<code>order</code>	<code>integer</code>	NOT NULL	<code>default(0)</code>	No PK, FK, or unique marker	<code>integer('order').notNull().default(0)</code>
<code>createdAt</code>	<code>created_at</code>	<code>timestamp</code>	NOT NULL	<code>defaultNow()</code>	No PK, FK, or unique marker	<code>timestamp('created_at').notNull().defaultNow()</code>
<code>updatedAt</code>	<code>updated_at</code>	<code>timestamp</code>	NOT NULL	<code>defaultNow()</code>	No PK, FK, or unique marker	<code>timestamp('updated_at').notNull().defaultNow()</code>

Indexes and table constraints.

- `moduleIdIdx: index('module_grading_scale_entries_module_id_idx').on( table.moduleId, )`
- `moduleOrderIdx: index('module_grading_scale_entries_module_order_idx').on( table.moduleId, table.order, )`

Relationship map.

- `module_id`: FK → `classModules.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `module_id`: FK → `classModules.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

uploaded_files

Purpose. File metadata, storage ownership, library classification, indexing state, and extraction linkage.

Drizzle declaration. `uploadedFiles` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
<code>id</code>	<code>id</code>	<code>uuid</code>	NOT NULL	<code>defaultRandom()</code>	PRIMARY KEY	<code>uuid('id').primaryKey().defaultRandom()</code>
<code>folderId</code>	<code>folder_id</code>	<code>uuid</code>	NULL allowed	No default	No PK, FK, or unique marker	<code>uuid('folder_id')</code>
<code>teacherId</code>	<code>teacher_id</code>	<code>uuid</code>	NOT NULL	No default	FK → <code>users.id</code> (onDelete: 'cascade')	<code>uuid('teacher_id').notNull().references(() => users.id, { onDelete: 'cascade' })</code>
<code>classId</code>	<code>class_id</code>	<code>uuid</code>	NULL allowed	No default	FK → <code>classes.id</code> (onDelete: 'cascade')	<code>uuid('class_id').references(() => classes.id, { onDelete: 'cascade' })</code>
<code>scope</code>	<code>scope</code>	<code>fileScopeEnum</code>	NOT NULL	<code>default('private')</code>	No PK, FK, or unique marker	<code>fileScopeEnum('scope').notNull().default('private')</code>
<code>aiEnabled</code>	<code>ai_enabled</code>	<code>boolean</code>	NOT NULL	<code>default(true)</code>	No PK, FK, or unique marker	<code>boolean('ai_enabled').notNull().default(true)</code>
<code>subjectKey</code>	<code>subject_key</code>	<code>librarySubjectKeyEnum</code>	NULL allowed	No default	No PK, FK, or unique marker	<code>librarySubjectKeyEnum('subject_key')</code>
<code>gradeLevel</code>	<code>grade_level</code>	<code>gradeLevelEnum</code>	NULL allowed	No default	No PK, FK, or unique marker	<code>gradeLevelEnum('grade_level')</code>
<code>teacherVisible</code>	<code>teacher_visible</code>	<code>boolean</code>	NOT NULL	<code>default(true)</code>	No PK, FK, or unique marker	<code>boolean('teacher_visible').notNull().default(true)</code>
<code>indexStatus</code>	<code>index_status</code>	<code>libraryIndexStatusEnum</code>	NOT NULL	<code>default('not_indexed')</code>	No PK, FK, or unique marker	<code>libraryIndexStatusEnum('index_status').notNull().default('not_indexed')</code>
<code>indexError</code>	<code>index_error</code>	<code>text</code>	NULL allowed	No default	No PK, FK, or unique marker	<code>text('index_error')</code>
<code>indexedAt</code>	<code>indexed_at</code>	<code>timestamp</code>	NULL allowed	No default	No PK, FK, or unique marker	<code>timestamp('indexed_at')</code>
<code>contentHash</code>	<code>content_hash</code>	<code>text</code>	NULL allowed	No default	No PK, FK, or unique marker	<code>text('content_hash')</code>
<code>fileKind</code>	<code>file_kind</code>	<code>libraryFileKindEnum</code>	NOT NULL	<code>default('pdf')</code>		

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
					No PK, FK, or unique marker	libraryFileKindEnum('file_kind').notNull().default('pdf')
originalName	original_name	varchar	NOT NULL	No default	No PK, FK, or unique marker	varchar('original_name', { length: 255 }).notNull()
storedName	stored_name	varchar	NOT NULL	No default	No PK, FK, or unique marker	varchar('stored_name', { length: 255 }).notNull()
mimeType	mime_type	varchar	NOT NULL	No default	No PK, FK, or unique marker	varchar('mime_type', { length: 100 }).notNull()
sizeBytes	size_bytes	bigint	NOT NULL	No default	No PK, FK, or unique marker	bigint('size_bytes', { mode: 'number' }).notNull()
filePath	file_path	text	NOT NULL	No default	No PK, FK, or unique marker	text('file_path').notNull()
storageKey	storage_key	text	NULL allowed	No default	No PK, FK, or unique marker	text('storage_key')
storageProvider	storage_provider	varchar	NULL allowed	default('local')	No PK, FK, or unique marker	varchar('storage_provider', { length: 50 }).default('local')
storageBucket	storage_bucket	varchar	NULL allowed	No default	No PK, FK, or unique marker	varchar('storage_bucket', { length: 100 })
uploadedAt	uploaded_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('uploaded_at').notNull().defaultNow()
deletedAt	deleted_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('deleted_at')

Indexes and table constraints.

- `folderIdx: index('uploaded_files_folder_idx').on(table.folderId)`
- `teacherIdx: index('uploaded_files_teacher_idx').on(table.teacherId)`
- `classIdx: index('uploaded_files_class_idx').on(table.classId)`
- `scopeIdx: index('uploaded_files_scope_idx').on(table.scope)`
- `teacherAiEnabledLookupIdx: index('uploaded_files_teacher_ai_enabled_lookup_idx', ).on( table.teacherId, table.aiEnabled, table.scope, table.subjectKey, table.gradeLevel, table.deletedAt, )`
- `generalPartitionIdx: index('uploaded_files_general_partition_idx').on( table.scope, table.subjectKey, table.gradeLevel, table.teacherVisible, table.deletedAt, )`
- `indexStatusIdx: index('uploaded_files_index_status_idx').on( table.indexStatus, )`
- `uploadedAtIdx: index('uploaded_files_uploaded_at_idx').on(table.uploadedAt)`

Relationship map.

- `teacher_id`: FK → `users.id` (onDelete: 'cascade')
- `class_id`: FK → `classes.id` (onDelete: 'cascade',)

Delete and lifecycle behavior.

- Cascading parent deletes affect: `teacher_id`: FK → `users.id` (onDelete: 'cascade'); `class_id`: FK → `classes.id` (onDelete: 'cascade',).
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

class_visibility_preferences

Purpose. Persisted class visibility preferences used to keep role and user presentation choices stable.

Drizzle declaration. `classVisibilityPreferences` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
userId	user_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('user_id').notNull().references(() => users.id, { onDelete: 'cascade' })
classId	class_id	uuid	NOT NULL	No default	FK → classes.id (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
isHidden	is_hidden	boolean	NOT NULL	default(true)	No PK, FK, or unique marker	boolean('is_hidden').notNull().default(true)
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `userIdx: index('class_visibility_preferences_user_idx').on(table.userId)`
- `classIdx: index('class_visibility_preferences_class_idx').on(table.classId)`
- `uniquePreference: unique('class_visibility_preferences_user_class_unique', ).on(table.userId, table.classId)`

Relationship map.

- `user_id`: FK → `users.id` (onDelete: 'cascade')
- `class_id`: FK → `classes.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `user_id`: FK → `users.id` (onDelete: 'cascade'); `class_id`: FK → `classes.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

student_class_presentation_preferences

Purpose. Persisted student class presentation preferences used to keep role and user presentation choices stable.

Drizzle declaration. `studentClassPresentationPreferences` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
userId	user_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('user_id').notNull().references(() => users.id,

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
						{ onDelete: 'cascade' }
classId	class_id	uuid	NOT NULL	No default	FK → classes.id (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
styleMode	style_mode	studentPresentationModeEnum	NOT NULL	default('gradient')	No PK, FK, or unique marker	studentPresentationModeEnum('style_mode').notNull().default('gradient')
styleToken	style_token	text	NOT NULL	No default	No PK, FK, or unique marker	text('style_token').notNull()
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `userId: index('student_class_presentation_preferences_user_idx').on( table.userId, )`
- `classId: index('student_class_presentation_preferences_class_idx').on( table.classId, )`
- `uniquePreference: unique( 'student_class_presentation_preferences_user_class_unique', ).on(table.userId, table.classId)`

Relationship map.

- `user_id`: FK → `users.id` (onDelete: 'cascade')
- `class_id`: FK → `classes.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `user_id`: FK → `users.id` (onDelete: 'cascade'); `class_id`: FK → `classes.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

student_course_view_preferences

Purpose. Persisted student course view preferences used to keep role and user presentation choices stable.

Drizzle declaration. `studentCourseViewPreferences` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
userId	user_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('user_id').notNull().references(() => users.id, { onDelete: 'cascade' })
viewMode	view_mode	studentCourseViewModeEnum	NOT NULL	default('card')	No PK, FK, or unique marker	studentCourseViewModeEnum('view_mode').notNull().default('card')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `userIdx: index('student_course_view_preferences_user_idx').on(table.userId)`
- `uniqueUserPreference: unique('student_course_view_preferences_user_unique', ).on(table.userId)`

Relationship map.

- `user_id: FK → users.id (onDelete: 'cascade')`

Delete and lifecycle behavior.

- Cascading parent deletes affect: `user_id: FK → users.id (onDelete: 'cascade')`.
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

section_visibility_preferences

Purpose. Persisted section visibility preferences used to keep role and user presentation choices stable.

Drizzle declaration. `sectionVisibilityPreferences` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
userId	user_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('user_id').notNull().references(() => users.id, { onDelete: 'cascade' })
sectionId	section_id	uuid	NOT NULL	No default	FK → sections.id (onDelete: 'cascade')	uuid('section_id').notNull().references(() => sections.id, { onDelete: 'cascade' })
isHidden	is_hidden	boolean	NOT NULL	default(true)	No PK, FK, or unique marker	boolean('is_hidden').notNull().default(true)
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `userIdx: index('section_visibility_preferences_user_idx').on(table.userId)`
- `sectionIdx: index('section_visibility_preferences_section_idx').on( table.sectionId, )`
- `uniquePreference: unique('section_visibility_preferences_user_section_unique', ).on(table.userId, table.sectionId)`

Relationship map.

- `user_id: FK → users.id (onDelete: 'cascade')`
- `section_id: FK → sections.id (onDelete: 'cascade')`

Delete and lifecycle behavior.

- Cascading parent deletes affect: `user_id`: FK → `users.id` (onDelete: 'cascade'); `section_id`: FK → `sections.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

library_folders

Purpose. Hierarchical user-owned or shared folders for the file library.

Drizzle declaration. `libraryFolders` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
<code>id</code>	<code>id</code>	<code>uuid</code>	NOT NULL	<code>defaultRandom()</code>	PRIMARY KEY	<code>uuid('id').primaryKey().defaultRandom()</code>
<code>name</code>	<code>name</code>	<code>varchar</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>varchar('name', { length: 255 }).notNull()</code>
<code>ownerId</code>	<code>owner_id</code>	<code>uuid</code>	NOT NULL	No default	FK → <code>users.id</code> (onDelete: 'cascade')	<code>uuid('owner_id').notNull().references(() => users.id, { onDelete: 'cascade' })</code>
<code>parentId</code>	<code>parent_id</code>	<code>uuid</code>	NULL allowed	No default	No PK, FK, or unique marker	<code>uuid('parent_id')</code>
<code>scope</code>	<code>scope</code>	<code>fileScopeEnum</code>	NOT NULL	<code>default('private')</code>	No PK, FK, or unique marker	<code>fileScopeEnum('scope').notNull().default('private')</code>
<code>createdAt</code>	<code>created_at</code>	<code>timestamp</code>	NOT NULL	<code>defaultNow()</code>	No PK, FK, or unique marker	<code>timestamp('created_at').notNull().defaultNow()</code>
<code>updatedAt</code>	<code>updated_at</code>	<code>timestamp</code>	NOT NULL	<code>defaultNow()</code>	No PK, FK, or unique marker	<code>timestamp('updated_at').notNull().defaultNow()</code>
<code>deletedAt</code>	<code>deleted_at</code>	<code>timestamp</code>	NULL allowed	No default	No PK, FK, or unique marker	<code>timestamp('deleted_at')</code>

Indexes and table constraints.

- `ownerIdx: index('library_folders_owner_idx').on(table.ownerId)`
- `parentIdx: index('library_folders_parent_idx').on(table.parentId)`
- `scopeIdx: index('library_folders_scope_idx').on(table.scope)`

Relationship map.

- `owner_id`: FK → `users.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `owner_id`: FK → `users.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

pending_roster

Purpose. Validated roster-import staging rows awaiting administrative application.

Drizzle declaration. `pendingRoster` in `backend/src/drizzle/schema/base.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
sectionId	section_id	uuid	NOT NULL	No default	FK → sections.id (onDelete: 'cascade')	uuid('section_id').references(() => sections.id, { onDelete: 'cascade' })
lastName	last_name	text	NOT NULL	No default	No PK, FK, or unique marker	text('last_name').notNull()
firstName	first_name	text	NOT NULL	No default	No PK, FK, or unique marker	text('first_name').notNull()
middleInitial	middle_initial	text	NULL allowed	No default	No PK, FK, or unique marker	text('middle_initial')
lrn	lrn	varchar	NOT NULL	No default	No PK, FK, or unique marker	varchar('lrn', { length: 12 }).notNull()
rosterEmail	roster_email	text	NOT NULL	No default	No PK, FK, or unique marker	text('roster_email').notNull()
resolvedAt	resolved_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('resolved_at')
resolvedUserId	resolved_user_id	uuid	NULL allowed	No default	FK → users.id (onDelete: 'set null')	uuid('resolved_user_id').references(() => users.id, { onDelete: 'set null', })
importedAt	imported_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('imported_at').notNull().defaultNow()

Indexes and table constraints.

- `sectionIdx: index('pending_roster_section_id_idx').on(table.sectionId)`
- `emailIdx: index('pending_roster_roster_email_idx').on(table.rosterEmail)`

Relationship map.

- `section_id`: FK → sections.id (onDelete: 'cascade')
- `resolved_user_id`: FK → users.id (onDelete: 'set null',)

Delete and lifecycle behavior.

- Cascading parent deletes affect: `section_id`: FK → sections.id (onDelete: 'cascade').
- Parent deletes set these references to NULL: `resolved_user_id`: FK → users.id (onDelete: 'set null',).
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

Source file: `backend/src/drizzle/schema/class-record.schema.ts`

This source file declares 5 active tables.

`class_records`

Purpose. Official class-record workbooks and lifecycle state for a class and grading period.

Drizzle declaration. `classRecords` in `backend/src/drizzle/schema/class-record.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
classId	class_id	uuid	NOT NULL	No default	FK → classes.id (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
teacherId	teacher_id	uuid	NULL allowed	No default	FK → users.id (onDelete: 'set null',)	uuid('teacher_id').references(() => users.id, { onDelete: 'set null', })
gradingPeriod	grading_period	gradingPeriodEnum	NOT NULL	No default	No PK, FK, or unique marker	gradingPeriodEnum('grading_period').notNull()
status	status	classRecord-StatusEnum	NOT NULL	default('draft')	No PK, FK, or unique marker	classRecord-StatusEnum('status').notNull().default('draft')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `classPeriodUnique: unique('class_records_class_period_unique').on( table.classId, table.gradingPeriod, )`
- `teacherIdx: index('class_records_teacher_idx').on(table.teacherId)`
- `classIdx: index('class_records_class_idx').on(table.classId)`

Relationship map.

- `class_id`: FK → `classes.id` (onDelete: 'cascade')
- `teacher_id`: FK → `users.id` (onDelete: 'set null',)

Delete and lifecycle behavior.

- Cascading parent deletes affect: `class_id`: FK → `classes.id` (onDelete: 'cascade').
- Parent deletes set these references to NULL: `teacher_id`: FK → `users.id` (onDelete: 'set null',).
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

class_record_categories

Purpose. Weighted grading categories inside a class record.

Drizzle declaration. `classRecordCategories` in `backend/src/drizzle/schema/class-record.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
classRecordId	gradebook_id	uuid	NOT NULL	No default	FK → classRecords.id (onDelete: 'cascade')	uuid('gradebook_id').notNull().references(() => classRecords.id, { onDelete: 'cascade' })
name	name	text	NOT NULL	No default	No PK, FK, or unique marker	text('name').notNull()
weightPercentage	weight_percentage	numeric	NOT NULL	No default		

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
					No PK, FK, or unique marker	numeric('weight_percentage', { precision: 5, scale: 2, }).notNull()
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()

Indexes and table constraints.

- `classRecordIdx: index('class_record_categories_class_record_idx').on( table.classRecordId, )`
- `classRecordNameUnique: unique('class_record_categories_name_unique').on( table.classRecordId, table.name, )`

Relationship map.

- `gradebook_id: FK → classRecords.id (onDelete: 'cascade')`

Delete and lifecycle behavior.

- Cascading parent deletes affect: `gradebook_id: FK → classRecords.id (onDelete: 'cascade')`.
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

class_record_items

Purpose. Scored activities or assessment-linked columns inside a class-record category.

Drizzle declaration. `classRecordItems` in `backend/src/drizzle/schema/class-record.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
classRecordId	gradebook_id	uuid	NOT NULL	No default	FK → classRecords.id (onDelete: 'cascade')	uuid('gradebook_id').notNull().references(() => classRecords.id, { onDelete: 'cascade' })
categoryId	category_id	uuid	NOT NULL	No default	FK → classRecordCategories.id (onDelete: 'cascade')	uuid('category_id').notNull().references(() => classRecordCategories.id, { onDelete: 'cascade' })
assessmentId	assessment_id	uuid	NULL allowed	No default	FK → assessments.id (onDelete: 'set null',)	uuid('assessment_id').references(() => assessments.id, { onDelete: 'set null', })
title	title	text	NOT NULL	No default	No PK, FK, or unique marker	text('title').notNull()
maxScore	max_score	numeric	NOT NULL	No default	No PK, FK, or unique marker	numeric('max_score', { precision: 8, scale: 2 }).notNull()
itemOrder	item_order	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('item_order').notNull().default(0)
dateGiven	date_given	date	NULL allowed	No default	No PK, FK, or unique marker	date('date_given')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
						timestamp('created_at').notNull().defaultNow()

Indexes and table constraints.

- `classRecordIdx: index('class_record_items_class_record_idx').on( table.classRecordId, )`
- `categoryIdx: index('class_record_items_category_idx').on(table.categoryId)`
- `assessmentIdx: index('class_record_items_assessment_idx').on( table.assessmentId, )`
- `orderId: index('class_record_items_order_idx').on(table.itemOrder)`

Relationship map.

- `gradebook_id`: FK → `classRecords.id` (onDelete: 'cascade')
- `category_id`: FK → `classRecordCategories.id` (onDelete: 'cascade')
- `assessment_id`: FK → `assessments.id` (onDelete: 'set null')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `gradebook_id`: FK → `classRecords.id` (onDelete: 'cascade'); `category_id`: FK → `classRecordCategories.id` (onDelete: 'cascade').
- Parent deletes set these references to NULL: `assessment_id`: FK → `assessments.id` (onDelete: 'set null').
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

class_record_scores

Purpose. Per-student score entries for class-record items.

Drizzle declaration. `classRecordScores` in `backend/src/drizzle/schema/class-record.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
classRecordItemId	gradebook_item_id	uuid	NOT NULL	No default	FK → <code>classRecordItems.id</code> (onDelete: 'cascade')	uuid('gradebook_item_id').notNull().references(() => classRecordItems.id, { onDelete: 'cascade' })
studentId	student_id	uuid	NOT NULL	No default	FK → <code>users.id</code> (onDelete: 'cascade')	uuid('student_id').notNull().references(() => users.id, { onDelete: 'cascade' })
score	score	numeric	NOT NULL	No default	No PK, FK, or unique marker	numeric('score', { precision: 8, scale: 2 }).notNull()
recordedAt	recorded_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('recorded_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `itemStudentUnique: unique('class_record_scores_item_student_unique').on( table.classRecordItemId, table.studentId, )`

- `studentIdx: index('class_record_scores_student_idx').on(table.studentId)`
- `itemIdx: index('class_record_scores_item_idx').on(table.classRecordItemId)`

Relationship map.

- `gradebook_item_id`: FK → `classRecordItems.id` (onDelete: 'cascade')
- `student_id`: FK → `users.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `gradebook_item_id`: FK → `classRecordItems.id` (onDelete: 'cascade'); `student_id`: FK → `users.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

class_record_final_grades

Purpose. Computed and finalized per-student grades for a class record.

Drizzle declaration. `classRecordFinalGrades` in `backend/src/drizzle/schema/class-record.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
<code>id</code>	<code>id</code>	<code>uuid</code>	NOT NULL	<code>defaultRandom()</code>	PRIMARY KEY	<code>uuid('id').primaryKey().defaultRandom()</code>
<code>classRecordId</code>	<code>gradebook_id</code>	<code>uuid</code>	NOT NULL	No default	FK → <code>classRecords.id</code> (onDelete: 'cascade')	<code>uuid('gradebook_id').notNull().references(() => classRecords.id, { onDelete: 'cascade' })</code>
<code>studentId</code>	<code>student_id</code>	<code>uuid</code>	NOT NULL	No default	FK → <code>users.id</code> (onDelete: 'cascade')	<code>uuid('student_id').notNull().references(() => users.id, { onDelete: 'cascade' })</code>
<code>finalPercentage</code>	<code>final_percentage</code>	<code>numeric</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>numeric('final_percentage', { precision: 6, scale: 3, }).notNull()</code>
<code>remarks</code>	<code>remarks</code>	<code>classRecordRemarksEnum</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>classRecordRemarksEnum('remarks').notNull()</code>
<code>computedAt</code>	<code>computed_at</code>	<code>timestamp</code>	NOT NULL	<code>defaultNow()</code>	No PK, FK, or unique marker	<code>timestamp('computed_at').notNull().defaultNow()</code>

Indexes and table constraints.

- `classRecordStudentUnique: unique('class_record_final_grades_record_student_unique', ).on(table.classRecordId, table.studentId)`
- `classRecordIdx: index('class_record_final_grades_record_idx').on( table.classRecordId, )`
- `studentIdx: index('class_record_final_grades_student_idx').on( table.studentId, )`
- `remarksIdx: index('class_record_final_grades_remarks_idx').on( table.remarks, )`

Relationship map.

- `gradebook_id`: FK → `classRecords.id` (onDelete: 'cascade')
- `student_id`: FK → `users.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: gradebook_id: FK → classRecords.id (onDelete: 'cascade'); student_id: FK → users.id (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

Source file: backend/src/drizzle/schema/class-templates.schema.ts

This source file declares 11 active tables.

class_templates

Purpose. Reusable class blueprints owned by administrators.

Drizzle declaration. `classTemplates` in `backend/src/drizzle/schema/class-templates.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
name	name	varchar	NOT NULL	No default	No PK, FK, or unique marker	varchar('name', { length: 180 }).notNull()
subjectCode	subject_code	varchar	NOT NULL	No default	No PK, FK, or unique marker	varchar('subject_code', { length: 64 }).notNull()
subjectGradeLevel	subject_grade_level	varchar	NOT NULL	No default	No PK, FK, or unique marker	varchar('subject_grade_level', { length: 10 }).notNull()
status	status	classTemplateStatusEnum	NOT NULL	default('draft')	No PK, FK, or unique marker	classTemplateStatusEnum('status').notNull().default('draft')
createdBy	created_by	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('created_by').notNull().references(() => users.id, { onDelete: 'cascade' })
publishedAt	published_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('published_at')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `subjectIdx: index('class_templates_subject_idx').on( table.subjectCode, table.subjectGradeLevel, )`
- `createdByIdx: index('class_templates_created_by_idx').on(table.createdBy)`
- `uniqueNameBySubject: unique('class_templates_unique_name_by_subject').on( table.name, table.subjectCode, table.subjectGradeLevel, )`

Relationship map.

- `created_by: FK → users.id (onDelete: 'cascade')`

Delete and lifecycle behavior.

- Cascading parent deletes affect: created_by: FK → users.id (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

class_template_modules

Purpose. Module definitions inside a reusable class template.

Drizzle declaration. `classTemplateModules` in `backend/src/drizzle/schema/class-templates.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
templateId	template_id	uuid	NOT NULL	No default	FK → classTemplates.id (onDelete: 'cascade')	uuid('template_id').notNull().references(() => classTemplates.id, { onDelete: 'cascade' })
title	title	text	NOT NULL	No default	No PK, FK, or unique marker	text('title').notNull()
description	description	text	NULL allowed	No default	No PK, FK, or unique marker	text('description')
order	order	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('order').notNull().default(0)
themeKind	theme_kind	text	NOT NULL	default('gradient')	No PK, FK, or unique marker	text('theme_kind').notNull().default('gradient')
gradientId	gradient_id	text	NOT NULL	default('oceanic-blue')	No PK, FK, or unique marker	text('gradient_id').notNull().default('oceanic-blue')
coverImageUrl	cover_image_url	text	NULL allowed	No default	No PK, FK, or unique marker	text('cover_image_url')
imagePositionX	image_position_x	integer	NOT NULL	default(50)	No PK, FK, or unique marker	integer('image_position_x').notNull().default(50)
imagePositionY	image_position_y	integer	NOT NULL	default(50)	No PK, FK, or unique marker	integer('image_position_y').notNull().default(50)
imageScale	image_scale	integer	NOT NULL	default(120)	No PK, FK, or unique marker	integer('image_scale').notNull().default(120)
isVisible	is_visible	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('is_visible').notNull().default(false)
isLocked	is_locked	boolean	NOT NULL	default(true)	No PK, FK, or unique marker	boolean('is_locked').notNull().default(true)
teacherNotes	teacher_notes	text	NULL allowed	No default	No PK, FK, or unique marker	text('teacher_notes')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `templateOrderIdx: index('class_template_modules_template_order_idx').on( table.templateId, table.order, )`

Relationship map.

- `template_id`: FK → `classTemplates.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `template_id`: FK → `classTemplates.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

class_template_module_sections

Purpose. Ordered section definitions inside a template module.

Drizzle declaration. `classTemplateModuleSections` in `backend/src/drizzle/schema/class-templates.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
<code>id</code>	<code>id</code>	<code>uuid</code>	NOT NULL	<code>defaultRandom()</code>	PRIMARY KEY	<code>uuid('id').primaryKey().defaultRandom()</code>
<code>templateModuleId</code>	<code>template_module_id</code>	<code>uuid</code>	NOT NULL	No default	FK → <code>classTemplateModules.id</code> (onDelete: 'cascade')	<code>uuid('template_module_id').notNull().references(() => classTemplateModules.id, { onDelete: 'cascade' })</code>
<code>title</code>	<code>title</code>	<code>text</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>text('title').notNull()</code>
<code>description</code>	<code>description</code>	<code>text</code>	NULL allowed	No default	No PK, FK, or unique marker	<code>text('description')</code>
<code>order</code>	<code>order</code>	<code>integer</code>	NOT NULL	<code>default(0)</code>	No PK, FK, or unique marker	<code>integer('order').notNull().default(0)</code>
<code>createdAt</code>	<code>created_at</code>	<code>timestamp</code>	NOT NULL	<code>defaultNow()</code>	No PK, FK, or unique marker	<code>timestamp('created_at').notNull().defaultNow()</code>
<code>updatedAt</code>	<code>updated_at</code>	<code>timestamp</code>	NOT NULL	<code>defaultNow()</code>	No PK, FK, or unique marker	<code>timestamp('updated_at').notNull().defaultNow()</code>

Indexes and table constraints.

- `sectionOrderIdx: index('class_template_module_sections_order_idx').on( table.templateModuleId, table.order, )`

Relationship map.

- `template_module_id`: FK → `classTemplateModules.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `template_module_id`: FK → `classTemplateModules.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

class_template_assessments

Purpose. Assessment definitions stored inside a class template.

Drizzle declaration. `classTemplateAssessments` in `backend/src/drizzle/schema/class-templates.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
templateId	template_id	uuid	NOT NULL	No default	FK → classTemplates.id (onDelete: 'cascade')	uuid('template_id').notNull().references(() => classTemplates.id, { onDelete: 'cascade' })
title	title	text	NOT NULL	No default	No PK, FK, or unique marker	text('title').notNull()
description	description	text	NULL allowed	No default	No PK, FK, or unique marker	text('description')
type	type	text	NOT NULL	default('quiz')	No PK, FK, or unique marker	text('type').notNull().default('quiz')
dueDateOffsetDays	due_date_offset_days	integer	NULL allowed	No default	No PK, FK, or unique marker	integer('due_date_offset_days')
settings	settings	json	NULL allowed	No default	No PK, FK, or unique marker	json('settings')
totalPoints	total_points	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('total_points').notNull().default(0)
order	order	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('order').notNull().default(0)
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `templateOrderIdx: index('class_template_assessments_template_order_idx').on( table.templateId, table.order, )`

Relationship map.

- `template_id`: FK → `classTemplates.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `template_id`: FK → `classTemplates.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

class_template_lessons

Purpose. Lesson definitions stored inside a class template.

Drizzle declaration. `classTemplateLessons` in `backend/src/drizzle/schema/class-templates.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
templateId	template_id	uuid	NOT NULL	No default	FK → classTemplates.id (onDelete: 'cascade')	uuid('template_id').notNull().references(() => classTemplates.id, { onDelete: 'cascade' })
title	title	text	NOT NULL	No default	No PK, FK, or unique marker	text('title').notNull()
summary	summary	text	NULL allowed	No default	No PK, FK, or unique marker	text('summary')
order	order	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('order').notNull().default(0)
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `templateOrderIdx: index('class_template_lessons_template_order_idx').on( table.templateId, table.order, )`

Relationship map.

- `template_id: FK → classTemplates.id (onDelete: 'cascade')`

Delete and lifecycle behavior.

- Cascading parent deletes affect: `template_id: FK → classTemplates.id (onDelete: 'cascade')`.
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

class_template_lesson_blocks

Purpose. Ordered content blocks for template lessons.

Drizzle declaration. `classTemplateLessonBlocks` in `backend/src/drizzle/schema/class-templates.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
templateLessonId	template_lesson_id	uuid	NOT NULL	No default	FK → classTemplateLessons.id (onDelete: 'cascade')	uuid('template_lesson_id').notNull().references(() => classTemplateLessons.id, { onDelete: 'cascade' })
blockType	block_type	text	NOT NULL	No default	No PK, FK, or unique marker	text('block_type').notNull()
blockVersion	block_version	integer	NOT NULL	default(1)	No PK, FK, or unique marker	integer('block_version').notNull().default(1)
payload	payload	json	NOT NULL	No default	No PK, FK, or unique marker	json('payload').notNull()
order	order	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('order').notNull().default(0)
createdAt	created_at	timestamp	NOT NULL	defaultNow()		

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
					No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `lessonOrderIdx: index('class_template_lesson_blocks_lesson_order_idx').on( table.templateLessonId, table.order, )`

Relationship map.

- `template_lesson_id: FK → classTemplateLessons.id (onDelete: 'cascade')`

Delete and lifecycle behavior.

- Cascading parent deletes affect: `template_lesson_id: FK → classTemplateLessons.id (onDelete: 'cascade')`.
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

class_template_assessment_questions

Purpose. Question definitions for template assessments.

Drizzle declaration. `classTemplateAssessmentQuestions` in `backend/src/drizzle/schema/class-templates.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
templateAssessmentId	template_assessment_id	uuid	NOT NULL	No default	FK → classTemplateAssessments.id (onDelete: 'cascade')	uuid('template_assessment_id').notNull().references(() => classTemplateAssessments.id, { onDelete: 'cascade' })
type	type	text	NOT NULL	default('multiple_choice')	No PK, FK, or unique marker	text('type').notNull().default('multiple_choice')
content	content	text	NOT NULL	No default	No PK, FK, or unique marker	text('content').notNull()
points	points	integer	NOT NULL	default(1)	No PK, FK, or unique marker	integer('points').notNull().default(1)
order	order	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('order').notNull().default(0)
isRequired	is_required	boolean	NOT NULL	default(true)	No PK, FK, or unique marker	boolean('is_required').notNull().default(true)
explanation	explanation	text	NULL allowed	No default	No PK, FK, or unique marker	text('explanation')
imageUrl	image_url	text	NULL allowed	No default	No PK, FK, or unique marker	text('image_url')
metadata	metadata	json	NULL allowed	No default	No PK, FK, or unique marker	json('metadata')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
						timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `assessmentOrderIdx: index( 'class_template_assessment_questions_assessment_order_idx', ).on(table.templateAssessmentId, table.order)`

Relationship map.

- `template_assessment_id: FK → classTemplateAssessments.id (onDelete: 'cascade')`

Delete and lifecycle behavior.

- Cascading parent deletes affect: `template_assessment_id: FK → classTemplateAssessments.id (onDelete: 'cascade')`.
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

class_template_assessment_question_options

Purpose. Objective options for template assessment questions.

Drizzle declaration. `classTemplateAssessmentQuestionOptions` in `backend/src/drizzle/schema/class-templates.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
templateAssessmentQuestionId	template_assessment_question_id	uuid	NOT NULL	No default	FK → classTemplateAssessmentQuestions.id (onDelete: 'cascade')	uuid('template_assessment_question_id').notNull().references(() => classTemplateAssessmentQuestions.id, { onDelete: 'cascade', })
text	text	text	NOT NULL	No default	No PK, FK, or unique marker	text('text').notNull()
isCorrect	is_correct	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('is_correct').notNull().default(false)
order	order	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('order').notNull().default(0)
metadata	metadata	json	NULL allowed	No default	No PK, FK, or unique marker	json('metadata')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `questionOrderIdx: index( 'class_template_assessment_question_options_question_order_idx', ).on(table.templateAssessmentQuestionId, table.order)`

Relationship map.

- `template_assessment_question_id: FK → classTemplateAssessmentQuestions.id (onDelete: 'cascade',)`

Delete and lifecycle behavior.

- Cascading parent deletes affect: `template_assessment_question_id: FK → classTemplateAssessmentQuestions.id (onDelete: 'cascade',)`.
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

class_template_engine_chunks

Purpose. Template-engine source chunks and checksums used during class instantiation.

Drizzle declaration. `classTemplateEngineChunks` in `backend/src/drizzle/schema/class-templates.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
<code>id</code>	<code>id</code>	<code>varchar</code>	NOT NULL	No default	PRIMARY KEY	<code>varchar('id', { length: 190 }).primaryKey()</code>
<code>templateId</code>	<code>template_id</code>	<code>uuid</code>	NOT NULL	No default	FK → <code>classTemplates.id</code> (onDelete: 'cascade')	<code>uuid('template_id').notNull().references(() => classTemplates.id, { onDelete: 'cascade' })</code>
<code>sourceType</code>	<code>source_type</code>	<code>text</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>text('source_type').notNull()</code>
<code>sourceId</code>	<code>source_id</code>	<code>text</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>text('source_id').notNull()</code>
<code>chunkOrder</code>	<code>chunk_order</code>	<code>integer</code>	NOT NULL	<code>default(0)</code>	No PK, FK, or unique marker	<code>integer('chunk_order').notNull().default(0)</code>
<code>content</code>	<code>content</code>	<code>text</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>text('content').notNull()</code>
<code>metadata</code>	<code>metadata</code>	<code>json</code>	NULL allowed	No default	No PK, FK, or unique marker	<code>json('metadata')</code>
<code>createdAt</code>	<code>created_at</code>	<code>timestamp</code>	NOT NULL	<code>defaultNow()</code>	No PK, FK, or unique marker	<code>timestamp('created_at').notNull().defaultNow()</code>
<code>updatedAt</code>	<code>updated_at</code>	<code>timestamp</code>	NOT NULL	<code>defaultNow()</code>	No PK, FK, or unique marker	<code>timestamp('updated_at').notNull().defaultNow()</code>

Indexes and table constraints.

- `templateOrderIdx: index( 'class_template_engine_chunks_template_order_idx', ).on(table.templateId, table.chunkOrder)`
- `sourceIdx: index('class_template_engine_chunks_source_idx').on( table.templateId, table.sourceType, table.sourceId, )`

Relationship map.

- `template_id: FK → classTemplates.id (onDelete: 'cascade')`

Delete and lifecycle behavior.

- Cascading parent deletes affect: `template_id`: FK → `classTemplates.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

class_template_module_items

Purpose. Ordered lesson, assessment, or file references inside template sections.

Drizzle declaration. `classTemplateModuleItems` in `backend/src/drizzle/schema/class-templates.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
templateSectionId	template_section_id	uuid	NOT NULL	No default	FK → classTemplateModuleSections.id (onDelete: 'cascade',)	uuid('template_section_id').notNull().references(() => classTemplateModuleSections.id, { onDelete: 'cascade', })
itemType	item_type	classTemplateItemTypeEnum	NOT NULL	default('assessment')	No PK, FK, or unique marker	classTemplateItemTypeEnum('item_type').notNull().default('assessment')
templateAssessmentId	template_assessment_id	uuid	NULL allowed	No default	FK → classTemplateAssessments.id (onDelete: 'set null')	uuid('template_assessment_id').references(() => classTemplateAssessments.id, { onDelete: 'set null' },)
templateLessonId	template_lesson_id	uuid	NULL allowed	No default	FK → classTemplateLessons.id (onDelete: 'set null')	uuid('template_lesson_id').references(() => classTemplateLessons.id, { onDelete: 'set null' },)
order	order	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('order').notNull().default(0)
isRequired	is_required	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('is_required').notNull().default(false)
metadata	metadata	json	NULL allowed	No default	No PK, FK, or unique marker	json('metadata')
points	points	integer	NULL allowed	No default	No PK, FK, or unique marker	integer('points')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `itemOrderIdx`: `index('class_template_module_items_order_idx').on( table.templateSectionId, table.order, )`

Relationship map.

- `template_section_id`: FK → `classTemplateModuleSections.id` (onDelete: 'cascade',)

- `template_assessment_id`: FK → `classTemplateAssessments.id` (onDelete: 'set null')
- `template_lesson_id`: FK → `classTemplateLessons.id` (onDelete: 'set null')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `template_section_id`: FK → `classTemplateModuleSections.id` (onDelete: 'cascade',).
- Parent deletes set these references to NULL: `template_assessment_id`: FK → `classTemplateAssessments.id` (onDelete: 'set null'); `template_lesson_id`: FK → `classTemplateLessons.id` (onDelete: 'set null').
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

class_template_announcements

Purpose. Announcement definitions stored in a template.

Drizzle declaration. `classTemplateAnnouncements` in `backend/src/drizzle/schema/class-templates.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
<code>id</code>	<code>id</code>	<code>uuid</code>	NOT NULL	<code>defaultRandom()</code>	PRIMARY KEY	<code>uuid('id').primaryKey().defaultRandom()</code>
<code>templateId</code>	<code>template_id</code>	<code>uuid</code>	NOT NULL	No default	FK → <code>classTemplates.id</code> (onDelete: 'cascade')	<code>uuid('template_id').notNull().references(() => classTemplates.id, { onDelete: 'cascade' })</code>
<code>title</code>	<code>title</code>	<code>varchar</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>varchar('title', { length: 255 }).notNull()</code>
<code>content</code>	<code>content</code>	<code>text</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>text('content').notNull()</code>
<code>isPinned</code>	<code>is_pinned</code>	<code>boolean</code>	NOT NULL	<code>default(false)</code>	No PK, FK, or unique marker	<code>boolean('is_pinned').notNull().default(false)</code>
<code>order</code>	<code>order</code>	<code>integer</code>	NOT NULL	<code>default(0)</code>	No PK, FK, or unique marker	<code>integer('order').notNull().default(0)</code>
<code>createdAt</code>	<code>created_at</code>	<code>timestamp</code>	NOT NULL	<code>defaultNow()</code>	No PK, FK, or unique marker	<code>timestamp('created_at').notNull().defaultNow()</code>
<code>updatedAt</code>	<code>updated_at</code>	<code>timestamp</code>	NOT NULL	<code>defaultNow()</code>	No PK, FK, or unique marker	<code>timestamp('updated_at').notNull().defaultNow()</code>

Indexes and table constraints.

- `templateOrderIdx`: `index( 'class_template_announcements_template_order_idx', ).on(table.templateId, table.order)`

Relationship map.

- `template_id`: FK → `classTemplates.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `template_id`: FK → `classTemplates.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

Source file: backend/src/drizzle/schema/discussion-board.schema.ts

This source file declares 5 active tables.

discussion_threads

Purpose. Class discussion topics with author, moderation, publication, and closure state.

Drizzle declaration. `discussionThreads` in `backend/src/drizzle/schema/discussion-board.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
classId	class_id	uuid	NOT NULL	No default	FK → classes.id (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
authorId	author_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('author_id').notNull().references(() => users.id, { onDelete: 'cascade' })
title	title	varchar	NOT NULL	No default	No PK, FK, or unique marker	varchar('title', { length: 255 }).notNull()
bodyHtml	body_html	text	NOT NULL	No default	No PK, FK, or unique marker	text('body_html').notNull()
themeId	theme_id	varchar	NOT NULL	default('classic')	No PK, FK, or unique marker	varchar('theme_id', { length: 64 }).notNull().default('classic')
commentLimitPerStudent	comment_limit_per_student	integer	NULL allowed	No default	No PK, FK, or unique marker	integer('comment_limit_per_student')
allowComments	allow_comments	boolean	NOT NULL	default(true)	No PK, FK, or unique marker	boolean('allow_comments').notNull().default(true)
isPinned	is_pinned	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('is_pinned').notNull().default(false)
status	status	discussionThread-StatusEnum	NOT NULL	default('draft')	No PK, FK, or unique marker	discussionThread-StatusEnum('status').notNull().default('draft')
publishedAt	published_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('published_at')
closedAt	closed_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('closed_at')
archivedAt	archived_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('archived_at')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `classStatusPublishedIdx: index('discussion_threads_class_status_pub_idx', ).on(table.classId, table.status, table.publishedAt)`
- `classCreatedIdx: index('discussion_threads_class_created_idx').on( table.classId, table.createdAt, )`

- `authorIdx: index('discussion_threads_author_idx').on(table.authorId)`

Relationship map.

- `class_id`: FK → `classes.id` (onDelete: 'cascade')
- `author_id`: FK → `users.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `class_id`: FK → `classes.id` (onDelete: 'cascade'); `author_id`: FK → `users.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

discussion_thread_attachments

Purpose. File attachments connected to discussion threads.

Drizzle declaration. `discussionThreadAttachments` in `backend/src/drizzle/schema/discussion-board.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
<code>id</code>	<code>id</code>	<code>uuid</code>	NOT NULL	<code>defaultRandom()</code>	PRIMARY KEY	<code>uuid('id').primaryKey().defaultRandom()</code>
<code>threadId</code>	<code>thread_id</code>	<code>uuid</code>	NOT NULL	No default	FK → <code>discussionThreads.id</code> (onDelete: 'cascade')	<code>uuid('thread_id').notNull().references(() => discussionThreads.id, { onDelete: 'cascade' })</code>
<code>attachmentType</code>	<code>attachment_type</code>	<code>discussionAttachmentTypeEnum</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>discussionAttachmentTypeEnum('attachment_type').notNull()</code>
<code>fileId</code>	<code>file_id</code>	<code>uuid</code>	NULL allowed	No default	FK → <code>uploadedFiles.id</code> (onDelete: 'set null',)	<code>uuid('file_id').references(() => uploadedFiles.id, { onDelete: 'set null', })</code>
<code>linkUrl</code>	<code>link_url</code>	<code>text</code>	NULL allowed	No default	No PK, FK, or unique marker	<code>text('link_url')</code>
<code>linkLabel</code>	<code>link_label</code>	<code>varchar</code>	NULL allowed	No default	No PK, FK, or unique marker	<code>varchar('link_label', { length: 255 })</code>
<code>createdAt</code>	<code>created_at</code>	<code>timestamp</code>	NOT NULL	<code>defaultNow()</code>	No PK, FK, or unique marker	<code>timestamp('created_at').notNull().defaultNow()</code>

Indexes and table constraints.

- `threadIdx: index('discussion_thread_attachments_thread_idx').on( table.threadId, )`
- `fileIdx: index('discussion_thread_attachments_file_idx').on(table.fileId)`

Relationship map.

- `thread_id`: FK → `discussionThreads.id` (onDelete: 'cascade')
- `file_id`: FK → `uploadedFiles.id` (onDelete: 'set null',)

Delete and lifecycle behavior.

- Cascading parent deletes affect: `thread_id`: FK → `discussionThreads.id` (onDelete: 'cascade').
- Parent deletes set these references to NULL: `file_id`: FK → `uploadedFiles.id` (onDelete: 'set null',).
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

discussion_comments

Purpose. Thread replies with author and moderation state.

Drizzle declaration. `discussionComments` in `backend/src/drizzle/schema/discussion-board.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
threadId	thread_id	uuid	NOT NULL	No default	FK → discussionThreads.id (onDelete: 'cascade')	uuid('thread_id').notNull().references(() => discussionThreads.id, { onDelete: 'cascade' })
authorId	author_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('author_id').notNull().references(() => users.id, { onDelete: 'cascade' })
bodyHtml	body_html	text	NULL allowed	No default	No PK, FK, or unique marker	text('body_html')
deletedAt	deleted_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('deleted_at')
deletedById	deleted_by_id	uuid	NULL allowed	No default	FK → users.id (onDelete: 'set null',)	uuid('deleted_by_id').references(() => users.id, { onDelete: 'set null', })
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `threadCreatedIdx: index('discussion_comments_thread_created_idx').on( table.threadId, table.createdAt, )`
- `authorIdx: index('discussion_comments_author_idx').on(table.authorId)`

Relationship map.

- thread_id: FK → discussionThreads.id (onDelete: 'cascade')
- author_id: FK → users.id (onDelete: 'cascade')
- deleted_by_id: FK → users.id (onDelete: 'set null',)

Delete and lifecycle behavior.

- Cascading parent deletes affect: thread_id: FK → discussionThreads.id (onDelete: 'cascade'); author_id: FK → users.id (onDelete: 'cascade').
- Parent deletes set these references to NULL: deleted_by_id: FK → users.id (onDelete: 'set null',).
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

discussion_comment_attachments

Purpose. File attachments connected to discussion comments.

Drizzle declaration. `discussionCommentAttachments` in `backend/src/drizzle/schema/discussion-board.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
commentId	comment_id	uuid	NOT NULL	No default	FK → discussionComments.id (onDelete: 'cascade')	uuid('comment_id').notNull().references(() => discussionComments.id, { onDelete: 'cascade' })
fileId	file_id	uuid	NOT NULL	No default	FK → uploadedFiles.id (onDelete: 'cascade')	uuid('file_id').notNull().references(() => uploadedFiles.id, { onDelete: 'cascade' })
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()

Indexes and table constraints.

- commentIdx: index('discussion_comment_attachments_comment_idx').on(table.commentId,)
- fileIdx: index('discussion_comment_attachments_file_idx').on(table.fileId)
- commentFileUniqueIdx: uniqueIndex('discussion_comment_attachments_comment_file_unique_idx',).on(table.commentId, table.fileId)

Relationship map.

- comment_id: FK → discussionComments.id (onDelete: 'cascade')
- file_id: FK → uploadedFiles.id (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: comment_id: FK → discussionComments.id (onDelete: 'cascade'); file_id: FK → uploadedFiles.id (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

discussion_comment_reactions

Purpose. Per-user reactions to discussion comments.

Drizzle declaration. `discussionCommentReactions` in `backend/src/drizzle/schema/discussion-board.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
commentId	comment_id	uuid	NOT NULL	No default	FK → discussionComments.id (onDelete: 'cascade')	uuid('comment_id').notNull().references(() => discussionComments.id, { onDelete: 'cascade' })
userId	user_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('user_id').notNull().references(() => users.id, { onDelete: 'cascade' })
reactionType	reaction_type	discussionReactionTypeEnum	NOT NULL	No default	No PK, FK, or unique marker	discussionReactionTypeEnum('reaction_type').notNull()
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
						timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- commentIdx: `index('discussion_comment_reactions_comment_idx').on( table.commentId, )`
- userId: `index('discussion_comment_reactions_user_idx').on(table.userId)`
- commentUserUniqueIdx: `uniqueIndex( 'discussion_comment_reactions_comment_user_unique_idx', ).on(table.commentId, table.userId)`

Relationship map.

- comment_id: FK → discussionComments.id (onDelete: 'cascade')
- user_id: FK → users.id (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: comment_id: FK → discussionComments.id (onDelete: 'cascade'); user_id: FK → users.id (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

Source file: backend/src/drizzle/schema/ja.schema.ts

This source file declares 9 active tables.

ja_sessions

Purpose. Durable Ja practice or review sessions for students.

Drizzle declaration. `jaSessions` in `backend/src/drizzle/schema/ja.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
studentId	student_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('student_id').notNull().references(() => users.id, { onDelete: 'cascade' })
classId	class_id	uuid	NOT NULL	No default	FK → classes.id (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
mode	mode	jaSessionModeEnum	NOT NULL	default('practice')	No PK, FK, or unique marker	jaSessionModeEnum('mode').notNull().default('practice')
status	status	jaSession-StatusEnum	NOT NULL	default('active')	No PK, FK, or unique marker	jaSession-StatusEnum('status').notNull().default('active')
questionCount	question_count	integer	NOT NULL	default(10)		integer('question_cou

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
					No PK, FK, or unique marker	nt').notNull().default(10)
currentIndex	current_index	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('current_index').notNull().default(0)
strikeCount	strike_count	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('strike_count').notNull().default(0)
rewardState	reward_state	jaRewardStateEnum	NOT NULL	default('pending')	No PK, FK, or unique marker	jaRewardStateEnum('reward_state').notNull().default('pending')
sourceSnapshotJson	source_snapshot_json	json	NULL allowed	No default	No PK, FK, or unique marker	json('source_snapshot_json')
groundingStatus	grounding_status	text	NOT NULL	default('grounded')	No PK, FK, or unique marker	text('grounding_status').notNull().default('grounded')
startedAt	started_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('started_at').notNull().defaultNow()
completedAt	completed_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('completed_at')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `studentStatusIdx: index('ja_sessions_student_status_idx').on( table.studentId, table.status, )`
- `classStatusIdx: index('ja_sessions_class_status_idx').on( table.classId, table.status, )`
- `startedAtIdx: index('ja_sessions_started_at_idx').on(table.startedAt)`

Relationship map.

- `student_id`: FK → `users.id` (onDelete: 'cascade')
- `class_id`: FK → `classes.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `student_id`: FK → `users.id` (onDelete: 'cascade'); `class_id`: FK → `classes.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

ja_session_items

Purpose. Ordered questions or activities within a Ja session.

Drizzle declaration. `jaSessionItems` in `backend/src/drizzle/schema/ja.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
sessionId	session_id	uuid	NOT NULL	No default	FK → <code>jaSessions.id</code> (onDelete: 'cascade')	uuid('session_id').notNull().references(() => jaSessions.id,

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
						{ onDelete: 'cascade' }
orderIndex	order_index	integer	NOT NULL	No default	No PK, FK, or unique marker	integer('order_index').notNull()
itemType	item_type	text	NOT NULL	No default	No PK, FK, or unique marker	text('item_type').notNull()
prompt	prompt	text	NOT NULL	No default	No PK, FK, or unique marker	text('prompt').notNull()
optionsJson	options_json	json	NULL allowed	No default	No PK, FK, or unique marker	json('options_json')
answerKeyJson	answer_key_json	json	NOT NULL	No default	No PK, FK, or unique marker	json('answer_key_json').notNull()
hint	hint	text	NULL allowed	No default	No PK, FK, or unique marker	text('hint')
explanation	explanation	text	NULL allowed	No default	No PK, FK, or unique marker	text('explanation')
citationsJson	citations_json	json	NULL allowed	No default	No PK, FK, or unique marker	json('citations_json')
validationJson	validation_json	json	NULL allowed	No default	No PK, FK, or unique marker	json('validation_json')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()

Indexes and table constraints.

- `sessionOrderUnique: uniqueIndex('ja_session_items_session_order_unique').on( table.sessionId, table.orderIndex, )`
- `sessionIdx: index('ja_session_items_session_idx').on(table.sessionId)`

Relationship map.

- session_id: FK → jaSessions.id (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: session_id: FK → jaSessions.id (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

ja_session_responses

Purpose. Student responses and evaluation detail for Ja session items.

Drizzle declaration. `jaSessionResponses` in `backend/src/drizzle/schema/ja.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
sessionItemId	session_item_id	uuid	NOT NULL	No default	FK → jaSessionItems.id (onDelete: 'cascade')	uuid('session_item_id').notNull().references(() => jaSessionItems.id, { onDelete: 'cascade' })
studentAnswerJson	student_answer_json	json	NOT NULL	No default	No PK, FK, or unique marker	json('student_answer_json').notNull()

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
isCorrect	is_correct	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('is_correct').notNull().default(false)
scoreDelta	score_delta	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('score_delta').notNull().default(0)
feedback	feedback	text	NULL allowed	No default	No PK, FK, or unique marker	text('feedback')
answeredAt	answered_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('answered_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `sessionItemUnique: uniqueIndex( 'ja_session_responses_session_item_unique', ).on(table.sessionItemId)`

Relationship map.

- `session_item_id: FK → jaSessionItems.id (onDelete: 'cascade')`

Delete and lifecycle behavior.

- Cascading parent deletes affect: `session_item_id: FK → jaSessionItems.id (onDelete: 'cascade')`.
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

ja_session_events

Purpose. Append-only Ja session lifecycle and interaction events.

Drizzle declaration. `jaSessionEvents` in `backend/src/drizzle/schema/ja.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
sessionId	session_id	uuid	NOT NULL	No default	FK → jaSessions.id (onDelete: 'cascade')	uuid('session_id').notNull().references(() => jaSessions.id, { onDelete: 'cascade' })
eventType	event_type	jaSessionEvent-TypeEnum	NOT NULL	No default	No PK, FK, or unique marker	jaSessionEvent-TypeEnum('event_type').notNull()
payloadJson	payload_json	json	NULL allowed	No default	No PK, FK, or unique marker	json('payload_json')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()

Indexes and table constraints.

- `sessionCreatedAtIdx: index('ja_session_events_session_created_at_idx').on( table.sessionId, table.createdAt, )`

Relationship map.

- `session_id: FK → jaSessions.id (onDelete: 'cascade')`

Delete and lifecycle behavior.

- Cascading parent deletes affect: session_id: FK → jaSessions.id (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

ja_progress

Purpose. Derived Ja progress state by student and class.

Drizzle declaration. `jaProgress` in `backend/src/drizzle/schema/ja.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
studentId	student_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('student_id').notNull().references(() => users.id, { onDelete: 'cascade' })
classId	class_id	uuid	NOT NULL	No default	FK → classes.id (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
xpTotal	xp_total	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('xp_total').notNull().default(0)
streakDays	streak_days	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('streak_days').notNull().default(0)
sessionsCompleted	sessions_completed	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('sessions_completed').notNull().default(0)
lastActivityAt	last_activity_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('last_activity_at')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `pk: primaryKey({ columns: [table.studentId, table.classId] })`
- `classIdx: index('ja_progress_class_idx').on(table.classId)`

Relationship map.

- student_id: FK → users.id (onDelete: 'cascade')
- class_id: FK → classes.id (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: student_id: FK → users.id (onDelete: 'cascade'); class_id: FK → classes.id (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

ja_xp_ledger

Purpose. Append-only Ja experience-point awards with deduplication context.

Drizzle declaration. `jaXpLedger` in `backend/src/drizzle/schema/ja.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
studentId	student_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('student_id').notNull().references(() => users.id, { onDelete: 'cascade' })
classId	class_id	uuid	NOT NULL	No default	FK → classes.id (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
sessionId	session_id	uuid	NULL allowed	No default	FK → jaSessions.id (onDelete: 'set null')	uuid('session_id').references(() => jaSessions.id, { onDelete: 'set null', })
eventType	event_type	jaXpEventTypeEnum	NOT NULL	default('session_completion')	No PK, FK, or unique marker	jaXpEventTypeEnum('event_type').notNull().default('session_completion')
xpDelta	xp_delta	integer	NOT NULL	No default	No PK, FK, or unique marker	integer('xp_delta').notNull()
metadataJson	metadata_json	json	NULL allowed	No default	No PK, FK, or unique marker	json('metadata_json')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()

Indexes and table constraints.

- `studentClassIdx: index('ja_xp_ledger_student_class_idx').on( table.studentId, table.classId, )`
- `sessionEventUnique: uniqueIndex('ja_xp_ledger_session_event_unique').on( table.sessionId, table.eventType, )`

Relationship map.

- `student_id`: FK → `users.id` (onDelete: 'cascade')
- `class_id`: FK → `classes.id` (onDelete: 'cascade')
- `session_id`: FK → `jaSessions.id` (onDelete: 'set null')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `student_id`: FK → `users.id` (onDelete: 'cascade'); `class_id`: FK → `classes.id` (onDelete: 'cascade').
- Parent deletes set these references to NULL: `session_id`: FK → `jaSessions.id` (onDelete: 'set null').
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

ja_threads

Purpose. Durable Ja ask-conversation threads.

Drizzle declaration. `jaThreads` in `backend/src/drizzle/schema/ja.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
studentId	student_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('student_id').notNull().references(() => users.id, { onDelete: 'cascade' })
classId	class_id	uuid	NOT NULL	No default	FK → classes.id (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
title	title	text	NOT NULL	default('JA Ask Thread')	No PK, FK, or unique marker	text('title').notNull().default('JA Ask Thread')
status	status	jaThread-StatusEnum	NOT NULL	default('active')	No PK, FK, or unique marker	jaThread-StatusEnum('status').notNull().default('active')
lastMessageAt	last_message_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('last_message_at')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `studentClassIdx: index('ja_threads_student_class_idx').on( table.studentId, table.classId, )`
- `classStatusIdx: index('ja_threads_class_status_idx').on( table.classId, table.status, )`

Relationship map.

- student_id: FK → users.id (onDelete: 'cascade')
- class_id: FK → classes.id (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: student_id: FK → users.id (onDelete: 'cascade'); class_id: FK → classes.id (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

ja_thread_messages

Purpose. Messages belonging to a Ja ask thread.

Drizzle declaration. `jaThreadMessages` in `backend/src/drizzle/schema/ja.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
threadId	thread_id	uuid	NOT NULL	No default	FK → jaThreads.id (onDelete: 'cascade')	uuid('thread_id').notNull().references(() => jaThreads.id, { onDelete: 'cascade' })
role	role	jaThreadMessageRoleEnum	NOT NULL	No default	No PK, FK, or unique marker	

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
						jaThreadMessageRoleEnum('role').notNull()
content	content	text	NOT NULL	No default	No PK, FK, or unique marker	text('content').notNull()
citationsJson	citations_json	json	NULL allowed	No default	No PK, FK, or unique marker	json('citations_json')
quickAction	quick_action	text	NULL allowed	No default	No PK, FK, or unique marker	text('quick_action')
blocked	blocked	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('blocked').notNull().default(false)
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()

Indexes and table constraints.

- `threadCreatedAtIdx: index('ja_thread_messages_thread_created_at_idx').on( table.threadId, table.createdAt, )`

Relationship map.

- `thread_id: FK → jaThreads.id (onDelete: 'cascade')`

Delete and lifecycle behavior.

- Cascading parent deletes affect: `thread_id: FK → jaThreads.id (onDelete: 'cascade')`.
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

ja_guardrail_events

Purpose. Recorded Ja safety or policy guardrail events.

Drizzle declaration. `jaGuardrailEvents` in `backend/src/drizzle/schema/ja.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
studentId	student_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('student_id').notNull().references(() => users.id, { onDelete: 'cascade' })
classId	class_id	uuid	NOT NULL	No default	FK → classes.id (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
threadId	thread_id	uuid	NULL allowed	No default	FK → jaThreads.id (onDelete: 'set null',)	uuid('thread_id').references(() => jaThreads.id, { onDelete: 'set null', })
messageId	message_id	uuid	NULL allowed	No default	FK → jaThreadMessages.id (onDelete: 'set null',)	uuid('message_id').references(() => jaThreadMessages.id, { onDelete: 'set null', })
eventType	event_type	jaGuardrailEvent-TypeEnum	NOT NULL	No default	No PK, FK, or unique marker	

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
						jaGuardrailEvent- TypeEnum('event_type').notNull()
payloadJson	payload_json	json	NULL allowed	No default	No PK, FK, or unique marker	json('payload_json')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').not- Null().defaultNow()

Indexes and table constraints.

- `studentClassIdx: index('ja_guardrail_events_student_class_idx').on( table.studentId, table.classId, )`
- `threadCreatedAtIdx: index('ja_guardrail_events_thread_created_at_idx').on( table.threadId, table.createdAt, )`

Relationship map.

- `student_id`: FK → `users.id` (onDelete: 'cascade')
- `class_id`: FK → `classes.id` (onDelete: 'cascade')
- `thread_id`: FK → `jaThreads.id` (onDelete: 'set null',)
- `message_id`: FK → `jaThreadMessages.id` (onDelete: 'set null',)

Delete and lifecycle behavior.

- Cascading parent deletes affect: `student_id`: FK → `users.id` (onDelete: 'cascade'); `class_id`: FK → `classes.id` (onDelete: 'cascade').
- Parent deletes set these references to NULL: `thread_id`: FK → `jaThreads.id` (onDelete: 'set null',); `message_id`: FK → `jaThreadMessages.id` (onDelete: 'set null',).
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

Source file: backend/src/drizzle/schema/lxp.schema.ts

This source file declares 12 active tables.

intervention_cases

Purpose. Teacher-governed LXP intervention cases for at-risk students.

Drizzle declaration. `interventionCases` in `backend/src/drizzle/schema/lxp.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
classId	class_id	uuid	NOT NULL	No default	FK → <code>classes.id</code> (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
studentId	student_id	uuid	NOT NULL	No default	FK → <code>users.id</code> (onDelete: 'cascade')	uuid('student_id').notNull().references(() => users.id, { onDelete: 'cascade' })
status	status	interventionCase- StatusEnum	NOT NULL	default('pending')	No PK, FK, or unique marker	interventionCase- StatusEnum('status').not- Null().default('pending')

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
triggerSource	trigger_source	text	NOT NULL	default('performance_event')	No PK, FK, or unique marker	text('trigger_source').notNull().default('performance_event')
triggerScore	trigger_score	numeric	NULL allowed	No default	No PK, FK, or unique marker	numeric('trigger_score', { precision: 6, scale: 3 })
thresholdApplied	threshold_applied	numeric	NOT NULL	No default	No PK, FK, or unique marker	numeric('threshold_applied', { precision: 6, scale: 3 }).notNull()
note	note	text	NULL allowed	No default	No PK, FK, or unique marker	text('note')
openedAt	opened_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('opened_at').notNull().defaultNow()
closedAt	closed_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('closed_at')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `classStudentStatusIdx: index( 'intervention_cases_class_student_status_idx', ).on(table.classId, table.studentId, table.status)`
- `studentStatusIdx: index('intervention_cases_student_status_idx').on( table.studentId, table.status, )`
- `classStatusIdx: index('intervention_cases_class_status_idx').on( table.classId, table.status, )`

Relationship map.

- class_id: FK → classes.id (onDelete: 'cascade')
- student_id: FK → users.id (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: class_id: FK → classes.id (onDelete: 'cascade'); student_id: FK → users.id (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

intervention_assignments

Purpose. Ordered lesson, assessment, or generated-content checkpoints assigned within an intervention.

Drizzle declaration. `interventionAssignments` in `backend/src/drizzle/schema/lxp.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
caseId	case_id	uuid	NOT NULL	No default	FK → intervention-Cases.id (onDelete: 'cascade')	uuid('case_id').notNull().references(() => intervention-Cases.id, { onDelete: 'cascade' })

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
assignmentType	assignment_type	IxpAssignment-TypeEnum	NOT NULL	No default	No PK, FK, or unique marker	IxpAssignment-TypeEnum('assignment_type').notNull()
lessonId	lesson_id	uuid	NULL allowed	No default	FK → lessons.id (onDelete: 'set null',)	uuid('lesson_id').references(() => lessons.id, { onDelete: 'set null', })
assessmentId	assessment_id	uuid	NULL allowed	No default	FK → assessments.id (onDelete: 'set null',)	uuid('assessment_id').references(() => assessments.id, { onDelete: 'set null', })
generatedRemedial-LessonId	generated_remedial_lesson_id	uuid	NULL allowed	No default	No PK, FK, or unique marker	uuid('generated_remedial_lesson_id')
generatedGuided-AssessmentId	generated_guided_assessment_id	uuid	NULL allowed	No default	No PK, FK, or unique marker	uuid('generated_guided_assessment_id')
checkpointLabel	checkpoint_label	text	NOT NULL	No default	No PK, FK, or unique marker	text('checkpoint_label').notNull()
orderIndex	order_index	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('order_index').notNull().default(0)
isCompleted	is_completed	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('is_completed').notNull().default(false)
completedAt	completed_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('completed_at')
xpAwarded	xp_awarded	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('xp_awarded').notNull().default(0)
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `caseOrderIdx: index('intervention_assignments_case_order_idx').on( table.caseId, table.orderIndex, )`
- `caseCompletedIdx: index('intervention_assignments_case_completed_idx').on( table.caseId, table.isCompleted, )`
- `lessonIdx: index('intervention_assignments_lesson_idx').on(table.lessonId)`
- `assessmentIdx: index('intervention_assignments_assessment_idx').on( table.assessmentId, )`
- `generatedLessonIdx: index( 'intervention_assignments_generated_lesson_idx', ).on(table.generatedRemedialLessonId)`
- `generatedAssessmentIdx: index( 'intervention_assignments_generated_assessment_idx', ).on(table.generatedGuidedAssessmentId)`

Relationship map.

- `case_id`: FK → `interventionCases.id` (onDelete: 'cascade')
- `lesson_id`: FK → `lessons.id` (onDelete: 'set null',)
- `assessment_id`: FK → `assessments.id` (onDelete: 'set null',)

Delete and lifecycle behavior.

- Cascading parent deletes affect: `case_id`: FK → `interventionCases.id` (onDelete: 'cascade').

- Parent deletes set these references to NULL: lesson_id: FK → lessons.id (onDelete: 'set null',); assessment_id: FK → assessments.id (onDelete: 'set null',).
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

lxp_generated_remedial_lessons

Purpose. AI-produced remedial lesson drafts with approval and publication lifecycle.

Drizzle declaration. `generatedRemedialLessons` in `backend/src/drizzle/schema/lxp.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
caseId	case_id	uuid	NOT NULL	No default	FK → intervention-Cases.id (onDelete: 'cascade')	uuid('case_id').notNull().references(() => intervention-Cases.id, { onDelete: 'cascade' })
classId	class_id	uuid	NOT NULL	No default	FK → classes.id (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
studentId	student_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('student_id').notNull().references(() => users.id, { onDelete: 'cascade' })
approvalStatus	approval_status	lxpGeneratedArtifactStatusEnum	NOT NULL	default('draft')	No PK, FK, or unique marker	lxpGeneratedArtifactStatusEnum('approval_status').notNull().default('draft')
title	title	text	NOT NULL	No default	No PK, FK, or unique marker	text('title').notNull()
summary	summary	text	NULL allowed	No default	No PK, FK, or unique marker	text('summary')
lessonBody	lesson_body	text	NOT NULL	No default	No PK, FK, or unique marker	text('lesson_body').notNull()
weakConcepts	weak_concepts	json	NOT NULL	No default	No PK, FK, or unique marker	json('weak_concepts').notNull()
sourceLessonIds	source_lesson_ids	json	NOT NULL	No default	No PK, FK, or unique marker	json('source_lesson_ids').notNull()
sourceReferences	source_references	json	NOT NULL	No default	No PK, FK, or unique marker	json('source_references').notNull()
approvedBy	approved_by	uuid	NULL allowed	No default	FK → users.id (onDelete: 'set null',)	uuid('approved_by').references(() => users.id, { onDelete: 'set null', })
approvedAt	approved_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('approved_at')
rejectedAt	rejected_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('rejected_at')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `caseIdx: index('lxp_generated_remedial_lessons_case_idx').on(table.caseId)`

- `classIdx: index('lxp_generated_remedial_lessons_class_idx').on( table.classId, )`
- `studentIdx: index('lxp_generated_remedial_lessons_student_idx').on( table.studentId, )`
- `statusIdx: index('lxp_generated_remedial_lessons_status_idx').on( table.approvalStatus, )`

Relationship map.

- `case_id`: FK → `interventionCases.id` (onDelete: 'cascade')
- `class_id`: FK → `classes.id` (onDelete: 'cascade')
- `student_id`: FK → `users.id` (onDelete: 'cascade')
- `approved_by`: FK → `users.id` (onDelete: 'set null',)

Delete and lifecycle behavior.

- Cascading parent deletes affect: `case_id`: FK → `interventionCases.id` (onDelete: 'cascade'); `class_id`: FK → `classes.id` (onDelete: 'cascade'); `student_id`: FK → `users.id` (onDelete: 'cascade').
- Parent deletes set these references to NULL: `approved_by`: FK → `users.id` (onDelete: 'set null',).
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

lxp_generated_guided_assessments

Purpose. AI-produced guided assessment drafts governed by teacher approval.

Drizzle declaration. `generatedGuidedAssessments` in `backend/src/drizzle/schema/lxp.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
<code>id</code>	<code>id</code>	<code>uuid</code>	NOT NULL	<code>defaultRandom()</code>	PRIMARY KEY	<code>uuid('id').primaryKey().defaultRandom()</code>
<code>caseId</code>	<code>case_id</code>	<code>uuid</code>	NOT NULL	No default	FK → <code>interventionCases.id</code> (onDelete: 'cascade')	<code>uuid('case_id').notNull().references(() => interventionCases.id, { onDelete: 'cascade' })</code>
<code>classId</code>	<code>class_id</code>	<code>uuid</code>	NOT NULL	No default	FK → <code>classes.id</code> (onDelete: 'cascade')	<code>uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })</code>
<code>studentId</code>	<code>student_id</code>	<code>uuid</code>	NOT NULL	No default	FK → <code>users.id</code> (onDelete: 'cascade')	<code>uuid('student_id').notNull().references(() => users.id, { onDelete: 'cascade' })</code>
<code>approvalStatus</code>	<code>approval_status</code>	<code>IxpGeneratedArtifactStatusEnum</code>	NOT NULL	<code>default('draft')</code>	No PK, FK, or unique marker	<code>IxpGeneratedArtifactStatusEnum('approval_status').notNull().default('draft')</code>
<code>sourceAssessmentId</code>	<code>source_assessment_id</code>	<code>uuid</code>	NULL allowed	No default	FK → <code>assessments.id</code> (onDelete: 'set null')	<code>uuid('source_assessment_id').references(() => assessments.id, { onDelete: 'set null' },)</code>
<code>title</code>	<code>title</code>	<code>text</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>text('title').notNull()</code>
<code>description</code>	<code>description</code>	<code>text</code>	NULL allowed	No default	No PK, FK, or unique marker	<code>text('description')</code>
<code>weakConcepts</code>	<code>weak_concepts</code>	<code>json</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>json('weak_concepts').notNull()</code>
<code>sourceReferences</code>	<code>source_references</code>	<code>json</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>json('source_references').notNull()</code>
<code>questions</code>	<code>questions</code>	<code>json</code>	NOT NULL	No default		

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
					No PK, FK, or unique marker	json('questions').notNull()
formativeSummary	formative_summary	text	NULL allowed	No default	No PK, FK, or unique marker	text('formative_summary')
approvedBy	approved_by	uuid	NULL allowed	No default	FK → users.id (onDelete: 'set null')	uuid('approved_by').references(() => users.id, { onDelete: 'set null', })
approvedAt	approved_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('approved_at')
rejectedAt	rejected_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('rejected_at')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `caseIdx: index('lxp_generated_guided_assessments_case_idx').on( table.caseId, )`
- `classIdx: index('lxp_generated_guided_assessments_class_idx').on( table.classId, )`
- `studentIdx: index('lxp_generated_guided_assessments_student_idx').on( table.studentId, )`
- `statusIdx: index('lxp_generated_guided_assessments_status_idx').on( table.approvalStatus, )`

Relationship map.

- `case_id`: FK → `interventionCases.id` (onDelete: 'cascade')
- `class_id`: FK → `classes.id` (onDelete: 'cascade')
- `student_id`: FK → `users.id` (onDelete: 'cascade')
- `source_assessment_id`: FK → `assessments.id` (onDelete: 'set null')
- `approved_by`: FK → `users.id` (onDelete: 'set null')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `case_id`: FK → `interventionCases.id` (onDelete: 'cascade'); `class_id`: FK → `classes.id` (onDelete: 'cascade'); `student_id`: FK → `users.id` (onDelete: 'cascade').
- Parent deletes set these references to NULL: `source_assessment_id`: FK → `assessments.id` (onDelete: 'set null'); `approved_by`: FK → `users.id` (onDelete: 'set null').
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

lxp_generated_guided_assessment_attempts

Purpose. Student attempts against approved guided assessments.

Drizzle declaration. `generatedGuidedAssessmentAttempts` in `backend/src/drizzle/schema/lxp.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
guidedAssessmentId	guided_assessment_id	uuid	NOT NULL	No default	FK → generatedGuidedAssessments.id (onDelete: 'cascade')	uuid('guided_assessment_id').notNull().references(() => generatedGuidedAssessments.id,

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
						{ onDelete: 'cascade' }
caseId	case_id	uuid	NOT NULL	No default	FK → intervention-Cases.id (onDelete: 'cascade')	uuid('case_id').notNull().references(() => intervention-Cases.id, { onDelete: 'cascade' })
classId	class_id	uuid	NOT NULL	No default	FK → classes.id (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
studentId	student_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('student_id').notNull().references(() => users.id, { onDelete: 'cascade' })
assignmentId	assignment_id	uuid	NOT NULL	No default	FK → interventionAssignments.id (onDelete: 'cascade')	uuid('assignment_id').notNull().references(() => interventionAssignments.id, { onDelete: 'cascade' })
attemptNumber	attempt_number	integer	NOT NULL	default(1)	No PK, FK, or unique marker	integer('attempt_number').notNull().default(1)
status	status	IxpGuidedAttemptStatusEnum	NOT NULL	default('in_progress')	No PK, FK, or unique marker	IxpGuidedAttemptStatusEnum('status').notNull().default('in_progress')
currentQuestionIndex	current_question_index	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('current_question_index').notNull().default(0)
responses	responses	json	NOT NULL	default([])	No PK, FK, or unique marker	json('responses').notNull().default([])
hintUsage	hint_usage	json	NOT NULL	default([])	No PK, FK, or unique marker	json('hint_usage').notNull().default([])
score	score	integer	NULL allowed	No default	No PK, FK, or unique marker	integer('score')
totalQuestions	total_questions	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('total_questions').notNull().default(0)
correctCount	correct_count	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('correct_count').notNull().default(0)
formativeSummary	formative_summary	json	NULL allowed	No default	No PK, FK, or unique marker	json('formative_summary')
startedAt	started_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('started_at').notNull().defaultNow()
submittedAt	submitted_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('submitted_at')
lastActivityAt	last_activity_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('last_activity_at')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `guidedAssessmentIdx: index('lxp_generated_guided_attempts_guided_assessment_idx', ).on(table.guidedAssessmentId)`
- `caseStudentIdx: index('lxp_generated_guided_attempts_case_student_idx').on( table.caseId, table.studentId, )`
- `assignmentIdx: index('lxp_generated_guided_attempts_assignment_idx').on( table.assignmentId, table.studentId, )`
- `assignmentAttemptIdx: uniqueIndex('lxp_generated_guided_attempts_assignment_attempt_unique', ).on(table.assignmentId, table.studentId, table.attemptNumber)`

Relationship map.

- `guided_assessment_id`: FK → `generatedGuidedAssessments.id` (onDelete: 'cascade')
- `case_id`: FK → `interventionCases.id` (onDelete: 'cascade')
- `class_id`: FK → `classes.id` (onDelete: 'cascade')
- `student_id`: FK → `users.id` (onDelete: 'cascade')
- `assignment_id`: FK → `interventionAssignments.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `guided_assessment_id`: FK → `generatedGuidedAssessments.id` (onDelete: 'cascade'); `case_id`: FK → `interventionCases.id` (onDelete: 'cascade'); `class_id`: FK → `classes.id` (onDelete: 'cascade'); `student_id`: FK → `users.id` (onDelete: 'cascade'); `assignment_id`: FK → `interventionAssignments.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

lxp_progress

Purpose. Derived progress through intervention checkpoints.

Drizzle declaration. `lxpProgress` in `backend/src/drizzle/schema/lxp.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
<code>studentId</code>	<code>student_id</code>	<code>uuid</code>	NOT NULL	No default	FK → <code>users.id</code> (onDelete: 'cascade')	<code>uuid('student_id').notNull().references(() => users.id, { onDelete: 'cascade' })</code>
<code>classId</code>	<code>class_id</code>	<code>uuid</code>	NOT NULL	No default	FK → <code>classes.id</code> (onDelete: 'cascade')	<code>uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })</code>
<code>xpTotal</code>	<code>xp_total</code>	<code>integer</code>	NOT NULL	<code>default(0)</code>	No PK, FK, or unique marker	<code>integer('xp_total').notNull().default(0)</code>
<code>streakDays</code>	<code>streak_days</code>	<code>integer</code>	NOT NULL	<code>default(0)</code>	No PK, FK, or unique marker	<code>integer('streak_days').notNull().default(0)</code>
<code>checkpointsCompleted</code>	<code>checkpoints_completed</code>	<code>integer</code>	NOT NULL	<code>default(0)</code>	No PK, FK, or unique marker	<code>integer('checkpoints_completed').notNull().default(0)</code>
<code>lastActivityAt</code>	<code>last_activity_at</code>	<code>timestamp</code>	NULL allowed	No default	No PK, FK, or unique marker	<code>timestamp('last_activity_at')</code>
<code>createdAt</code>	<code>created_at</code>	<code>timestamp</code>	NOT NULL	<code>defaultNow()</code>		

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
					No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `pk: primaryKey({ columns: [table.studentId, table.classId] })`
- `classIdx: index('lxp_progress_class_idx').on(table.classId)`

Relationship map.

- `student_id`: FK → `users.id` (onDelete: 'cascade')
- `class_id`: FK → `classes.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `student_id`: FK → `users.id` (onDelete: 'cascade'); `class_id`: FK → `classes.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

system_evaluations

Purpose. Submitted evaluation answers and scores for system modules or campaigns.

Drizzle declaration. `systemEvaluations` in `backend/src/drizzle/schema/lxp.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
campaignId	campaign_id	uuid	NULL allowed	No default	FK → systemEvaluationCampaigns.id (onDelete: 'set null')	uuid('campaign_id').references(() => systemEvaluationCampaigns.id, { onDelete: 'set null' },)
submittedBy	submitted_by	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('submitted_by').notNull().references(() => users.id, { onDelete: 'cascade' })
targetModule	target_module	systemEvaluationTargetEnum	NOT NULL	No default	No PK, FK, or unique marker	systemEvaluationTargetEnum('target_module').notNull()
usabilityScore	usability_score	integer	NOT NULL	No default	No PK, FK, or unique marker	integer('usability_score').notNull()
functionalityScore	functionality_score	integer	NOT NULL	No default	No PK, FK, or unique marker	integer('functionality_score').notNull()
performanceScore	performance_score	integer	NOT NULL	No default	No PK, FK, or unique marker	integer('performance_score').notNull()
satisfactionScore	satisfaction_score	integer	NOT NULL	No default	No PK, FK, or unique marker	integer('satisfaction_score').notNull()
overallScore	overall_score	integer	NULL allowed	No default		

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
					No PK, FK, or unique marker	integer('overall_score')
questionRatingsJson	question_ratings_json	json	NULL allowed	No default	No PK, FK, or unique marker	json('question_ratings_json')
feedback	feedback	text	NULL allowed	No default	No PK, FK, or unique marker	text('feedback')
aiContextMetadata	ai_context_metadata	json	NULL allowed	No default	No PK, FK, or unique marker	json('ai_context_metadata')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()

Indexes and table constraints.

- `moduleCreatedIdx: index('system_evaluations_module_created_idx').on( table.targetModule, table.createdAt, )`
- `userIdx: index('system_evaluations_submitted_by_idx').on(table.submittedBy)`
- `campaignIdx: index('system_evaluations_campaign_idx').on(table.campaignId)`

Relationship map.

- `campaign_id`: FK → `systemEvaluationCampaigns.id` (onDelete: 'set null')
- `submitted_by`: FK → `users.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `submitted_by`: FK → `users.id` (onDelete: 'cascade').
- Parent deletes set these references to NULL: `campaign_id`: FK → `systemEvaluationCampaigns.id` (onDelete: 'set null').
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

system_evaluation_campaigns

Purpose. Administrative evaluation campaigns, audience, class scope, and lifecycle.

Drizzle declaration. `systemEvaluationCampaigns` in `backend/src/drizzle/schema/lxp.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
createdBy	created_by	uuid	NOT NULL	No default	FK → <code>users.id</code> (onDelete: 'cascade')	uuid('created_by').notNull().references(() => users.id, { onDelete: 'cascade' })
formType	form_type	systemEvaluationFormTypeEnum	NOT NULL	No default	No PK, FK, or unique marker	systemEvaluationFormTypeEnum('form_type').notNull()
targetModule	target_module	systemEvaluationTargetEnum	NOT NULL	No default	No PK, FK, or unique marker	systemEvaluationTargetEnum('target_module').notNull()
audienceRole	audience_role	systemEvaluationAudienceRoleEnum	NOT NULL	No default	No PK, FK, or unique marker	systemEvaluationAudienceRoleEnum('audience_role').notNull()
classId	class_id	uuid	NULL allowed	No default	FK → <code>classes.id</code> (onDelete: 'cascade',)	uuid('class_id').references(() => classes.id,

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
						{ onDelete: 'cascade', })
title	title	text	NOT NULL	No default	No PK, FK, or unique marker	text('title').notNull()
startsAt	starts_at	timestamp	NOT NULL	No default	No PK, FK, or unique marker	timestamp('starts_at').notNull()
endsAt	ends_at	timestamp	NOT NULL	No default	No PK, FK, or unique marker	timestamp('ends_at').notNull()
status	status	systemEvaluation-Campaign-StatusEnum	NOT NULL	default('draft')	No PK, FK, or unique marker	systemEvaluation-Campaign-StatusEnum('status').notNull().default('draft')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `statusIdx: index('system_evaluation_campaigns_status_idx').on(table.status)`
- `formAudienceIdx: index('system_evaluation_campaigns_form_audience_idx').on( table.formType, table.audienceRole, )`
- `classIdx: index('system_evaluation_campaigns_class_idx').on(table.classId)`
- `createdByIdx: index('system_evaluation_campaigns_created_by_idx').on( table.createdBy, )`

Relationship map.

- `created_by`: FK → `users.id` (onDelete: 'cascade')
- `class_id`: FK → `classes.id` (onDelete: 'cascade',)

Delete and lifecycle behavior.

- Cascading parent deletes affect: `created_by`: FK → `users.id` (onDelete: 'cascade'); `class_id`: FK → `classes.id` (onDelete: 'cascade',).
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

system_evaluation_assignments

Purpose. Per-respondent campaign assignments and completion status.

Drizzle declaration. `systemEvaluationAssignments` in `backend/src/drizzle/schema/lxp.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
campaignId	campaign_id	uuid	NOT NULL	No default	FK → systemEvaluationCampaigns.id (onDelete: 'cascade')	uuid('campaign_id').notNull().references(() => systemEvaluationCampaigns.id, { onDelete: 'cascade' })
respondentId	respondent_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('respondent_id').notNull().references(() => users.id,

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
						{ onDelete: 'cascade' }
respondentRole	respondent_role	systemEvaluationAudienceRoleEnum	NOT NULL	No default	No PK, FK, or unique marker	systemEvaluationAudienceRoleEnum('respondent_role').notNull()
status	status	systemEvaluationAssignmentStatusEnum	NOT NULL	default('pending')	No PK, FK, or unique marker	systemEvaluationAssignmentStatusEnum('status').notNull().default('pending')
submittedEvaluationId	submitted_evaluation_id	uuid	NULL allowed	No default	FK → systemEvaluations.id (onDelete: 'set null')	uuid('submitted_evaluation_id').references(() => systemEvaluations.id, { onDelete: 'set null' },)
submittedAt	submitted_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('submitted_at')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `campaignRespondentUnique: uniqueIndex( 'system_evaluation_assignments_campaign_respondent_unique', ).on( table.campaignId, table.respondentId )`
- `respondentIdx: index('system_evaluation_assignments_respondent_idx').on( table.respondentId, table.status, )`
- `campaignIdx: index('system_evaluation_assignments_campaign_idx').on( table.campaignId, )`

Relationship map.

- `campaign_id`: FK → `systemEvaluationCampaigns.id` (onDelete: 'cascade')
- `respondent_id`: FK → `users.id` (onDelete: 'cascade')
- `submitted_evaluation_id`: FK → `systemEvaluations.id` (onDelete: 'set null')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `campaign_id`: FK → `systemEvaluationCampaigns.id` (onDelete: 'cascade'); `respondent_id`: FK → `users.id` (onDelete: 'cascade').
- Parent deletes set these references to NULL: `submitted_evaluation_id`: FK → `systemEvaluations.id` (onDelete: 'set null').
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

class_ai_policies

Purpose. Teacher-controlled class policy for AI availability and source scope.

Drizzle declaration. `classAiPolicies` in `backend/src/drizzle/schema/lxp.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
classId	class_id	uuid	NOT NULL	No default	FK → <code>classes.id</code> (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id,

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
						{ onDelete: 'cascade' }
mentorExplainEnabled	mentor_explain_enabled	boolean	NOT NULL	default(true)	No PK, FK, or unique marker	boolean('mentor_explain_enabled').notNull().default(true)
maxFollowUpTurns	max_follow_up_turns	integer	NOT NULL	default(3)	No PK, FK, or unique marker	integer('max_follow_up_turns').notNull().default(3)
sourceScope	source_scope	aiPolicySourceScopeEnum	NOT NULL	default('class_materials')	No PK, FK, or unique marker	aiPolicySourceScopeEnum('source_scope').notNull().default('class_materials')
strictGrounding	strict_grounding	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('strict_grounding').notNull().default(false)
updatedAt	updated_by	uuid	NULL allowed	No default	FK → users.id (onDelete: 'set null',)	uuid('updated_by').references(() => users.id, { onDelete: 'set null', })
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `pk: primaryKey({ columns: [table.classId] })`
- `updatedAtIdx: index('class_ai_policies_updated_by_idx').on(table.updatedBy)`

Relationship map.

- `class_id`: FK → `classes.id` (onDelete: 'cascade')
- `updated_by`: FK → `users.id` (onDelete: 'set null',)

Delete and lifecycle behavior.

- Cascading parent deletes affect: `class_id`: FK → `classes.id` (onDelete: 'cascade').
- Parent deletes set these references to NULL: `updated_by`: FK → `users.id` (onDelete: 'set null',).
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

teacher_evaluation_windows

Purpose. Class and grading-period windows during which students may evaluate teachers.

Drizzle declaration. `teacherEvaluationWindows` in `backend/src/drizzle/schema/lxp.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
classId	class_id	uuid	NOT NULL	No default	FK → <code>classes.id</code> (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
teacherId	teacher_id	uuid	NOT NULL	No default	FK → <code>users.id</code> (onDelete: 'cascade')	uuid('teacher_id').notNull().references(() => users.id,

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
						{ onDelete: 'cascade' }
schoolYear	school_year	text	NOT NULL	No default	No PK, FK, or unique marker	text('school_year').notNull()
gradingPeriod	grading_period	gradingPeriodEnum	NOT NULL	No default	No PK, FK, or unique marker	gradingPeriodEnum('grading_period').notNull()
evaluationType	evaluation_type	teacherEvaluationTypeEnum	NOT NULL	No default	No PK, FK, or unique marker	teacherEvaluationTypeEnum('evaluation_type').notNull()
status	status	teacherEvaluationWindowStatusEnum	NOT NULL	default('active')	No PK, FK, or unique marker	teacherEvaluationWindowStatusEnum('status').notNull().default('active')
eligibleCount	eligible_count	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('eligible_count').notNull().default(0)
opensAt	opens_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('opens_at').notNull().defaultNow()
closesAt	closes_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('closes_at')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `classPeriodTypeUnique`: `uniqueIndex( 'teacher_evaluation_windows_class_period_type_unique', ).on( table.classId, table.schoolYear, table.gradingPeriod, table.evaluationType, )`
- `teacherIdx`: `index('teacher_evaluation_windows_teacher_idx').on( table.teacherId, )`
- `classIdx`: `index('teacher_evaluation_windows_class_idx').on(table.classId)`
- `periodTypeIdx`: `index('teacher_evaluation_windows_period_type_idx').on( table.gradingPeriod, table.evaluationType, )`

Relationship map.

- `class_id`: FK → `classes.id` (onDelete: 'cascade')
- `teacher_id`: FK → `users.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `class_id`: FK → `classes.id` (onDelete: 'cascade'); `teacher_id`: FK → `users.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

teacher_evaluation_submissions

Purpose. Student teacher-evaluation responses with uniqueness by class and period.

Drizzle declaration. `teacherEvaluationSubmissions` in `backend/src/drizzle/schema/lxp.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
windowId	window_id	uuid	NOT NULL	No default	FK → teacherEvaluationWindows.id (onDelete: 'cascade')	uuid('window_id').notNull().references(() => teacherEvaluationWindows.id, { onDelete: 'cascade' })
classId	class_id	uuid	NOT NULL	No default	FK → classes.id (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
teacherId	teacher_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('teacher_id').notNull().references(() => users.id, { onDelete: 'cascade' })
studentId	student_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('student_id').notNull().references(() => users.id, { onDelete: 'cascade' })
schoolYear	school_year	text	NOT NULL	No default	No PK, FK, or unique marker	text('school_year').notNull()
gradingPeriod	grading_period	gradingPeriodEnum	NOT NULL	No default	No PK, FK, or unique marker	gradingPeriodEnum('grading_period').notNull()
evaluationType	evaluation_type	teacherEvaluationTypeEnum	NOT NULL	No default	No PK, FK, or unique marker	teacherEvaluationTypeEnum('evaluation_type').notNull()
ratingsJson	ratings_json	json	NOT NULL	No default	No PK, FK, or unique marker	json('ratings_json').notNull()
comment	comment	text	NULL allowed	No default	No PK, FK, or unique marker	text('comment')
submittedAt	submitted_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('submitted_at').notNull().defaultNow()
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `studentScopeUnique: uniqueIndex( 'teacher_evaluation_submissions_student_scope_unique', ).on( table.studentId, table.classId, table.schoolYear, table.gradingPeriod, table.evaluationType, )`
- `windowIdx: index('teacher_evaluation_submissions_window_idx').on( table.windowId, )`
- `teacherIdx: index('teacher_evaluation_submissions_teacher_idx').on( table.teacherId, )`
- `classPeriodIdx: index('teacher_evaluation_submissions_class_period_idx').on( table.classId, table.gradingPeriod, table.evaluationType, )`

Relationship map.

- `window_id: FK → teacherEvaluationWindows.id (onDelete: 'cascade')`
- `class_id: FK → classes.id (onDelete: 'cascade')`
- `teacher_id: FK → users.id (onDelete: 'cascade')`
- `student_id: FK → users.id (onDelete: 'cascade')`

Delete and lifecycle behavior.

- Cascading parent deletes affect: `window_id`: FK → `teacherEvaluationWindows.id` (onDelete: 'cascade'); `class_id`: FK → `classes.id` (onDelete: 'cascade'); `teacher_id`: FK → `users.id` (onDelete: 'cascade'); `student_id`: FK → `users.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

Source file: `backend/src/drizzle/schema/otp.schema.ts`

This source file declares 1 active table.

otp_verifications

Purpose. One-time verification challenges, purpose, attempt count, expiry, and consumption state.

Drizzle declaration. `otpVerifications` in `backend/src/drizzle/schema/otp.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
<code>id</code>	<code>id</code>	<code>uuid</code>	NOT NULL	<code>defaultRandom()</code>	PRIMARY KEY	<code>uuid('id').primaryKey().defaultRandom()</code>
<code>userId</code>	<code>user_id</code>	<code>uuid</code>	NOT NULL	No default	FK → <code>users.id</code> (onDelete: 'cascade')	<code>uuid('user_id').notNull().references(() => users.id, { onDelete: 'cascade' })</code>
<code>codeHash</code>	<code>code_hash</code>	<code>text</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>text('code_hash').notNull()</code>
<code>purpose</code>	<code>purpose</code>	<code>otpPurposeEnum</code>	NOT NULL	<code>default('email_verification')</code>	No PK, FK, or unique marker	<code>otpPurposeEnum('purpose').notNull().default('email_verification')</code>
<code>expiresAt</code>	<code>expires_at</code>	<code>timestamp</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>timestamp('expires_at').notNull()</code>
<code>isUsed</code>	<code>is_used</code>	<code>boolean</code>	NOT NULL	<code>default(false)</code>	No PK, FK, or unique marker	<code>boolean('is_used').notNull().default(false)</code>
<code>usedAt</code>	<code>used_at</code>	<code>timestamp</code>	NULL allowed	No default	No PK, FK, or unique marker	<code>timestamp('used_at')</code>
<code>createdAt</code>	<code>created_at</code>	<code>timestamp</code>	NOT NULL	<code>defaultNow()</code>	No PK, FK, or unique marker	<code>timestamp('created_at').notNull().defaultNow()</code>
<code>attemptCount</code>	<code>attempt_count</code>	<code>integer</code>	NOT NULL	<code>default(0)</code>	No PK, FK, or unique marker	<code>integer('attempt_count').notNull().default(0)</code>

Indexes and table constraints.

- `userIdIdx: index('otp_verifications_user_id_idx').on(table.userId)`
- `expiresAtIdx: index('otp_verifications_expires_at_idx').on(table.expiresAt)`
- `purposeIdx: index('otp_verifications_purpose_idx').on(table.purpose)`
- `isUsedIdx: index('otp_verifications_is_used_idx').on(table.isUsed)`
- `activeUniqueIdx: uniqueIndex('otp_active_unique_idx').on(table.userId, table.purpose).where(sql.isUsed = false)`

Relationship map.

- user_id: FK → users.id (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: user_id: FK → users.id (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

Source file: backend/src/drizzle/schema/performance.schema.ts

This source file declares 2 active tables.

performance_snapshots

Purpose. Current derived student performance and risk snapshot per class.

Drizzle declaration. `performanceSnapshots` in `backend/src/drizzle/schema/performance.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
classId	class_id	uuid	NOT NULL	No default	FK → classes.id (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
studentId	student_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('student_id').notNull().references(() => users.id, { onDelete: 'cascade' })
assessmentAverage	assessment_average	numeric	NULL allowed	No default	No PK, FK, or unique marker	numeric('assessment_average', { precision: 6, scale: 3 })
classRecordAverage	class_record_average	numeric	NULL allowed	No default	No PK, FK, or unique marker	numeric('class_record_average', { precision: 6, scale: 3 })
blendedScore	blended_score	numeric	NULL allowed	No default	No PK, FK, or unique marker	numeric('blended_score', { precision: 6, scale: 3 })
assessment-SampleSize	assessment_sample_size	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('assessment_sample_size').notNull().default(0)
classRecord-SampleSize	class_record_sample_size	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('class_record_sample_size').notNull().default(0)
hasData	has_data	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('has_data').notNull().default(false)
isAtRisk	is_at_risk	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('is_at_risk').notNull().default(false)
thresholdApplied	threshold_applied	numeric	NOT NULL	default('74')	No PK, FK, or unique marker	numeric('threshold_applied', { precision: 6,

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
						scale: 3, }) .notNull() .default('74')
lastComputedAt	last_computed_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('last_computed_at').notNull().defaultNow()
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `classStudentUnique: unique('performance_snapshots_class_student_unique').on( table.classId, table.studentId, )`
- `classRiskIdx: index('performance_snapshots_class_risk_idx').on( table.classId, table.isAtRisk, )`
- `classStudentIdx: index('performance_snapshots_class_student_idx').on( table.classId, table.studentId, )`
- `classIdx: index('performance_snapshots_class_idx').on(table.classId)`
- `studentIdx: index('performance_snapshots_student_idx').on(table.studentId)`

Relationship map.

- class_id: FK → classes.id (onDelete: 'cascade')
- student_id: FK → users.id (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: class_id: FK → classes.id (onDelete: 'cascade'); student_id: FK → users.id (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

performance_logs

Purpose. Historical performance recomputation and signal records.

Drizzle declaration. `performanceLogs` in `backend/src/drizzle/schema/performance.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
classId	class_id	uuid	NOT NULL	No default	FK → classes.id (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
studentId	student_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('student_id').notNull().references(() => users.id, { onDelete: 'cascade' })
previousIsAtRisk	previous_is_at_risk	boolean	NULL allowed	No default	No PK, FK, or unique marker	boolean('previous_is_at_risk')
currentIsAtRisk	current_is_at_risk	boolean	NOT NULL	No default	No PK, FK, or unique marker	boolean('current_is_at_risk').notNull()
assessmentAverage	assessment_average	numeric	NULL allowed	No default	No PK, FK, or unique marker	numeric('assessment-

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
						ment_average', { precision: 6, scale: 3, })
classRecordAverage	class_record_average	numeric	NULL allowed	No default	No PK, FK, or unique marker	numeric('class_record_average', { precision: 6, scale: 3, })
blendedScore	blended_score	numeric	NULL allowed	No default	No PK, FK, or unique marker	numeric('blended_score', { precision: 6, scale: 3, })
thresholdApplied	threshold_applied	numeric	NOT NULL	No default	No PK, FK, or unique marker	numeric('threshold_applied', { precision: 6, scale: 3, }).notNull()
triggerSource	trigger_source	text	NOT NULL	No default	No PK, FK, or unique marker	text('trigger_source').notNull()
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()

Indexes and table constraints.

- `classCreatedAtIdx: index('performance_logs_class_created_at_idx').on( table.classId, table.createdAt, )`
- `classStudentIdx: index('performance_logs_class_student_idx').on( table.classId, table.studentId, )`
- `studentCreatedAtIdx: index('performance_logs_student_created_at_idx').on( table.studentId, table.createdAt, )`

Relationship map.

- `class_id: FK → classes.id (onDelete: 'cascade')`
- `student_id: FK → users.id (onDelete: 'cascade')`

Delete and lifecycle behavior.

- Cascading parent deletes affect: `class_id: FK → classes.id (onDelete: 'cascade')`; `student_id: FK → users.id (onDelete: 'cascade')`.
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

Source file: backend/src/drizzle/schema/rag.schema.ts

This source file declares 5 active tables.

content_chunks

Purpose. Normalized retrieval chunks with source lineage, class and library scope, hashes, and metadata.

Drizzle declaration. `contentChunks` in `backend/src/drizzle/schema/rag.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
sourceType	source_type	contentSource-TypeEnum	NOT NULL	No default	No PK, FK, or unique marker	contentSource-TypeEnum('source_type').notNull()
sourceId	source_id	uuid	NOT NULL	No default		

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
					No PK, FK, or unique marker	uuid('source_id').notNull()
classId	class_id	uuid	NULL allowed	No default	FK → classes.id (onDelete: 'cascade',)	uuid('class_id').references(() => classes.id, { onDelete: 'cascade', })
libraryFileId	library_file_id	uuid	NULL allowed	No default	FK → uploaded-Files.id (onDelete: 'cascade',)	uuid('library_file_id').references(() => uploaded-Files.id, { onDelete: 'cascade', })
subjectKey	subject_key	librarySubjectKeyEnum	NULL allowed	No default	No PK, FK, or unique marker	librarySubjectKeyEnum('subject_key')
gradeLevel	grade_level	gradeLevelEnum	NULL allowed	No default	No PK, FK, or unique marker	gradeLevelEnum('grade_level')
lessonId	lesson_id	uuid	NULL allowed	No default	FK → lessons.id (onDelete: 'cascade',)	uuid('lesson_id').references(() => lessons.id, { onDelete: 'cascade', })
assessmentId	assessment_id	uuid	NULL allowed	No default	FK → assessments.id (onDelete: 'cascade',)	uuid('assessment_id').references(() => assessments.id, { onDelete: 'cascade', })
questionId	question_id	uuid	NULL allowed	No default	FK → assessmentQuestions.id (onDelete: 'cascade',)	uuid('question_id').references(() => assessmentQuestions.id, { onDelete: 'cascade', })
extractionId	extraction_id	uuid	NULL allowed	No default	FK → extractedModules.id (onDelete: 'cascade',)	uuid('extraction_id').references(() => extractedModules.id, { onDelete: 'cascade', })
chunkText	chunk_text	text	NOT NULL	No default	No PK, FK, or unique marker	text('chunk_text').notNull()
chunkOrder	chunk_order	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('chunk_order').notNull().default(0)
tokenCount	token_count	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('token_count').notNull().default(0)
contentHash	content_hash	text	NOT NULL	No default	No PK, FK, or unique marker	text('content_hash').notNull()
metadataJson	metadata_json	json	NULL allowed	No default	No PK, FK, or unique marker	json('metadata_json')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `classIdx: index('content_chunks_class_id_idx').on(table.classId)`
- `sourceIdx: index('content_chunks_source_type_source_id_idx').on( table.sourceType, table.sourceId, )`
- `lessonIdx: index('content_chunks_lesson_id_idx').on(table.lessonId)`
- `assessmentIdx: index('content_chunks_assessment_id_idx').on( table.assessmentId, )`
- `questionIdx: index('content_chunks_question_id_idx').on(table.questionId)`
- `extractionIdx: index('content_chunks_extraction_id_idx').on( table.extractionId, )`

- `libraryFileIdx: index('content_chunks_library_file_idx').on( table.sourceType, table.libraryFileId, table.subjectKey, table.gradeLevel, )`

Relationship map.

- `class_id`: FK → `classes.id` (onDelete: 'cascade',)
- `library_file_id`: FK → `uploadedFiles.id` (onDelete: 'cascade',)
- `lesson_id`: FK → `lessons.id` (onDelete: 'cascade',)
- `assessment_id`: FK → `assessments.id` (onDelete: 'cascade',)
- `question_id`: FK → `assessmentQuestions.id` (onDelete: 'cascade',)
- `extraction_id`: FK → `extractedModules.id` (onDelete: 'cascade',)

Delete and lifecycle behavior.

- Cascading parent deletes affect: `class_id`: FK → `classes.id` (onDelete: 'cascade',); `library_file_id`: FK → `uploadedFiles.id` (onDelete: 'cascade',); `lesson_id`: FK → `lessons.id` (onDelete: 'cascade',); `assessment_id`: FK → `assessments.id` (onDelete: 'cascade',); `question_id`: FK → `assessmentQuestions.id` (onDelete: 'cascade',); `extraction_id`: FK → `extractedModules.id` (onDelete: 'cascade',).
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

content_chunk_embeddings

Purpose. One 768-dimensional embedding per content chunk with model and generation time.

Drizzle declaration. `contentChunkEmbeddings` in `backend/src/drizzle/schema/rag.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
<code>chunkId</code>	<code>chunk_id</code>	<code>uuid</code>	NOT NULL	No default	PRIMARY KEY; FK → <code>contentChunks.id</code> (onDelete: 'cascade')	<code>uuid('chunk_id').primaryKey().references(() => contentChunks.id, { onDelete: 'cascade' })</code>
<code>embedding</code>	<code>embedding</code>	<code>vector(768)</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>vector('embedding').notNull()</code>
<code>embeddingModel</code>	<code>embedding_model</code>	<code>text</code>	NOT NULL	No default	No PK, FK, or unique marker	<code>text('embedding_model').notNull()</code>
<code>embeddedAt</code>	<code>embedded_at</code>	<code>timestamp</code>	NOT NULL	<code>defaultNow()</code>	No PK, FK, or unique marker	<code>timestamp('embedded_at').notNull().defaultNow()</code>

Indexes and table constraints.

- `modelIdx: index('content_chunk_embeddings_model_idx').on( table.embeddingModel, )`

Relationship map.

- `chunk_id`: FK → `contentChunks.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `chunk_id`: FK → `contentChunks.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

student_concept_mastery

Purpose. Derived per-student concept evidence and mastery indicators used for assistive retrieval and tutoring.

Drizzle declaration. `studentConceptMastery` in `backend/src/drizzle/schema/rag.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
studentId	student_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('student_id').notNull().references(() => users.id, { onDelete: 'cascade' })
classId	class_id	uuid	NOT NULL	No default	FK → classes.id (onDelete: 'cascade')	uuid('class_id').notNull().references(() => classes.id, { onDelete: 'cascade' })
conceptKey	concept_key	text	NOT NULL	No default	No PK, FK, or unique marker	text('concept_key').notNull()
evidenceCount	evidence_count	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('evidence_count').notNull().default(0)
errorCount	error_count	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('error_count').notNull().default(0)
masteryScore	mastery_score	integer	NOT NULL	default(0)	No PK, FK, or unique marker	integer('mastery_score').notNull().default(0)
lastSeenAt	last_seen_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('last_seen_at').notNull().defaultNow()
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `studentClassConceptIdx: uniqueIndex( 'student_concept_mastery_student_class_concept_idx', ).on(table.studentId, table.classId, table.conceptKey)`
- `classIdx: index('student_concept_mastery_class_id_idx').on(table.classId)`

Relationship map.

- `student_id`: FK → `users.id` (onDelete: 'cascade')
- `class_id`: FK → `classes.id` (onDelete: 'cascade')

Delete and lifecycle behavior.

- Cascading parent deletes affect: `student_id`: FK → `users.id` (onDelete: 'cascade'); `class_id`: FK → `classes.id` (onDelete: 'cascade').
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

ai_generation_jobs

Purpose. Durable teacher AI job state, filters, ownership, failure detail, and timestamps.

Drizzle declaration. `aiGenerationJobs` in `backend/src/drizzle/schema/rag.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
jobType	job_type	aiGenerationJobTypeEnum	NOT NULL	No default	No PK, FK, or unique marker	aiGenerationJobTypeEnum('job_type').notNull()
classId	class_id	uuid	NULL allowed	No default	FK → classes.id (onDelete: 'cascade',)	uuid('class_id').references(() => classes.id, { onDelete: 'cascade', })
teacherId	teacher_id	uuid	NULL allowed	No default	FK → users.id (onDelete: 'cascade',)	uuid('teacher_id').references(() => users.id, { onDelete: 'cascade', })
status	status	aiGeneration-StatusEnum	NOT NULL	default('pending')	No PK, FK, or unique marker	aiGeneration-StatusEnum('status').notNull().default('pending')
sourceFilters	source_filters	json	NULL allowed	No default	No PK, FK, or unique marker	json('source_filters')
errorMessage	error_message	text	NULL allowed	No default	No PK, FK, or unique marker	text('error_message')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `classIdx: index('ai_generation_jobs_class_id_idx').on(table.classId)`
- `teacherIdx: index('ai_generation_jobs_teacher_id_idx').on(table.teacherId)`
- `statusIdx: index('ai_generation_jobs_status_idx').on(table.status)`

Relationship map.

- `class_id`: FK → `classes.id` (onDelete: 'cascade',)
- `teacher_id`: FK → `users.id` (onDelete: 'cascade',)

Delete and lifecycle behavior.

- Cascading parent deletes affect: `class_id`: FK → `classes.id` (onDelete: 'cascade',); `teacher_id`: FK → `users.id` (onDelete: 'cascade',).
- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

ai_generation_outputs

Purpose. Structured AI drafts linked to jobs with approval state and target ownership.

Drizzle declaration. `aiGenerationOutputs` in `backend/src/drizzle/schema/rag.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
jobId	job_id	uuid	NOT NULL	No default	FK → aiGenerationJobs.id (onDelete: 'cascade')	uuid('job_id').notNull().references(() => aiGenerationJobs.id, { onDelete: 'cascade' })
outputType	output_type	aiGenerationOutputTypeEnum	NOT NULL	No default	No PK, FK, or unique marker	aiGenerationOutputTypeEnum('output_type').notNull()
targetClassId	target_class_id	uuid	NULL allowed	No default	FK → classes.id (onDelete: 'cascade',)	uuid('target_class_id').references(() => classes.id, { onDelete: 'cascade', })
targetTeacherId	target_teacher_id	uuid	NULL allowed	No default	FK → users.id (onDelete: 'cascade',)	uuid('target_teacher_id').references(() => users.id, { onDelete: 'cascade', })
sourceFilters	source_filters	json	NULL allowed	No default	No PK, FK, or unique marker	json('source_filters')
structuredOutput	structured_output	json	NOT NULL	No default	No PK, FK, or unique marker	json('structured_output').notNull()
status	status	aiGenerationStatusEnum	NOT NULL	default('completed')	No PK, FK, or unique marker	aiGenerationStatusEnum('status').notNull().default('completed')
approvedBy	approved_by	uuid	NULL allowed	No default	FK → users.id (onDelete: 'set null',)	uuid('approved_by').references(() => users.id, { onDelete: 'set null', })
approvedAt	approved_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('approved_at')
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()

Indexes and table constraints.

- `jobIdx: index('ai_generation_outputs_job_id_idx').on(table.jobId)`
- `classIdx: index('ai_generation_outputs_target_class_id_idx').on( table.targetClassId, )`
- `teacherIdx: index('ai_generation_outputs_target_teacher_id_idx').on( table.targetTeacherId, )`
- `statusIdx: index('ai_generation_outputs_status_idx').on(table.status)`

Relationship map.

- `job_id`: FK → `aiGenerationJobs.id` (onDelete: 'cascade')
- `target_class_id`: FK → `classes.id` (onDelete: 'cascade',)
- `target_teacher_id`: FK → `users.id` (onDelete: 'cascade',)
- `approved_by`: FK → `users.id` (onDelete: 'set null',)

Delete and lifecycle behavior.

- Cascading parent deletes affect: `job_id`: FK → `aiGenerationJobs.id` (onDelete: 'cascade'); `target_class_id`: FK → `classes.id` (onDelete: 'cascade',); `target_teacher_id`: FK → `users.id` (onDelete: 'cascade',).
- Parent deletes set these references to NULL: `approved_by`: FK → `users.id` (onDelete: 'set null',).

- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

Source file: backend/src/drizzle/schema/refresh-tokens.schema.ts

This source file declares 1 active table.

refresh_tokens

Purpose. Hashed opaque refresh-token records used for rotation, revocation, device context, and expiry checks.

Drizzle declaration. `refreshTokens` in `backend/src/drizzle/schema/refresh-tokens.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
userId	user_id	uuid	NOT NULL	No default	FK → users.id (onDelete: 'cascade')	uuid('user_id').notNull().references(() => users.id, { onDelete: 'cascade' })
tokenHash	token_hash	text	NOT NULL	No default	UNIQUE	text('token_hash').notNull().unique()
userAgent	user_agent	text	NULL allowed	No default	No PK, FK, or unique marker	text('user_agent')
ip	ip	text	NULL allowed	No default	No PK, FK, or unique marker	text('ip')
revoked	revoked	boolean	NOT NULL	default(false)	No PK, FK, or unique marker	boolean('revoked').notNull().default(false)
replacedByTokenHash	replaced_by_token_hash	text	NULL allowed	No default	No PK, FK, or unique marker	text('replaced_by_token_hash')
graceExpiresAt	grace_expires_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('grace_expires_at')
rotatedAt	rotated_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('rotated_at')
expiresAt	expires_at	timestamp	NOT NULL	No default	No PK, FK, or unique marker	timestamp('expires_at').notNull()
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()

Indexes and table constraints.

- `userIdIdx: index('refresh_tokens_user_id_idx').on(table.userId)`
- `tokenHashIdx: index('refresh_tokens_token_hash_idx').on(table.tokenHash)`
- `revokedIdx: index('refresh_tokens_revoked_idx').on(table.revoked)`
- `expiresAtIdx: index('refresh_tokens_expires_at_idx').on(table.expiresAt)`

Relationship map.

- `user_id: FK → users.id (onDelete: 'cascade')`

Delete and lifecycle behavior.

- Cascading parent deletes affect: `user_id: FK → users.id (onDelete: 'cascade')`.
- No outgoing foreign key declares SET NULL behavior.

- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

Source file: backend/src/drizzle/schema/school-events.schema.ts

This source file declares 1 active table.

school_events

Purpose. School calendar events with visibility and archival lifecycle.

Drizzle declaration. `schoolEvents` in `backend/src/drizzle/schema/school-events.schema.ts`.

Property	PostgreSQL column	Drizzle or SQL type	Nullability	Default	Constraints	Exact builder chain
id	id	uuid	NOT NULL	defaultRandom()	PRIMARY KEY	uuid('id').primaryKey().defaultRandom()
eventType	event_type	schoolEvent-TypeEnum	NOT NULL	default('school_event')	No PK, FK, or unique marker	schoolEvent-TypeEnum('event_type').notNull().default('school_event')
schoolYear	school_year	text	NOT NULL	No default	No PK, FK, or unique marker	text('school_year').notNull()
title	title	text	NOT NULL	No default	No PK, FK, or unique marker	text('title').notNull()
description	description	text	NULL allowed	No default	No PK, FK, or unique marker	text('description')
location	location	text	NULL allowed	No default	No PK, FK, or unique marker	text('location')
startsAt	starts_at	timestamp	NOT NULL	No default	No PK, FK, or unique marker	timestamp('starts_at').notNull()
endsAt	ends_at	timestamp	NOT NULL	No default	No PK, FK, or unique marker	timestamp('ends_at').notNull()
allDay	all_day	boolean	NOT NULL	default(true)	No PK, FK, or unique marker	boolean('all_day').notNull().default(true)
createdAt	created_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('created_at').notNull().defaultNow()
updatedAt	updated_at	timestamp	NOT NULL	defaultNow()	No PK, FK, or unique marker	timestamp('updated_at').notNull().defaultNow()
archivedAt	archived_at	timestamp	NULL allowed	No default	No PK, FK, or unique marker	timestamp('archived_at')

Indexes and table constraints.

- `schoolYearIdx: index('school_events_school_year_idx').on(table.schoolYear)`
- `startsAtIdx: index('school_events_starts_at_idx').on(table.startsAt)`
- `endsAtIdx: index('school_events_ends_at_idx').on(table.endsAt)`
- `archivedAtIdx: index('school_events_archived_at_idx').on(table.archivedAt)`

Relationship map.

- No outgoing foreign key is declared on this table.

Delete and lifecycle behavior.

- No outgoing foreign key declares cascading deletion.

- No outgoing foreign key declares SET NULL behavior.
- Deletes without an explicit `onDelete` option use the PostgreSQL foreign-key default and must be tested before destructive maintenance.

Drizzle relation declarations

Exhaustive inventory rule. The 73 relation declarations below were extracted from `backend/src/drizzle/schema/*.ts` at commit `3d0c93e`. A later source change requires regenerating or manually reconciling this chapter.

Relation symbol	Owner table symbol	Exact relation callback	Source
<code>aiInteractionLogsRelations</code>	<code>aiInteractionLogs</code>	<code>{ one } => { user: one(users, { fields: [aiInteractionLogs.userId], references: [users.id], }), }</code>	<code>backend/src/drizzle/schema/ai-mentor.schema.ts</code>
<code>extractedModulesRelations</code>	<code>extractedModules</code>	<code>{ one } => { file: one(uploadedFiles, { fields: [extractedModules.fileId], references: [uploadedFiles.id], }), class: one(classes, { fields: [extractedModules.classId], references: [classes.id], }), teacher: one(users, { fields: [extractedModules.teacherId], references: [users.id], }), }</code>	<code>backend/src/drizzle/schema/ai-mentor.schema.ts</code>
<code>announcementsRelations</code>	<code>announcements</code>	<code>{ one } => { class: one(classes, { fields: [announcements.classId], references: [classes.id], }), author: one(users, { fields: [announcements.authorId], references: [users.id], }), }</code>	<code>backend/src/drizzle/schema/announcements-notifications.schema.ts</code>
<code>notificationsRelations</code>	<code>notifications</code>	<code>{ one } => { user: one(users, { fields: [notifications.userId], references: [users.id], }), }</code>	<code>backend/src/drizzle/schema/announcements-notifications.schema.ts</code>
<code>usersRelations</code>	<code>users</code>	<code>{ many, one } => { userRoles: many(userRoles), // Relationship to Roles classesTaught: many(classes, { relationName: 'teacherClasses' }), advisedSections: many(sections), enrollments: many(enrollments), auditLogs: many(auditLogs), lessonCompletions: many(lessonCompletions), assessmentAttempts: many(assessmentAttempts), classVisibilityPreferences: many(classVisibilityPreferences), studentClassPresentationPreferences: many(studentClassPresentationPreferences), studentCourseViewPreference: one(studentCourseViewPreferences, { fields: [users.id], references: [studentCourseViewPreferences.userId], }), // Keep the property name profile for compatibility but point to student_profiles profile: one(studentProfiles, { fields: [users.id], references: [studentProfiles.userId], }), teacherProfile: one(teacherProfiles, { fields: [users.id], references: [teacherProfiles.userId], }), }</code>	<code>backend/src/drizzle/schema/base.schema.ts</code>
<code>studentProfilesRelations</code>	<code>studentProfiles</code>	<code>{ one } => { user: one(users, { fields: [studentProfiles.userId], references: [users.id], }), }</code>	<code>backend/src/drizzle/schema/base.schema.ts</code>
<code>teacherProfilesRelations</code>	<code>teacherProfiles</code>	<code>{ one } => { user: one(users, { fields: [teacherProfiles.userId], references: [users.id], }), }</code>	<code>backend/src/drizzle/schema/base.schema.ts</code>
<code>rolesRelations</code>	<code>roles</code>	<code>{ many } => { userRoles: many(userRoles), }</code>	<code>backend/src/drizzle/schema/base.schema.ts</code>
<code>userRolesRelations</code>	<code>userRoles</code>	<code>{ one } => { user: one(users, { fields: [userRoles.userId], references: [users.id], }), role: one(roles, { fields: [userRoles.roleId], references: [roles.id], }), }</code>	<code>backend/src/drizzle/schema/base.schema.ts</code>

Relation symbol	Owner table symbol	Exact relation callback	Source
sectionsRelations	sections	{ { many, one } } => { { classes: many(classes), enrollments: many(enrollments), adviser: one(users, { fields: [sections.adviserId], references: [users.id], }), })	backend/src/drizzle/schema/base.schema.ts
classesRelations	classes	{ { one, many } } => { { schedules: many(classSchedules), section: one(sections, { fields: [classes.sectionId], references: [sections.id], }), teacher: one(users, { fields: [classes.teacherId], references: [users.id], relationName: 'teacherClasses', }), enrollments: many(enrollments), lessons: many(lessons), assessments: many(assessments), modules: many(classModules), visibilityPreferences: many(classVisibilityPreferences), studentPresentationPreferences: many(studentClassPresentationPreferences), })	backend/src/drizzle/schema/base.schema.ts
classSchedulesRelations	classSchedules	{ { one } } => { { class: one(classes, { fields: [classSchedules.classId], references: [classes.id], }), })	backend/src/drizzle/schema/base.schema.ts
enrollmentsRelations	enrollments	{ { one } } => { { student: one(users, { fields: [enrollments.studentId], references: [users.id], }), class: one(classes, { fields: [enrollments.classId], references: [classes.id], }), section: one(sections, { fields: [enrollments.sectionId], references: [sections.id], }), })	backend/src/drizzle/schema/base.schema.ts
auditLogsRelations	auditLogs	{ { one } } => { { actor: one(users, { fields: [auditLogs.actorId], references: [users.id], }), })	backend/src/drizzle/schema/base.schema.ts
lessonsRelations	lessons	{ { one, many } } => { { class: one(classes, { fields: [lessons.classId], references: [classes.id], }), contentBlocks: many(lessonContentBlocks), completions: many(lessonCompletions), versions: many(lessonVersions), moduleItems: many(moduleItems), })	backend/src/drizzle/schema/base.schema.ts
lessonContentBlocksRelations	lessonContentBlocks	{ { one } } => { { lesson: one(lessons, { fields: [lessonContentBlocks.lessonId], references: [lessons.id], }), })	backend/src/drizzle/schema/base.schema.ts
lessonCompletionsRelations	lessonCompletions	{ { one } } => { { student: one(users, { fields: [lessonCompletions.studentId], references: [users.id], }), lesson: one(lessons, { fields: [lessonCompletions.lessonId], references: [lessons.id], }), })	backend/src/drizzle/schema/base.schema.ts
lessonVersionsRelations	lessonVersions	{ { one } } => { { lesson: one(lessons, { fields: [lessonVersions.lessonId], references: [lessons.id], }), creator: one(users, { fields: [lessonVersions.createdBy], references: [users.id], }), })	backend/src/drizzle/schema/base.schema.ts
assessmentsRelations	assessments	{ { one, many } } => { { class: one(classes, { fields: [assessments.classId], references: [classes.id], }), questions: many(assessmentQuestions), attempts: many(assessmentAttempts), moduleItems: many(moduleItems), })	backend/src/drizzle/schema/base.schema.ts
classModulesRelations	classModules	{ { one, many } } => { { class: one(classes, { fields: [classModules.classId], references: [classes.id], }), sections: many(moduleSections), gradingScaleEntries: many(moduleGradingScaleEntries), })	backend/src/drizzle/schema/base.schema.ts
moduleSectionsRelations	moduleSections	{ { one, many } } => { { module: one(classModules, { fields: [moduleSections.moduleId], references: [classModules.id], }), items: many(moduleItems), })	backend/src/drizzle/schema/base.schema.ts

Relation symbol	Owner table symbol	Exact relation callback	Source
moduleItemsRelations	moduleItems	<code>{ one } => { section: one(moduleSections, { fields: [moduleItems.moduleSectionId], references: [moduleSections.id], }), lesson: one(lessons, { fields: [moduleItems.lessonId], references: [lessons.id], }), assessment: one(assessments, { fields: [moduleItems.assessmentId], references: [assessments.id], }) }</code>	backend/src/drizzle/schema/base.schema.ts
moduleGradingScaleEntriesRelations	moduleGradingScaleEntries	<code>{ one } => { module: one(classModules, { fields: [moduleGradingScaleEntries.moduleId], references: [classModules.id], }) }</code>	backend/src/drizzle/schema/base.schema.ts
assessmentQuestionsRelations	assessmentQuestions	<code>{ one, many } => { assessment: one(assessments, { fields: [assessmentQuestions.assessmentId], references: [assessments.id], }), options: many(assessmentQuestionOptions, responses: many(assessmentResponses), }) }</code>	backend/src/drizzle/schema/base.schema.ts
assessmentQuestionOptionsRelations	assessmentQuestionOptions	<code>{ one } => { question: one(assessmentQuestions, { fields: [assessmentQuestionOptions.questionId], references: [assessmentQuestions.id], }) }</code>	backend/src/drizzle/schema/base.schema.ts
assessmentAttemptsRelations	assessmentAttempts	<code>{ one, many } => { student: one(users, { fields: [assessmentAttempts.studentId], references: [users.id], }), assessment: one(assessments, { fields: [assessmentAttempts.assessmentId], references: [assessments.id], }), responses: many(assessmentResponses), }) }</code>	backend/src/drizzle/schema/base.schema.ts
assessmentResponsesRelations	assessmentResponses	<code>{ one } => { attempt: one(assessmentAttempts, { fields: [assessmentResponses.attemptId], references: [assessmentAttempts.id], }), question: one(assessmentQuestions, { fields: [assessmentResponses.questionId], references: [assessmentQuestions.id], }), selectedOption: one(assessmentQuestionOptions, { fields: [assessmentResponses.selectedOptionId], references: [assessmentQuestionOptions.id], }) }</code>	backend/src/drizzle/schema/base.schema.ts
uploadedFilesRelations	uploadedFiles	<code>{ one } => { teacher: one(users, { fields: [uploadedFiles.teacherId], references: [users.id], }), class: one(classes, { fields: [uploadedFiles.classId], references: [classes.id], }), folder: one(libraryFolders, { fields: [uploadedFiles.folderId], references: [libraryFolders.id], }) }</code>	backend/src/drizzle/schema/base.schema.ts
classVisibilityPreferencesRelations	classVisibilityPreferences	<code>{ one } => { user: one(users, { fields: [classVisibilityPreferences.userId], references: [users.id], }), class: one(classes, { fields: [classVisibilityPreferences.classId], references: [classes.id], }) }</code>	backend/src/drizzle/schema/base.schema.ts
studentClassPresentationPreferencesRelations	studentClassPresentationPreferences	<code>{ one } => { user: one(users, { fields: [studentClassPresentationPreferences.userId], references: [users.id], }), class: one(classes, { fields: [studentClassPresentationPreferences.classId], references: [classes.id], }) }</code>	backend/src/drizzle/schema/base.schema.ts
studentCourseViewPreferencesRelations	studentCourseViewPreferences	<code>{ one } => { user: one(users, { fields: [studentCourseViewPreferences.userId], references: [users.id], }) }</code>	backend/src/drizzle/schema/base.schema.ts
sectionVisibilityPreferencesRelations	sectionVisibilityPreferences	<code>{ one } => { user: one(users, { fields: [sectionVisibilityPreferences.userId], references: [users.id], }), section: one(sections, { fields: [sectionVisibilityPreferences.userId], }) }</code>	backend/src/drizzle/schema/base.schema.ts

Relation symbol	Owner table symbol	Exact relation callback	Source
		ences.sectionId], references: [sections.id], }, }	
libraryFoldersRelations	libraryFolders	({ one, many }) => ({ owner: one(users, { fields: [libraryFolders.ownerId], references: [users.id], }, parent: one(libraryFolders, { fields: [libraryFolders.parentId], references: [libraryFolders.id], relationName: 'libraryFolderTree', }, children: many(libraryFolders, { relationName: 'libraryFolderTree', }, files: many(uploadedFiles, })	backend/src/drizzle/schema/base.schema.ts
pendingRosterRelations	pendingRoster	({ one }) => ({ section: one(sections, { fields: [pendingRoster.sectionId], references: [sections.id], }, resolvedUser: one(users, { fields: [pendingRoster.resolvedUserId], references: [users.id], }, })	backend/src/drizzle/schema/base.schema.ts
classRecordsRelations	classRecords	({ one, many }) => ({ class: one(classes, { fields: [classRecords.classId], references: [classes.id], }, teacher: one(users, { fields: [classRecords.teacherId], references: [users.id], }, categories: many(classRecordCategories), items: many(classRecordItems), finalGrades: many(classRecordFinalGrades), })	backend/src/drizzle/schema/class-record.schema.ts
classRecordCategoriesRelations	classRecordCategories	({ one, many }) => ({ classRecord: one(classRecords, { fields: [classRecordCategories.classRecordId], references: [classRecords.id], }, items: many(classRecordItems), })	backend/src/drizzle/schema/class-record.schema.ts
classRecordItemsRelations	classRecordItems	({ one, many }) => ({ classRecord: one(classRecords, { fields: [classRecordItems.classRecordId], references: [classRecords.id], }, category: one(classRecordCategories, { fields: [classRecordItems.categoryId], references: [classRecordCategories.id], }, assessment: one(assessments, { fields: [classRecordItems.assessmentId], references: [assessments.id], }, scores: many(classRecordScores), })	backend/src/drizzle/schema/class-record.schema.ts
classRecordScoresRelations	classRecordScores	({ one }) => ({ item: one(classRecordItems, { fields: [classRecordScores.classRecordItemId], references: [classRecordItems.id], }, student: one(users, { fields: [classRecordScores.studentId], references: [users.id], }, })	backend/src/drizzle/schema/class-record.schema.ts
classRecordFinalGradesRelations	classRecordFinalGrades	({ one }) => ({ classRecord: one(classRecords, { fields: [classRecordFinalGrades.classRecordId], references: [classRecords.id], }, student: one(users, { fields: [classRecordFinalGrades.studentId], references: [users.id], }, })	backend/src/drizzle/schema/class-record.schema.ts
discussionThreadsRelations	discussionThreads	({ one, many }) => ({ class: one(classes, { fields: [discussionThreads.classId], references: [classes.id], }, author: one(users, { fields: [discussionThreads.authorId], references: [users.id], }, attachments: many(discussionThreadAttachments), comments: many(discussionComments), })	backend/src/drizzle/schema/discussion-board.schema.ts
discussionThreadAttachmentsRelations	discussionThreadAttachments	({ one }) => ({ thread: one(discussionThreads, { fields: [discussionThreadAttachments.threadId], references: [discussionThreads.id], }, file: one(uploadedFiles, { fields: [discussionThreadAttachments.fileId], references: [uploadedFiles.id], }, })	backend/src/drizzle/schema/discussion-board.schema.ts

Relation symbol	Owner table symbol	Exact relation callback	Source
discussionCommentsRelations	discussionComments	{ { one, many } } => { { thread: one(discussionThreads, { fields: [discussionComments.threadId], references: [discussionThreads.id], }), author: one(users, { fields: [discussionComments.authorId], references: [users.id], }), deletedBy: one(users, { fields: [discussionComments.deletedById], references: [users.id], }), attachments: many(discussionCommentAttachments), reactions: many(discussionCommentReactions), })	backend/src/drizzle/schema/discussion-board.schema.ts
discussionCommentAttachmentsRelations	discussionCommentAttachments	{ { one } } => { { comment: one(discussionComments, { fields: [discussionCommentAttachments.commentId], references: [discussionComments.id], }), file: one(uploadedFiles, { fields: [discussionCommentAttachments.fileId], references: [uploadedFiles.id], }), })	backend/src/drizzle/schema/discussion-board.schema.ts
discussionCommentReactionsRelations	discussionCommentReactions	{ { one } } => { { comment: one(discussionComments, { fields: [discussionCommentReactions.commentId], references: [discussionComments.id], }), user: one(users, { fields: [discussionCommentReactions.userId], references: [users.id], }), })	backend/src/drizzle/schema/discussion-board.schema.ts
jaSessionsRelations	jaSessions	{ { one, many } } => { { student: one(users, { fields: [jaSessions.studentId], references: [users.id], }), class: one(classes, { fields: [jaSessions.classId], references: [classes.id], }), items: many(jaSessionItems), events: many(jaSessionEvents), xpLedger: many(jaXpLedger), })	backend/src/drizzle/schema/ja.schema.ts
jaSessionItemsRelations	jaSessionItems	{ { one, many } } => { { session: one(jaSessions, { fields: [jaSessionItems.sessionId], references: [jaSessions.id], }), responses: many(jaSessionResponses), })	backend/src/drizzle/schema/ja.schema.ts
jaSessionResponsesRelations	jaSessionResponses	{ { one } } => { { sessionItem: one(jaSessionItems, { fields: [jaSessionResponses.sessionItemId], references: [jaSessionItems.id], }), })	backend/src/drizzle/schema/ja.schema.ts
jaSessionEventsRelations	jaSessionEvents	{ { one } } => { { session: one(jaSessions, { fields: [jaSessionEvents.sessionId], references: [jaSessions.id], }), })	backend/src/drizzle/schema/ja.schema.ts
jaProgressRelations	jaProgress	{ { one } } => { { student: one(users, { fields: [jaProgress.studentId], references: [users.id], }), class: one(classes, { fields: [jaProgress.classId], references: [classes.id], }), })	backend/src/drizzle/schema/ja.schema.ts
jaXpLedgerRelations	jaXpLedger	{ { one } } => { { student: one(users, { fields: [jaXpLedger.studentId], references: [users.id], }), class: one(classes, { fields: [jaXpLedger.classId], references: [classes.id], }), session: one(jaSessions, { fields: [jaXpLedger.sessionId], references: [jaSessions.id], }), })	backend/src/drizzle/schema/ja.schema.ts
jaThreadsRelations	jaThreads	{ { one, many } } => { { student: one(users, { fields: [jaThreads.studentId], references: [users.id], }), class: one(classes, { fields: [jaThreads.classId], references: [classes.id], }), messages: many(jaThreadMessages), guardrailEvents: many(jaGuardrailEvents), })	backend/src/drizzle/schema/ja.schema.ts
jaThreadMessagesRelations	jaThreadMessages	{ { one, many } } => { { thread: one(jaThreads, { fields: [jaThreadMessages.threadId], references: [jaThreads.id], }), guardrailEvents: many(jaGuardrailEvents), })	backend/src/drizzle/schema/ja.schema.ts
jaGuardrailEventsRelations	jaGuardrailEvents		

Relation symbol	Owner table symbol	Exact relation callback	Source
		<pre>{ one } => { student: one(users, { fields: [jaGuardrailEvents.studentId], references: [users.id], }), class: one(classes, { fields: [jaGuardrailEvents.classId], references: [classes.id], }), thread: one(jaThreads, { fields: [jaGuardrailEvents.threadId], references: [jaThreads.id], }), message: one(jaThreadMessages, { fields: [jaGuardrailEvents.messageId], references: [jaThreadMessages.id], }), })</pre>	backend/src/drizzle/schema/ja.schema.ts
interventionCasesRelations	interventionCases	<pre>{ one, many } => { class: one(classes, { fields: [interventionCases.classId], references: [classes.id], }), student: one(users, { fields: [interventionCases.studentId], references: [users.id], }), assignments: many(interventionAssignments), generatedLessons: many(generatedRemedialLessons), generatedAssessments: many(generatedGuidedAssessments), })</pre>	backend/src/drizzle/schema/lxp.schema.ts
interventionAssignmentsRelations	interventionAssignments	<pre>{ one } => { interventionCase: one(interventionCases, { fields: [interventionAssignments.caseId], references: [interventionCases.id], }), lesson: one(lessons, { fields: [interventionAssignments.lessonId], references: [lessons.id], }), assessment: one(assessments, { fields: [interventionAssignments.assessmentId], references: [assessments.id], }), generatedRemedialLesson: one(generatedRemedialLessons, { fields: [interventionAssignments.generatedRemedialLessonId], references: [generatedRemedialLessons.id], }), generatedGuidedAssessment: one(generatedGuidedAssessments, { fields: [interventionAssignments.generatedGuidedAssessmentId], references: [generatedGuidedAssessments.id], }), })</pre>	backend/src/drizzle/schema/lxp.schema.ts
generatedRemedialLessonsRelations	generatedRemedialLessons	<pre>{ one, many } => { interventionCase: one(interventionCases, { fields: [generatedRemedialLessons.caseId], references: [interventionCases.id], }), class: one(classes, { fields: [generatedRemedialLessons.classId], references: [classes.id], }), student: one(users, { fields: [generatedRemedialLessons.studentId], references: [users.id], }), assignments: many(interventionAssignments), })</pre>	backend/src/drizzle/schema/lxp.schema.ts
generatedGuidedAssessmentsRelations	generatedGuidedAssessments	<pre>{ one, many } => { interventionCase: one(interventionCases, { fields: [generatedGuidedAssessments.caseId], references: [interventionCases.id], }), class: one(classes, { fields: [generatedGuidedAssessments.classId], references: [classes.id], }), student: one(users, { fields: [generatedGuidedAssessments.studentId], references: [users.id], }), sourceAssessment: one(assessments, { fields: [generatedGuidedAssessments.sourceAssessmentId], references: [assessments.id], }), assignments: many(interventionAssignments), attempts: many(generatedGuidedAssessmentAttempts), })</pre>	backend/src/drizzle/schema/lxp.schema.ts
generatedGuidedAssessmentAttemptsRelations	generatedGuidedAssessmentAttempts	<pre>{ one } => { guidedAssessment: one(generatedGuidedAssessments, { fields: [generatedGuidedAssessmentAttempts.guidedAssessmentId], references: [generatedGuidedAssessments.id], }), interventionCase: one(interventionCases, { fields: [generatedGuidedAssessmentAttempts.caseId], references: [interventionCases.id], }), class: one(classes,</pre>	backend/src/drizzle/schema/lxp.schema.ts

Relation symbol	Owner table symbol	Exact relation callback	Source
		{ fields: [generatedGuidedAssessmentAttempts.classId], references: [classes.id,] }, student: one(users, { fields: [generatedGuidedAssessmentAttempts.studentId], references: [users.id,] }, assignment: one(interventionAssignments, { fields: [generatedGuidedAssessmentAttempts.assignmentId], references: [interventionAssignments.id,] }, })	
lxpProgressRelations	lxpProgress	({ one }) => ({ student: one(users, { fields: [lxpProgress.studentId], references: [users.id,] }, class: one(classes, { fields: [lxpProgress.classId], references: [classes.id,] }, })	backend/src/drizzle/schema/lxp.schema.ts
systemEvaluationsRelations	systemEvaluations	({ one }) => ({ submitter: one(users, { fields: [systemEvaluations.submittedBy], references: [users.id,] }, campaign: one(systemEvaluationCampaigns, { fields: [systemEvaluations.campaignId], references: [systemEvaluationCampaigns.id,] }, })	backend/src/drizzle/schema/lxp.schema.ts
systemEvaluationCampaignsRelations	systemEvaluationCampaigns	({ one, many }) => ({ creator: one(users, { fields: [systemEvaluationCampaigns.createdBy], references: [users.id,] }, class: one(classes, { fields: [systemEvaluationCampaigns.classId], references: [classes.id,] }, assignments: many(systemEvaluationAssignments), evaluations: many(systemEvaluations), })	backend/src/drizzle/schema/lxp.schema.ts
systemEvaluationAssignmentsRelations	systemEvaluationAssignments	({ one }) => ({ campaign: one(systemEvaluationCampaigns, { fields: [systemEvaluationAssignments.campaignId], references: [systemEvaluationCampaigns.id,] }, respondent: one(users, { fields: [systemEvaluationAssignments.respondentId], references: [users.id,] }, submittedEvaluation: one(systemEvaluations, { fields: [systemEvaluationAssignments.submittedEvaluationId], references: [systemEvaluations.id,] }, })	backend/src/drizzle/schema/lxp.schema.ts
classAiPoliciesRelations	classAiPolicies	({ one }) => ({ class: one(classes, { fields: [classAiPolicies.classId], references: [classes.id,] }, updatedByUser: one(users, { fields: [classAiPolicies.updatedBy], references: [users.id,] }, })	backend/src/drizzle/schema/lxp.schema.ts
teacherEvaluationWindowsRelations	teacherEvaluationWindows	({ one, many }) => ({ class: one(classes, { fields: [teacherEvaluationWindows.classId], references: [classes.id,] }, teacher: one(users, { fields: [teacherEvaluationWindows.teacherId], references: [users.id,] }, submissions: many(teacherEvaluationSubmissions), })	backend/src/drizzle/schema/lxp.schema.ts
teacherEvaluationSubmissionsRelations	teacherEvaluationSubmissions	({ one }) => ({ window: one(teacherEvaluationWindows, { fields: [teacherEvaluationSubmissions.windowId], references: [teacherEvaluationWindows.id,] }, class: one(classes, { fields: [teacherEvaluationSubmissions.classId], references: [classes.id,] }, teacher: one(users, { fields: [teacherEvaluationSubmissions.teacherId], references: [users.id,] }, student: one(users, { fields: [teacherEvaluationSubmissions.studentId], references: [users.id,] }, })	backend/src/drizzle/schema/lxp.schema.ts
otpVerificationsRelations	otpVerifications	({ one }) => ({ user: one(users, { fields: [otpVerifications.userId], references: [users.id,] }, })	backend/src/drizzle/schema/otp.schema.ts
performanceSnapshotsRelations	performanceSnapshots		

Relation symbol	Owner table symbol	Exact relation callback	Source
		<code>{ one } => { class: one(classes, { fields: [performanceSnapshots.classId], references: [classes.id], }), student: one(users, { fields: [performanceSnapshots.studentId], references: [users.id], }), }</code>	<code>backend/src/drizzle/schema/performance.schema.ts</code>
<code>performanceLogsRelations</code>	<code>performanceLogs</code>	<code>{ one } => { class: one(classes, { fields: [performanceLogs.classId], references: [classes.id], }), student: one(users, { fields: [performanceLogs.studentId], references: [users.id], }), }</code>	<code>backend/src/drizzle/schema/performance.schema.ts</code>
<code>contentChunksRelations</code>	<code>contentChunks</code>	<code>{ one } => { class: one(classes, { fields: [contentChunks.classId], references: [classes.id], }), lesson: one(lessons, { fields: [contentChunks.lessonId], references: [lessons.id], }), assessment: one(assessments, { fields: [contentChunks.assessmentId], references: [assessments.id], }), question: one(assessmentQuestions, { fields: [contentChunks.questionId], references: [assessmentQuestions.id], }), extraction: one(extractedModules, { fields: [contentChunks.extractionId], references: [extractedModules.id], }), libraryFile: one(uploadedFiles, { fields: [contentChunks.libraryFileId], references: [uploadedFiles.id], }), }</code>	<code>backend/src/drizzle/schema/rag.schema.ts</code>
<code>contentChunkEmbeddingsRelations</code>	<code>contentChunkEmbeddings</code>	<code>{ one } => { chunk: one(contentChunks, { fields: [contentChunkEmbeddings.chunkId], references: [contentChunks.id], }), }</code>	<code>backend/src/drizzle/schema/rag.schema.ts</code>
<code>aiGenerationJobsRelations</code>	<code>aiGenerationJobs</code>	<code>{ one, many } => { class: one(classes, { fields: [aiGenerationJobs.classId], references: [classes.id], }), teacher: one(users, { fields: [aiGenerationJobs.teacherId], references: [users.id], }), outputs: many(aiGenerationOutputs,) }</code>	<code>backend/src/drizzle/schema/rag.schema.ts</code>
<code>aiGenerationOutputsRelations</code>	<code>aiGenerationOutputs</code>	<code>{ one } => { job: one(aiGenerationJobs, { fields: [aiGenerationOutputs.jobId], references: [aiGenerationJobs.id], }), targetClass: one(classes, { fields: [aiGenerationOutputs.targetClassId], references: [classes.id], }), targetTeacher: one(users, { fields: [aiGenerationOutputs.targetTeacherId], references: [users.id], }), approver: one(users, { fields: [aiGenerationOutputs.approvedBy], references: [users.id], }), }</code>	<code>backend/src/drizzle/schema/rag.schema.ts</code>
<code>refreshTokensRelations</code>	<code>refreshTokens</code>	<code>{ one } => { user: one(users, { fields: [refreshTokens.userId], references: [users.id], }), }</code>	<code>backend/src/drizzle/schema/refresh-tokens.schema.ts</code>

Safe schema maintenance checklist

1. Change the owning file in `backend/src/drizzle/schema/`.
2. Trace each renamed field into DTOs, services, web types, mobile types, FastAPI SQL, seeds, reports, and tests.
3. Generate a new forward migration; never rewrite an already-applied production migration.
4. Check extension prerequisites before creating a type that depends on an extension.
5. Review every foreign key for intended delete behavior and every query predicate for an index.
6. Run `npm --prefix backend run check:migrations`, the targeted tests, and `npm --prefix backend run build`.
7. Exercise the migration against a disposable database before touching persistent developer or deployed volumes.

Chapter 03 — BullMQ Async Queue and Job Orchestration

Snapshot authority. This chapter describes commit `3d0c93e5270d44b9912deee0218e95c9a311dd5` on branch `developement`. Source paths named below are the authority if the implementation changes after 2026-07-13.

This chapter is the asynchronous service manual for Redis-backed work. It documents every registered queue, producer call, job contract, worker, concurrency value, retry policy, retention rule, cancellation boundary, and recovery procedure.

Source map

- `backend/src/app.module.ts`
- `backend/src/modules/ai-mentor/`
- `backend/src/modules/announcements/`
- `backend/src/modules/discussion-board/`
- `backend/src/modules/file-upload/`
- `backend/src/modules/notifications/`
- `backend/src/modules/performance/`
- `backend/src/modules/rag/`

Orchestration authority

- NestJS owns durable job orchestration and Redis queue contracts.
- PostgreSQL owns authoritative academic state, AI job state, extraction state, files, and audit history.
- FastAPI performs AI and indexing execution but does not replace BullMQ retry ownership.
- Redis queue state is operational state. Never treat a completed BullMQ record as the sole proof that a durable business transition occurred.

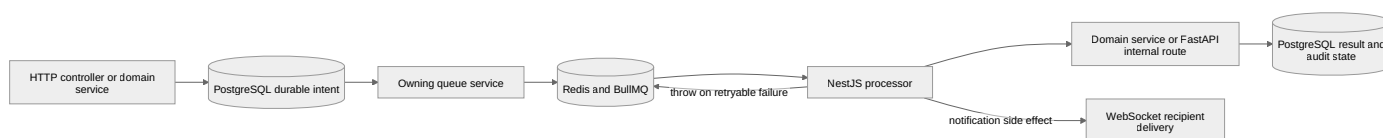


Diagram 6

Global BullMQ defaults

Option	Value	Meaning
Connection	redis.url from ConfigService	All registered queues use the configured Redis connection.
Attempts	3	Applies when a producer does not override attempts.
Backoff	Exponential, initial delay 2,000 ms	Applies when a producer does not override backoff.
Completed retention	Age 3,600 seconds and count 100	The global default bounds completed operational history.

Option	Value	Meaning
Failed retention	Age 86,400 seconds and count 50	The global default bounds failed operational history.

Per-job options override these defaults. The queue-specific tables below are authoritative for each enqueue site.

Complete queue register

Exhaustive inventory rule. The 7 queue processors below were extracted from `backend/src/modules/**/*processor.ts` at commit `3d0c93e`. A later source change requires regenerating or manually reconciling this chapter.

Queue	Concurrency	Processor	Producer class	Registered by
ai-teacher-generation	2	AiGenerationProcessor	AiGenerationQueueService	backend/src/modules/ai-mentor/ai-mentor.module.ts
announcements	3	AnnouncementFanOutProcessor	AnnouncementsService	backend/src/modules/announcements/announcements.module.ts; backend/src/modules/notifications/notifications.module.ts
discussion-board	3	DiscussionBoardProcessor	DiscussionBoardService	backend/src/modules/discussion-board/discussion-board.module.ts
library-indexing	2	LibraryIndexingProcessor	LibraryIndexingService	backend/src/modules/file-upload/file-upload.module.ts
notifications	3	AssessmentNotificationProcessor	AssessmentNotificationDispatchService	backend/src/modules/notifications/notifications.module.ts
performance-recompute	3	PerformanceRecomputeProcessor	PerformanceRecomputeQueueService	backend/src/modules/performance/performance.module.ts
rag-indexing	1	RagIndexingProcessor	RagIndexingService	backend/src/modules/rag/rag.module.ts

Queue: ai-teacher-generation

Executes durable teacher lesson-plan, quiz, intervention, and module-extraction work through the FastAPI internal execution boundary.

Service item	Current value
Registration	backend/src/modules/ai-mentor/ai-mentor.module.ts
Producer boundary	AiGenerationQueueService in backend/src/modules/ai-mentor/ai-generation-queue.service.ts
Processor	AiGenerationProcessor in backend/src/modules/ai-mentor/processors/ai-generation.processor.ts
Worker concurrency	2
Job names	lesson-plan-generation, quiz-generation, intervention-recommendation-generation, module-extraction
Success effect	The processor updates <code>ai_generation_jobs</code> and <code>ai_generation_outputs</code> , or <code>extracted_modules</code> for extraction work.
Failure effect	The processor records the attempt and terminal failure in durable PostgreSQL state, then rethrows so BullMQ owns retry timing.

Exact enqueue contracts.

Job name	Payload expression	Options expression	Producer source
lesson-plan-generation	{ jobId, requestedByUserId: userId, queuedAt: new Date().toISOString() }	{ SHARED_JOB_OPTIONS merged with, jobId: <code>lesson-plan:\${jobId}</code> , }	backend/src/modules/ai-mentor/ai-generation-queue.service.ts
quiz-generation	{ jobId, requestedByUserId: userId, queuedAt: new Date().toISOString() }	{ SHARED_JOB_OPTIONS merged with, jobId: <code>quiz:\${jobId}</code> , }	backend/src/modules/ai-mentor/ai-generation-queue.service.ts
intervention-recommendation-generation	{ jobId, requestedByUserId: userId, queuedAt: new Date().toISOString() }	{ SHARED_JOB_OPTIONS merged with, jobId: <code>intervention:\${jobId}</code> , }	backend/src/modules/ai-mentor/ai-generation-queue.service.ts
module-extraction	{ extractionId, requestedByUserId: userId, queuedAt: new Date().toISOString(), }	{ SHARED_JOB_OPTIONS merged with, jobId: <code>extraction-\${extractionId}</code> , }	backend/src/modules/ai-mentor/ai-generation-queue.service.ts

Maintenance boundary. A job payload is a versioned internal contract even though it is not an HTTP DTO. Update producer, processor, tests, durable state transition, and this table together.

Queue: announcements

Fans a published or scheduled class announcement out into per-user notification records and real-time notification events.

Service item	Current value
Registration	backend/src/modules/announcements/announcements.module.ts; backend/src/modules/notifications/notifications.module.ts
Producer boundary	AnnouncementsService in backend/src/modules/announcements/announcements.service.ts
Processor	AnnouncementFanOutProcessor in backend/src/modules/notifications/processors/announcement-fan-out.processor.ts
Worker concurrency	3
Job names	fan-out, fan-out
Success effect	Notification rows are deduplicated by the notification service and emitted to connected recipients.
Failure effect	Failed jobs remain in Redis because removeOnFail is false; the producer config permits three attempts.

Exact enqueue contracts.

Job name	Payload expression	Options expression	Producer source
fan-out	{ announcementId: announcement.id, classId: announcement.title, content: announcement.content, }	{ attempts: 3, backoff: { type: 'exponential', delay: 2000 }, removeOnComplete: true, removeOnFail: false, }	backend/src/modules/announcements/announcements.service.ts
fan-out	{ announcementId: ann.id, classId: ann.classId, title: ann.title, content: ann.content, }	{ attempts: 3, backoff: { type: 'exponential', delay: 2000 }, removeOnComplete: true, removeOnFail: false, }	backend/src/modules/announcements/announcements.service.ts

Maintenance boundary. A job payload is a versioned internal contract even though it is not an HTTP DTO. Update producer, processor, tests, durable state transition, and this table together.

Queue: discussion-board

Creates notification side effects for newly published threads and newly created comments without holding the HTTP request open.

Service item	Current value
Registration	backend/src/modules/discussion-board/discussion-board.module.ts
Producer boundary	DiscussionBoardService in backend/src/modules/discussion-board/discussion-board.service.ts
Processor	DiscussionBoardProcessor in backend/src/modules/discussion-board/discussion-board.processor.ts
Worker concurrency	3

Service item	Current value
Job names	thread-published, comment-created
Success effect	The worker strips unsafe presentation text, writes deduplicated notifications, and emits recipient updates.
Failure effect	Failed jobs remain in Redis because removeOnFail is false; three exponential-backoff attempts are configured.

Exact enqueue contracts.

Job name	Payload expression	Options expression	Producer source
thread-published	{ classId, threadId: thread.id, title: thread.title, bodyHtml: thread.bodyHtml, }	{ attempts: 3, backoff: { type: 'exponential', delay: 2000 }, removeOnComplete: true, removeOnFail: false, }	backend/src/modules/discussion-board/discussion-board.service.ts
comment-created	{ classId, threadId: thread.id, commentId: comment.id, threadTitle: thread.title, commenterId: actorId, classTeacherId: access.classTeacherId, }	{ attempts: 3, backoff: { type: 'exponential', delay: 2000 }, removeOnComplete: true, removeOnFail: false, }	backend/src/modules/discussion-board/discussion-board.service.ts

Maintenance boundary. A job payload is a versioned internal contract even though it is not an HTTP DTO. Update producer, processor, tests, durable state transition, and this table together.

Queue: library-indexing

Indexes one uploaded library file through the FastAPI internal file-indexing endpoint and reconciles indexing state and audit evidence.

Service item	Current value
Registration	backend/src/modules/file-upload/file-upload.module.ts
Producer boundary	LibraryIndexingService in backend/src/modules/file-upload/library-indexing.service.ts
Processor	LibraryIndexingProcessor in backend/src/modules/file-upload/processors/library-indexing.processor.ts
Worker concurrency	2
Job names	index-library-file
Success effect	uploaded_files indexing status and related audit data reflect the FastAPI outcome.
Failure effect	The worker records an error state for the file and rethrows; the job is retained within the configured failed-job count.

Exact enqueue contracts.

Job name	Payload expression	Options expression	Producer source
index-library-file	{ fileId, actorId: options.actorId ?? null, reason: options.reason, queuedAt: new Date().toISOString(), }	{ jobId: `library-file:\${fileId}:\${Date.now()}`, removeOnComplete: 100, removeOnFail: 200, attempts: 3, backoff: { type: 'exponential', delay: 5000, }, }	backend/src/modules/file-upload/library-indexing.service.ts

Maintenance boundary. A job payload is a versioned internal contract even though it is not an HTTP DTO. Update producer, processor, tests, durable state transition, and this table together.

Queue: notifications

Creates assessment-assigned notifications immediately and assessment-due reminders at a calculated delayed time.

Service item	Current value
Registration	backend/src/modules/notifications/notifications.module.ts

Service item	Current value
Producer boundary	AssessmentNotificationDispatchService in backend/src/modules/notifications/assessment-notification-dispatch.service.ts
Processor	AssessmentNotificationProcessor in backend/src/modules/notifications/processors/assessment-notification.processor.ts
Worker concurrency	3
Job names	ASSESSMENT_ASSIGNED_JOB, ASSESSMENT_DUE_REMINDER_JOB
Success effect	Eligible enrolled students receive durable and socket-delivered notifications; submitted students are excluded from due reminders.
Failure effect	Failed jobs remain in Redis because removeOnFail is false and may be inspected or retried operationally.

Exact enqueue contracts.

Job name	Payload expression	Options expression	Producer source
ASSESSMENT_ASSIGNED_JOB	this.toJobData(assessment)	{ jobId: <code>`\${ASSESSMENT_ASSIGNED_JOB}:\${assessment.id}`</code> , attempts: 3, backoff: { type: 'exponential', delay: 5_000 }, removeOnComplete: true, removeOnFail: false, }	backend/src/modules/notifications/assessment-notification-dispatch.service.ts
ASSESSMENT_DUE_REMINDER_JOB	this.toJobData(assessment)	{ jobId: <code>`\${ASSESSMENT_DUE_REMINDER_JOB}:\${assessment.id}`</code> , delay, attempts: 3, backoff: { type: 'exponential', delay: 5_000 }, removeOnComplete: true, removeOnFail: false, }	backend/src/modules/notifications/assessment-notification-dispatch.service.ts

Maintenance boundary. A job payload is a versioned internal contract even though it is not an HTTP DTO. Update producer, processor, tests, durable state transition, and this table together.

Queue: performance-recompute

Recomputes derived assessment performance or class-record score snapshots after authoritative academic mutations.

Service item	Current value
Registration	backend/src/modules/performance/performance.module.ts
Producer boundary	PerformanceRecomputeQueueService in backend/src/modules/performance/performance-recompute-queue.service.ts
Processor	PerformanceRecomputeProcessor in backend/src/modules/performance/performance-recompute.processor.ts
Worker concurrency	3
Job names	recompute-assessment, recompute-class-scores
Success effect	PerformanceService refreshes derived state; the queue is not an authority for official scores.
Failure effect	The producer catches enqueue failures and logs them. Runtime jobs inherit the application default attempts and backoff.

Exact enqueue contracts.

Job name	Payload expression	Options expression	Producer source
recompute-assessment	{ assessmentId, studentId }	{ jobId: <code>assessment-\${assessmentId}-\${studentId}</code> , <code>{Math.floor(Date.now() / 15000)}</code> , removeOnComplete: true, removeOnFail: { age: 86400, count: 50 }, }	backend/src/modules/performance/performance-recompute-queue.service.ts
recompute-class-scores	{ classId, studentIds, triggerSource }	{ jobId: <code>class-\${classId}</code> , <code>{Math.floor(Date.now() / 15000)}</code> , removeOnComplete: true, removeOnFail: { age: 86400, count: 50 }, }	backend/src/modules/performance/performance-recompute-queue.service.ts

Maintenance boundary. A job payload is a versioned internal contract even though it is not an HTTP DTO. Update producer, processor, tests, durable state transition, and this table together.

Queue: rag-indexing

Serializes class-wide retrieval reindexing through the FastAPI internal class-index endpoint.

Service item	Current value
Registration	backend/src/modules/rag/rag.module.ts
Producer boundary	RagIndexingService in backend/src/modules/rag/rag-indexing.service.ts
Processor	RagIndexingProcessor in backend/src/modules/rag/processors/rag-indexing.processor.ts
Worker concurrency	1
Job names	reindex-class
Success effect	The AI service rebuilds retrieval chunks and embeddings for the requested class scope.
Failure effect	The worker rethrows indexing failures; up to three attempts use five-second exponential backoff and retained failure history.

Exact enqueue contracts.

Job name	Payload expression	Options expression	Producer source
reindex-class	{ classId, reason: options.reason, actorId: options.actorId ?? null, source: options.source ?? null, queuedAt: new Date().toISOString(), }	{ jobId, removeOnComplete: 100, removeOnFail: 200, attempts: 3, backoff: { type: 'exponential', delay: 5000, }, }	backend/src/modules/rag/rag-indexing.service.ts

Maintenance boundary. A job payload is a versioned internal contract even though it is not an HTTP DTO. Update producer, processor, tests, durable state transition, and this table together.

Canonical job data contracts

```

type TeacherAiJobData = {
  jobId?: string;
  extractionId?: string;
  requestedByUserId: string;
  queuedAt?: string;
};

type AnnouncementFanOutData = {
  announcementId: string;
  classId: string;
  title: string;
  content: string;
};

type DiscussionThreadPublishedData = {
  classId: string;
  threadId: string;
  title: string;
  bodyHtml: string;
};

type DiscussionCommentCreatedData = {
  classId: string;
  threadId: string;
  commentId: string;
  threadTitle: string;
  commenterId: string;
  classTeacherId: string;
};

type LibraryIndexingData = {
  fileId: string;
  actorId: string | null;
  reason: 'upload' | 'retry' | 'backfill' | 'metadata_update';
  queuedAt: string;
};

```

```
type AssessmentNotificationJobData = {
  assessmentId: string;
  classId: string;
  title: string;
  dueDate: string | null;
};

type AssessmentPerformanceData = { assessmentId: string; studentId: string };
type ClassPerformanceData = { classId: string; studentIds?: string[]; triggerSource?: string };

type RagIndexingData = {
  classId: string;
  reason: string;
  actorId: string | null;
  source: string | null;
  queuedAt: string;
};
```

Job identity, deduplication, and scheduling

Queue	Job identity rule	Operational effect
ai-teacher-generation	lesson-plan:, quiz:, intervention:, or extraction-	A second enqueue with the same durable work identifier cannot silently create another active identity.
announcements	No explicit jobId	Each publish or scheduled-publish call creates a new fan-out job; downstream notification deduplication is required.
discussion-board	No explicit jobId	Each thread or comment event is queued independently; durable notification uniqueness prevents duplicate inbox rows.
library-indexing	library-file::	Manual retries and metadata-triggered reindex operations are distinct executions.
notifications	assessment-assigned: and assessment-due-reminder:	One queued identity exists for each assessment event kind.
performance-recompute	Entity identifier plus a 15-second time bucket	Rapid duplicate mutations coalesce while later mutations can schedule another recomputation.
rag-indexing	reindex:	Only one active class reindex identity is intended at a time.

- Assessment due reminders use BullMQ delay calculated from the due date, targeting one day before the deadline.
- Cancellation is explicitly implemented only for queued teacher AI work. It removes jobs in waiting, delayed, or prioritized state and refuses active or completed work.
- A cancellation request must also reconcile the related PostgreSQL job or extraction state. Removing Redis work alone is not a durable business cancellation.

Teacher AI execution sequence

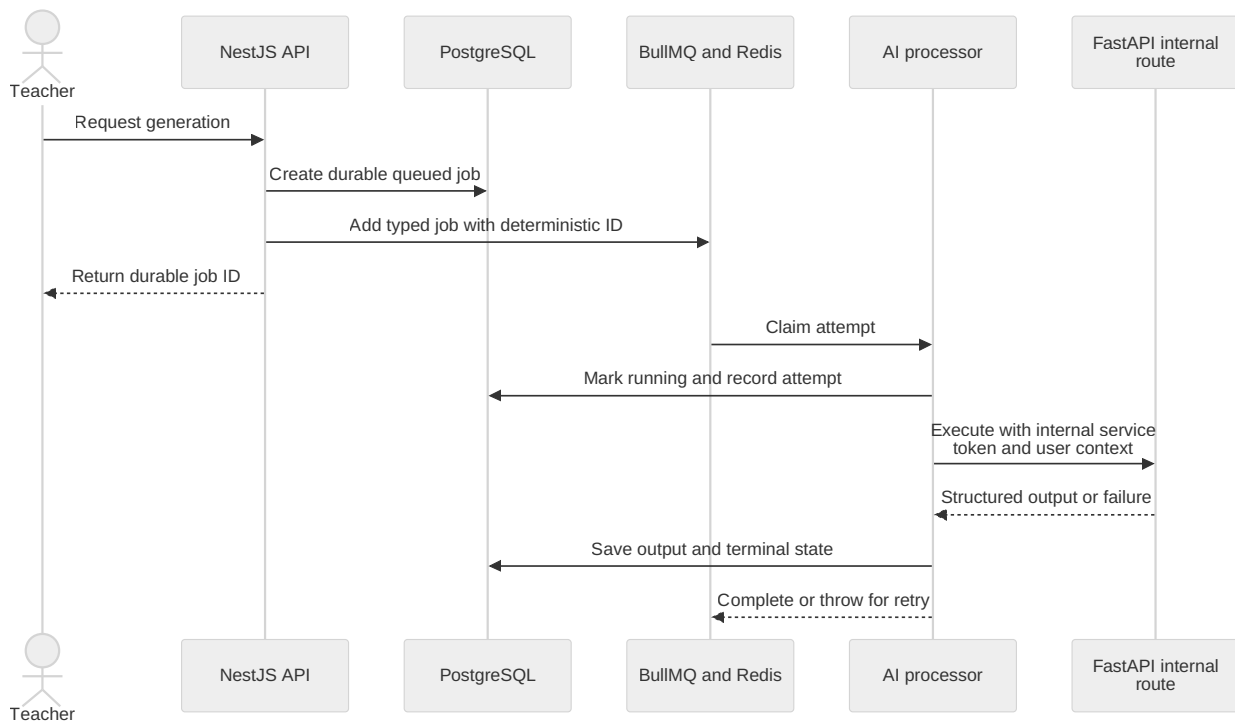


Diagram 7

Failure handling and dead-letter truth

- No dedicated dead-letter queue, QueueEvents failure bridge, or application-wide failed-job event handler was found in the active source.
- Failed work is retained according to per-job `removeOnFail` or the global age-and-count policy. That retained failed set is the operational inspection surface, not a separate DLQ.
- Announcement, discussion, and assessment-notification jobs set `removeOnFail` to false. Operators must bound Redis growth and explicitly decide whether a failure is safe to retry.
- The performance producer logs and absorbs enqueue failures. Therefore a missing performance job can require a manual recomputation even when the originating HTTP mutation succeeded.
- Chapter 09 documents the BullMQ backlog alert and the current metric-export caveat.

Recovery runbook

1. Identify the queue, BullMQ job ID, job name, attempts made, failure reason, timestamp, and durable entity ID.
2. Verify PostgreSQL state before retrying. A timed-out processor may have completed the durable write even if Redis recorded a failure.
3. Verify Redis and the worker process are healthy, then verify the downstream database, FastAPI route, storage path, or socket dependency named by the worker.
4. For teacher AI and extraction jobs, compare `ai_generation_jobs` or `extracted_modules` state with the BullMQ state and inspect the associated output or error fields.
5. Retry through the owning application service or an approved BullMQ administrative tool. Do not edit raw Redis keys.
6. For RAG or library work, use the supported retry or reindex endpoint so file and audit state are updated with the new execution.

7. For performance work, call the owning recompute workflow after confirming the official assessment or class-record write is committed.
8. Record the operator, reason, affected entity, previous state, new job ID, and result in the maintenance log.

Safe change checklist

- Add or change a payload type at both the producer and processor boundary.
- Make processor side effects idempotent before increasing attempts or concurrency.
- Keep official academic mutations transactional and separate from derived recomputation.
- Use a deterministic job ID only when duplicate executions represent the same durable intent.
- Test success, retryable failure, terminal failure, duplicate enqueue, cancellation, Redis outage, and downstream timeout.
- Verify retained-job behavior and Redis memory impact before changing removal rules.
- Update Chapter 09 alerts if a new queue requires distinct backlog or failure monitoring.

Chapter 04 — Authentication, RBAC, Security, and Session Lifecycle

Snapshot authority. This chapter describes commit `3d0c93e5270d44b9912deeeae0218e95c9a311dd5` on branch `developement`. Source paths named below are the authority if the implementation changes after 2026-07-13.

This chapter documents Nexora identity and trust boundaries: credential login, web cookies, mobile bearer tokens, access-token validation, refresh rotation, session revocation, role guards, throttling, inter-service authentication, and operator-safe failure handling.

Source map

- `backend/src/modules/auth/`
- `backend/src/common/guards/`
- `backend/src/common/decorators/`
- `backend/src/config/jwt.config.ts`
- `backend/src/app.module.ts`
- `backend/src/main.ts`
- `next-frontend/src/lib/api-client.ts`
- `next-frontend/src/providers/AuthProvider.tsx`
- `next-frontend/proxy.ts`
- `mobile/src/services/api.ts`
- `ai-service/app/dependencies/auth.py`

Trust model

- The backend is the public authentication and RBAC authority.
- Web and mobile clients call backend routes under `/api`; neither client calls FastAPI directly.
- FastAPI receives an internal service secret plus forwarded user identity only from the backend boundary.
- AI output is assistive and does not become an official record without the backend-owned approval or mutation workflow.
- Access control is enforced server-side. Client route gating improves navigation but is not an authorization boundary.

Token and session inventory

Artifact	Transport and storage	Default lifetime	Authority and behavior
Access JWT	Returned in JSON; web keeps it in memory, mobile uses bearer auth	15 minutes	Signed by the backend access secret and validated by the global JWT guard.
Web refresh token	Opaque value in an httpOnly cookie	7 days	Hashed with SHA-256 before PostgreSQL storage; rotated on every accepted refresh.
Mobile refresh token		7 days	

Artifact	Transport and storage	Default lifetime	Authority and behavior
	Opaque value returned in JSON and held through the mobile secure-storage layer		Sent in the mobile refresh and logout request body; stored server-side only as a hash.
Refresh record	refresh_tokens table	Until expiry or revocation	Tracks user, hash, expiry, revocation, replacement, device context, and rotation grace.
Rotation grace	Redis auth:grace key with database fallback	45 seconds	Allows a benign concurrent refresh race without treating it as theft.

Web login, refresh, and logout

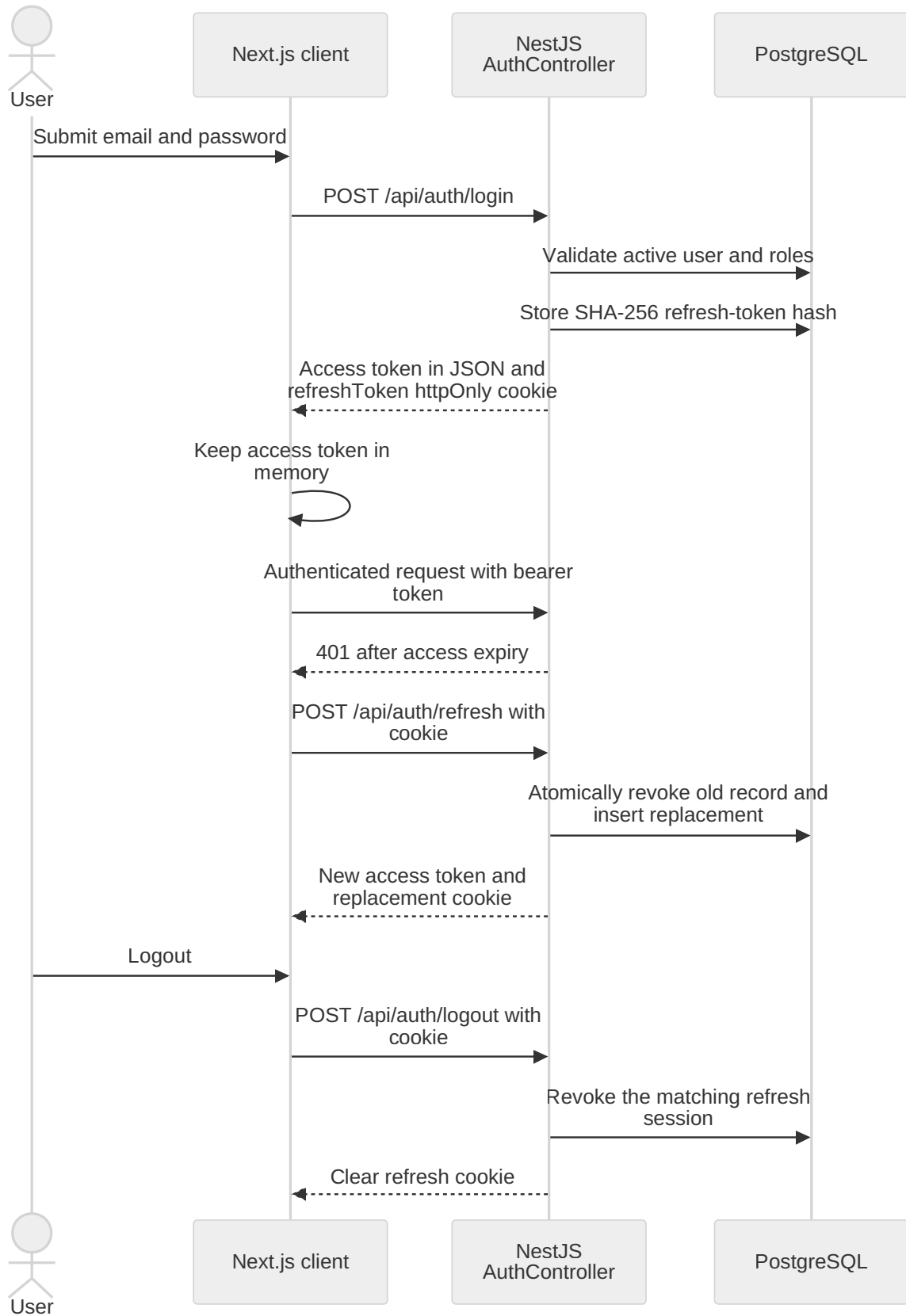


Diagram 8

Cookie controls

Attribute	Current rule	Reason
httpOnly	Always true	Client JavaScript cannot read the refresh token.
secure	True in production	Production browsers send the cookie only over HTTPS.
sameSite	lax by default; none when cross-site configuration requires it	Balances CSRF resistance with an explicitly cross-site deployment.
domain	Optional environment-controlled value	Supports an intentional shared parent domain only.
path	/	Makes refresh and logout routes receive the cookie.
maxAge	Derived from configured refresh-token expiry	Browser retention follows the server token lifetime.

Mobile login, refresh, and logout

- `POST /api/auth/mobile/login` returns access and refresh tokens in JSON; no browser cookie is required.
- The mobile API layer attaches the access token as a bearer credential and serializes refresh so concurrent 401 responses do not rotate the same token independently.
- The mobile refresh token is sent to `POST /api/auth/mobile/refresh`. Logout sends it to `POST /api/auth/mobile/logout` for server-side revocation.
- Expo SecureStore is the primary credential store. AsyncStorage is a compatibility fallback and has weaker at-rest protection, so production devices should keep SecureStore available.
- Logout-all is authenticated and revokes all active refresh sessions for the current user.

Refresh rotation and reuse detection

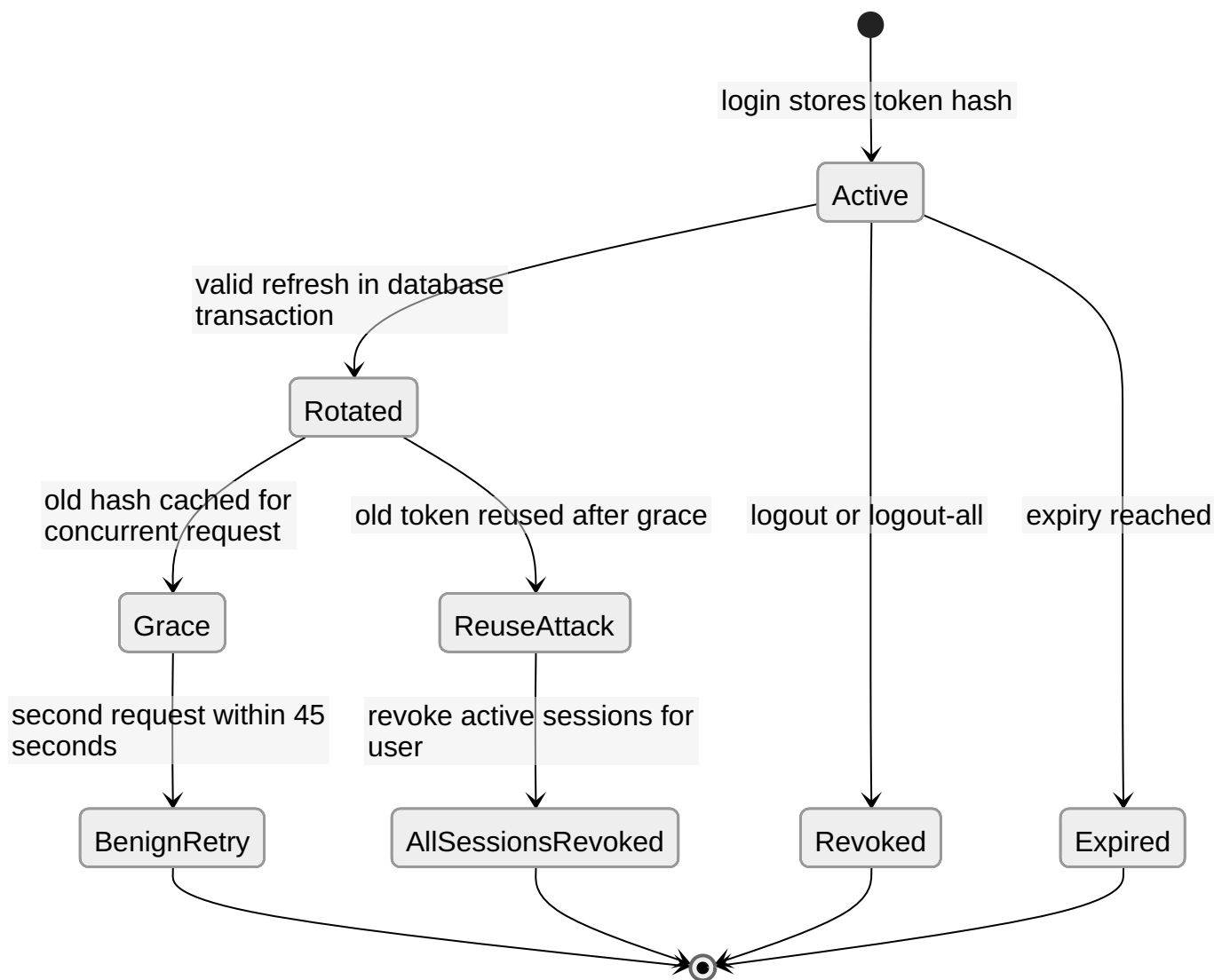


Diagram 9

1. The raw opaque token is SHA-256 hashed; only the hash is queried and stored.
2. Rotation runs in one database transaction: the old record is revoked and linked to a newly inserted record.
3. The old hash receives a 45-second grace marker in Redis. The database `graceExpiresAt` field is a fallback if Redis is unavailable.
4. Reuse within grace is treated as a benign concurrent client race and returns a retry-oriented failure.
5. Reuse outside grace is treated as possible theft and revokes all active sessions for that user.

Backend guard pipeline

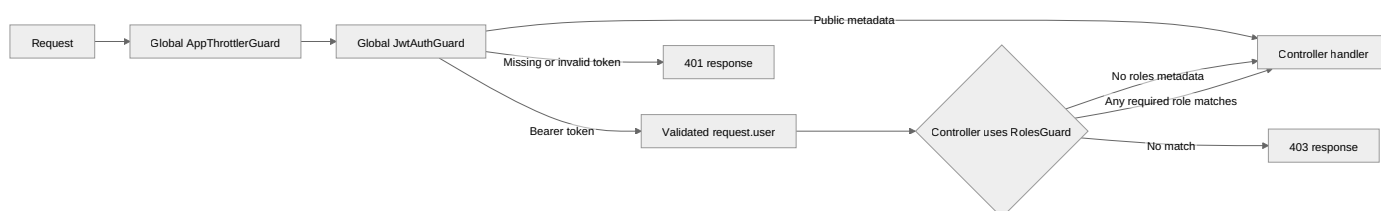


Diagram 10

Control	Scope	Current behavior
AppThrottlerGuard	Global APP_GUARD	Tracks by authenticated user, normalized login email, then forwarded or socket IP.
JwtAuthGuard	Global APP_GUARD	Allows Public metadata; otherwise requires a valid access JWT and populates request.user.
RolesGuard	Controller or handler opt-in	Allows a route with no role metadata; otherwise accepts any one required role present on the principal.
CurrentUser decorator	Handler parameter	Reads the validated principal from the request.
Public decorator	Controller or handler metadata	Bypasses access-token authentication only; throttling and handler validation still apply.
Global ValidationPipe	All controller inputs	Whitelists declared properties, forbids unknown properties, and transforms typed input.
Helmet and CORS	Application bootstrap	Sets security headers and constrains credentialed browser origins.
Global exception filter	Unhandled and HTTP failures	Formats sanitized API failures and prevents raw internal errors from becoming the public contract.

Controller access inventory

Exhaustive inventory rule. The 385 controller routes below were extracted from every `backend/src/**/*.*.controller.ts` source at commit `3d0c93e`. A later source change requires regenerating or manually reconciling this chapter.

Controller	Routes	Public routes	Effective access forms	Source
AcademicStateController	3	0	JWT plus ADMIN	backend/src/modules/academic-state/academic-state.controller.ts
AdminController	5	0	JWT plus ADMIN	backend/src/modules/admin/admin.controller.ts
AiMentorController	41	2	JWT plus STUDENT; JWT plus STUDENT or ADMIN; JWT plus STUDENT or TEACHER or ADMIN; JWT plus TEACHER or ADMIN; Public through @Public	backend/src/modules/ai-mentor/ai-mentor.controller.ts
JaHubController	11	0	JWT plus STUDENT	backend/src/modules/ja/ja-hub.controller.ts
JaController	7	0	JWT plus STUDENT	backend/src/modules/ja/ja.controller.ts
AnalyticsController	4	0	JWT plus ADMIN; JWT plus TEACHER or ADMIN	backend/src/modules/analytics/analytics.controller.ts
AppVersionController	1	1	Public through @Public	backend/src/modules/app-version/app-version.controller.ts
AssessmentsController	36	1	JWT plus ADMIN or STUDENT; JWT plus ADMIN or TEACHER; JWT plus ADMIN or TEACHER or STUDENT; JWT plus STUDENT; Public through @Public	backend/src/modules/assessments/assessments.controller.ts
AssessmentsPublicController	1	1	Public through @Public	backend/src/modules/assessments/assessments-public.controller.ts
AuthController	15	11	JWT authenticated; Public through @Public	backend/src/modules/auth/auth.controller.ts
ClassRecordController	18	0	JWT plus ADMIN or TEACHER; JWT plus TEACHER or ADMIN; JWT plus TEACHER or ADMIN or STUDENT	backend/src/modules/class-record/class-record.controller.ts
ClassTemplatesController	14	1	JWT plus ADMIN; Public through @Public	backend/src/modules/class-templates/class-templates.controller.ts

Controller	Routes	Public routes	Effective access forms	Source
ClassesController	28	0	JWT plus ADMIN; JWT plus ADMIN or STUDENT; JWT plus ADMIN or TEACHER; JWT plus ADMIN or TEACHER or STUDENT; JWT plus STUDENT	backend/src/modules/classes/classes.controller.ts
AnnouncementsController	6	0	JWT plus TEACHER or ADMIN; JWT plus TEACHER or STUDENT or ADMIN	backend/src/modules/announcements/announcements.controller.ts
DiscussionBoardController	19	0	JWT plus ADMIN or STUDENT; JWT plus ADMIN or TEACHER; JWT plus ADMIN or TEACHER or STUDENT	backend/src/modules/discussion-board/discussion-board.controller.ts
ClassesPublicController	1	1	Public through @Public	backend/src/modules/classes/classes.controller.ts
FileUploadController	13	0	JWT plus ADMIN; JWT plus ADMIN or TEACHER; JWT plus ADMIN or TEACHER or STUDENT; JWT plus TEACHER or ADMIN	backend/src/modules/file-upload/file-upload.controller.ts
HealthController	3	3	Public through @Public	backend/src/modules/health/health.controller.ts
MetricsController	1	1	Public through class-level @Public	backend/src/monitoring/metrics.controller.ts
InternalUploadsController	1	1	Public through @Public	backend/src/modules/file-upload/internal-uploads.controller.ts
LessonsController	20	0	JWT plus ADMIN or TEACHER; JWT plus ADMIN or TEACHER or STUDENT; JWT plus STUDENT	backend/src/modules/lessons/lessons.controller.ts
LxpController	32	0	JWT plus ADMIN; JWT plus STUDENT; JWT plus STUDENT or TEACHER; JWT plus STUDENT or TEACHER or ADMIN; JWT plus TEACHER or ADMIN	backend/src/modules/lxp/lxp.controller.ts
ContentModulesController	20	1	JWT plus ADMIN or TEACHER; JWT plus ADMIN or TEACHER or STUDENT; Public through @Public	backend/src/modules/content-modules/content-modules.controller.ts
NotificationsController	4	0	JWT plus TEACHER or STUDENT or ADMIN	backend/src/modules/notifications/notifications.controller.ts
OtpController	2	2	Public through @Public	backend/src/modules/otp/otp.controller.ts
PerformanceController	11	0	JWT plus ADMIN; JWT plus STUDENT; JWT plus TEACHER or ADMIN	backend/src/modules/performance/performance.controller.ts
ProfilesController	9	1	JWT plus ADMIN; JWT plus ADMIN or STUDENT or TEACHER; JWT plus STUDENT; JWT plus STUDENT or TEACHER or ADMIN; Public through @Public	backend/src/modules/profiles/profiles.controller.ts
ReportsController	6	0	JWT plus ADMIN or TEACHER	backend/src/modules/reports/reports.controller.ts
RosterImportController	4	0	JWT plus ADMIN or TEACHER	backend/src/modules/roster-import/roster-import.controller.ts
SchoolEventsController	4	0	JWT plus ADMIN; JWT plus ADMIN or TEACHER or STUDENT	backend/src/modules/school-events/school-events.controller.ts
SectionsController	24	0	JWT plus ADMIN; JWT plus ADMIN or TEACHER; JWT plus ADMIN or TEACHER or STUDENT	backend/src/modules/sections/sections.controller.ts
SectionsPublicController	1	1	Public through @Public	backend/src/modules/sections/sections.controller.ts
TeacherProfilesController	4	0	JWT plus TEACHER; JWT plus TEACHER or ADMIN	backend/src/modules/teacher-profiles/teacher-profiles.controller.ts

Controller	Routes	Public routes	Effective access forms	Source
TeacherController	3	0	JWT plus TEACHER or ADMIN	backend/src/modules/teacher/teacher.controller.ts
UsersController	13	0	JWT plus ADMIN	backend/src/modules/users/users.controller.ts

Public route register

These are the routes explicitly marked public at this snapshot. Public means no access JWT is required; validation, throttling, secrets on internal FastAPI routes, and business checks still apply.

Method	Path	Handler	Input	Source
POST	/api/ai/demo/intervention-plan	AiMentorController.generateDemoInterventionPlan	Body() dto: DemoInterventionPlanDto	backend/src/modules/ai-mentor/ai-mentor.controller.ts
GET	/api/ai/health	AiMentorController.health	No handler parameters	backend/src/modules/ai-mentor/ai-mentor.controller.ts
GET	/api/app-version/check	AppVersionController.check	Query() query: CheckAppVersionDto	backend/src/modules/app-version/app-version.controller.ts
GET	/api/assessments/questions/images-private/:filename	AssessmentsController.serveQuestionImage	Param('filename') filename: string; Res() res: Response	backend/src/modules/assessments/assessments.controller.ts
GET	/api/assessments/questions/images/:filename	AssessmentsPublicController.serveQuestionImage	Param('filename') filename: string; Res() res: Response	backend/src/modules/assessments/assessments-public.controller.ts
POST	/api/auth/forgot-password	AuthController.forgotPassword	Body() dto: ForgotPasswordDto	backend/src/modules/auth/auth.controller.ts
POST	/api/auth/login	AuthController.login	Body() loginDto: LoginDto; Req() request: express.Request; Res({ passthrough: true }) response: express.Response	backend/src/modules/auth/auth.controller.ts
POST	/api/auth/logout	AuthController.logout	Req() request: express.Request; Res({ passthrough: true }) response: express.Response	backend/src/modules/auth/auth.controller.ts
POST	/api/auth/mobile/login	AuthController.mobileLogin	Body() loginDto: LoginDto; Req() request: express.Request	backend/src/modules/auth/auth.controller.ts
POST	/api/auth/mobile/logout	AuthController.mobileLogout	Body() dto: MobileLogoutDto	backend/src/modules/auth/auth.controller.ts
POST	/api/auth/mobile/refresh	AuthController.mobileRefresh	Body() dto: MobileRefreshDto; Req() request: express.Request	backend/src/modules/auth/auth.controller.ts
POST	/api/auth/refresh	AuthController.refresh	Req() request: express.Request; Res({ passthrough: true }) response: express.Response	backend/src/modules/auth/auth.controller.ts
POST	/api/auth/reset-password	AuthController.resetPassword	Body() dto: ResetPasswordDto	backend/src/modules/auth/auth.controller.ts
POST	/api/auth/set-activation-password	AuthController.setActivationPassword	Body() dto: SetActivationPasswordDto	backend/src/modules/auth/auth.controller.ts
POST	/api/auth/set-initial-password	AuthController.setInitialPassword	Body() dto: SetInitialPasswordDto	backend/src/modules/auth/auth.controller.ts
POST	/api/auth/validate-credentials	AuthController.validateCredentials	Body() dto: ValidateCredentialsDto	backend/src/modules/auth/auth.controller.ts
GET	/api/class-templates/images/:filename	ClassTemplatesController.serveAssessmentImage	Param('filename') filename: string; Res() res: Response	backend/src/modules/class-templates/class-templates.controller.ts
GET	/api/classes/banners/:filename	ClassesPublicController.serveClassBanner	Param('filename') filename: string; Res() res: any	backend/src/modules/classes/classes.controller.ts
GET	/api/health	HealthController.aliasCheck	No handler parameters	backend/src/modules/health/health.controller.ts
GET	/api/health/live	HealthController.check	No handler parameters	backend/src/modules/health/health.controller.ts

Method	Path	Handler	Input	Source
GET	/api/health/ready	HealthController.readiness	No handler parameters	backend/src/modules/health/health.controller.ts
GET	/api/internal/uploads/raw	InternalUploadsController.readUpload	Query('path') requestedPath: string; Headers('x-internal-service-token') token: string undefined; Res() res: Response	backend/src/modules/file-upload/internal-uploads.controller.ts
GET	/api/modules/covers/:filename	ContentModulesController.serveModuleCover	Param('filename') filename: string; Res() res: Response	backend/src/modules/content-modules/content-modules.controller.ts
POST	/api/otp/resend	OtpController.resendOtp	Body() resendOtpDto: ResendOtpDto	backend/src/modules/otp/otp.controller.ts
POST	/api/otp/verify	OtpController.verifyOtp	Body() verifyOtpDto: VerifyOtpDto	backend/src/modules/otp/otp.controller.ts
GET	/api/profiles/images/:filename	ProfilesController.serveProfileImage	Param('filename') filename: string; Res() res: Response	backend/src/modules/profiles/profiles.controller.ts
GET	/api/sections/banners/:filename	SectionsPublicController.serveSectionBanner	Param('filename') filename: string; Res() res: any	backend/src/modules/sections/sections.controller.ts

Role capability matrix

Capability	Admin	Teacher	Student	Enforcement notes
Identity, own profile, own sessions	Yes	Yes	Yes	JWT plus ownership checks in auth and profile services.
User, role, roster, and account administration	Yes	No	No	Admin controller role metadata and service checks.
Sections and academic system state administration	Yes	Limited viewing where exposed	Limited viewing where exposed	Route-specific roles and class membership rules apply.
Create and manage assigned classes	Administrative scope	Yes for owned classes	No	Role checks are supplemented by teacher ownership in services.
Author lessons, assessments, announcements, and class records	Administrative scope where exposed	Yes for owned classes	No	Ownership and class access are not replaced by role membership.
Submit assessments and consume published learning content	No normal learner flow	Preview or governance paths	Yes for enrolled classes	Enrollment, publication, schedule, and attempt rules are enforced in services.
Teacher AI generation and approval	Administrative visibility where exposed	Yes	No	Teacher role, class ownership, quotas, and approval state govern durable use.
Student Ja tutor and remedial experiences	Oversight paths	Oversight paths	Yes	Class AI policy, enrollment, rate limits, and guardrails apply.
Official grade finalization	Administrative governance	Yes for owned class records	No	AI has no direct authority to finalize official academic records.
Observability endpoints	Deployment controlled	Deployment controlled	Deployment controlled	Metrics and health exposure is a network/deployment concern, not a school role capability.

Rate limiting

Surface	Limit	Tracker
Global fallback	300 requests per 60 seconds	Authenticated user ID, then login email, then client IP.
Web and mobile login	15 attempts per 60 seconds	Normalized account email when present.
Web and mobile refresh	60 attempts per 60 seconds	Request tracker selected by the global guard.
Forgot password	3 attempts per 5 minutes	Email or IP fallback.
Reset, initial password, and activation password	5 attempts per 5 minutes	Email or IP fallback.

Surface	Limit	Tracker
Credential validation and change password	5 attempts per 60 seconds	Email or authenticated user.
Logout and current-user reads	Explicitly skip throttling	Authentication and revocation rules still apply.

Backend-to-FastAPI authentication

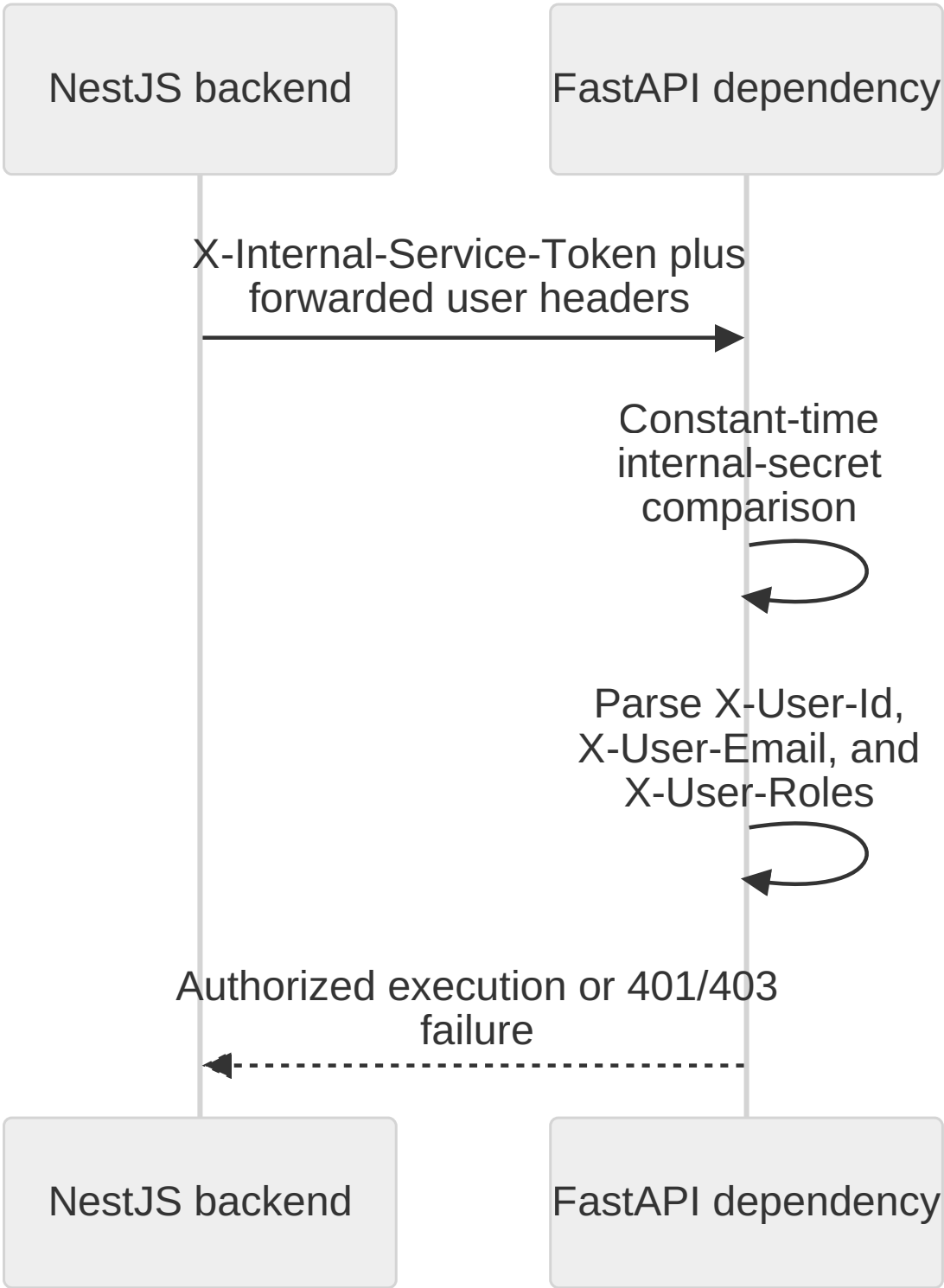


Diagram 11

Header	Owner	Use
X-Internal-Service-Token	Backend service client	Authenticates privileged internal execution routes; a nonempty matching configured secret is required.
X-User-Id	Backend from validated principal	Scopes ownership, auditing, and teacher or student work.
X-User-Email	Backend from validated principal	Carries human-readable identity context where required.
X-User-Roles	Backend from validated principal	Carries role context; FastAPI does not become the public RBAC authority.

- FastAPI current-user dependency enforces the internal secret when one is configured. Privileged require-internal-service routes always require a nonempty matching secret.
- Do not expose FastAPI directly to browsers or mobile devices. Network isolation and the secret are both required controls.
- Rotate the internal secret through coordinated backend and AI-service deployment. Never log the secret or the Authorization header.

Password, OTP, and account lifecycle

- Login accepts only a valid active account with a verified password flow and resolves current roles from backend-owned data.
- Forgot-password and reset-password routes are public but tightly throttled and use one-time verification state.
- Initial-password and activation-password routes support provisioned accounts without weakening normal login requirements.
- Password change is authenticated; session revocation behavior must be verified whenever password policy changes.
- OTP records track purpose, attempts, expiry, and consumption. An OTP is not reusable after successful consumption.
- Account status, role assignment, archived identity, and audit logs must be considered together during disablement or removal.

Client route gating

- Next.js proxy checks for the refresh cookie before allowing dashboard navigation. It does not decode or authorize roles.
- AuthProvider bootstraps the session through the refresh endpoint, stores the access token in memory, and uses the dashboard route-access map for user experience gating.
- The API client permits credentials, uses a 30-second request timeout, serializes refresh with one shared promise, and retries a failed request once after successful rotation.
- Mobile chooses role-specific navigation after resolving the authenticated user. Screen visibility remains subordinate to backend authorization.

Security failure matrix

Failure or threat	Expected behavior	Operator evidence
Invalid credentials	Generic authentication failure; no password detail or account enumeration signal.	Throttled request metadata and sanitized application log.
Expired access token	401; client may perform one serialized refresh.	Request status and refresh result.
Revoked or expired refresh token	Refresh rejected and local session cleared.	refresh_tokens state and auth log.

Failure or threat	Expected behavior	Operator evidence
Refresh-token reuse after grace	All active sessions for the user are revoked.	Security log and revoked refresh records.
Role mismatch	403 from RolesGuard or a domain access check.	Controller route, principal roles, and ownership data.
Cross-class object access	Service rejects even if the role itself is allowed.	Class ownership or enrollment query and audit trail.
Missing internal service secret	FastAPI privileged internal route rejects.	Backend downstream error and FastAPI authentication log without secret material.
Redis unavailable during refresh grace	Database grace field provides the race-detection fallback.	Redis health plus refresh record timestamps.
Client-side route bypass	Backend still rejects unauthorized API calls.	401 or 403 response from NestJS.

Security maintenance checklist

1. Trace every new route through Public metadata, JWT behavior, role metadata, ownership checks, validation, throttling, and audit requirements.
2. Never return or log password hashes, refresh tokens, OTP values, internal service secrets, cookies, or full Authorization headers.
3. Preserve atomic refresh rotation and the reuse-detection tests when changing sessions.
4. Test web cookie mode and mobile body-token mode independently.
5. Test role alone, ownership, enrollment, publication state, and lifecycle state; role membership is not sufficient for object access.
6. Confirm CORS origins and cookie SameSite mode together before any cross-site deployment.
7. Keep FastAPI on a private network path and reject direct client access.
8. Run targeted auth, guard, controller, frontend auth-provider, API-client, and mobile session tests before release.

Chapter 05 — NestJS Backend Module and API Catalog

Snapshot authority. This chapter describes commit `3d0c93e5270d44b9912deeeae0218e95c9a311dd5` on branch `development`. Source paths named below are the authority if the implementation changes after 2026-07-13.

This chapter is the backend service catalog. It records the application pipeline, all active Nest modules, every controller route, effective access, handler inputs, declared outcomes, primary delegates, every DTO class, and the non-class contract declarations used at the API boundary.

Source map

- `backend/src/app.module.ts`
- `backend/src/main.ts`
- `backend/src/database/`
- `backend/src/monitoring/`
- `backend/src/modules/`
- `backend/src/common/`
- `backend/src/config/`
- `backend/test/`

Application service pipeline

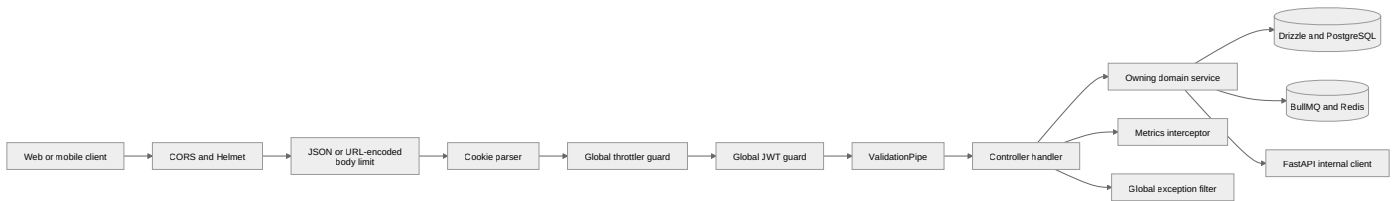


Diagram 12

Bootstrap item	Current contract
Bind address	0.0.0.0:3000
Global API prefix	/api
Request body limit	REQUEST_BODY_LIMIT, default 1mb; file upload limits are handled separately by Multer.
Validation	whitelist true, forbidNonWhitelisted true, transform true
Security headers	Helmet; content-security-policy disabled only in development for Swagger compatibility.
CORS	Credentialed allowlist from frontend, Next frontend, mobile, and explicit origin environment values; localhost additions only outside production.
Proxy trust	TRUST_PROXY_HOPS; production default one hop, development default zero.
Swagger	Generated at /api only outside production.
Sockets	Socket.IO adapter supports backend-owned real-time notifications.
Shutdown	Nest shutdown hooks enabled so in-flight work can drain during controlled termination.

Module graph

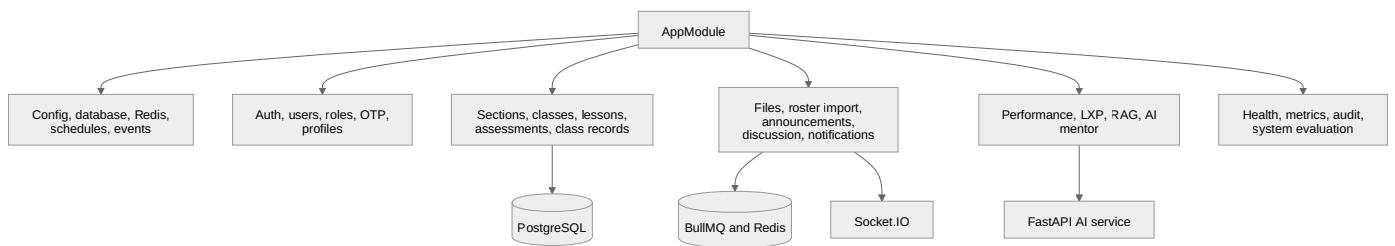


Diagram 13

Complete module register

Exhaustive inventory rule. The 37 Nest module declarations below were extracted from `backend/src/**/*.module.ts` at commit `3d0c93e`. A later source change requires regenerating or manually reconciling this chapter.

Module class	Primary slice	Route count in slice	Extracted imports	Extracted controllers	Extracted providers	Extracted exports	Source
AppModule	application-root	385	ConfigModule.forRoot({ isGlobal: true, load: [databaseConfig, jwtConfig, redisConfig, olamaConfig] })	No direct controller array extracted	{ provide: APP_FILTER, useClass: GlobalExceptionHandler, // Sanitise all unhandled errors }, { provide: APP_GUARD, useClass: JwtAuthGuard, // Global auth guard }, { provide: APP_GUARD, useClass: AppThrottlerGuard, // Global rate-limit guard },	No direct export array extracted	backend/src/app.module.ts
DatabaseModule	database	0	No explicit imports extracted	No direct controller array extracted	DatabaseService	DatabaseService	backend/src/database/database.module.ts
MetricsModule	monitoring	1	No explicit imports extracted	MetricsController	{ provide: PROM_CLIENT_REGISTRY, useValue: register }	PROM_CLIENT_REGISTRY	backend/src/monitoring/metrics.module.ts
AcademicStateModule	academic-state	3	DatabaseModule, AuditModule	AcademicStateController	AcademicStateService	AcademicStateService	backend/src/modules/academic-state/academic-state.module.ts
AdminModule	admin	5	AuditModule, ReportsModule, AnalyticsModule, HealthModule	AdminController	AdminService	AdminService	backend/src/modules/admin/admin.module.ts
AiMentorModule	ai-mentor	41	DatabaseModule, AuditModule, AdminModule, ReportsModule, AnalyticsModule, PerformanceModule, LxpModule, BullModule.registerQueue({ name: 'ai-teacher-generation' })),	AiMentorController	AiProxyService, AdminAnalyticsChatService, AiGenerationQueueService, AiGenerationProcessor,	AiProxyService, AdminAnalyticsChatService	backend/src/modules/ai-mentor/ai-mentor.module.ts
AnalyticsModule	analytics	4	DatabaseModule	AnalyticsController	AnalyticsService	AnalyticsService	backend/src/modules/

Module class	Primary slice	Route count in slice	Extracted imports	Extracted controllers	Extracted providers	Extracted exports	Source
							analytics/analytics.module.ts
Announcements-Module	announcements	6	AuditModule, BullModule.registerQueue({ name: 'announcements', }),	AnnouncementsController	AnnouncementsService, AnnouncementsScheduler	AnnouncementsService	backend/src/modules/announcements/announcements.module.ts
AppVersionModule	app-version	1	DatabaseModule	AppVersionController	AppVersionService	AppVersionService	backend/src/modules/app-version/app-version.module.ts
Assessments-Module	assessments	37	DatabaseModule, EventEmitterModule, AuditModule, NotificationsModule,	AssessmentsController, AssessmentsPublicController	AssessmentsService, AssessmentAccessService, FeedbackService	AssessmentsService, AssessmentAccessService	backend/src/modules/assessments/assessments.module.ts
AuditModule	audit	0	DatabaseModule	No direct controller array extracted	AuditService	AuditService	backend/src/modules/audit/audit.module.ts
AuthModule	auth	15	UsersModule, OTPModule, DatabaseModule, AuditModule, PassportModule.register({ defaultStrategy: 'jwt' }), JwtModule.registerAsync({ imports: [ConfigModule	AuthController	AuthService, TokenService, TokenCleanupService, JwtStrategy	AuthService, TokenService, JwtModule	backend/src/modules/auth/auth.module.ts
ClassRecordModule	class-record	18	DatabaseModule, EventEmitterModule, AuditModule	ClassRecordController	ClassRecordService, ClassRecordComputationService, ClassRecordSyncService, AdviserSectionGuard,	ClassRecordService	backend/src/modules/class-record/class-record.module.ts
ClassTemplatesModule	class-templates	14	DatabaseModule, AuditModule	ClassTemplatesController	ClassTemplatesService	ClassTemplatesService	backend/src/modules/class-templates/class-templates.module.ts
ClassesModule	classes	29	DatabaseModule, AuditModule, ClassRecordModule, AcademicStateModule, AiMentorModule,	ClassesController, ClassesPublicController	ClassesService	ClassesService	backend/src/modules/classes/classes.module.ts
ContentModulesModule	content-modules	20	DatabaseModule, AuditModule	ContentModulesController	ContentModulesService	ContentModulesService	backend/src/modules/content-modules/content-modules.module.ts
DiscussionBoardModule	discussion-board	19	AuditModule, NotificationsModule, BullModule.registerQueue({ name: 'discussion-board', }),	DiscussionBoardController	DiscussionBoardService, DiscussionBoardProcessor	DiscussionBoardService	backend/src/modules/discussion-board/discussion-board.module.ts
FileUploadModule	file-upload	14	DatabaseModule, AuditModule, StorageModule, BullModule.registerQueue({ name: 'library-indexing', }),	FileUploadController, InternalUploadsController	FileUploadService, LibraryIndexingService, LibraryIndexingProcessor,	FileUploadService, LibraryIndexingService, StorageModule	backend/src/modules/file-upload/file-upload.module.ts
StorageModule	file-upload	14	No explicit imports extracted	No direct controller array extracted	{ provide: STORAGE_PROVIDER_TOKEN, useFactory: () => { const driver = (process.env.STORAGE_DRIVER	STORAGE_PROVIDER_TOKEN, StorageService	backend/src/modules/file-upload/storage/storage.module.ts

Module class	Primary slice	Route count in slice	Extracted imports	Extracted controllers	Extracted providers	Extracted exports	Source
					process.env.STORAGE_PROVIDER 'local').toLowerCase(); if (driver === 's3' driver === 'r2') { return new S3StorageProvider(); } return new LocalStorageProvider(); }, }, StorageService,		
HealthModule	health	3	ConfigModule, DatabaseModule	HealthController	HealthService	HealthService	backend/src/modules/health/health.module.ts
JaModule	ja	18	DatabaseModule, AuditModule, AiMentorModule, LxpModule	JaController, JaHubController	JaService	JaService	backend/src/modules/ja/ja.module.ts
LessonsModule	lessons	20	DatabaseModule, AuditModule	LessonsController	LessonsService	LessonsService	backend/src/modules/lessons/lessons.module.ts
LxpModule	lxp	32	DatabaseModule, NotificationsModule, EventEmitterModule, AuditModule,	LxpController	LxpService, SystemEvaluationService, LxpPerformanceListener	LxpService, SystemEvaluationService	backend/src/modules/lxp/lxp.module.ts
MailModule	mail	0	No explicit imports extracted	No direct controller array extracted	MailService	MailService	backend/src/modules/mail/mail.module.ts
NotificationsModule	notifications	4	// Consume the same 'announcements' queue that AnnouncementsModule enqueues into BullModule.registerQueue({ name: 'announcements', }), BullModule.registerQueue({ name: 'notifications', }), // JwtService needed by the WebSocket gateway for token verification JwtModule.registerAsync({ inject: [ConfigService	NotificationsController	NotificationsService, NotificationsGateway, AssessmentNotificationDispatchService, AnnouncementFanOutProcessor, AssessmentNotificationProcessor,	NotificationsService, NotificationsGateway, AssessmentNotificationDispatchService,	backend/src/modules/notifications/notifications.module.ts
OtpModule	otp	2	DatabaseModule, MailModule	OtpController	OtpService	OtpService	backend/src/modules/otp/otp.module.ts
PerformanceModule	performance	11	DatabaseModule, EventEmitterModule, AuditModule, BullModule.registerQueue({ name: 'performance-recompute', }),	PerformanceController	PerformanceService, PerformanceSnapshotReadService, PerformanceEventsListener, PerformanceRecomputeQueueService, PerformanceRecomputeProcessor,	PerformanceService, PerformanceSnapshotReadService, PerformanceRecomputeQueueService,	backend/src/modules/performance/performance.module.ts
ProfilesModule	profiles	9	DatabaseModule, AuditModule	ProfilesController	ProfilesService	ProfilesService	backend/src/modules/profiles/profiles.module.ts
RagModule	rag	0	ConfigModule, BullModule.register-	No direct controller array extracted	RagIndexingService, RagIndexingProcessor	RagIndexingService	backend/src/modules/rag/rag.module.ts

Module class	Primary slice	Route count in slice	Extracted imports	Extracted controllers	Extracted providers	Extracted exports	Source
			Queue({ name: 'rag-indexing', }),				
ReportsModule	reports	6	DatabaseModule, AuditModule	ReportsController	ReportsService	ReportsService	backend/src/modules/reports/reports.module.ts
RolesModule	roles	0	No explicit imports extracted	No direct controller array extracted	RolesService	RolesService	backend/src/modules/roles/roles.module.ts
RosterImportModule	roster-import	4	DatabaseModule	RosterImportController	RosterImportService	RosterImportService	backend/src/modules/roster-import/roster-import.module.ts
SchoolEventsModule	school-events	4	AuditModule	SchoolEventsController	SchoolEventsService	SchoolEventsService	backend/src/modules/school-events/school-events.module.ts
SectionsModule	sections	25	DatabaseModule, AuditModule, ClassRecordModule	SectionsController, SectionsPublicController	SectionsService	SectionsService	backend/src/modules/sections/sections.module.ts
TeacherProfilesModule	teacher-profiles	4	DatabaseModule, AuditModule	TeacherProfileController	TeacherProfilesService	TeacherProfilesService	backend/src/modules/teacher-profiles/teacher-profiles.module.ts
TeacherModule	teacher	3	LessonsModule, AssessmentsModule, ClassesModule	TeacherController	TeacherService	No direct export array extracted	backend/src/modules/teacher/teacher.module.ts
UsersModule	users	13	RolesModule, OTPModule, MailModule, AuditModule	UsersController	UsersService, UserEventsListener	UsersService	backend/src/modules/users/users.module.ts

Module ownership rules

- Controllers translate HTTP transport into typed domain calls; business invariants belong in services.
- Drizzle schema and database providers remain the persistence boundary. Clients never query PostgreSQL directly.
- Queue producer services own asynchronous payload construction. Processors call domain or internal execution services.
- Backend AI clients attach the internal secret and forwarded user context. Web and mobile are not permitted to perform that call.
- Official scores, roles, enrollment, publication state, audit history, and session state remain backend-owned even when another process computes assistive output.

Response and error contract

- Successful response bodies are handler-specific. The application does not install a global success-envelope interceptor.
- Nest defaults to HTTP 200 for most handlers and HTTP 201 for POST handlers unless `HttpCode` or a response decorator overrides it.
- Controller decorators in the catalog record explicit status or Swagger response declarations where present.
- The global exception filter returns a sanitized failure body with success false, statusCode, message, timestamp, and path. Optional data is preserved only when an `HttpException` explicitly supplies it.
- Unexpected exceptions are logged with stack detail internally and return the generic message `An unexpected error occurred`.
- File-size overflow returns 413 with the 100 MB upload-limit message; oversized JSON or form bodies return 413 with a generic payload-size message.


```
{
  "success": false,
  "statusCode": 403,
  "message": "Forbidden resource",
  "timestamp": "2026-07-13T00:00:00.000Z",
  "path": "/api/classes/7f9c4e84-37b8-4e7d-beb9-2bd77e2c4081"
}
```

Complete controller and endpoint catalog

Exhaustive inventory rule. The 385 HTTP routes below were extracted from every `backend/src/**/*.controller.ts` source at commit `3d0c93e`. A later source change requires regenerating or manually reconciling this chapter.

Each full path already includes the global `/api` prefix. “Primary delegates” lists direct `this.<service>.<method>` calls extracted from the handler body; handler-local orchestration may perform additional work.

AcademicStateController

Base path: `academic-state`. Source: `backend/src/modules/academic-state/academic-state.controller.ts`. Route count: 3.

Method	Full path	Access	Handler	Inputs	Declared re- sponse and errors	Primary delegates
GET	<code>/api/academic-state/current</code>	JWT plus ADMIN	<code>getCurrent</code>	No handler parameters	200 by Nest default; global filter formats failures	<code>this.academic-StateService.getCurrentState</code>
GET	<code>/api/academic-state/impact-preview</code>	JWT plus ADMIN	<code>getImpactPreview</code>	Query() query: ImpactPreviewQuery-Dto	200 by Nest default; global filter formats failures	<code>this.academic-StateService.getImpactPreview</code>
POST	<code>/api/academic-state/transition</code>	JWT plus ADMIN	<code>transition</code>	Body() dto: TransitionAcademic-StateDto; CurrentUser() user: any	201 by Nest default unless overridden; global filter formats failures	<code>this.academic-StateService.transition</code>

AdminController

Base path: `admin`. Source: `backend/src/modules/admin/admin.controller.ts`. Route count: 5.

Method	Full path	Access	Handler	Inputs	Declared re- sponse and errors	Primary delegates
GET	<code>/api/admin/activity-export</code>	JWT plus ADMIN	<code>exportActivity</code>	Query('dateFrom') dateFrom: string undefined; Query('dateTo') dateTo: string undefined; Res() res: Response	200 by Nest default; global filter formats failures	<code>this.adminService.getUsageSummary</code>
GET	<code>/api/admin/audit-logs</code>	JWT plus ADMIN	<code>getAuditLogs</code>	Query('page') page: string optional; Query('limit') limit: string optional; Query('action') action: string optional; Query('actorId') actorId: string optional; Query('dateFrom') dateFrom: string optional; Query('dateTo') dateTo: string optional	200 by Nest default; global filter formats failures	<code>this.adminService.getAuditLogs</code>
GET	<code>/api/admin/dashboard/stats</code>	JWT plus ADMIN	<code>getDashboardStats</code>	No handler parameters		

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
					200 by Nest default; global filter formats failures	this.adminService.getDashboardStats
GET	/api/admin/overview	JWT plus ADMIN	getOverview	No handler parameters	200 by Nest default; global filter formats failures	this.adminService.getDashboardOverview
GET	/api/admin/usage-summary	JWT plus ADMIN	getUsageSummary	Query('dateFrom') dateFrom: string optional; Query('dateTo') dateTo: string optional	200 by Nest default; global filter formats failures	this.adminService.getUsageSummary

AiMentorController

Base path: `ai`. Source: `backend/src/modules/ai-mentor/ai-mentor.controller.ts`. Route count: 41.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
POST	/api/ai/admin/chat	JWT plus STUDENT or TEACHER or ADMIN	adminChat	Body() dto: AdminAnalyticsChatRequestDto; CurrentUser() user:	HttpCode HttpStatus.OK	this.adminAnalyticsChatService.chat
GET	/api/ai/admin/history	JWT plus STUDENT or TEACHER or ADMIN	adminHistory	CurrentUser() user:	200 by Nest default; global filter formats failures	this.adminAnalyticsChatService.history
GET	/api/ai/admin/sessions/:sessionId	JWT plus STUDENT or TEACHER or ADMIN	getAdminSession	Param('sessionId', ParseUUIDPipe) sessionId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.adminAnalyticsChatService.getSession
POST	/api/ai/chat	JWT plus STUDENT or ADMIN	chat	Body() dto: ChatRequestDto; CurrentUser() user:	HttpCode HttpStatus.OK; 200 Ja	this.proxy.forward
POST	/api/ai/demo/intervention-plan	Public through @Public	generateDemoInterventionPlan	Body() dto: DemoInterventionPlanDto	HttpCode HttpStatus.OK; 200 Demo intervention plan generated; Throws ServiceUnavailableException	this.proxy.forward; this.logger.warn
POST	/api/ai/extract-module	JWT plus TEACHER or ADMIN	extractModule	Body() dto: ExtractModuleDto; CurrentUser() user:	HttpCode HttpStatus.ACCEPTED; 202 Extraction queued - poll for status; Throws ServiceUnavailableException	this.proxy.forward; this.aiGenerationQueueService.enqueueExtractionJob; this.proxy.markInternalExtractionFailed; this.logger.error; this.logger.warn
GET	/api/ai/extractions	JWT plus TEACHER or ADMIN	listExtractions	Query('classId', ParseUUIDPipe) classId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.proxy.forward; this.logger.warn; this.db.query.extractedModules.findMany
DELETE	/api/ai/extractions/:id	JWT plus TEACHER or ADMIN	deleteExtraction	Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	Throws ServiceUnavailableException	this.proxy.forward; this.logger.warn
GET	/api/ai/extractions/:id	JWT plus TEACHER or ADMIN	getExtraction	Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	Throws NotFoundException	this.proxy.forward; this.logger.warn; this.db.query.extractedModules.findFirst
PATCH	/api/ai/extractions/:id	JWT plus TEACHER or ADMIN	updateExtraction	Param('id', ParseUUIDPipe) id: string; Body() dto: UpdateExtractionDto; CurrentUser() user:	200 Extraction updated; Throws ServiceUnavailableException	this.proxy.forward; this.logger.warn
POST	/api/ai/extractions/:id/apply	JWT plus TEACHER or ADMIN	applyExtraction	Param('id', ParseUUIDPipe) id: string; Body() dto: ApplyExtractionDto; CurrentUser() user:	HttpCode HttpStatus.CREATED; 201 Module sections with draft lessons/assessments created from extraction; Throws NotFoundException,	this.proxy.forward; this.logger.warn; this.db.query.extractedModules.findFirst

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
					ServiceUnavailableException	
POST	/api/ai/extractions/:id/apply/preview	JWT plus TEACHER or ADMIN	previewApplyExtraction	Param('id', ParseUUIDPipe) id: string; Body() dto: ApplyExtractionDto; CurrentUser() user:	Throws ServiceUnavailableException	this.proxy.forward; this.logger.warn
POST	/api/ai/extractions/:id/cancel	JWT plus TEACHER or ADMIN	cancelExtraction	Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	Throws ServiceUnavailableException	this.proxy.forward; this.aiGenerationQueueService.cancelQueuedExtractionJob; this.logger.warn
POST	/api/ai/extractions/:id/retry	JWT plus TEACHER or ADMIN	retryExtraction	Param('id', ParseUUIDPipe) id: string; Body() dto: RetryExtractionDto; CurrentUser() user:	HttpCode HttpStatus.ACCEPTED; Throws ServiceUnavailableException	this.proxy.forward; this.aiGenerationQueueService.enqueueExtractionJob; this.proxy.markInternalExtractionFailed; this.logger.error; this.logger.warn
GET	/api/ai/extractions/:id/status	JWT plus TEACHER or ADMIN	getExtractionStatus	Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	Throws NotFoundException	this.proxy.forward; this.logger.warn; this.db.query.extractedModules.findFirst
GET	/api/ai/health	Public through @Public	health	No handler parameters	200 by Nest default; global filter formats failures	this.proxy.forward; this.logger.warn
GET	/api/ai/history	JWT plus STUDENT or TEACHER or ADMIN	history	CurrentUser() user:	200 by Nest default; global filter formats failures	this.proxy.forward; this.logger.warn
POST	/api/ai/index/classes/:classId	JWT plus TEACHER or ADMIN	reindexClass	Param('classId', ParseUUIDPipe) classId: string; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.proxy.forward
GET	/api/ai/index/classes/:classId/status	JWT plus TEACHER or ADMIN	getIndexClassStatus	Param('classId', ParseUUIDPipe) classId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.proxy.forward
POST	/api/ai/mentor/explain	JWT plus STUDENT or ADMIN	explainMistake	Body() dto: MentorExplainDto; CurrentUser() user:	HttpCode HttpStatus.OK	this.proxy.forward
GET	/api/ai/student/tutor/bootstrap	JWT plus STUDENT	studentTutorBootstrap	Query() query: StudentTutorBootstrapQueryDto; CurrentUser() user:	200 by Nest default; global filter formats failures	this.proxy.forward
POST	/api/ai/student/tutor/session	JWT plus STUDENT	startStudentTutorSession	Body() dto: StudentTutorStartDto; CurrentUser() user:	Throws ForbiddenException	this.proxy.forward
GET	/api/ai/student/tutor/session/:sessionId	JWT plus STUDENT	getStudentTutorSession	Param('sessionId', ParseUUIDPipe) sessionId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.proxy.forward
POST	/api/ai/student/tutor/session/:sessionId/answers	JWT plus STUDENT	answerStudentTutorSession	Param('sessionId', ParseUUIDPipe) sessionId: string; Body() dto: StudentTutorAnswersDto; CurrentUser() user:	Throws NotFoundException	this.proxy.forward
POST	/api/ai/student/tutor/session/:sessionId/message	JWT plus STUDENT	messageStudentTutorSession	Param('sessionId', ParseUUIDPipe) sessionId: string; Body() dto: StudentTutorMessageDto; CurrentUser() user:	Throws NotFoundException	this.proxy.forward
GET	/api/ai/teacher/classes/:classId/policy	JWT plus TEACHER or ADMIN	getTeacherClassAiPolicy	Param('classId', ParseUUIDPipe) classId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	Handler-local composition or no direct this-call extracted
PATCH	/api/ai/teacher/classes/:classId/policy	JWT plus TEACHER or ADMIN	updateTeacherClassAiPolicy	Param('classId', ParseUUIDPipe) classId: string; Body() dto: Update-	Throws BadRequestException	this.db.insert(classAiPolicies).values({ classId, mentorEx-

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
				ClassAiPolicyDto; CurrentUser() user:		plainEnabled: dto.mentorExplainEnabled ?? this.defaultClassAiPolicy.mentorExplainEnabled, maxFollowUpTurns: dto.maxFollowUpTurns ?? this.defaultClassAiPolicy.maxFollowUpTurns, sourceScope: dto.sourceScope ?? this.defaultClassAiPolicy.sourceScope, strictGrounding: dto.strictGrounding ?? this.defaultClassAiPolicy.strictGrounding, updatedBy: user.id, }) .onConflictDoUpdate({ target: classAiPolicies.classId, set: { mentorExplainEnabled: dto.mentorExplainEnabled ?? sql \${classAiPolicies.mentorExplainEnabled}, maxFollowUpTurns: dto.maxFollowUpTurns ?? sql \${classAiPolicies.maxFollowUpTurns}, sourceScope: dto.sourceScope ?? sql \${classAiPolicies.sourceScope}, strictGrounding: dto.strictGrounding ?? sql \${classAiPolicies.strictGrounding}, updatedBy: user.id, updatedAt: new Date(), }, },).returning; this.db .insert(classAiPolicies) .values({ classId, mentorExplainEnabled: dto.mentorExplainEnabled ?? this.defaultClassAiPolicy.mentorExplainEnabled, maxFollowUpTurns: dto.maxFollowUpTurns ?? this.defaultClassAiPolicy.maxFollowUpTurns, sourceScope: dto.sourceScope ?? this.defaultClassAiPolicy.sourceScope, strictGrounding: dto.strictGrounding ?? this.defaultClassAiPolicy.strictGrounding, updatedBy: user.id, },).onConflictDoUpdate; this.db .insert(classAiPolicies) .values; this.db.insert
POST	/api/ai/teacher/interventions/caseld/jobs	JWT plus TEACHER or ADMIN	queueInterventionRecommendation	Param('caseld', ParseUUIDPipe) caseld: string; Body() dto: InterventionRecommendationDto; CurrentUser() user:	HttpCode HttpStatus.ACCEPTED	this.proxy.forward; this.aiGenerationQueueService.enqueueInterventionJob

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
POST	/api/ai/teacher/interventions/:caseld/recommend	JWT plus TEACHER or ADMIN	recommendIntervention	Param('caseld', ParseUUIDPipe) caseld: string; Body() dto: InterventionRecommendationDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.proxy.forward
DELETE	/api/ai/teacher/jobs/:jobld	JWT plus TEACHER or ADMIN	deleteTeacherJob	Param('jobld', ParseUUIDPipe) jobld: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.aiGenerationQueueService.cancelQueuedTeacherAJob; this.proxy.forward
GET	/api/ai/teacher/jobs/:jobld	JWT plus TEACHER or ADMIN	getTeacherJobStatus	Param('jobld', ParseUUIDPipe) jobld: string; CurrentUser() user:	Throws NotFoundException	this.proxy.forward; this.logger.warn; this.db.query.aiGenerationJobs.findFirst; this.db.query.aiGenerationOutputs.findFirst
GET	/api/ai/teacher/jobs/:jobld/result	JWT plus TEACHER or ADMIN	getTeacherJobResult	Param('jobld', ParseUUIDPipe) jobld: string; CurrentUser() user:	Throws NotFoundException	this.proxy.forward; this.logger.warn; this.db.query.aiGenerationJobs.findFirst; this.db.query.aiGenerationOutputs.findFirst
POST	/api/ai/teacher/lesson-plans/jobs	JWT plus TEACHER or ADMIN	queueLessonPlanJob	Body() dto: GenerateLessonPlanDto; CurrentUser() user:	HttpCode HttpStatus.ACCEPTED	this.proxy.forward; this.aiGenerationQueueService.enqueueLessonPlanJob
PATCH	/api/ai/teacher/lesson-plans/jobs/:jobld/draft	JWT plus TEACHER or ADMIN	updateLessonPlanDraft	Param('jobld', ParseUUIDPipe) jobld: string; Body() dto: UpdateLessonPlanDraftDto; CurrentUser() user:	200 by Nest default; global filter formats failures	this.proxy.forward
POST	/api/ai/teacher/quizzes/generate-draft	JWT plus TEACHER or ADMIN	generateQuizDraft	Body() dto: GenerateQuizDraftDto; CurrentUser() user:	HttpCode HttpStatus.CREATED	this.proxy.forward
POST	/api/ai/teacher/quizzes/jobs	JWT plus TEACHER or ADMIN	queueQuizDraftJob	Body() dto: GenerateQuizDraftDto; CurrentUser() user:	HttpCode HttpStatus.ACCEPTED	this.proxy.forward; this.aiGenerationQueueService.enqueueQuizJob
POST	/api/ai/teacher/quizzes/jobs/:jobld/apply	JWT plus TEACHER or ADMIN	applyQuizDraft	Param('jobld', ParseUUIDPipe) jobld: string; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.proxy.forward
POST	/api/ai/teacher/quizzes/jobs/:jobld/apply/preview	JWT plus TEACHER or ADMIN	previewQuizDraftApply	Param('jobld', ParseUUIDPipe) jobld: string; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.proxy.forward
POST	/api/ai/teacher/quizzes/jobs/:jobld/cancel	JWT plus TEACHER or ADMIN	cancelQuizDraftJob	Param('jobld', ParseUUIDPipe) jobld: string; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.aiGenerationQueueService.cancelQueuedTeacherAJob; this.proxy.forward
PATCH	/api/ai/teacher/quizzes/jobs/:jobld/draft	JWT plus TEACHER or ADMIN	updateQuizDraft	Param('jobld', ParseUUIDPipe) jobld: string; Body() dto: UpdateQuizDraftDto; CurrentUser() user:	200 by Nest default; global filter formats failures	this.proxy.forward
POST	/api/ai/teacher/quizzes/jobs/:jobld/retry	JWT plus TEACHER or ADMIN	retryQuizDraftJob	Param('jobld', ParseUUIDPipe) jobld: string; CurrentUser() user:	HttpCode HttpStatus.ACCEPTED	this.proxy.forward; this.aiGenerationQueueService.enqueueQuizJob

JaHubController

Base path: `ai/student/ja` . Source: `backend/src/modules/ja/ja-hub.controller.ts` . Route count: 11.

Method	Full path	Access	Handler	Inputs	Declared re- sponse and errors	Primary delegates
GET	/api/ai/student/ja/ask/bootstrap	JWT plus STUDENT	askBootstrap	Query() query: JaAskBootstrapQueryDto; CurrentUser() user:	200 by Nest default; global filter formats failures	this.jaService.askBootstrap
POST	/api/ai/student/ja/ask/threads	JWT plus STUDENT	createAskThread	Body() dto: CreateJaAskThreadDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.jaService.createAskThread
GET	/api/ai/student/ja/ask/threads/:threadId	JWT plus STUDENT	getAskThread	Param('threadId', ParseUUIDPipe) threadId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.jaService.getAskThread
POST	/api/ai/student/ja/ask/threads/:threadId/messages	JWT plus STUDENT	sendAskMessage	Param('threadId', ParseUUIDPipe) threadId: string; Body() dto: SendJaAskMessageDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.jaService.sendAskMessage
GET	/api/ai/student/ja/hub	JWT plus STUDENT	hub	Query() query: JaHubQueryDto; CurrentUser() user:	200 by Nest default; global filter formats failures	this.jaService.hub
GET	/api/ai/student/ja/review/bootstrap	JWT plus STUDENT	reviewBootstrap	Query() query: JaReviewBootstrapQueryDto; CurrentUser() user:	200 by Nest default; global filter formats failures	this.jaService.reviewBootstrap
POST	/api/ai/student/ja/review/sessions	JWT plus STUDENT	createReviewSession	Body() dto: CreateJaReviewSessionDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.jaService.createReviewSession
GET	/api/ai/student/ja/review/sessions/:sessionId	JWT plus STUDENT	getReviewSession	Param('sessionId', ParseUUIDPipe) sessionId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.jaService.getSession
POST	/api/ai/student/ja/review/sessions/:sessionId/complete	JWT plus STUDENT	completeReviewSession	Param('sessionId', ParseUUIDPipe) sessionId: string; Body() _dto: CompleteJaPracticeSessionDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.jaService.completeSession
POST	/api/ai/student/ja/review/sessions/:sessionId/events	JWT plus STUDENT	logReviewEvent	Param('sessionId', ParseUUIDPipe) sessionId: string; Body() dto: JaReviewEventDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.jaService.addEvent
POST	/api/ai/student/ja/review/sessions/:sessionId/responses	JWT plus STUDENT	submitReviewResponse	Param('sessionId', ParseUUIDPipe) sessionId: string; Body() dto: JaReviewSubmitResponseDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.jaService.submitResponse

JaController

Base path: `ai/student/ja/practice` . Source: `backend/src/modules/ja/ja.controller.ts` . Route count: 7.

Method	Full path	Access	Handler	Inputs	Declared re- sponse and errors	Primary delegates
GET	/api/ai/student/ja/practice/bootstrap	JWT plus STUDENT	bootstrap	Query() query: JaPracticeBootstrapQueryDto; CurrentUser() user:	200 by Nest default; global filter formats failures	this.jaService.bootstrap
POST	/api/ai/student/ja/practice/sessions	JWT plus STUDENT	createSession	Body() dto: CreateJaPracticeSessionDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.jaService.createSession
DELETE	/api/ai/student/ja/practice/sessions/:sessionId	JWT plus STUDENT	deleteSession	Param('sessionId', ParseUUIDPipe) sessionId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.jaService.deleteSession

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
GET	/api/ai/student/ja/practice/sessions/:sessionId	JWT plus STUDENT	getSession	Param('sessionId', ParseUUIDPipe) sessionId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.jaService.getSession
POST	/api/ai/student/ja/practice/sessions/:sessionId/complete	JWT plus STUDENT	completeSession	Param('sessionId', ParseUUIDPipe) sessionId: string; Body() _dto: CompleteJaPracticeSessionDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.jaService.completeSession
POST	/api/ai/student/ja/practice/sessions/:sessionId/events	JWT plus STUDENT	logEvent	Param('sessionId', ParseUUIDPipe) sessionId: string; Body() dto: LogJaPracticeEventDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.jaService.addEvent
POST	/api/ai/student/ja/practice/sessions/:sessionId/responses	JWT plus STUDENT	submitResponse	Param('sessionId', ParseUUIDPipe) sessionId: string; Body() dto: SubmitJaPracticeResponseDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.jaService.submitResponse

AnalyticsController

Base path: `analytics` . Source: `backend/src/modules/analytics/analytics.controller.ts` . Route count: 4.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
GET	/api/analytics/admin/overview	JWT plus ADMIN	getAdminOverview	No handler parameters	200 by Nest default; global filter formats failures	this.analyticsService.getAdminOverview
GET	/api/analytics/classes/:classId/intervention-outcomes	JWT plus TEACHER or ADMIN	getInterventionOutcomes	Param('classId') classId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.analyticsService.getInterventionOutcomes
GET	/api/analytics/classes/:classId/trends	JWT plus TEACHER or ADMIN	getClassTrends	Param('classId') classId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.analyticsService.getClassTrends
GET	/api/analytics/teachers/:teacherId/workload	JWT plus TEACHER or ADMIN	getTeacherWorkload	Param('teacherId') teacherId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.analyticsService.getTeacherWorkload

AppVersionController

Base path: `app-version` . Source: `backend/src/modules/app-version/app-version.controller.ts` . Route count: 1.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
GET	/api/app-version/check	Public through @Public	check	Query() query: CheckAppVersionDto	200 App version check evaluated successfully	this.appVersionService.checkVersion

AssessmentsController

Base path: `assessments` . Source: `backend/src/modules/assessments/assessments.controller.ts` . Route count: 36.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
POST	/api/assessments	JWT plus ADMIN or TEACHER	createAssessment	Body() createAssessmentDto: CreateAssessmentDto; CurrentUser() user: any	HttpCode HttpStatus.CREATED	this.assessmentsService.createAssessment
GET				Param('assessmentId') assessmentId:		this.assessmentsSer-

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
	/api/assessments/:assessmentId/all-attempts	JWT plus ADMIN or TEACHER	getAssessmentAttempts	string; CurrentUser() user: any	200 by Nest default; global filter formats failures	vice.getAssessmentAttempts
GET	/api/assessments/:assessmentId/ongoing-attempt	JWT plus ADMIN or STUDENT	getOngoingAttempt	Param('assessmentId') assessmentId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.assessmentsService.getOngoingAttempt
GET	/api/assessments/:assessmentId/question-analytics	JWT plus ADMIN or TEACHER	getQuestionAnalytics	Param('assessmentId') assessmentId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.assessmentsService.getQuestionAnalytics
POST	/api/assessments/:assessmentId/return-all	JWT plus ADMIN or TEACHER	returnAllGrades	Param('assessmentId') assessmentId: string; Body() returnGradeDto: ReturnGradeDto; CurrentUser() user: any	HttpCode HttpStatus.OK	this.assessmentsService.returnAllGrades
PUT	/api/assessments/:assessmentId/rubric-review	JWT plus ADMIN or TEACHER	reviewRubric	Param('assessmentId') assessmentId: string; CurrentUser() user: any; Body() dto: UpdateAssessmentDto	200 by Nest default; global filter formats failures	this.assessmentsService.reviewRubric
POST	/api/assessments/:assessmentId/rubric-source	JWT plus ADMIN or TEACHER	uploadRubricSource	Param('assessmentId') assessmentId: string; CurrentUser() user: any; UploadedFile() file: Express.Multer.File	Throws BadRequestException	this.assessmentsService.uploadRubricSource
POST	/api/assessments/:assessmentId/start	JWT plus ADMIN or STUDENT	startAttempt	Param('assessmentId') assessmentId: string; CurrentUser() user: any	HttpCode HttpStatus.CREATED	this.assessmentsService.startAttempt
GET	/api/assessments/:assessmentId/stats	JWT plus ADMIN or TEACHER	getAssessmentStats	Param('assessmentId') assessmentId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.assessmentsService.getAssessmentStats
GET	/api/assessments/:assessmentId/student-attempts	JWT plus ADMIN or TEACHER or STUDENT	getStudentAttempts	Param('assessmentId') assessmentId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.assessmentsService.getStudentAttempts
POST	/api/assessments/:assessmentId/submission-file	JWT plus STUDENT	uploadSubmissionFile	Param('assessmentId') assessmentId: string; CurrentUser() user: any; UploadedFile() file: Express.Multer.File	Throws BadRequestException	this.assessmentsService.uploadStudentSubmissionFile
DELETE	/api/assessments/:assessmentId/submission-files/:fileId	JWT plus STUDENT	removeSubmissionFile	Param('assessmentId') assessmentId: string; Param('fileId') fileId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.assessmentsService.removeStudentSubmissionFile
GET	/api/assessments/:assessmentId/submissions	JWT plus ADMIN or TEACHER	getAssessmentSubmissions	Param('assessmentId') assessmentId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.assessmentsService.getAssessmentSubmissions
POST	/api/assessments/:assessmentId/teacher-attachment	JWT plus ADMIN or TEACHER	uploadTeacherAttachment	Param('assessmentId') assessmentId: string; CurrentUser() user: any; UploadedFile() file: Express.Multer.File	Throws BadRequestException	this.assessmentsService.uploadTeacherAttachment
GET	/api/assessments/:assessmentId/teacher-attachment/download	JWT plus ADMIN or TEACHER or STUDENT	downloadTeacherAttachment	Param('assessmentId') assessmentId: string; CurrentUser() user: any; Res() res: Response	Throws BadRequestException	this.assessmentsService.getTeacherAttachmentDownload
POST	/api/assessments/:assessmentId/unsubmit-file-upload	JWT plus ADMIN or STUDENT	unsubmitFileUploadAssessment	Param('assessmentId') assessmentId: string; CurrentUser() user: any	HttpCode HttpStatus.OK	this.assessmentsService.unsubmitFileUploadAssessment
DELETE	/api/assessments/:id		deleteAssessment			this.assessmentsSer-

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
		JWT plus ADMIN or TEACHER		Param('id') id: string; CurrentUser() user: any	HttpCode HttpStatusCode.OK	vice.deleteAssessment
GET	/api/assessments/:id	JWT plus ADMIN or TEACHER or STUDENT	getAssessmentById	Param('id') id: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.assessmentsService.getAssessmentById
PUT	/api/assessments/:id	JWT plus ADMIN or TEACHER	updateAssessment	Param('id') id: string; Body() updateAssessmentDto: UpdateAssessmentDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.assessmentsService.updateAssessment
PATCH	/api/assessments/:id/core-release	JWT plus ADMIN or TEACHER	releaseCoreAssessment	Param('id') id: string; Body() dto: ReleaseCoreAssessmentDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.assessmentsService.releaseCoreAssessment
PATCH	/api/assessments/attempts/:attemptId/progress	JWT plus ADMIN or STUDENT	updateAttemptProgress	Param('attemptId') attemptId: string; Body() updateAttemptProgressDto: UpdateAttemptProgressDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.assessmentsService.updateAttemptProgress
GET	/api/assessments/attempts/:attemptId/results	JWT plus ADMIN or TEACHER or STUDENT	getAttemptResults	Param('attemptId') attemptId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.assessmentsService.getAttemptResults
POST	/api/assessments/attempts/:attemptId/return	JWT plus ADMIN or TEACHER	returnGrade	Param('attemptId') attemptId: string; Body() returnGradeDto: ReturnGradeDto; CurrentUser() user: any	HttpCode HttpStatusCode.OK	this.assessmentsService.returnGrade
GET	/api/assessments/attempts/:attemptId/submission-file/download	JWT plus ADMIN or TEACHER or STUDENT	downloadAttemptSubmissionFile	Param('attemptId') attemptId: string; CurrentUser() user: any; Res() res: Response	Throws BadRequestException	this.assessmentsService.getAttemptSubmissionDownload
GET	/api/assessments/attempts/:attemptId/submission-files/:fileId/download	JWT plus ADMIN or TEACHER or STUDENT	downloadAttemptSubmissionAttachmentFile	Param('attemptId') attemptId: string; Param('fileId') fileId: string; CurrentUser() user: any; Res() res: Response	Throws BadRequestException	this.assessmentsService.getAttemptSubmissionDownload
POST	/api/assessments/attempts/:attemptId/unreturn	JWT plus ADMIN or TEACHER	unreturnGrade	Param('attemptId') attemptId: string; CurrentUser() user: any	HttpCode HttpStatusCode.OK	this.assessmentsService.unreturnGrade
POST	/api/assessments/attempts/bulk-return	JWT plus ADMIN or TEACHER	bulkReturnGrades	Body() bulkReturnGradesDto: BulkReturnGradesDto; CurrentUser() user: any	HttpCode HttpStatusCode.OK	this.assessmentsService.bulkReturnGrades
GET	/api/assessments/attempts/ongoing	JWT plus ADMIN or STUDENT	getOngoingAttempts	CurrentUser() user: any	200 by Nest default; global filter formats failures	this.assessmentsService.getOngoingAttempts
GET	/api/assessments/class/:classId	JWT plus ADMIN or TEACHER or STUDENT	getAssessmentsByClass	Param('classId') classId: string; Query('page') pageQuery: string undefined; Query('limit') limitQuery: string undefined; Query('status') statusQuery: 'all' 'upcoming' 'past_due' 'completed' undefined; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.assessmentsService.getAssessmentsByClass
POST	/api/assessments/options/:id/image	JWT plus ADMIN or TEACHER	uploadOptionImage	Param('id') id: string; CurrentUser() user: any; Up-	Throws BadRequestException	this.assessmentsSer-

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
				loadedFile() file: Express.Multer.File		vice.updateQuestionOptionImage
POST	/api/assessments/questions	JWT plus ADMIN or TEACHER	createQuestion	Body() createQuestionDto: CreateQuestionDto; CurrentUser() user: any	HttpCode HttpStatus.CREATED	this.assessmentsService.createQuestion
DELETE	/api/assessments/questions/:id	JWT plus ADMIN or TEACHER	deleteQuestion	Param('id') id: string; CurrentUser() user: any	HttpCode HttpStatus.OK	this.assessmentsService.deleteQuestion
PUT	/api/assessments/questions/:id	JWT plus ADMIN or TEACHER	updateQuestion	Param('id') id: string; Body() updateQuestionDto: UpdateQuestionDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.assessmentsService.updateQuestion
POST	/api/assessments/questions/:id/image	JWT plus ADMIN or TEACHER	uploadQuestionImage	Param('id') id: string; CurrentUser() user: any; UploadedFile() file: Express.Multer.File	Throws BadRequestException	this.assessmentsService.updateQuestion
GET	/api/assessments/questions/images-private/:filename	Public through @Public	serveQuestionImage	Param('filename') filename: string; Res() res: Response	Throws BadRequestException	Handler-local composition or no direct this-call extracted
POST	/api/assessments/submit	JWT plus ADMIN or STUDENT	submitAssessment	Body() submitAssessmentDto: SubmitAssessmentDto; CurrentUser() user: any	HttpCode HttpStatus.OK	this.assessmentsService.submitAssessment

AssessmentsPublicController

Base path: `assessments` . Source: `backend/src/modules/assessments/assessments-public.controller.ts` . Route count: 1.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
GET	/api/assessments/questions/images/:filename	Public through @Public	serveQuestionImage	Param('filename') filename: string; Res() res: Response	Throws BadRequestException	Handler-local composition or no direct this-call extracted

AuthController

Base path: `auth` . Source: `backend/src/modules/auth/auth.controller.ts` . Route count: 15.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
POST	/api/auth/change-password	JWT authenticated	changePassword	CurrentUser() user: any; Body() dto: ChangePasswordDto	HttpCode HttpStatus.OK; 200 Password changed successfully; 400 Current password is incorrect; Throws UnauthorizedException	this.authService.changePassword
POST	/api/auth/forgot-password	Public through @Public	forgotPassword	Body() dto: ForgotPasswordDto	200 Reset code sent; 404 Account does not exist	this.authService.requestPasswordReset
POST	/api/auth/login	Public through @Public	login	Body() loginDto: LoginDto; Req() request: express.Request; Res({ passthrough: true }) response: express.Response	HttpCode HttpStatus.OK; 200 Login successful; 401 Invalid credentials; 429 Too many requests	this.authService.login
POST	/api/auth/logout	Public through @Public	logout	Req() request: express.Request; Res({ passthrough: true }) response: express.Response	HttpCode HttpStatus.OK; 200 Logout successful	this.authService.logout

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
POST	/api/auth/logout-all	JWT authenticated	logoutAll	CurrentUser() user: any; Res({ passthrough: true }) response: express.Response	HttpCode HttpStatus.OK; 200 All sessions revoked; Throws UnauthorizedException	this.tokenService.revokeAllForUser
GET	/api/auth/me	JWT authenticated	getCurrentUser	CurrentUser() user: any	200 Current user data	Handler-local composition or no direct this-call extracted
POST	/api/auth/mobile/login	Public through @Public	mobileLogin	Body() loginDto: LoginDto; Req() request: express.Request	HttpCode HttpStatus.OK; 200 Mobile login successful	this.authService.login
POST	/api/auth/mobile/logout	Public through @Public	mobileLogout	Body() dto: MobileLogoutDto	HttpCode HttpStatus.OK; 200 Mobile logout successful	this.authService.logout
POST	/api/auth/mobile/refresh	Public through @Public	mobileRefresh	Body() dto: MobileRefreshDto; Req() request: express.Request	200 Mobile token refreshed	this.authService.refreshToken
PATCH	/api/auth/profile	JWT authenticated	updateProfile	CurrentUser() user: any; Body() dto: UpdateProfileDto	200 Profile updated; Throws UnauthorizedException, InternalServerErrorException	this.logger.warn; this.logger.debug; this.authService.updateProfile; this.logger.error
POST	/api/auth/refresh	Public through @Public	refresh	Req() request: express.Request; Res({ passthrough: true }) response: express.Response	200 Token refreshed; 401 Invalid or expired refresh token; Throws UnauthorizedException	this.authService.refreshToken
POST	/api/auth/reset-password	Public through @Public	resetPassword	Body() dto: ResetPasswordDto	200 Password reset successful	this.authService.resetPassword
POST	/api/auth/set-activation-password	Public through @Public	setActivationPassword	Body() dto: SetActivationPasswordDto	HttpCode HttpStatus.OK; 200 Password set successfully	this.authService.setActivationPassword
POST	/api/auth/set-initial-password	Public through @Public	setInitialPassword	Body() dto: SetInitialPasswordDto	HttpCode HttpStatus.OK; 200 Password set successfully	this.authService.setInitialPassword
POST	/api/auth/validate-credentials	Public through @Public	validateCredentials	Body() dto: ValidateCredentialsDto	HttpCode HttpStatus.OK; 200 Credentials valid; 401 Invalid credentials; 429 Too many requests	this.authService.validateCredentials

ClassRecordController

Base path: `class-record`. Source: `backend/src/modules/class-record/class-record.controller.ts`. Route count: 18.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
POST	/api/class-record	JWT plus TEACHER or ADMIN	generateClassRecord	Body() dto: CreateClassRecordDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.classRecordService.generateClassRecord
GET	/api/class-record/:classRecordId/final-grades/:studentId	JWT plus TEACHER or ADMIN or STUDENT	getStudentGrade	Param('classRecordId', ParseUUIDPipe) classRecordId: string; Param('studentId', ParseUUIDPipe) studentId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.classRecordService.getStudentGrade
GET	/api/class-record/:id	JWT plus TEACHER or ADMIN	getClassRecord	Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.classRecordService.getClassRecord
GET			getFinalGrades	Param('id', ParseUUIDPipe) id:		

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
	/api/class-record/:id/final-grades	JWT plus TEACHER or ADMIN		string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.classRecordService.getFinalGrades
POST	/api/class-record/:id/finalize	JWT plus TEACHER or ADMIN	finalizeClassRecord	Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.classRecordService.finalizeClassRecord
GET	/api/class-record/:id/preview-grades	JWT plus TEACHER or ADMIN	previewGrades	Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.classRecordService.previewGrades
POST	/api/class-record/:id/reopen	JWT plus TEACHER or ADMIN	reopenClassRecord	Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.classRecordService.reopenClassRecord
GET	/api/class-record/:id/reports/class-average	JWT plus TEACHER or ADMIN	classAverage	Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.classRecordService.getClassAverage
GET	/api/class-record/:id/reports/distribution	JWT plus TEACHER or ADMIN	gradeDistribution	Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.classRecordService.getGradeDistribution
GET	/api/class-record/:id/reports/intervention	JWT plus TEACHER or ADMIN	interventionList	Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.classRecordService.getInterventionList
GET	/api/class-record/:id/spreadsheet	JWT plus TEACHER or ADMIN	getSpreadsheet	Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.classRecordService.getSpreadsheet
GET	/api/class-record/adviser/section/:sectionId	JWT plus ADMIN or TEACHER	listAdviserSection	Param('sectionId', ParseUUIDPipe) sectionId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.classRecordService.listAdviserSection
GET	/api/class-record/by-class/:classId	JWT plus TEACHER or ADMIN	listForClass	Param('classId', ParseUUIDPipe) classId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.classRecordService.listForClass
GET	/api/class-record/by-class/:classId/slot-overview	JWT plus TEACHER or ADMIN	getSlotOverview	Param('classId', ParseUUIDPipe) classId: string; Query('gradingPeriod') gradingPeriod: CreateClassRecordDto['gradingPeriod']; Query('assessmentId') assessmentId: string undefined; CurrentUser() user:	200 by Nest default; global filter formats failures	this.classRecordService.getSlotOverview
PATCH	/api/class-record/items/:itemId	JWT plus TEACHER or ADMIN	update-ClassRecordItem	Param('itemId', ParseUUIDPipe) itemId: string; Body() dto: UpdateClassRecordItemDto; CurrentUser() user:	200 by Nest default; global filter formats failures	this.classRecordService.updateClassRecordItem
POST	/api/class-record/items/:itemId/scores	JWT plus TEACHER or ADMIN	recordScore	Param('itemId', ParseUUIDPipe) itemId: string; Body() dto: RecordScoreDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.classRecordService.recordScore
POST	/api/class-record/items/:itemId/scores/bulk	JWT plus TEACHER or ADMIN	bulkRecordScores	Param('itemId', ParseUUIDPipe) itemId: string; Body() dto: BulkRecordScoresDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.classRecordService.bulkRecordScores
POST	/api/class-record/items/:itemId/sync-scores	JWT plus TEACHER or ADMIN	syncScores	Param('itemId', ParseUUIDPipe) itemId: string; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.classRecordService.syncScoresFromAssessment

ClassTemplatesController

Base path: `class-templates` . Source: `backend/src/modules/class-templates/class-templates.controller.ts` .
Route count: 14.

Method	Full path	Access	Handler	Inputs	Declared re- sponse and errors	Primary delegates
GET	/api/class-templates	JWT plus ADMIN	getAll	CurrentUser() user: any; Query('subjectCode') subjectCode: string optional; Query('subjectGradeLevel') subjectGradeLevel: string optional	200 by Nest default; global filter formats failures	this.classTemplatesService.findAll
POST	/api/class-templates	JWT plus ADMIN	create	Body() dto: CreateClassTemplateDto; CurrentUser() user: any	201 by Nest default unless overridden; global filter formats failures	this.classTemplatesService.create
DELETE	/api/class-templates/:id	JWT plus ADMIN	remove	Param('id') id: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.classTemplatesService.remove
GET	/api/class-templates/:id	JWT plus ADMIN	findOne	Param('id') id: string	200 by Nest default; global filter formats failures	this.classTemplatesService.findOne
PATCH	/api/class-templates/:id	JWT plus ADMIN	update	Param('id') id: string; Body() dto: UpdateClassTemplateDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.classTemplatesService.update
POST	/api/class-templates/:id/assessment-images	JWT plus ADMIN	uploadAssessmentImage	UploadedFile() file: Express.Multer.File	Throws BadRequestException	Handler-local composition or no direct this-call extracted
GET	/api/class-templates/:id/content	JWT plus ADMIN	getContent	Param('id') id: string	200 by Nest default; global filter formats failures	this.classTemplatesService.getContent
PUT	/api/class-templates/:id/content	JWT plus ADMIN	updateContent	Param('id') id: string; Body() dto: UpdateClassTemplateContentDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.classTemplatesService.updateContent
GET	/api/class-templates/:id/engine-export	JWT plus ADMIN	getEngineExport	Param('id') id: string	200 by Nest default; global filter formats failures	this.classTemplatesService.getEngineExport
POST	/api/class-templates/:id/publish	JWT plus ADMIN	publish	Param('id') id: string; Body() dto: PublishClassTemplateDto; CurrentUser() user: any	201 by Nest default unless overridden; global filter formats failures	this.classTemplatesService.publish
GET	/api/class-templates/compatible	JWT plus ADMIN	getCompatible	Query('subjectCode') subjectCode: string; Query('subjectGradeLevel') subjectGradeLevel: string	200 by Nest default; global filter formats failures	this.classTemplatesService.getPublishedByCompatibility
POST	/api/class-templates/engine-import	JWT plus ADMIN	importEngine	Body() dto: EngineImportDto; CurrentUser() user: any	201 by Nest default unless overridden; global filter formats failures	this.classTemplatesService.importEngine
POST	/api/class-templates/engine-import/validate	JWT plus ADMIN	validateEngineImport	Body() dto: EngineImportValidateDto	201 by Nest default unless overridden; global filter formats failures	this.classTemplatesService.validateEngineImport
GET	/api/class-templates/images/:filename	Public through @Public	serveAssessmentImage	Param('filename') filename: string; Res() res: Response	Throws BadRequestException	Handler-local composition or no direct this-call extracted

ClassesController

Base path: `classes` . Source: `backend/src/modules/classes/classes.controller.ts` . Route count: 28.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
GET	/api/classes	JWT plus ADMIN or TEACHER	getAllClassesLegacy	Query('subjectId') subjectId: string optional; Query('sectionId') sectionId: string optional; Query('teacherId') teacherId: string optional; Query('schoolYear') schoolYear: string optional; Query('subjectGradeLevel') subjectGradeLevel: string optional; Query('isActive') isActive: string optional; Query('search') search: string optional; Query('room') room: string optional; Query('page') page: string optional; Query('limit') limit: string optional	200 by Nest default; global filter formats failures	Handler-local composition or no direct this-call extracted
POST	/api/classes	JWT plus ADMIN	createClass	Body() createClassDto: CreateClassDto; CurrentUser() user: any	HttpCode HttpStatus.CREATED	this.classesService.create; this.aiProxy.forward('POST', /index/classes/\${newClass.id} , { id: user?.userId, email: user?.email ?? "", roles: user?.roles ?? [], }, undefined,).catch; this.aiProxy.forward; this.logger.warn
GET	/api/classes/:classId/candidates	JWT plus ADMIN or TEACHER	getCandidates	Param('classId') classId: string	200 by Nest default; global filter formats failures	this.classesService.getCandidates
GET	/api/classes/:classId/enrollments	JWT plus ADMIN or TEACHER	getEnrollments	Param('classId') classId: string	200 by Nest default; global filter formats failures	this.classesService.getEnrollments
POST	/api/classes/:classId/enrollments	JWT plus ADMIN or TEACHER	enrollStudent	Param('classId') classId: string; Body() dto: EnrollStudentDto; CurrentUser() user: any	HttpCode HttpStatus.CREATED	this.classesService.enrollStudent
DELETE	/api/classes/:classId/enrollments/:studentId	JWT plus ADMIN or TEACHER	removeStudent	Param('classId') classId: string; Param('studentId') studentId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.classesService.removeStudent
GET	/api/classes/:classId/students/:studentId/overview	JWT plus ADMIN or TEACHER	getStudentOverviewForClass	Param('classId') classId: string; Param('studentId') studentId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.classesService.getStudentOverviewForClass
GET	/api/classes/:classId/students/:studentId/profile	JWT plus ADMIN or TEACHER	getStudentProfileForClass	Param('classId') classId: string; Param('studentId') studentId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.classesService.getStudentProfileForClass
GET	/api/classes/:classId/students/masterlist	JWT plus ADMIN or TEACHER	getStudentsMasterlistForClass	Param('classId') classId: string; CurrentUser() user: any; Query('gradeLevel') gradeLevel: string optional; Query('sectionId') sectionId: string optional; Query('search')	200 by Nest default; global filter formats failures	this.classesService.getStudentsMasterlistForClass

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
				search: string optional; Query('eligibility') eligibility: 'all' 'eligible' 'mismatch' optional; Query('sortBy') sortBy: 'lastName' 'firstName' 'email' 'gradeLevel' 'lrn' 'eligibility' optional; Query('sortDirection') sortDirection: 'asc' 'desc' optional; Query('prioritizeEligible') prioritizeEligible: string optional; Query('page') page: string optional; Query('limit') limit: string optional		
DELETE	/api/classes/:id	JWT plus ADMIN	deleteClass	Param('id') id: string; CurrentUser() user: any	HttpCode HttpStatus.NO_CONTENT	this.classesService.delete
GET	/api/classes/:id	JWT plus ADMIN or TEACHER or STUDENT	getClassById	Param('id') id: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.classesService.findById
PUT	/api/classes/:id	JWT plus ADMIN	updateClass	Param('id') id: string; Body() updateClassDto: UpdateClassDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.classesService.update
POST	/api/classes/:id/banner	JWT plus ADMIN or TEACHER	uploadClassBanner	Param('id') id: string; UploadedFile() file: Express.Multer.File; CurrentUser() user: any	HttpCode HttpStatus.CREATED	this.classesService.updatePresentation
PATCH	/api/classes/:id/hide	JWT plus ADMIN or TEACHER or STUDENT	hideClass	Param('id') id: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.classesService.setClassHiddenState
PATCH	/api/classes/:id/presentation	JWT plus ADMIN or TEACHER	updateClassPresentation	Param('id') id: string; Body() dto: UpdateClassPresentationDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.classesService.updatePresentation
DELETE	/api/classes/:id/purge	JWT plus ADMIN	purgeClass	Param('id') id: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.classesService.purge
PUT	/api/classes/:id/student-presentation	JWT plus STUDENT	updateStudentClassPresentation	Param('id') id: string; Body() dto: UpdateStudentClassPresentationDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.classesService.updateStudentClassPresentationPreference
PUT	/api/classes/:id/toggle-status	JWT plus ADMIN	toggleClassStatus	Param('id') id: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.classesService.toggleActive
PATCH	/api/classes/:id/unhide	JWT plus ADMIN or TEACHER or STUDENT	unhideClass	Param('id') id: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.classesService.setClassHiddenState
GET	/api/classes/all	JWT plus ADMIN or TEACHER	getAllClasses	Query('subjectId') subjectId: string optional; Query('sectionId') sectionId: string optional; Query('teacherId') teacherId: string optional; Query('schoolYear') schoolYear: string optional; Query('subjectGradeLevel') subjectGradeLevel: string optional;	200 by Nest default; global filter formats failures	this.classesService.findAll

Method	Full path	Access	Handler	Inputs	Declared re- sponse and errors	Primary delegates
				Query('isActive') isActive: string optional; Query('search') search: string optional; Query('room') room: string optional; Query('page') page: string optional; Query('limit') limit: string optional		
POST	/api/classes/bulk/lifecycle	JWT plus ADMIN	bulkLifecycle	Body() dto: Bulk-ClassLifecycleDto; CurrentUser() user: any	201 by Nest default unless overridden; global filter formats failures	this.classesService.bulkLifecycleAction
GET	/api/classes/section/:sectionId	JWT plus ADMIN or TEACHER	getClassesBySection	Param('sectionId') sectionId: string	200 by Nest default; global filter formats failures	this.classesService.getClassesBySection
GET	/api/classes/student/:studentId	JWT plus ADMIN or TEACHER or STUDENT	getClassesByStudent	Param('studentId') studentId: string; Query('status') statusQuery: 'active' 'archived' 'hidden' 'all' undefined; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.classesService.getClassesByStudent
GET	/api/classes/student/:studentId/preferences/presentation	JWT plus ADMIN or STUDENT	getStudentClassPresentationPreferences	Param('studentId') studentId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.classesService.getStudentClassPresentationPreferences
GET	/api/classes/student/:studentId/preferences/view	JWT plus ADMIN or STUDENT	getStudentCourseViewPreference	Param('studentId') studentId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.classesService.getStudentCourseViewPreference
PUT	/api/classes/student/:studentId/preferences/view	JWT plus ADMIN or STUDENT	setStudentCourseViewPreference	Param('studentId') studentId: string; Body() dto: UpdateStudentCourseViewDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.classesService.setStudentCourseViewPreference
GET	/api/classes/subject/:subjectCode	JWT plus ADMIN or TEACHER	getClassesBySubject	Param('subjectCode') subjectCode: string	200 by Nest default; global filter formats failures	this.classesService.getClassesBySubject
GET	/api/classes/teacher/:teacherId	JWT plus ADMIN or TEACHER	getClassesByTeacher	Param('teacherId') teacherId: string; Query('status') statusQuery: 'active' 'archived' 'hidden' 'all' undefined; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.classesService.getClassesByTeacher

AnnouncementsController

Base path: `classes/:classId/announcements` . Source: `backend/src/modules/announcements/announcements.controller.ts` . Route count: 6.

Method	Full path	Access	Handler	Inputs	Declared re- sponse and errors	Primary delegates
GET	/api/classes/:classId/announcements	JWT plus TEACHER or STUDENT or ADMIN	findAll	Param('classId', ParseUUIDPipe) classId: string; Query() query: QueryAnnouncementsDto; CurrentUser() user:	200 by Nest default; global filter formats failures	this.announcementsService.findAllByClass
POST	/api/classes/:classId/announcements	JWT plus TEACHER or ADMIN	create	Param('classId', ParseUUIDPipe) classId: string; Body() dto: CreateAnnouncementDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.announcementsService.create
DELETE			remove			

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
	/api/classes/:classId/announcements/:id	JWT plus TEACHER or ADMIN		Param('classId', ParseUUIDPipe) classId: string; Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	HttpCode HttpStatus.OK	this.announcementsService.remove
GET	/api/classes/:classId/announcements/:id	JWT plus TEACHER or STUDENT or ADMIN	findOne	Param('classId', ParseUUIDPipe) classId: string; Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.announcementsService.findOne
PATCH	/api/classes/:classId/announcements/:id	JWT plus TEACHER or ADMIN	update	Param('classId', ParseUUIDPipe) classId: string; Param('id', ParseUUIDPipe) id: string; Body() dto: UpdateAnnouncementDto; CurrentUser() user:	200 by Nest default; global filter formats failures	this.announcementsService.update
PATCH	/api/classes/:classId/announcements/:id/core-release	JWT plus TEACHER or ADMIN	releaseCore	Param('classId', ParseUUIDPipe) classId: string; Param('id', ParseUUIDPipe) id: string; Body() dto: ReleaseCoreAnnouncementDto; CurrentUser() user:	200 by Nest default; global filter formats failures	this.announcementsService.releaseCoreAnnouncement

DiscussionBoardController

Base path: `classes/:classId/discussion-threads` . Source: `backend/src/modules/discussion-board/discussion-board.controller.ts` . Route count: 19.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
GET	/api/classes/:classId/discussion-threads	JWT plus ADMIN or TEACHER or STUDENT	listThreads	Param('classId', ParseUUIDPipe) classId: string; CurrentUser() user: { userId: string; roles: string[] }; Query() query: QueryDiscussionThreadsDto	200 by Nest default; global filter formats failures	this.discussionBoardService.listThreads
POST	/api/classes/:classId/discussion-threads	JWT plus ADMIN or TEACHER	createThread	Param('classId', ParseUUIDPipe) classId: string; CurrentUser() user: { userId: string; roles: string[] }; Body() dto: CreateDiscussionThreadDto	HttpCode HttpStatus.CREATED	this.discussionBoardService.createThread
DELETE	/api/classes/:classId/discussion-threads/:threadId	JWT plus ADMIN or TEACHER	archiveThread	Param('classId', ParseUUIDPipe) classId: string; Param('threadId', ParseUUIDPipe) threadId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.discussionBoardService.archiveThread
GET	/api/classes/:classId/discussion-threads/:threadId	JWT plus ADMIN or TEACHER or STUDENT	getThread	Param('classId', ParseUUIDPipe) classId: string; Param('threadId', ParseUUIDPipe) threadId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.discussionBoardService.getThread
PATCH	/api/classes/:classId/discussion-threads/:threadId	JWT plus ADMIN or TEACHER	updateThread	Param('classId', ParseUUIDPipe) classId: string; Param('threadId', ParseUUIDPipe) threadId: string;	200 by Nest default; global filter formats failures	this.discussionBoardService.updateThread

Method	Full path	Access	Handler	Inputs	Declared re- sponse and errors	Primary delegates
				CurrentUser() user: { userId: string; roles: string[] }; Body() dto: UpdateDiscussionThreadDto		
GET	/api/classes/:classId/discussion-threads/:threadId/attachments/:attachmentId/download	JWT plus ADMIN or TEACHER or STUDENT	downloadThreadAttachment	Param('classId', ParseUUIDPipe) classId: string; Param('threadId', ParseUUIDPipe) threadId: string; Param('attachmentId', ParseUUIDPipe) attachmentId: string; CurrentUser() user: { userId: string; roles: string[] }; Res() res: Response	200 by Nest default; global filter formats failures	this.discussion-BoardService.getThreadAttachmentFile
GET	/api/classes/:classId/discussion-threads/:threadId/attachments/:attachmentId/inline	JWT plus ADMIN or TEACHER or STUDENT	openThreadAttachmentInline	Param('classId', ParseUUIDPipe) classId: string; Param('threadId', ParseUUIDPipe) threadId: string; Param('attachmentId', ParseUUIDPipe) attachmentId: string; CurrentUser() user: { userId: string; roles: string[] }; Res() res: Response	200 by Nest default; global filter formats failures	this.discussion-BoardService.getThreadAttachmentFile
POST	/api/classes/:classId/discussion-threads/:threadId/close	JWT plus ADMIN or TEACHER	closeThread	Param('classId', ParseUUIDPipe) classId: string; Param('threadId', ParseUUIDPipe) threadId: string; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.discussion-BoardService.closeThread
POST	/api/classes/:classId/discussion-threads/:threadId/comments	JWT plus ADMIN or STUDENT	createComment	Param('classId', ParseUUIDPipe) classId: string; Param('threadId', ParseUUIDPipe) threadId: string; CurrentUser() user: { userId: string; roles: string[] }; Body() dto: CreateDiscussionCommentDto	HttpCode HttpStatus.CREATED	this.discussion-BoardService.createComment
DELETE	/api/classes/:classId/discussion-threads/:threadId/comments/:commentId	JWT plus ADMIN or TEACHER or STUDENT	deleteComment	Param('classId', ParseUUIDPipe) classId: string; Param('threadId', ParseUUIDPipe) threadId: string; Param('commentId', ParseUUIDPipe) commentId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.discussion-BoardService.deleteComment
GET	/api/classes/:classId/discussion-threads/:threadId/comments/:commentId/attachments/:attachmentId/download	JWT plus ADMIN or TEACHER or STUDENT	downloadCommentAttachment	Param('classId', ParseUUIDPipe) classId: string; Param('threadId', ParseUUIDPipe) threadId: string; Param('commentId', ParseUUIDPipe) commentId: string; Param('attachmentId', ParseUUIDPipe) attachmentId: string; CurrentUser() user: { userId: string; roles: string[] }; Res() res: Response	200 by Nest default; global filter formats failures	this.discussion-BoardService.getCommentAttachmentFile
GET	/api/classes/:classId/discussion-threads/:threadId/	JWT plus ADMIN or TEACHER or STUDENT	openCommentAttachmentInline	Param('classId', ParseUUIDPipe) classId: string;	200 by Nest default; global filter formats failures	this.discussion-BoardService-

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
	com-ments/:commentId/attach-ments:attachmentId/inline			Param('threadId', ParseUUIDPipe) threadId: string; Param('commentId', ParseUUIDPipe) commentId: string; Param('attachmentId', ParseUUIDPipe) attachmentId: string; CurrentUser() user: { userId: string; roles: string[] }; Res() res: Response		vice.getCommentAttachmentFile
DELETE	/api/classes/:classId/discussion-threads/:threadId/com-ments/:commentId/reaction	JWT plus ADMIN or STUDENT	removeReaction	Param('classId', ParseUUIDPipe) classId: string; Param('threadId', ParseUUIDPipe) threadId: string; Param('commentId', ParseUUIDPipe) commentId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.discussion-BoardService.removeCommentReaction
PUT	/api/classes/:classId/discussion-threads/:threadId/com-ments/:commentId/reaction	JWT plus ADMIN or STUDENT	setReaction	Param('classId', ParseUUIDPipe) classId: string; Param('threadId', ParseUUIDPipe) threadId: string; Param('commentId', ParseUUIDPipe) commentId: string; CurrentUser() user: { userId: string; roles: string[] }; Body() dto: SetDiscussionReactionDto	200 by Nest default; global filter formats failures	this.discussion-BoardService.setCommentReaction
POST	/api/classes/:classId/discussion-threads/:threadId/com-ments/:commentId/report	JWT plus ADMIN or TEACHER	reportComment	Param('classId', ParseUUIDPipe) classId: string; Param('threadId', ParseUUIDPipe) threadId: string; Param('commentId', ParseUUIDPipe) commentId: string; CurrentUser() user: { userId: string; roles: string[] }; Body() dto: ReportDiscussionCommentDto	201 by Nest default unless overridden; global filter formats failures	this.discussion-BoardService.reportComment
POST	/api/classes/:classId/discussion-threads/:threadId/comments/uploads	JWT plus ADMIN or STUDENT	uploadCommentImage	Param('classId', ParseUUIDPipe) classId: string; Param('threadId', ParseUUIDPipe) threadId: string; CurrentUser() user: { userId: string; roles: string[] }; UploadedFile() file: Express.Multer.File	Throws BadRequestException	this.discussion-BoardService.uploadCommentImageFile
POST	/api/classes/:classId/discussion-threads/:threadId/publish	JWT plus ADMIN or TEACHER	publishThread	Param('classId', ParseUUIDPipe) classId: string; Param('threadId', ParseUUIDPipe) threadId: string; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.discussion-BoardService.publishThread
POST	/api/classes/:classId/discussion-threads/:threadId/reopen	JWT plus ADMIN or TEACHER	reopenThread	Param('classId', ParseUUIDPipe) classId: string; Param('threadId', ParseUUIDPipe) threadId: string; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.discussion-BoardService.reopenThread
POST	/api/classes/:classId/discussion-threads/uploads	JWT plus ADMIN or TEACHER	uploadThreadAttachment	Param('classId', ParseUUIDPipe) classId: string; CurrentUser() user: { userId: string;	Throws BadRequestException	this.discussion-BoardService.uploadThreadAttachmentFile

Method	Full path	Access	Handler	Inputs	Declared re- sponse and errors	Primary delegates
				roles: string[] }; UploadedFile() file: Express.Multer.File		

ClassesPublicController

Base path: `classes`. Source: `backend/src/modules/classes/classes.controller.ts`. Route count: 1.

Method	Full path	Access	Handler	Inputs	Declared re- sponse and errors	Primary delegates
GET	/api/classes/banners/:filename	Public through @Public	serveClassBanner	Param('filename') filename: string; Res() res: any	200 by Nest default; global filter formats failures	Handler-local composition or no direct this-call extracted

FileUploadController

Base path: `files`. Source: `backend/src/modules/file-upload/file-upload.controller.ts`. Route count: 13.

Method	Full path	Access	Handler	Inputs	Declared re- sponse and errors	Primary delegates
GET	/api/files	JWT plus ADMIN or TEACHER or STUDENT	listFiles	CurrentUser() user: any; Query() query: FileQueryDto optional	200 by Nest default; global filter formats failures	this.fileUploadService.findAll
DELETE	/api/files/:id	JWT plus ADMIN or TEACHER	deleteFile	Param('id', ParseUUIDPipe) id: string; CurrentUser() user: any	HttpCode HttpStatus.OK	this.fileUploadService.softDelete
GET	/api/files/:id	JWT plus ADMIN or TEACHER or STUDENT	getFile	Param('id', ParseUUIDPipe) id: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.fileUploadService.findOne
PATCH	/api/files/:id	JWT plus ADMIN or TEACHER	updateFile	Param('id', ParseUUIDPipe) id: string; Body() dto: UpdateFileMetadataDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.fileUploadService.updateFileMetadata
GET	/api/files/:id/download	JWT plus ADMIN or TEACHER or STUDENT	downloadFile	Param('id', ParseUUIDPipe) id: string; CurrentUser() user: any; Res() res: Response	200 by Nest default; global filter formats failures	this.fileUploadService.getFileForDownload; this.storageService.serveOrRedirect
POST	/api/files/:id/index/retry	JWT plus ADMIN or TEACHER	retryIndex	Param('id', ParseUUIDPipe) id: string; CurrentUser() user: any	201 by Nest default unless overridden; global filter formats failures	this.fileUploadService.retryIndex
POST	/api/files/admin/backfill-storage	JWT plus ADMIN	backfillStorage	No handler parameters	201 by Nest default unless overridden; global filter formats failures	this.fileUploadService.migrateLocalFilesToStorage
GET	/api/files/folders	JWT plus ADMIN or TEACHER	listFolders	CurrentUser() user: any; Query() query: FileQueryDto	200 by Nest default; global filter formats failures	this.fileUploadService.listFolders
POST	/api/files/folders	JWT plus ADMIN or TEACHER	createFolder	Body() dto: CreateLibraryFolderDto; CurrentUser() user: any	201 by Nest default unless overridden; global filter formats failures	this.fileUploadService.createFolder
DELETE	/api/files/folders/:id	JWT plus ADMIN or TEACHER	deleteFolder	Param('id', ParseUUIDPipe) id: string; CurrentUser() user: any	HttpCode HttpStatus.OK	this.fileUploadService.deleteFolder
PATCH	/api/files/folders/:id	JWT plus ADMIN or TEACHER	updateFolder	Param('id', ParseUUIDPipe) id: string; Body() dto: UpdateLibraryFolderDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.fileUploadService.updateFolder
GET		JWT plus ADMIN				

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
	/api/files/storage-summary		getStorageSummary	No handler parameters	200 by Nest default; global filter formats failures	this.fileUploadService.getStorageSummary
POST	/api/files/upload	JWT plus TEACHER or ADMIN	uploadFile	UploadedFile(new LibraryFileValidationPipe()) file: Express.Multer.File; Query() query: UploadFileDto; CurrentUser() user: any	HttpCode HttpStatus.CREATED	this.fileUploadService.saveFileRecord

HealthController

Base path: `health`. Source: `backend/src/modules/health/health.controller.ts`. Route count: 3.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
GET	/api/health	Public through @Public	aliasCheck	No handler parameters	200 by Nest default; global filter formats failures	Handler-local composition or no direct this-call extracted
GET	/api/health/live	Public through @Public	check	No handler parameters	200 Server is healthy	this.healthService.getServiceMetadata
GET	/api/health/ready	Public through @Public	readiness	No handler parameters	200 Server is ready to receive traffic; 503 One or more dependencies are not ready; Throws ServiceUnavailableException	this.healthService.getReadiness

MetricsController

Base path: root controller. Source: `backend/src/monitoring/metrics.controller.ts`. Route count: 1.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
GET	/api/metrics	Public through class-level @Public	metrics	Res() res: Response	200 with the shared Prometheus registry content type; global filter formats failures	this.databaseService.getPoolDiagnostics; updates dbPoolTotal, dbPoolIdle, and dbPoolWaiting gauges; this.register.metrics

InternalUploadsController

Base path: `internal/uploads`. Source: `backend/src/modules/file-upload/internal-uploads.controller.ts`. Route count: 1.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
GET	/api/internal/uploads/raw	Public through @Public	readUpload	Query('path') requestedPath: string; Headers('x-internal-service-token') token: string undefined; Res() res: Response	Throws NotFoundException, ForbiddenException	this.storageService.getSignedDownloadUrl

LessonsController

Base path: `lessons`. Source: `backend/src/modules/lessons/lessons.controller.ts`. Route count: 20.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
POST	/api/lessons		createLesson			

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
		JWT plus ADMIN or TEACHER		Body() createLessonDto: CreateLessonDto; CurrentUser() user: any	HttpCode HttpStatus.CREATED	this.lessonsService.createLesson
DELETE	/api/lessons/:id	JWT plus ADMIN or TEACHER	deleteLesson	Param('id') id: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.lessonsService.deleteLesson
GET	/api/lessons/:id	JWT plus ADMIN or TEACHER or STUDENT	getLessonById	Param('id') id: string	200 by Nest default; global filter formats failures	this.lessonsService.getLessonById
PUT	/api/lessons/:id	JWT plus ADMIN or TEACHER	updateLesson	Param('id') id: string; Body() updateLessonDto: UpdateLessonDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.lessonsService.updateLesson
PUT	/api/lessons/:id/publish	JWT plus ADMIN or TEACHER	publishLesson	Param('id') id: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.lessonsService.publishLesson
GET	/api/lessons/:id/versions	JWT plus ADMIN or TEACHER	getLessonVersions	Param('id') lessonId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.lessonsService.getLessonVersions
POST	/api/lessons/:id/versions	JWT plus ADMIN or TEACHER	createLessonVersion	Param('id') lessonId: string; Body() dto: CreateLessonVersionDto; CurrentUser() user: any	201 by Nest default unless overridden; global filter formats failures	this.lessonsService.createManualVersion
POST	/api/lessons/:id/versions/:versionId/restore	JWT plus ADMIN or TEACHER	restoreLessonVersion	Param('id') lessonId: string; Param('versionId') versionId: string; CurrentUser() user: any	201 by Nest default unless overridden; global filter formats failures	this.lessonsService.restoreLessonVersion
POST	/api/lessons/:lessonId/blocks	JWT plus ADMIN or TEACHER	addContentBlock	Param('lessonId') lessonId: string; Body() createBlockDto: CreateContentBlockDto; CurrentUser() user: any	HttpCode HttpStatus.CREATED	this.lessonsService.addContentBlock
POST	/api/lessons/:lessonId/complete	JWT plus STUDENT	markLessonComplete	Param('lessonId') lessonId: string; CurrentUser() user: any	201 by Nest default unless overridden; global filter formats failures	this.lessonsService.markLessonComplete
GET	/api/lessons/:lessonId/completion-status	JWT plus STUDENT	getCompletionStatus	Param('lessonId') lessonId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.lessonsService.isLessonCompleted
PUT	/api/lessons/:lessonId/reorder-blocks	JWT plus ADMIN or TEACHER	reorderBlocks	Param('lessonId') lessonId: string; Body() reorderDto: ReorderBlocksDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.lessonsService.reorderBlocks
DELETE	/api/lessons/blocks/:blockId	JWT plus ADMIN or TEACHER	deleteContentBlock	Param('blockId') blockId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.lessonsService.deleteContentBlock
PUT	/api/lessons/blocks/:blockId	JWT plus ADMIN or TEACHER	updateContentBlock	Param('blockId') blockId: string; Body() updateBlockDto: UpdateContentBlockDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.lessonsService.updateContentBlock
GET	/api/lessons/class/:classId	JWT plus ADMIN or TEACHER or STUDENT	getLessonsByClass	Param('classId') classId: string; Query('page') pageQuery: string undefined; Query('pageSize') pageSizeQuery: string undefined; Query('status') statusQuery: string undefined; Query('includeBlocks') includeBlocksQuery: string un-	200 by Nest default; global filter formats failures	this.lessonsService.getLessonsByClass

Method	Full path	Access	Handler	Inputs	Declared re- sponse and errors	Primary delegates
				defined; CurrentUser() user: any		
POST	/api/lessons/class/:classId/bulk-delete	JWT plus ADMIN or TEACHER	bulkDeleteLessons	Param('classId') classId: string; Body() dto: BulkLessonIdsDto; CurrentUser() user: any	201 by Nest default unless overridden; global filter formats failures	this.lessonsService.bulkDeleteLessons
PUT	/api/lessons/class/:classId/bulk-status	JWT plus ADMIN or TEACHER	bulkUpdateLessonDraftState	Param('classId') classId: string; Body() dto: BulkLessonDraftStateDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.lessonsService.bulkUpdateLessonDraftState
GET	/api/lessons/class/:classId/completed	JWT plus STUDENT	getCompletedLessons	Param('classId') classId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.lessonsService.getCompletedLessonsForClass
GET	/api/lessons/class/:classId/drafts	JWT plus ADMIN or TEACHER	getDraftLessons	Param('classId') classId: string	200 by Nest default; global filter formats failures	this.lessonsService.getDraftLessons
PUT	/api/lessons/class/:classId/reorder	JWT plus ADMIN or TEACHER	reorderLessons	Param('classId') classId: string; Body() reorderDto: ReorderLessonsDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.lessonsService.reorderLessons

LxpController

Base path: `/lxp`. Source: `backend/src/modules/lxp/lxp.controller.ts`. Route count: 32.

Method	Full path	Access	Handler	Inputs	Declared re- sponse and errors	Primary delegates
GET	/api/lxp/evaluations	JWT plus ADMIN	listEvaluations	CurrentUser() user: { userId: string; roles: string[] }; Query() query: ListSystemEvaluationQueryDto optional	200 by Nest default; global filter formats failures	this.lxpService.listSystemEvaluations
POST	/api/lxp/evaluations	JWT plus STUDENT or TEACHER or ADMIN	submitEvaluation	CurrentUser() user: { userId: string; roles: string[] }; Body() dto: SubmitSystemEvaluationDto	201 by Nest default unless overridden; global filter formats failures	this.lxpService.submitSystemEvaluation
GET	/api/lxp/me/eligibility	JWT plus STUDENT	getEligibility	CurrentUser() user:	200 by Nest default; global filter formats failures	this.lxpService.getStudentEligibility
GET	/api/lxp/me/intervention-alerts	JWT plus STUDENT	getInterventionAlerts	CurrentUser() user:	200 by Nest default; global filter formats failures	this.lxpService.getStudentInterventionAlerts
GET	/api/lxp/me/overview/:classId	JWT plus STUDENT	getOverview	Param('classId', ParseUUIDPipe) classId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.lxpService.getStudentOverview
GET	/api/lxp/me/playlist/:classId	JWT plus STUDENT	getPlaylist	Param('classId', ParseUUIDPipe) classId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.lxpService.getStudentPlaylist
POST	/api/lxp/me/playlist/:classId/checkpoints/:assignmentId/complete	JWT plus STUDENT	completeCheckpoint	Param('classId', ParseUUIDPipe) classId: string; Param('assignmentId', ParseUUIDPipe) assignmentId: string; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.lxpService.completeCheckpoint
GET	/api/lxp/me/playlist/:classId/generated-lessons/:assignmentId	JWT plus STUDENT	getGeneratedLesson	Param('classId', ParseUUIDPipe) classId: string; Param('assignmentId', ParseUUIDPipe) assignmentId:	200 by Nest default; global filter formats failures	this.lxpService.getGeneratedLesson

Method	Full path	Access	Handler	Inputs	Declared re- sponse and errors	Primary delegates
				string; CurrentUser() user:		
PATCH	/api/lxp/me/playlist/:classId/guided-assessments/:assignmentId/progress	JWT plus STUDENT	updateGuidedAssessmentProgress	Param('classId', ParseUUIDPipe) classId: string; Param('assignmentId', ParseUUIDPipe) assignmentId: string; Body() dto: UpdateGuidedAssessmentProgressDto; CurrentUser() user:	200 by Nest default; global filter formats failures	this.IxpService.updateGuidedAssessmentProgress
GET	/api/lxp/me/playlist/:classId/guided-assessments/:assignmentId/result	JWT plus STUDENT	getGuidedAssessmentResult	Param('classId', ParseUUIDPipe) classId: string; Param('assignmentId', ParseUUIDPipe) assignmentId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.IxpService.getGuidedAssessmentResult
POST	/api/lxp/me/playlist/:classId/guided-assessments/:assignmentId/start	JWT plus STUDENT	startGuidedAssessment	Param('classId', ParseUUIDPipe) classId: string; Param('assignmentId', ParseUUIDPipe) assignmentId: string; CurrentUser() user: { userId: string; roles: string[] }; Body() body: { forceNewAttempt?: boolean } optional	201 by Nest default unless overridden; global filter formats failures	this.IxpService.startGuidedAssessment
POST	/api/lxp/me/playlist/:classId/guided-assessments/:assignmentId/submit	JWT plus STUDENT	submitGuidedAssessment	Param('classId', ParseUUIDPipe) classId: string; Param('assignmentId', ParseUUIDPipe) assignmentId: string; Body() dto: SubmitGuidedAssessmentDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.IxpService.submitGuidedAssessment
GET	/api/lxp/me/system-evaluations	JWT plus STUDENT or TEACHER	getMySystemEvaluations	CurrentUser() user:	200 by Nest default; global filter formats failures	this.IxpService.getMySystemEvaluationDashboard
POST	/api/lxp/me/system-evaluations/:assignmentId/submit	JWT plus STUDENT or TEACHER	submitAssignedSystemEvaluation	Param('assignmentId', ParseUUIDPipe) assignmentId: string; CurrentUser() user: { userId: string; roles: string[] }; Body() dto: SubmitAssignedSystemEvaluationDto	201 by Nest default unless overridden; global filter formats failures	this.IxpService.submitAssignedSystemEvaluation
GET	/api/lxp/me/teacher-evaluations	JWT plus STUDENT	getTeacherEvaluationDashboard	CurrentUser() user:	200 by Nest default; global filter formats failures	this.IxpService.getStudentTeacherEvaluationDashboard
POST	/api/lxp/me/teacher-evaluations	JWT plus STUDENT	submitTeacherEvaluation	CurrentUser() user: { userId: string; roles: string[] }; Body() dto: SubmitTeacherEvaluationDto	201 by Nest default unless overridden; global filter formats failures	this.IxpService.submitTeacherEvaluation
GET	/api/lxp/system-evaluation-campaigns	JWT plus TEACHER or ADMIN	listSystemEvaluationCampaigns	CurrentUser() user: { userId: string; roles: string[] }; Query() query: ListSystemEvaluationCampaignsQueryDto optional	200 by Nest default; global filter formats failures	this.IxpService.listSystemEvaluationCampaigns
POST	/api/lxp/system-evaluation-campaigns	JWT plus TEACHER or ADMIN	createSystemEvaluationCampaign	CurrentUser() user: { userId: string; roles: string[] }; Body() dto: CreateSystemEvaluationCampaignDto	201 by Nest default unless overridden; global filter formats failures	this.IxpService.createSystemEvaluationCampaign

Method	Full path	Access	Handler	Inputs	Declared re- sponse and errors	Primary delegates
PATCH	/api/lxp/system-evaluation-campaigns/:campaignId/status	JWT plus TEACHER or ADMIN	updateSystemEvaluationCampaignStatus	Param('campaignId', ParseUUIDPipe) campaignId: string; CurrentUser() user: { userId: string; roles: string[] }; Body() dto: UpdateSystemEvaluationCampaignStatusDto	200 by Nest default; global filter formats failures	this.IxpService.updateSystemEvaluationCampaignStatus
GET	/api/lxp/teacher/classes/:classId/interventions	JWT plus TEACHER or ADMIN	getTeacherQueue	Param('classId', ParseUUIDPipe) classId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.IxpService.getTeacherQueue
GET	/api/lxp/teacher/classes/:classId/interventions/history	JWT plus TEACHER or ADMIN	getTeacherInterventionHistory	Param('classId', ParseUUIDPipe) classId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.IxpService.getTeacherInterventionHistory
GET	/api/lxp/teacher/classes/:classId/reports/summary	JWT plus TEACHER or ADMIN	getClassReport	Param('classId', ParseUUIDPipe) classId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.IxpService.getClassReport
GET	/api/lxp/teacher/evaluations/summary	JWT plus TEACHER or ADMIN	getTeacherEvaluationSummary	CurrentUser() user: { userId: string; roles: string[] }; Query() query: ListTeacherEvaluationSummaryQueryDto	200 by Nest default; global filter formats failures	this.IxpService.getTeacherEvaluationSummary
GET	/api/lxp/teacher/interventions/:caseId	JWT plus TEACHER or ADMIN	getTeacherInterventionCase	Param('caseId', ParseUUIDPipe) caseId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.IxpService.getTeacherInterventionCase
POST	/api/lxp/teacher/interventions/:caseId/activate	JWT plus TEACHER or ADMIN	activateIntervention	Param('caseId', ParseUUIDPipe) caseId: string; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.IxpService.activateIntervention
POST	/api/lxp/teacher/interventions/:caseId/assign	JWT plus TEACHER or ADMIN	assignIntervention	Param('caseId', ParseUUIDPipe) caseId: string; Body() dto: AssignInterventionDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.IxpService.assignIntervention
GET	/api/lxp/teacher/interventions/:caseId/detail	JWT plus TEACHER or ADMIN	getTeacherInterventionCaseDetail	Param('caseId', ParseUUIDPipe) caseId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.IxpService.getTeacherInterventionCaseDetail
POST	/api/lxp/teacher/interventions/:caseId/generated-content/approve	JWT plus TEACHER or ADMIN	approveGeneratedArtifacts	Param('caseId', ParseUUIDPipe) caseId: string; Body() dto: ApproveGeneratedArtifactsDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.IxpService.approveGeneratedArtifacts
POST	/api/lxp/teacher/interventions/:caseId/generated-content/reject	JWT plus TEACHER or ADMIN	rejectGeneratedArtifacts	Param('caseId', ParseUUIDPipe) caseId: string; Body() dto: ApproveGeneratedArtifactsDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.IxpService.rejectGeneratedArtifacts
POST	/api/lxp/teacher/interventions/:caseId/regenerate	JWT plus TEACHER or ADMIN	regenerateInterventionPath	Param('caseId', ParseUUIDPipe) caseId: string; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.IxpService.regenerateInterventionPath
POST	/api/lxp/teacher/interventions/:caseId/resolve	JWT plus TEACHER or ADMIN	resolveIntervention	Param('caseId', ParseUUIDPipe) caseId: string; Body() dto: ResolveInterventionDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.IxpService.resolveIntervention
GET	/api/lxp/teacher/interventions/pending-count	JWT plus TEACHER or ADMIN	getTeacherPendingInterventionCount	CurrentUser() user:	200 by Nest default; global filter formats failures	this.IxpService.getTeacherPendingInterventionCount

ContentModulesController

Base path: `modules` . Source: `backend/src/modules/content-modules/content-modules.controller.ts` . Route count: 20.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
POST	/api/modules	JWT plus ADMIN or TEACHER	create	Body() dto: CreateModuleDto; CurrentUser() user: any	201 by Nest default unless overridden; global filter formats failures	this.contentModulesService.createModule
DELETE	/api/modules/:moduleId	JWT plus ADMIN or TEACHER	delete	Param('moduleId') moduleId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.contentModulesService.deleteModule
PATCH	/api/modules/:moduleId	JWT plus ADMIN or TEACHER	update	Param('moduleId') moduleId: string; Body() dto: UpdateModuleDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.contentModulesService.updateModule
PATCH	/api/modules/:moduleId/core-release	JWT plus ADMIN or TEACHER	releaseCoreModule	Param('moduleId') moduleId: string; Body() dto: ReleaseCoreModuleDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.contentModulesService.releaseCoreModule
POST	/api/modules/:moduleId/cover	JWT plus ADMIN or TEACHER	uploadModuleCover	Param('moduleId') moduleId: string; UploadedFile() file: Express.Multer.File; CurrentUser() user: any	HttpCode HttpStatus.CREATED; Throws BadRequestException	this.contentModulesService.updateModule
PUT	/api/modules/:moduleId/grading-scale	JWT plus ADMIN or TEACHER	replaceGradingScale	Param('moduleId') moduleId: string; Body() dto: ReplaceModuleGradingScaleDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.contentModulesService.replaceGradingScale
POST	/api/modules/:moduleId/sections	JWT plus ADMIN or TEACHER	createSection	Param('moduleId') moduleId: string; Body() dto: CreateModuleSectionDto; CurrentUser() user: any	201 by Nest default unless overridden; global filter formats failures	this.contentModulesService.createSection
PUT	/api/modules/:moduleId/sections/reorder	JWT plus ADMIN or TEACHER	reorderSections	Param('moduleId') moduleId: string; Body() dto: ReorderModuleSectionsDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.contentModulesService.reorderSections
GET	/api/modules/class/:classId	JWT plus ADMIN or TEACHER or STUDENT	getClass	Param('classId') classId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.contentModulesService.getClassByClass
GET	/api/modules/class/:classId/moduleId	JWT plus ADMIN or TEACHER or STUDENT	getClassAndModule	Param('classId') classId: string; Param('moduleId') moduleId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.contentModulesService.getClassByClass
PUT	/api/modules/class/:classId/reorder	JWT plus ADMIN or TEACHER	reorderModules	Param('classId') classId: string; Body() dto: ReorderModulesDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.contentModulesService.reorderModules
GET	/api/modules/covers/:filename	Public through @Public	serveModuleCover	Param('filename') filename: string; Res() res: Response	Throws BadRequestException	Handler-local composition or no direct this-call extracted
DELETE	/api/modules/items/:itemId	JWT plus ADMIN or TEACHER	deleteItem	Param('itemId') itemId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.contentModulesService.deleteItem
PATCH	/api/modules/items/:itemId	JWT plus ADMIN or TEACHER	updateItem	Param('itemId') itemId: string; Body() dto: UpdateModuleItemDto;	200 by Nest default; global filter formats failures	this.contentModulesService.updateItem

Method	Full path	Access	Handler	Inputs	Declared re- sponse and errors	Primary delegates
				CurrentUser() user: any		
PATCH	/api/modules/items/:itemId/core-release	JWT plus ADMIN or TEACHER	releaseCoreModuleItem	Param('itemId') itemId: string; Body() dto: ReleaseCoreModuleItemDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.contentModulesService.releaseCoreModuleItem
GET	/api/modules/items/:itemId/file/download	JWT plus ADMIN or TEACHER or STUDENT	downloadAttachedFile	Param('itemId', ParseUUIDPipe) itemId: string; CurrentUser() user: any; Res() res: Response	200 by Nest default; global filter formats failures	this.contentModulesService.getAttachedFileForDownload
DELETE	/api/modules/sections/:sectionId	JWT plus ADMIN or TEACHER	deleteSection	Param('sectionId') sectionId: string; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.contentModulesService.deleteSection
PATCH	/api/modules/sections/:sectionId	JWT plus ADMIN or TEACHER	updateSection	Param('sectionId') sectionId: string; Body() dto: UpdateModuleSectionDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.contentModulesService.updateSection
POST	/api/modules/sections/:sectionId/items	JWT plus ADMIN or TEACHER	attachItem	Param('sectionId') sectionId: string; Body() dto: AttachModuleItemDto; CurrentUser() user: any	201 by Nest default unless overridden; global filter formats failures	this.contentModulesService.attachItem
PUT	/api/modules/sections/:sectionId/items/reorder	JWT plus ADMIN or TEACHER	reorderItems	Param('sectionId') sectionId: string; Body() dto: ReorderModuleItemsDto; CurrentUser() user: any	200 by Nest default; global filter formats failures	this.contentModulesService.reorderItems

NotificationsController

Base path: `notifications`. Source: `backend/src/modules/notifications/notifications.controller.ts`. Route count: 4.

Method	Full path	Access	Handler	Inputs	Declared re- sponse and errors	Primary delegates
GET	/api/notifications	JWT plus TEACHER or STUDENT or ADMIN	findAll	CurrentUser() user: { userId: string }; Query() query: QueryNotificationsDto	200 by Nest default; global filter formats failures	this.notificationsService.findByUser
PATCH	/api/notifications/:id/read	JWT plus TEACHER or STUDENT or ADMIN	markRead	Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	HttpCode HttpStatus.OK	this.notificationsService.markRead
PATCH	/api/notifications/read-all	JWT plus TEACHER or STUDENT or ADMIN	markAllRead	CurrentUser() user:	HttpCode HttpStatus.OK	this.notificationsService.markAllRead
GET	/api/notifications/unread-count	JWT plus TEACHER or STUDENT or ADMIN	getUnreadCount	CurrentUser() user:	200 by Nest default; global filter formats failures	this.notificationsService.getUnreadCount

OtpController

Base path: `otp`. Source: `backend/src/modules/otp/otp.controller.ts`. Route count: 2.

Method	Full path	Access	Handler	Inputs	Declared re- sponse and errors	Primary delegates
POST	/api/otp/resend	Public through @Public	resendOtp	Body() resendOtpDto: ResendOtpDto	HttpCode HttpStatus.OK; 200 If the address is registered and unveri-	this.otpService.resendOTP

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
					fied, a new code has been sent.	
POST	/api/otp/verify	Public through @Public	verifyOtp	Body() verifyOtpDto: VerifyOtpDto	HttpCode HttpStatus.OK; 200 Email verified successfully; 400 Invalid or expired OTP	this.otpService.verifyOTP

PerformanceController

Base path: `performance` . Source: `backend/src/modules/performance/performance.controller.ts` . Route count: 11.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
GET	/api/performance/admin/analytics	JWT plus ADMIN	getAdminAnalytics	CurrentUser() user:	200 by Nest default; global filter formats failures	this.performanceService.getAdminAnalytics
GET	/api/performance/analysis/jobs/:jobId	JWT plus TEACHER or ADMIN	getPerformanceAnalysisJobStatus	Param('jobId', ParseUUIDPipe) jobId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.performanceService.getPerformanceAnalysisJobStatus
GET	/api/performance/analysis/jobs/:jobId/result	JWT plus TEACHER or ADMIN	getPerformanceAnalysisJobResult	Param('jobId', ParseUUIDPipe) jobId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.performanceService.getPerformanceAnalysisJobResult
POST	/api/performance/classes/:classId/analysis/jobs	JWT plus TEACHER or ADMIN	createPerformanceAnalysisJob	Param('classId', ParseUUIDPipe) classId: string; Body() dto: CreatePerformanceAnalysisJobDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.performanceService.createPerformanceAnalysisJob
GET	/api/performance/classes/:classId/at-risk	JWT plus TEACHER or ADMIN	getAtRiskStudents	Param('classId', ParseUUIDPipe) classId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.performanceService.getAtRiskStudents
GET	/api/performance/classes/:classId/diagnostics	JWT plus TEACHER or ADMIN	getClassDiagnostics	Param('classId', ParseUUIDPipe) classId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.performanceService.getClassDiagnostics
GET	/api/performance/classes/:classId/intervention-quiz-comparison	JWT plus TEACHER or ADMIN	getInterventionQuizComparison	Param('classId', ParseUUIDPipe) classId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.performanceService.getInterventionQuizComparison
GET	/api/performance/classes/:classId/logs	JWT plus TEACHER or ADMIN	getClassLogs	Param('classId', ParseUUIDPipe) classId: string; Query() query: QueryPerformanceLogsDto; CurrentUser() user:	200 by Nest default; global filter formats failures	this.performanceService.getClassLogs
POST	/api/performance/classes/:classId/recompute	JWT plus TEACHER or ADMIN	recomputeClass	Param('classId', ParseUUIDPipe) classId: string; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.performanceService.recomputeClass
GET	/api/performance/classes/:classId/summary	JWT plus TEACHER or ADMIN	getClassSummary	Param('classId', ParseUUIDPipe) classId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.performanceService.getClassSummary
GET	/api/performance/students/me/summary	JWT plus STUDENT	getStudentSummary	CurrentUser() user:	200 by Nest default; global filter formats failures	this.performanceService.getStudentOwnSummary

ProfilesController

Base path: `profiles` . Source: `backend/src/modules/profiles/profiles.controller.ts` . Route count: 9.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
GET	/api/profiles/:userId	JWT plus ADMIN or STUDENT or TEACHER	getProfileByUserId	Param('userId') userId: string; CurrentUser() user: any	Throws ForbiddenException	this.profilesService.findByUserId
POST	/api/profiles/create	JWT plus ADMIN	createProfile	Body() dto: UpdateProfileDto & { userId: string }; CurrentUser() user: any	201 by Nest default unless overridden; global filter formats failures	this.profilesService.createProfile
GET	/api/profiles/images/:filename	Public through @Public	serveProfileImage	Param('filename') filename: string; Res() res: Response	Throws BadRequestException	Handler-local composition or no direct this-call extracted
GET	/api/profiles/me	JWT plus STUDENT or TEACHER or ADMIN	getMyProfile	CurrentUser() user: any	200 by Nest default; global filter formats failures	this.profilesService.findByUserId
GET	/api/profiles/me/academic-summary	JWT plus STUDENT	getMyAcademicSummary	CurrentUser() user: any	200 by Nest default; global filter formats failures	this.profilesService.getAcademicSummary
GET	/api/profiles/me/assessment-history	JWT plus STUDENT	getMyAssessmentHistory	CurrentUser() user: any; Query('page') page: string optional; Query('limit') limit: string optional; Query('submission') submission: 'all' 'submitted' 'in-progress' optional; Query('search') search: string optional	200 by Nest default; global filter formats failures	this.profilesService.getAssessmentHistory
POST	/api/profiles/me/avatar	JWT plus STUDENT	uploadMyAvatar	CurrentUser() user: any; UploadedFile() file: Express.Multer.File	Throws BadRequestException	this.profilesService.updateProfile
GET	/api/profiles/me/transcript	JWT plus STUDENT	getMyTranscript	CurrentUser() user: any; Query('page') page: string optional; Query('limit') limit: string optional; Query('status') status: 'all' 'enrolled' 'dropped' 'completed' optional; Query('search') search: string optional	200 by Nest default; global filter formats failures	this.profilesService.getTranscript
PUT	/api/profiles/update/:userId	JWT plus STUDENT or TEACHER or ADMIN	updateProfile	Param('userId') userId: string; Body() dto: UpdateProfileDto; CurrentUser() user: any	Throws ForbiddenException	this.profilesService.updateProfile

ReportsController

Base path: `reports` . Source: `backend/src/modules/reports/reports.controller.ts` . Route count: 6.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
GET	/api/reports/assessment-summary	JWT plus ADMIN or TEACHER	getAssessmentSummary	Query() query: any; CurrentUser() user: AuthUser; Res() res: Response	200 by Nest default; global filter formats failures	this.reportsService.getAssessmentSummary
GET	/api/reports/class-enrollment	JWT plus ADMIN or TEACHER	getClassEnrollment	Query() query: any; CurrentUser() user: AuthUser; Res() res: Response	200 by Nest default; global filter formats failures	this.reportsService.getClassEnrollment
GET	/api/reports/intervention-participation	JWT plus ADMIN or TEACHER	getInterventionParticipation	Query() query: any; CurrentUser() user: AuthUser; Res() res: Response	200 by Nest default; global filter formats failures	this.reportsService.getInterventionParticipation
GET	/api/reports/student-master-list	JWT plus ADMIN or TEACHER	getStudentMasterList	Query() query: any; CurrentUser() user: AuthUser; Res() res: Response	200 by Nest default; global filter formats failures	this.reportsService.getStudentMasterList

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
GET	/api/reports/student-performance	JWT plus ADMIN or TEACHER	getStudentPerformance	Query() query: any; CurrentUser() user: AuthUser; Res() res: Response	200 by Nest default; global filter formats failures	this.reportsService.getStudentPerformance
GET	/api/reports/system-usage	JWT plus ADMIN or TEACHER	getSystemUsage	Query() query: any; CurrentUser() user: AuthUser; Res() res: Response	200 by Nest default; global filter formats failures	this.reportsService.getSystemUsage

RosterImportController

Base path: `roster-import`. Source: `backend/src/modules/roster-import/roster-import.controller.ts`. Route count: 4.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
POST	/api/roster-import/:sectionId/commit	JWT plus ADMIN or TEACHER	commit	Param('sectionId', ParseUUIDPipe) sectionId: string; Body() dto: RosterImportCommitDto; CurrentUser() user:	HttpCode HttpStatus.CREATED	this.rosterImportService.commitRoster
GET	/api/roster-import/:sectionId/pending	JWT plus ADMIN or TEACHER	getPending	Param('sectionId', ParseUUIDPipe) sectionId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.rosterImportService.getPendingRoster
POST	/api/roster-import/:sectionId/preview	JWT plus ADMIN or TEACHER	preview	Param('sectionId', ParseUUIDPipe) sectionId: string; UploadedFile(new RosterFileValidationPipe()) file: Express.Multer.File; CurrentUser() user:	HttpCode HttpStatus.OK	this.rosterImportService.parseAndPreview
PATCH	/api/roster-import/pending/:id/resolve	JWT plus ADMIN or TEACHER	resolvePending	Param('id', ParseUUIDPipe) id: string; Body() dto: ResolvePendingRowDto; CurrentUser() user:	200 by Nest default; global filter formats failures	this.rosterImportService.resolvePendingRow

SchoolEventsController

Base path: `school-events`. Source: `backend/src/modules/school-events/school-events.controller.ts`. Route count: 4.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
GET	/api/school-events	JWT plus ADMIN or TEACHER or STUDENT	findAll	Query() query: QuerySchoolEventsDto	200 by Nest default; global filter formats failures	this.schoolEventsService.findAll
POST	/api/school-events	JWT plus ADMIN	create	Body() dto: CreateSchoolEventDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.schoolEventsService.create
DELETE	/api/school-events/:id	JWT plus ADMIN	remove	Param('id') id: string; CurrentUser() user:	HttpCode HttpStatus.OK	this.schoolEventsService.remove
PATCH	/api/school-events/:id	JWT plus ADMIN	update	Param('id') id: string; Body() dto: UpdateSchoolEventDto; CurrentUser() user:	200 by Nest default; global filter formats failures	this.schoolEventsService.update

SectionsController

Base path: `sections`. Source: `backend/src/modules/sections/sections.controller.ts`. Route count: 24.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
GET	/api/sections/:id	JWT plus ADMIN or TEACHER	getSectionById	Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.sectionsService.findById
POST	/api/sections/:id/banner	JWT plus ADMIN or TEACHER	uploadSectionBanner	Param('id', ParseUUIDPipe) id: string; UploadedFile() file: Express.Multer.File; CurrentUser() user:	HttpCode HttpStatus.CREATED	this.sectionsService.updatePresentation
GET	/api/sections/:id/candidates	JWT plus ADMIN or TEACHER	getCandidates	Param('id', ParseUUIDPipe) id: string; CurrentUser() user: { userId: string; roles: string[] }; Query('gradeLevel') gradeLevel: string optional; Query('search') search: string optional; Query('assignedSectionId') assignedSectionId: string optional; Query('eligibility') eligibility: 'all' 'eligible' 'mismatch' optional; Query('sortBy') sortBy: 'lastName' 'firstName' 'email' 'gradeLevel' 'Irn' 'eligibility' optional; Query('sortDirection') sortDirection: 'asc' 'desc' optional; Query('prioritizeEligible') prioritizeEligible: string optional; Query('page') page: string optional; Query('limit') limit: string optional	200 by Nest default; global filter formats failures	this.sectionsService.getCandidates
PATCH	/api/sections/:id/hidden	JWT plus ADMIN or TEACHER	hideSection	Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.sectionsService.setSectionHiddenState
PATCH	/api/sections/:id/presentation	JWT plus ADMIN or TEACHER	updateSectionPresentation	Param('id', ParseUUIDPipe) id: string; Body() dto: UpdateSectionPresentationDto; CurrentUser() user:	200 by Nest default; global filter formats failures	this.sectionsService.updatePresentation
PUT	/api/sections/:id/restore	JWT plus ADMIN	restoreSection	Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.sectionsService.restoreSection
GET	/api/sections/:id/roster	JWT plus ADMIN or TEACHER	getRoster	Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.sectionsService.getRoster
POST	/api/sections/:id/roster	JWT plus ADMIN or TEACHER	addStudentsToSection	Param('id', ParseUUIDPipe) id: string; CurrentUser() user: { userId: string; roles: string[] }; Body() dto: BulkStudentsDto	201 by Nest default unless overridden; global filter formats failures	this.sectionsService.addStudentsToSection
DELETE	/api/sections/:id/roster/:studentId	JWT plus ADMIN or TEACHER	removeStudentFromSection	Param('id', ParseUUIDPipe) id: string; Param('studentId', ParseUUIDPipe) studentId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.sectionsService.removeStudentFromSection
GET			getSectionSchedule	Param('id', ParseUUIDPipe) id:		

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
	/api/sections/:id/schedule	JWT plus ADMIN or TEACHER or STUDENT		string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.sectionsService.getSectionSchedule
GET	/api/sections/:id/students/:studentId/profile	JWT plus ADMIN or TEACHER	getStudentProfileForSection	Param('id', ParseUUIDPipe) id: string; Param('studentId', ParseUUIDPipe) studentId: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.sectionsService.getStudentProfileForSection
PATCH	/api/sections/:id/unhide	JWT plus ADMIN or TEACHER	unhideSection	Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.sectionsService.setSectionHiddenState
POST	/api/sections/access-students/fail	JWT plus ADMIN	failStudents	Body() dto: FailStudentsDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.sectionsService.failStudents
POST	/api/sections/access-students/finalize-grades	JWT plus ADMIN	finalizeAccessStudentGrades	Body() dto: FinalizeAccessStudentGradesDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.sectionsService.finalizeAccessStudentGrades
POST	/api/sections/access-students/move-up	JWT plus ADMIN	moveUpStudents	Body() dto: MoveUpStudentsDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.sectionsService.moveUpStudents
GET	/api/sections/access-students/overview	JWT plus ADMIN	getAccessStudentsOverview	Query() query: AccessStudentsOverviewQueryDto	200 by Nest default; global filter formats failures	this.sectionsService.getAccessStudentsOverview
GET	/api/sections/access-students/target-sections	JWT plus ADMIN	getAccessStudentsTargetSections	Query() query: AccessStudentsTargetSectionsQueryDto	200 by Nest default; global filter formats failures	this.sectionsService.getAccessStudentsTargetSections
GET	/api/sections/all	JWT plus ADMIN or TEACHER	getAllSections	Query('gradeLevel') gradeLevel: string optional; Query('schoolYear') schoolYear: string optional; Query('isActive') isActive: string optional; Query('search') search: string optional; Query('page') page: string optional; Query('limit') limit: string optional	200 by Nest default; global filter formats failures	this.sectionsService.findAll
POST	/api/sections/bulk/lifecycle	JWT plus ADMIN	bulkLifecycle	Body() dto: BulkSectionLifecycleDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.sectionsService.bulkLifecycleAction
POST	/api/sections/create	JWT plus ADMIN	createSection	Body() createSectionDto: CreateSectionDto; CurrentUser() user:	201 by Nest default unless overridden; global filter formats failures	this.sectionsService.createSection
DELETE	/api/sections/delete/:id	JWT plus ADMIN	deleteSection	Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.sectionsService.archiveSection
GET	/api/sections/my	JWT plus ADMIN or TEACHER	getMySections	Query('status') statusQuery: 'active' 'archived' 'hidden' 'all' undefined; CurrentUser() user:	200 by Nest default; global filter formats failures	this.sectionsService.findAll
DELETE	/api/sections/permanent/:id	JWT plus ADMIN	permanentlyDeleteSection	Param('id', ParseUUIDPipe) id: string; CurrentUser() user:	200 by Nest default; global filter formats failures	this.sectionsService.permanentlyDeleteSection
PUT	/api/sections/update/:id	JWT plus ADMIN	updateSection	Param('id', ParseUUIDPipe) id: string; Body() updateSectionDto: UpdateSectionDto; CurrentUser() user:	200 by Nest default; global filter formats failures	this.sectionsService.updateSection

SectionsPublicController

Base path: `sections` . Source: `backend/src/modules/sections/sections.controller.ts` . Route count: 1.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
GET	/api/sections/banners/:filename	Public through @Public	serveSectionBanner	Param('filename') filename: string; Res() res: any	200 by Nest default; global filter formats failures	Handler-local composition or no direct this-call extracted

TeacherProfilesController

Base path: `teacher-profiles` . Source: `backend/src/modules/teacher-profiles/teacher-profiles.controller.ts` . Route count: 4.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
GET	/api/teacher-profiles/:userId	JWT plus TEACHER or ADMIN	getByUserId	Param('userId') userId: string; CurrentUser() user: AuthUser	Throws ForbiddenException	this.teacherProfilesService.findById
PUT	/api/teacher-profiles/:userId	JWT plus TEACHER or ADMIN	updateByUserId	Param('userId') userId: string; Body() dto: UpdateTeacherProfileDto; CurrentUser() user: AuthUser	Throws ForbiddenException	this.teacherProfilesService.updateProfile
GET	/api/teacher-profiles/me	JWT plus TEACHER or ADMIN	getMyProfile	CurrentUser() user: AuthUser	200 by Nest default; global filter formats failures	this.teacherProfilesService.findById
POST	/api/teacher-profiles/me/avatar	JWT plus TEACHER	uploadMyAvatar	CurrentUser() user: AuthUser; UploadedFile() file: Express.Multer.File	Throws BadRequestException	this.teacherProfilesService.updateProfile

TeacherController

Base path: `teacher` . Source: `backend/src/modules/teacher/teacher.controller.ts` . Route count: 3.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
GET	/api/teacher/assessments	JWT plus TEACHER or ADMIN	getAssessments	CurrentUser() user: any	200 by Nest default; global filter formats failures	this.teacherService.getTeacherAssessments
GET	/api/teacher/classes	JWT plus TEACHER or ADMIN	getClasses	CurrentUser() user: any	200 by Nest default; global filter formats failures	this.teacherService.getTeacherClasses
GET	/api/teacher/lessons	JWT plus TEACHER or ADMIN	getLessons	CurrentUser() user: any	200 by Nest default; global filter formats failures	this.teacherService.getTeacherLessons

UsersController

Base path: `users` . Source: `backend/src/modules/users/users.controller.ts` . Route count: 13.

Method	Full path	Access	Handler	Inputs	Declared response and errors	Primary delegates
GET	/api/users/:id	JWT plus ADMIN	getUserById	Param('id') id: string	200 by Nest default; global filter formats failures	this.userService.findPublicById
GET	/api/users/:id/export	JWT plus ADMIN	exportUserData	Param('id') id: string; Res() res: Response; CurrentUser() admin: any	200 by Nest default; global filter formats failures	this.userService.exportUserData
DELETE	/api/users/:id/purge	JWT plus ADMIN	purgeUser	Param('id') id: string; CurrentUser() admin: any	200 by Nest default; global filter formats failures	this.userService.purgeUser

Method	Full path	Access	Handler	Inputs	Declared re- sponse and errors	Primary delegates
PATCH	/api/users/:id/reactivate	JWT plus ADMIN	reactivateUser	Param('id') id: string; CurrentUser() admin: any	200 by Nest default; global filter formats failures	this.usersService.reactivateUser
POST	/api/users/:id/reset-password	JWT plus ADMIN	resetUserPassword	Param('id') id: string; CurrentUser() admin: any	201 by Nest default unless overridden; global filter formats failures	this.usersService.adminResetPassword
DELETE	/api/users/:id/soft-delete	JWT plus ADMIN	softDeleteUser	Param('id') id: string; CurrentUser() admin: any	200 by Nest default; global filter formats failures	this.usersService.softDeleteUser
PATCH	/api/users/:id/suspend	JWT plus ADMIN	suspendUser	Param('id') id: string; CurrentUser() admin: any	200 by Nest default; global filter formats failures	this.usersService.suspendUser
GET	/api/users/all	JWT plus ADMIN	getAllUsers	Query('role') role: string optional; Query('status') status: string optional; Query('page') page: number optional; Query('limit') limit: number optional; Query('includeStatusCounts') includeStatusCountsQuery: string optional	200 by Nest default; global filter formats failures	this.usersService.findAll
POST	/api/users/bulk/lifecycle	JWT plus ADMIN	bulkLifecycle	Body() dto: BulkUserLifecycleDto; CurrentUser() admin: any	201 by Nest default unless overridden; global filter formats failures	this.usersService.bulkLifecycleAction
POST	/api/users/create	JWT plus ADMIN	createUser	Body() createUserDto: CreateUserDto; CurrentUser() admin: any	201 by Nest default unless overridden; global filter formats failures	this.usersService.createUser
DELETE	/api/users/delete/:id	JWT plus ADMIN	deleteUser	Param('id') id: string; CurrentUser() admin: any	200 by Nest default; global filter formats failures	this.usersService.deleteUser
GET	/api/users/reports/monitoring	JWT plus ADMIN	getMonitoringReport	Query('role') role: string optional; Query('status') status: string optional; Query('search') search: string optional; Query('page') page: string optional; Query('limit') limit: string optional	200 by Nest default; global filter formats failures	this.usersService.getMonitoringReports
PUT	/api/users/update/:id	JWT plus ADMIN	updateUser	Param('id') id: string; Body() updateUserDto: UpdateUserDto; CurrentUser() admin: any	200 by Nest default; global filter formats failures	this.usersService.updateUser

Complete DTO dictionary

Exhaustive inventory rule. The 208 DTO classes below were extracted from `backend/src/modules/**/*dto.ts` at commit `3d0c93e`. A later source change requires regenerating or manually reconciling this chapter.

- Required means the TypeScript property lacks a question mark. Conditional validation can still change runtime requirements.
- Validation decorators are listed exactly enough to identify format, size, enum, nested-object, array, transform, and conditional rules.

- Global whitelisting rejects an input property that is absent from the DTO contract.

DTO source: backend/src/modules/academic-state/DTO/impact-preview-query.dto.ts

ImpactPreviewQueryDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
schoolYear	string	Yes	No explicit default	IsString(); Matches(/^\d{4}-\d{4}\$/, { message: 'schoolYear must be in YYYY-YYYY format', })

DTO source: backend/src/modules/academic-state/DTO/transition-academic-state.dto.ts

TransitionAcademicStateDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
schoolYear	string	Yes	No explicit default	IsString(); Matches(/^\d{4}-\d{4}\$/, { message: 'schoolYear must be in YYYY-YYYY format', })
currentPassword	string	Yes	No explicit default	IsString(); IsNotEmpty()
confirmationText	string	Yes	No explicit default	IsString(); IsNotEmpty()

DTO source: backend/src/modules/ai-mentor/DTO/admin-chat.dto.ts

AdminAnalyticsChartSeriesDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
name	string	Yes	No explicit default	ApiProperty({ example: 'At-risk students' }); IsString(); IsNotEmpty()
data	number[]	Yes	No explicit default	ApiProperty({ type: [Number], example: [2, 1, 0] }); IsArray()

AdminAnalyticsChartDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
type	'bar' 'line' 'pie' 'donut'	Yes	No explicit default	ApiProperty({ enum: ['bar', 'line', 'pie', 'donut'], example: 'bar' }); IsString(); IsIn(['bar', 'line', 'pie', 'donut'])
title	string	Yes	No explicit default	ApiProperty({ example: 'At-risk students by class' }); IsString(); IsNotEmpty()
labels	string[]	Yes	No explicit default	ApiProperty({ type: [String], example: ['MATH-7', 'SCI-7'] }); IsArray()

Property	Type	Required	Default	Validation and transform decorators
series	AdminAnalyticsChartSeriesDto[]	Yes	No explicit default	ApiProperty({ type: [AdminAnalyticsChartSeriesDto] }); isArray(); ValidateNested({ each: true }); Type() => AdminAnalyticsChartSeriesDto)
yAxisLabel	string	No	No explicit default	ApiPropertyOptional({ example: 'Students' }); isOptional(); isString()
xAxisLabel	string	No	No explicit default	ApiPropertyOptional({ example: 'Classes' }); isOptional(); isString()

AdminAnalyticsSourceDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
source	string	Yes	No explicit default	ApiProperty({ example: 'student-performance-report' }); isString(); isEmpty()
filters	Record	Yes	No explicit default	ApiProperty({ example: { window: 'latest', classId: null } }); isObject()
window	string	No	No explicit default	ApiPropertyOptional({ example: 'latest snapshot' }); isOptional(); isString()

AdminAnalyticsChatRequestDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
message	string	Yes	No explicit default	ApiProperty({ description: 'The analytics question to ask the admin assistant.', example: 'Show me current at-risk trends across the platform.', }); isString(); isEmpty(); MaxLength(2000, { message: 'Message must be 2000 characters or fewer' })
sessionId	string	No	No explicit default	ApiPropertyOptional({ description: 'Session ID from a previous admin analytics response. Omit to start a new conversation.', example: '11111111-1111-1111-1111-111111111111-assistant-0' }); isOptional(); isUUID()

AdminAnalyticsChatMessageDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
id	string	Yes	No explicit default	ApiProperty({ example: '11111111-1111-1111-1111-111111111111-assistant-0' }); isString()
role	'user' 'assistant'	Yes	No explicit default	ApiProperty({ enum: ['user', 'assistant'], example: 'assistant' }); isString(); isIn(['user', 'assistant'])
content	string	Yes	No explicit default	

Property	Type	Required	Default	Validation and transform decorators
				ApiProperty({ example: '2 students are currently flagged as at risk.' }); IsString()
createdAt	string	Yes	No explicit default	ApiProperty({ example: '2026-04-13T00:00:00.000Z' }); IsString()
chart	AdminAnalyticsChartDto null	No	No explicit default	ApiPropertyOptional({ type: () => AdminAnalyticsChartDto }); IsOptional(); ValidateNested(); Type(() => AdminAnalyticsChartDto)
sources	AdminAnalyticsSourceDto[]	No	No explicit default	ApiPropertyOptional({ type: [AdminAnalyticsSourceDto] }); IsOptional(); IsArray(); ValidateNested({ each: true }); Type(() => AdminAnalyticsSourceDto)

AdminAnalyticsSessionSummaryDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
sessionId	string	Yes	No explicit default	ApiProperty({ example: '11111111-1111-1111-1111-111111111111' }); IsString()
sessionType	string	Yes	No explicit default	ApiProperty({ example: 'admin_analytics_chat' }); IsString()
title	string	Yes	No explicit default	ApiProperty({ example: 'At-risk trends across classes' }); IsString()
preview	string	Yes	No explicit default	ApiProperty({ example: '2 students are currently flagged as at risk.' }); IsString()
updatedAt	string	Yes	No explicit default	ApiProperty({ example: '2026-04-13T00:00:00.000Z' }); IsString()

AdminAnalyticsSessionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
sessionId	string	Yes	No explicit default	ApiProperty({ example: '11111111-1111-1111-1111-111111111111' }); IsString()
title	string	Yes	No explicit default	ApiProperty({ example: 'At-risk trends across classes' }); IsString()
updatedAt	string	Yes	No explicit default	ApiProperty({ example: '2026-04-13T00:00:00.000Z' }); IsString()
messages	AdminAnalyticsChatMessageDto[]	Yes	No explicit default	ApiProperty({ type: [AdminAnalyticsChatMessageDto] }); IsArray(); ValidateNested({ each: true }); Type(() => AdminAnalyticsChatMessageDto)

AdminAnalyticsChatResponseDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
reply	string	Yes	No explicit default	ApiProperty({ example: '2 students are currently flagged as at risk.' }); IsString()
sessionId	string	Yes	No explicit default	ApiProperty({ example: '11111111-1111-1111-1111-111111111111' }); IsString()
chart	AdminAnalyticsChartDto null	No	No explicit default	ApiPropertyOptional({ type: () => AdminAnalyticsChartDto }); IsOptional(); ValidateNested(); Type() => AdminAnalyticsChartDto)
sources	AdminAnalyticsSourceDto[]	Yes	No explicit default	ApiProperty({ type: [AdminAnalyticsSourceDto] }); IsArray(); ValidateNested({ each: true }); Type() => AdminAnalyticsSourceDto)

DTO source: backend/src/modules/ai-mentor/DTO/chat.dto.ts

ChatRequestDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
message	string	Yes	No explicit default	ApiProperty({ description: 'The message to send to Ja (JAKIPR AI Mentor)', example: 'Hi Ja, can you help me understand fractions?', }); IsString(); IsNotEmpty(); MaxLength(2000, { message: 'Message must be 2000 characters or fewer' })
sessionId	string	No	No explicit default	ApiPropertyOptional({ description: 'Session ID from a previous chat response. Omit to start a new conversation.', example: 'a1b2c3d4-e5f6-7890-abcdef1234567890', }); IsUUID(); IsOptional()
attachments	ImageAttachmentDto[]	No	No explicit default	ApiPropertyOptional({ description: 'Optional image attachments for multimodal chat', type: [ImageAttachmentDto], }); IsOptional(); IsArray(); ValidateNested({ each: true }); Type() => ImageAttachmentDto)

DTO source: backend/src/modules/ai-mentor/DTO/class-ai-policy.dto.ts

UpdateClassAiPolicyDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
mentorExplainEnabled	boolean	No	No explicit default	ApiPropertyOptional({ description: 'Enable/disable AI mistake explanations and tutor interactions for this class.', example: true, }); IsOptional(); IsBoolean()
maxFollowUpTurns	number	No	No explicit default	ApiPropertyOptional({ description: 'Maximum allowed follow-up turns for AI assistance per context.', example: 3, minimum: 0, max-

Property	Type	Required	Default	Validation and transform decorators
				imum: 10, }); IsOptional(); IsInt(); Min(0); Max(10)
sourceScope	AiPolicySourceScope	No	No explicit default	ApiPropertyOptional({ description: 'How broad the AI source retrieval scope should be.', enum: aiPolicySourceScopeEnum.enumValues, example: 'class_materials', }); IsOptional(); IsIn(aiPolicySourceScopeEnum.enumValues)
strictGrounding	boolean	No	No explicit default	ApiPropertyOptional({ description: 'When true, AI should return guarded fall-back if grounding is weak.', example: false, }); IsOptional(); IsBoolean()

DTO source: backend/src/modules/ai-mentor/DTO/demo-intervention-plan.dto.ts

DemoInterventionPlanDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
subjectId	'english' 'science'	Yes	No explicit default	ApiProperty({ description: 'Selected demo subject track', enum: ['english', 'science'], example: 'science', }); IsString(); IsIn(['english', 'science'])
quarterExamScore	number	Yes	No explicit default	ApiProperty({ description: 'Quarter exam score from demo run', example: 68, }); IsNumber(); Min(0)
weakConcepts	string[]	No	No explicit default	ApiPropertyOptional({ description: 'Weak concepts detected during demo progression', type: [String], example: ['Cell structures and functions', 'Scientific method and variable control',], }); IsOptional(); IsArray(); ArrayMaxSize(8); IsString({ each: true }); MaxLength(180, { each: true })
moduleScores	number[]	No	No explicit default	ApiPropertyOptional({ description: 'Per-module score signals from demo assessments', type: [Number], example: [62, 70, 65], }); IsOptional(); IsArray(); ArrayMinSize(0); ArrayMaxSize(6); IsNumber({}, { each: true })

DTO source: backend/src/modules/ai-mentor/DTO/extract-module.dto.ts

ExtractModuleDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
fileId	string	Yes	No explicit default	ApiProperty({ description: 'UUID of the uploaded PDF file (from uploaded_files table)', example: 'a1b2c3d4-e5f6-7890-abcd-ef1234567890', }); IsUUID()
targetSectionCount	3 4 5	Yes	No explicit default	ApiProperty({ description: 'Teacher-selected target sec-

Property	Type	Required	Default	Validation and transform decorators
				tion count for structured extraction.', example: 4, enum: [3, 4, 5], }); IsInt(); IsIn([3, 4, 5])
extractionStyle	'faithful' 'clean' 'student_friendly'	No	No explicit default	ApiPropertyOptional({ description: 'Extraction output style.', enum: ['faithful', 'clean', 'student_friendly'], default: 'clean', }); IsOptional(); IsString(); IsIn(['faithful', 'clean', 'student_friendly'])

RetryExtractionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
targetSectionCount	3 4 5	No	No explicit default	ApiPropertyOptional({ description: 'Optional target section count override for retry.', enum: [3, 4, 5], }); IsOptional(); IsInt(); IsIn([3, 4, 5])
extractionStyle	'faithful' 'clean' 'student_friendly'	No	No explicit default	ApiPropertyOptional({ description: 'Optional extraction style override for retry.', enum: ['faithful', 'clean', 'student_friendly'], }); IsOptional(); IsString(); IsIn(['faithful', 'clean', 'student_friendly'])

ApplyExtractionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
sectionIndices	number[]	No	No explicit default	ApiPropertyOptional({ description: 'Array of section indices (0-based) to apply. If omitted, all sections are applied.', example: [0, 2, 3], type: [Number], }); IsOptional(); IsArray(); IsNumber({}, { each: true })
lessonIndices	number[]	No	No explicit default	ApiPropertyOptional({ description: 'Legacy alias of sectionIndices for backward compatibility.', example: [0, 2, 3], type: [Number], }); IsOptional(); IsArray(); IsNumber({}, { each: true })

ExtractionBlockDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
type	string	Yes	No explicit default	ApiProperty({ description: 'Block type', example: 'text' }); IsString()
order	number	Yes	No explicit default	ApiProperty({ description: 'Display order', example: 0 }); IsNumber()
content	Record string	Yes	No explicit default	ApiProperty({ description: 'Block content object or plain string', oneOf: [{ type: 'object' }, { type: 'string' }], })

Property	Type	Required	Default	Validation and transform decorators
metadata	Record	No	No explicit default	ApiPropertyOptional({ description: 'Block metadata object' }); IsOptional(); IsObject()

ExtractionLessonDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
title	string	Yes	No explicit default	ApiProperty({ description: 'Lesson title' }); IsString()
description	string	No	No explicit default	ApiPropertyOptional({ description: 'Lesson description' }); IsOptional(); IsString()
blocks	ExtractionBlockDto[]	Yes	No explicit default	ApiProperty({ description: 'Content blocks for this lesson', type: [ExtractionBlockDto], }); IsArray(); ValidateNested({ each: true }); Type(() => ExtractionBlockDto)

ExtractionAssessmentOptionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
text	string	Yes	No explicit default	ApiProperty({ description: 'Option text' }); IsString()
isCorrect	boolean	No	No explicit default	ApiPropertyOptional({ description: 'Whether this option is correct' }); IsOptional(); IsBoolean()
order	number	No	No explicit default	ApiPropertyOptional({ description: 'Display order' }); IsOptional(); IsNumber()

ExtractionAssessmentQuestionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
content	string	Yes	No explicit default	ApiProperty({ description: 'Question content' }); IsString()
type	string	No	No explicit default	ApiPropertyOptional({ description: 'Question type' }); IsOptional(); IsString()
points	number	No	No explicit default	ApiPropertyOptional({ description: 'Points' }); IsOptional(); IsNumber()
order	number	No	No explicit default	ApiPropertyOptional({ description: 'Display order' }); IsOptional(); IsNumber()
explanation	string	No	No explicit default	ApiPropertyOptional({ description: 'Explanation' }); IsOptional(); IsString()
imageUrl	string	No	No explicit default	ApiPropertyOptional({ description: 'Optional image URL (supports data

Property	Type	Required	Default	Validation and transform decorators
				URL', }); IsOptional(); IsString()
conceptTags	string[]	No	No explicit default	ApiPropertyOptional({ description: 'Optional concept tags', type: [String] }); IsOptional(); IsArray(); IsString({ each: true })
options	ExtractionAssessmentOptionDto[]	No	No explicit default	ApiPropertyOptional({ description: 'Options', type: [ExtractionAssessmentOptionDto], }); IsOptional(); IsArray(); ValidateNested({ each: true }); Type() => ExtractionAssessmentOptionDto)

ExtractionAssessmentDraftDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
title	string	No	No explicit default	ApiPropertyOptional({ description: 'Draft assessment title' }); IsOptional(); IsString()
description	string	No	No explicit default	ApiPropertyOptional({ description: 'Draft assessment description' }); IsOptional(); IsString()
type	string	No	No explicit default	ApiPropertyOptional({ description: 'Assessment type' }); IsOptional(); IsString()
passingScore	number	No	No explicit default	ApiPropertyOptional({ description: 'Passing score' }); IsOptional(); IsNumber()
feedbackLevel	string	No	No explicit default	ApiPropertyOptional({ description: 'Feedback level' }); IsOptional(); IsString()
questions	ExtractionAssessmentQuestionDto[]	No	No explicit default	ApiPropertyOptional({ description: 'Draft questions', type: [ExtractionAssessmentQuestionDto], }); IsOptional(); IsArray(); ValidateNested({ each: true }); Type() => ExtractionAssessmentQuestionDto)

ExtractionMediaCandidateDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
sectionIndex	number	Yes	No explicit default	ApiProperty({ description: 'Candidate section index', example: 0 }); IsNumber()
score	number	Yes	No explicit default	ApiProperty({ description: 'Assignment score', example: 0.88 }); IsNumber()
explicitMatch	boolean	No	No explicit default	ApiPropertyOptional({ description: 'Whether the candidate was an explicit citation match', }); IsOptional(); IsBoolean()
scoreBreakdown	Record	No	No explicit default	ApiPropertyOptional({ description: 'Assignment score breakdown object' }); IsOptional(); IsObject()

ExtractionMediaAssetDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
id	string	Yes	No explicit default	ApiProperty({ description: 'Stable media asset id' }); IsString()
url	string	Yes	No explicit default	ApiProperty({ description: 'Media URL or data URL' }); IsString()
pageNumber	number	No	No explicit default	ApiPropertyOptional({ description: 'Page number' }); IsOptional(); IsNumber()
caption	string	No	No explicit default	ApiPropertyOptional({ description: 'Caption text' }); IsOptional(); IsString()
anchorText	string	No	No explicit default	ApiPropertyOptional({ description: 'Anchor text found near the image' }); IsOptional(); IsString()
keywords	string[]	No	No explicit default	ApiPropertyOptional({ description: 'Image keywords', type: [String] }); IsOptional(); IsArray(); IsString({ each: true })
figureReferences	string[]	No	No explicit default	ApiPropertyOptional({ description: 'Figure references', type: [String] }); IsOptional(); IsArray(); IsString({ each: true })
selectedSectionIndex	number null	No	No explicit default	ApiPropertyOptional({ description: 'Selected section index for this image' }); IsOptional(); IsNumber()
assignmentConfidence	number	No	No explicit default	ApiPropertyOptional({ description: 'Assignment confidence score' }); IsOptional(); IsNumber()
assignmentBreakdown	Record	No	No explicit default	ApiPropertyOptional({ description: 'Assignment score breakdown object' }); IsOptional(); IsObject()
candidateSections	ExtractionMediaCandidateDto[]	No	No explicit default	ApiPropertyOptional({ description: 'Candidate section scores', type: [ExtractionMediaCandidateDto] }); IsOptional(); IsArray(); ValidateNested({ each: true }); Type(() => ExtractionMediaCandidateDto)
teacherReviewed	boolean	No	No explicit default	ApiPropertyOptional({ description: 'Whether a teacher has reviewed this image placement' }); IsOptional(); IsBoolean()
reviewState	string	No	No explicit default	ApiPropertyOptional({ description: 'Review state label' }); IsOptional(); IsString()

ExtractionSectionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
title	string	Yes	No explicit default	ApiProperty({ description: 'Section title' }); IsString()
description	string	No	No explicit default	

Property	Type	Required	Default	Validation and transform decorators
				ApiPropertyOptional({ description: 'Section description' }); IsOptional(); IsString()
order	number	No	No explicit default	ApiPropertyOptional({ description: 'Display order' }); IsOptional(); IsNumber()
lessonBlocks	ExtractionBlockDto[]	Yes	No explicit default	ApiProperty({ description: 'Lesson blocks for this section', type: [ExtractionBlockDto], }); IsArray(); ValidateNested({ each: true }); Type() => ExtractionBlockDto)
assessmentDraft	ExtractionAssessmentDraftDto	No	No explicit default	ApiPropertyOptional({ description: 'Optional draft assessment for this section', type: ExtractionAssessmentDraftDto, }); IsOptional(); ValidateNested(); Type() => ExtractionAssessmentDraftDto)
confidence	number	No	No explicit default	ApiPropertyOptional({ description: 'Section confidence score' }); IsOptional(); IsNumber()
graphKeywords	string[]	No	No explicit default	ApiPropertyOptional({ description: 'Graph keywords', type: [String] }); IsOptional(); IsArray(); IsString({ each: true })
figureReferences	string[]	No	No explicit default	ApiPropertyOptional({ description: 'Figure references', type: [String] }); IsOptional(); IsArray(); IsString({ each: true })

UpdateExtractionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
title	string	No	No explicit default	ApiPropertyOptional({ description: 'Module title' }); IsOptional(); IsString()
description	string	No	No explicit default	ApiPropertyOptional({ description: 'Module description' }); IsOptional(); IsString()
sections	ExtractionSectionDto[]	No	No explicit default	ApiPropertyOptional({ description: 'Section-based extraction payload (canonical)', type: [ExtractionSectionDto], }); IsOptional(); IsArray(); ValidateNested({ each: true }); Type() => ExtractionSectionDto)
reviewIssues	Record[]	No	No explicit default	ApiPropertyOptional({ description: 'Teacher review issue state from the extraction review workspace', type: [Object], }); IsOptional(); IsArray(); IsObject({ each: true })
reviewState	string	No	No explicit default	ApiPropertyOptional({ description: 'Teacher review state after local issue resolution', }); IsOptional(); IsString()
lessons	ExtractionLessonDto[]	No	No explicit default	ApiPropertyOptional({ description: 'Legacy lessons payload alias (accepted for compatibility)', type: [Ex-

Property	Type	Required	Default	Validation and transform decorators
				tractionLessonDto], }); IsOptional(); IsArray(); ValidateNested({ each: true }); Type() => ExtractionLessonDto)
mediaAssets	ExtractionMediaAssetDto[]	No	No explicit default	ApiPropertyOptional({ description: 'Extracted image assets and teacher review state', type: [ExtractionMediaAssetDto], }); IsOptional(); IsArray(); ValidateNested({ each: true }); Type() => ExtractionMediaAssetDto)

DTO source: backend/src/modules/ai-mentor/DTO/image-attachment.dto.ts

ImageAttachmentDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
filePath	string	No	No explicit default	ApiPropertyOptional({ description: 'Absolute or server-local file path for an image attachment', example: 'C:\uploads\question-images\diagram.png', }); IsOptional(); IsString(); MaxLength(500)
base64Data	string	No	No explicit default	ApiPropertyOptional({ description: 'Base64-encoded image content', }); IsOptional(); IsBase64()
mimeType	string	No	No explicit default	ApiPropertyOptional({ description: 'Image MIME type', example: 'image/png', }); IsOptional(); IsString(); MaxLength(100)

DTO source: backend/src/modules/ai-mentor/DTO/intervention-recommendation.dto.ts

InterventionRecommendationDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
note	string	No	No explicit default	ApiPropertyOptional({ description: 'Optional teacher instruction to bias the recommendation', example: 'Focus on foundational remediation before retrying assessments.', }); IsOptional(); IsString(); MaxLength(500)

DTO source: backend/src/modules/ai-mentor/DTO/lesson-plan.dto.ts

LessonPlanHeaderDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
instructionalFormat	string	No	No explicit default	ApiPropertyOptional({ description: 'Instructional

Property	Type	Required	Default	Validation and transform decorators
				format label shown in the PDF header', example: 'Detailed Lesson Plan', }); IsOptional(); IsString(); MaxLength(120)
schoolName	string	No	No explicit default	ApiPropertyOptional({ description: 'School name shown in the PDF header', example: 'Gat Andres Bonifacio High School', }); IsOptional(); IsString(); MaxLength(160)
quarter	string	No	No explicit default	ApiPropertyOptional({ example: 'Q1' }); IsOptional(); IsString(); MaxLength(40)
date	string	No	No explicit default	ApiPropertyOptional({ example: '2026-05-03' }); IsOptional(); IsString(); MaxLength(40)
startTime	string	No	No explicit default	ApiPropertyOptional({ example: '08:00' }); IsOptional(); IsString(); MaxLength(40)
endTime	string	No	No explicit default	ApiPropertyOptional({ example: '09:00' }); IsOptional(); IsString(); MaxLength(40)

GenerateLessonPlanDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
classId	string	Yes	No explicit default	ApiProperty({ description: 'Class where the lesson plan belongs', example: '7c6b6047-f8ef-483b-8d51-c4bac7ed13d2', }); IsUUID()
anchorType	'module' 'lesson'	Yes	No explicit default	ApiProperty({ description: 'Whether the teacher selected a module or lesson as the anchor', enum: ['module', 'lesson'], }); IsString(); IsIn(['module', 'lesson'])
anchorId	string	Yes	No explicit default	ApiProperty({ description: 'Selected module or lesson id', }); IsUUID()
teacherNote	string	No	No explicit default	ApiPropertyOptional({ description: 'Optional teacher note to guide the generated plan', example: 'Focus on mixed-readiness support around decimal operations.', }); IsOptional(); IsString(); MaxLength(1000)
header	LessonPlanHeaderDto	No	No explicit default	ApiPropertyOptional({ description: 'Optional header overrides for the generated DLP', type: LessonPlanHeaderDto, }); IsOptional(); ValidateNested(); Type() => LessonPlanHeaderDto)

UpdateLessonPlanDraftDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
structuredOutput	Record	Yes	No explicit default	ApiProperty({ description: 'Reviewed structured lesson

Property	Type	Required	Default	Validation and transform decorators
				plan payload to persist', type: Object, }); IsObject()

DTO source: backend/src/modules/ai-mentor/DTO/mentor-explain.dto.ts

MentorExplainDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
attemptId	string	Yes	No explicit default	ApiProperty({ description: 'Assessment attempt to explain', example: '0d8703f0-8249-4b14-9df9-94db137e0fd1', }); IsUUID()
questionId	string	Yes	No explicit default	ApiProperty({ description: 'Specific question to explain', example: '4af90fe7-7e0a-4d5c-8bd1-9ea38b9aab6e', }); IsUUID()
message	string	No	No explicit default	ApiPropertyOptional({ description: 'Optional follow-up prompt from the student', example: 'Can you explain why my answer is wrong without giving me the answer directly?', }); IsOptional(); IsString(); IsNotEmpty(); MaxLength(1000)
attachments	ImageAttachmentDto[]	No	No explicit default	ApiPropertyOptional({ description: 'Optional image attachments for image-based explanations', type: [ImageAttachmentDto], }); IsOptional(); IsArray(); ValidateNested({ each: true }); Type() => ImageAttachmentDto)

DTO source: backend/src/modules/ai-mentor/DTO/quiz-generation.dto.ts

GenerateQuizDraftDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
classId	string	Yes	No explicit default	ApiProperty({ description: 'Class where the assessment draft will be created', example: '7c6b6047-f8ef-483b-8d51-c4bac7ed13d2', }); IsUUID()
lessonIds	string[]	No	No explicit default	ApiPropertyOptional({ description: 'Limit generation to specific lessons', type: [String], }); IsOptional(); IsArray(); IsUUID('4', { each: true })
extractionIds	string[]	No	No explicit default	ApiPropertyOptional({ description: 'Optionally include a reviewed extraction as source material', type: [String], }); IsOptional(); IsArray(); IsUUID('4', { each: true })
title	string	No	No explicit default	ApiPropertyOptional({ description: 'Title for the generated draft assessment', example: 'Fractions Diagnost-

Property	Type	Required	Default	Validation and transform decorators
				ic Quiz', }); IsOptional(); IsString(); MaxLength(200)
questionCount	number	No	5	ApiPropertyOptional({ description: 'How many questions to generate', default: 5, }); IsOptional(); IsInt(); Min(1); Max(15)
questionType	QuestionType	No	Question-Type.MULTIPLE_CHOICE	ApiPropertyOptional({ description: 'Default question type to bias generation toward', enum: Question-Type, default: Question-Type.MULTIPLE_CHOICE, }); IsOptional(); IsEnum(QuestionType)
assessmentType	AssessmentType	No	AssessmentType QUIZ	ApiPropertyOptional({ description: 'Assessment type to create', enum: AssessmentType, default: AssessmentType QUIZ, }); IsOptional(); IsEnum(AssessmentType)
passingScore	number	No	60	ApiPropertyOptional({ description: 'Passing score for the draft assessment', default: 60, }); IsOptional(); IsInt(); Min(1); Max(100)
teacherNote	string	No	No explicit default	ApiPropertyOptional({ description: 'Optional teacher instruction to guide generation', example: 'Prioritize conceptual understanding and avoid purely memorization-based items.', }); IsOptional(); IsString(); MaxLength(1000)
feedbackLevel	FeedbackLevel	No	FeedbackLevel.STANDARD	ApiPropertyOptional({ enum: FeedbackLevel, default: FeedbackLevel.STANDARD, }); IsOptional(); IsEnum(FeedbackLevel)
classRecordCategory	ClassRecordCategory	No	No explicit default	ApiPropertyOptional({ enum: ClassRecordCategory, }); IsOptional(); IsEnum(ClassRecordCategory)
quarter	Quarter	No	No explicit default	ApiPropertyOptional({ enum: Quarter, }); IsOptional(); IsEnum(Quarter)
sourcePolicy	string	No	'published_default'	ApiPropertyOptional({ description: 'Source policy used by the AI service for quiz evidence selection', enum: ['published_default', 'published_only', 'any_indexed'], default: 'published_default', }); IsOptional(); IsString()
allowDraftSources	boolean	No	false	ApiPropertyOptional({ description: 'Teacher acknowledgement that explicitly selected draft sources may be used when indexed', default: false, }); IsOptional(); IsBoolean()
retryOfJobId	string	No	No explicit default	ApiPropertyOptional({ description: 'Previous quiz generation job that this request retries', }); IsOptional(); IsUUID()

UpdateQuizDraftDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
structuredOutput	Record	Yes	No explicit default	ApiProperty({ description: 'Reviewed structured quiz draft payload to persist', type: Object, }); IsObject()

DTO source: backend/src/modules/ai-mentor/DTO/student-tutor.dto.ts

TutorRecommendationPayloadDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
id	string	Yes	No explicit default	ApiProperty(); IsString(); IsNotEmpty()
title	string	Yes	No explicit default	ApiProperty(); IsString(); IsNotEmpty()
reason	string	Yes	No explicit default	ApiProperty(); IsString(); IsNotEmpty()
focusText	string	Yes	No explicit default	ApiProperty(); IsString(); IsNotEmpty()
lessonId	string null	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsUUID()
assessmentId	string null	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsUUID()
questionId	string null	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsUUID()
sourceChunkId	string null	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsString()

StudentTutorStartDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
classId	string	Yes	No explicit default	ApiProperty(); IsUUID()
recommendation	TutorRecommendationPayloadDto	Yes	No explicit default	ApiProperty({ type: TutorRecommendationPayloadDto }); ValidateNested(); Type() => TutorRecommendationPayloadDto)

StudentTutorMessageDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
sessionId	string	Yes	No explicit default	ApiProperty(); IsUUID()
message	string	Yes	No explicit default	ApiProperty(); IsString(); IsNotEmpty(); MaxLength(2000)
attachments	ImageAttachmentDto[]	No	No explicit default	ApiPropertyOptional({ type: ImageAttachmentDto }); IsOptional(); IsArray(); ValidateNested({ each: true }); Type() => ImageAttachmentDto)

StudentTutorAnswersDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
sessionId	string	Yes	No explicit default	ApiProperty(); IsUUID()
answers	string[]	Yes	No explicit default	ApiProperty({ type: [String] }); IsArray(); ArrayMinSize(1); ArrayMaxSize(3); IsString({ each: true })
attachments	ImageAttachmentDto[]	No	No explicit default	ApiPropertyOptional({ type: [ImageAttachmentDto] }); IsOptional(); IsArray(); ValidateNested({ each: true }); Type(() => ImageAttachmentDto)

StudentTutorBootstrapQueryDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
classId	string	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsUUID()

DTO source: backend/src/modules/announcements/DTO/create-announcement.dto.ts

CreateAnnouncementDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
title	string	Yes	No explicit default	ApiProperty({ example: 'Reminder: Project due tomorrow' }); IsString(); MinLength(1); MaxLength(255)
content	string	Yes	No explicit default	ApiProperty({ example: 'Please submit your projects by 11:59 PM.' }); IsString(); MinLength(1)
isPinned	boolean	No	No explicit default	ApiPropertyOptional({ default: false }); IsOptional(); IsBoolean()
scheduledAt	string	No	No explicit default	ApiPropertyOptional({ description: 'ISO 8601 future date to schedule the announcement', example: '2026-03-01T08:00:00.000Z' }); IsOptional(); IsISO8601()
fields	string[]	No	No explicit default	ApiPropertyOptional({ description: 'UUIDs of uploaded files to attach', type: [String] }); IsOptional(); IsArray(); IsUUID('4', { each: true })

DTO source: backend/src/modules/announcements/DTO/query-announcements.dto.ts

QueryAnnouncementsDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
page	number	No	1	ApiPropertyOptional({ default: 1 }); IsOptional()

Property	Type	Required	Default	Validation and transform decorators
				al(); Type() => Number; IsInt(); Min(1)
limit	number	No	20	ApiPropertyOptional({ default: 20 }); IsOptional(); Type() => Number; IsInt(); Min(1); Max(100)

DTO source: backend/src/modules/announcements/DTO/update-announcement.dto.ts

UpdateAnnouncementDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
title	string	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsString(); MinLength(1); MaxLength(255)
content	string	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsString(); MinLength(1)
isPinned	boolean	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsBoolean()
scheduledAt	string	No	No explicit default	ApiPropertyOptional(); IsOptional(); ISO8601()
fields	string[]	No	No explicit default	ApiPropertyOptional({ type: [String] }); IsOptional(); IsArray(); IsUUID('4', { each: true })

ReleaseCoreAnnouncementDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
isVisible	boolean	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsBoolean()
isPublished	boolean	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsBoolean()

DTO source: backend/src/modules/app-version/dto/check-app-version.dto.ts

CheckAppVersionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
platform	string	No	'android'	ApiPropertyOptional({ example: 'android', enum: ['android', 'ios'] }); IsOptional(); IsString(); IsIn(['android', 'ios'])
currentNativeVersion	string	No	No explicit default	ApiPropertyOptional({ example: '0.1.0' }); IsOptional(); IsString()
currentOtaVersion	string	No	No explicit default	ApiPropertyOptional({ example: '1' }); IsOptional(); IsString()
currentVersionCode	number	No	No explicit default	ApiPropertyOptional({ example: 1 }); IsOptional(); Type() => Number; IsInt()

DTO source: backend/src/modules/assessments/DTO/assessment.dto.ts**CreateAssessmentDto**

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
title	string	Yes	No explicit default	IsString()
description	string	No	No explicit default	IsOptional(); IsString()
classId	string	Yes	No explicit default	IsUUID()
type	AssessmentType	No	AssessmentType.QUIZ	IsOptional(); IsEnum(AssessmentType)
dueDate	string	No	No explicit default	IsOptional(); IsDateString()
closeWhenDue	boolean	No	true	IsOptional(); IsBoolean()
randomizeQuestions	boolean	No	false	IsOptional(); IsBoolean()
timedQuestionsEnabled	boolean	No	false	IsOptional(); IsBoolean()
questionTimeLimitSeconds	number null	No	No explicit default	IsOptional(); ValidateIf((o) => o.questionTimeLimitSeconds !== null); IsInt(); Min(5)
strictMode	boolean	No	false	IsOptional(); IsBoolean()
fileUploadInstructions	string	No	No explicit default	IsOptional(); IsString()
teacherAttachmentFileId	string	No	No explicit default	IsOptional(); IsUUID()
rubricSourceFileId	string	No	No explicit default	IsOptional(); IsUUID()
allowedUploadMimeTypes	string[]	No	No explicit default	IsOptional(); IsArray(); IsString({ each: true })
allowedUploadExtensions	string[]	No	No explicit default	IsOptional(); IsArray(); IsString({ each: true })
maxUploadSizeBytes	number	No	No explicit default	IsOptional(); IsInt(); Min(1); Max(104857600)
passingScore	number	No	60	IsOptional(); IsInt(); Min(1)
maxAttempts	number	No	1	IsOptional(); IsInt(); Min(1)
timeLimitMinutes	number null	No	No explicit default	IsOptional(); ValidateIf((o) => o.timeLimitMinutes !== null); IsInt(); Min(1)
feedbackLevel	FeedbackLevel	No	FeedbackLevel.STANDARD	IsOptional(); IsEnum(FeedbackLevel)
feedbackDelayHours	number	No	24	IsOptional(); IsInt()
classRecordCategory	ClassRecordCategory	No	No explicit default	IsOptional(); IsEnum(ClassRecordCategory)
quarter	Quarter	No	No explicit default	IsOptional(); IsEnum(Quarter)
classRecordItemId	string	No	No explicit default	IsOptional(); IsUUID()

RubricCriterionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
id	string	Yes	No explicit default	IsString(); MinLength(1)
title	string	Yes	No explicit default	IsString(); MinLength(1)
description	string	No	No explicit default	IsOptional(); IsString()
points	number	Yes	No explicit default	IsInt(); Min(0)

UpdateAssessmentDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
title	string	No	No explicit default	IsOptional(); IsString()
description	string	No	No explicit default	IsOptional(); IsString()
type	AssessmentType	No	No explicit default	IsOptional(); IsEnum(AssessmentType)
dueDate	string	No	No explicit default	IsOptional(); IsDateString()
closeWhenDue	boolean	No	No explicit default	IsOptional(); IsBoolean()
randomizeQuestions	boolean	No	No explicit default	IsOptional(); IsBoolean()
timedQuestionsEnabled	boolean	No	No explicit default	IsOptional(); IsBoolean()
questionTimeLimitSeconds	number null	No	No explicit default	IsOptional(); ValidateIf((o) => o.questionTimeLimitSeconds !== null); IsInt(); Min(5)
strictMode	boolean	No	No explicit default	IsOptional(); IsBoolean()
fileUploadInstructions	string	No	No explicit default	IsOptional(); IsString()
teacherAttachmentFileId	string null	No	No explicit default	IsOptional(); ValidateIf((o) => o.teacherAttachmentFileId !== null); IsUUID()
rubricSourceFileId	string null	No	No explicit default	IsOptional(); ValidateIf((o) => o.rubricSourceFileId !== null); IsUUID()
rubricCriteria	RubricCriterionDto[]	No	No explicit default	IsOptional(); IsArray(); ValidateNested({ each: true }); Type(() => RubricCriterionDto)
allowedUploadMimeTypes	string[]	No	No explicit default	IsOptional(); IsArray(); IsString({ each: true })
allowedUploadExtensions	string[]	No	No explicit default	IsOptional(); IsArray(); IsString({ each: true })
maxUploadSizeBytes	number	No	No explicit default	IsOptional(); IsInt(); Min(1); Max(104857600)
passingScore	number	No	No explicit default	IsOptional(); IsInt(); Min(1)
maxAttempts	number	No	No explicit default	IsOptional(); IsInt(); Min(1)
timeLimitMinutes	number null	No	No explicit default	IsOptional(); ValidateIf((o) => o.timeLimitMinutes !== null); IsInt(); Min(1)
isPublished	boolean	No	No explicit default	IsOptional(); IsBoolean()
feedbackLevel	FeedbackLevel	No	No explicit default	IsOptional(); IsEnum(FeedbackLevel)
feedbackDelayHours	number	No	No explicit default	IsOptional(); IsInt()
classRecordCategory	ClassRecordCategory	No	No explicit default	IsOptional(); IsEnum(ClassRecordCategory)
quarter	Quarter	No	No explicit default	IsOptional(); IsEnum(Quarter)
classRecordItemId	string null	No	No explicit default	IsOptional(); ValidateIf((o) => o.classRecordItemId !== null); IsUUID()

OptionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
text	string	Yes	No explicit default	IsString()
isCorrect	boolean	Yes	No explicit default	IsBoolean()
order	number	Yes	No explicit default	IsInt()
imageUrl	string	No	No explicit default	IsOptional(); IsString()
imageDisplayMode	'default' 'expanded'	No	No explicit default	IsOptional(); IsString()

Property	Type	Required	Default	Validation and transform decorators
imageZoom	number	No	No explicit default	IsOptional(); IsInt(); Min(50); Max(200)
imagePositionX	number	No	No explicit default	IsOptional(); IsInt(); Min(0); Max(100)
imagePositionY	number	No	No explicit default	IsOptional(); IsInt(); Min(0); Max(100)

CreateQuestionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
assessmentId	string	Yes	No explicit default	IsUUID()
type	QuestionType	Yes	No explicit default	IsEnum(QuestionType)
content	string	Yes	No explicit default	IsString()
points	number	Yes	1	IsInt()
order	number	Yes	No explicit default	IsInt()
isRequired	boolean	No	true	IsOptional(); IsBoolean()
explanation	string	No	No explicit default	IsOptional(); IsString()
imageUrl	string	No	No explicit default	IsOptional(); IsString()
imageDisplayMode	'default' 'expanded'	No	No explicit default	IsOptional(); IsString()
imageZoom	number	No	No explicit default	IsOptional(); IsInt(); Min(50); Max(200)
imagePositionX	number	No	No explicit default	IsOptional(); IsInt(); Min(0); Max(100)
imagePositionY	number	No	No explicit default	IsOptional(); IsInt(); Min(0); Max(100)
conceptTags	string[]	No	No explicit default	IsOptional(); IsArray(); IsString({ each: true })
options	OptionDto[]	No	No explicit default	IsOptional(); IsArray(); ValidateNested({ each: true }); Type(() => OptionDto)

UpdateQuestionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
content	string	No	No explicit default	IsOptional(); IsString()
points	number	No	No explicit default	IsOptional(); IsInt()
order	number	No	No explicit default	IsOptional(); IsInt()
isRequired	boolean	No	No explicit default	IsOptional(); IsBoolean()
explanation	string	No	No explicit default	IsOptional(); IsString()
imageUrl	string	No	No explicit default	IsOptional(); IsString()
imageDisplayMode	'default' 'expanded'	No	No explicit default	IsOptional(); IsString()
imageZoom	number	No	No explicit default	IsOptional(); IsInt(); Min(50); Max(200)
imagePositionX	number	No	No explicit default	IsOptional(); IsInt(); Min(0); Max(100)
imagePositionY	number	No	No explicit default	IsOptional(); IsInt(); Min(0); Max(100)
conceptTags	string[]	No	No explicit default	IsOptional(); IsArray(); IsString({ each: true })
options	OptionDto[]	No	No explicit default	

Property	Type	Required	Default	Validation and transform decorators
				IsOptional(); IsArray(); ValidateNested({ each: true }); Type() => OptionDto

ResponseAnswerDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
questionId	string	Yes	No explicit default	IsUUID()
studentAnswer	string	No	No explicit default	IsOptional(); IsString()
selectedOptionId	string	No	No explicit default	IsOptional(); IsUUID()
selectedOptionIds	string[]	No	No explicit default	IsOptional(); IsArray()

ProgressResponseAnswerDto

Inheritance or composition: `extends ResponseAnswerDto`.

Property	Type	Required	Default	Validation and transform decorators
questionIndex	number	No	No explicit default	IsOptional(); IsInt(); Min(0)

SubmitAssessmentDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
assessmentId	string	Yes	No explicit default	IsUUID()
responses	ResponseAnswerDto[]	Yes	No explicit default	IsArray(); Type() => ResponseAnswerDto
timeSpentSeconds	number	Yes	No explicit default	IsInt()

StartAssessmentDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
assessmentId	string	Yes	No explicit default	IsUUID()

UpdateAttemptProgressDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
currentQuestionIndex	number	No	No explicit default	IsOptional(); IsInt(); Min(0)
responses	ProgressResponseAnswerDto[]	No	No explicit default	IsOptional(); IsArray(); Type() => ProgressResponseAnswerDto
registerViolation	boolean	No	No explicit default	IsOptional(); IsBoolean()

ReturnedRubricScoreDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
criterionId	string	Yes	No explicit default	IsString(); MinLength(1)
pointsEarned	number	Yes	No explicit default	IsInt(); Min(0)
feedback	string	No	No explicit default	IsOptional(); IsString()

ManualResponseScoreDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
questionId	string	Yes	No explicit default	IsUUID('4')
pointsEarned	number	Yes	No explicit default	IsInt(); Min(0)

ReturnGradeDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
teacherFeedback	string	No	No explicit default	IsOptional(); IsString()
directScore	number	No	No explicit default	IsOptional(); IsInt(); Min(0); Max(100)
rubricScores	ReturnedRubricScoreDto[]	No	No explicit default	IsOptional(); IsArray(); ValidateNested({ each: true }); Type() => ReturnedRubricScoreDto)
manualResponseScores	ManualResponseScoreDto[]	No	No explicit default	IsOptional(); IsArray(); ValidateNested({ each: true }); Type() => ManualResponseScoreDto)

BulkReturnGradesDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
attemptIds	string[]	Yes	No explicit default	IsArray(); ArrayNotEmpty(); IsUUID('4', { each: true })
teacherFeedback	string	No	No explicit default	IsOptional(); IsString()

ReleaseCoreAssessmentDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
isPublished	boolean	Yes	No explicit default	IsBoolean()

DTO source: backend/src/modules/auth/DTO/change-password.dto.ts

ChangePasswordDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
oldPassword	string	Yes	No explicit default	ApiProperty({ example: 'OldP@ss1', description: 'Current password' }); IsString({ message: 'Old password must be a string' }); Is-

Property	Type	Required	Default	Validation and transform decorators
				NotEmpty({ message: 'Old password is required' })
newPassword	string	Yes	No explicit default	IsNotEmpty({ message: 'New password is required' }); IsStrongPassword('NewP@ss1!', 'New password (min 8 chars, uppercase, lowercase, digit, special character)',)
confirmPassword	string	Yes	No explicit default	IsNotEmpty({ message: 'Confirm password is required' }); IsStrongPassword('NewP@ss1!', 'Confirm password (min 8 chars, uppercase, lowercase, digit, special character)',)

DTO source: backend/src/modules/auth/DTO/forgot-password.dto.ts

ForgotPasswordDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
email	string	Yes	No explicit default	ApiProperty({ example: 'student@school.edu' }); IsEmail({}, { message: 'Must be a valid email address' }); Transform(({ value }) => value?.toLowerCase().trim())

DTO source: backend/src/modules/auth/DTO/login.dto.ts

LoginDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
email	string	Yes	No explicit default	ApiProperty({ example: 'admin@lms.local', description: 'User email address' }); IsEmail({}, { message: 'Must be a valid email address' }); Transform(({ value }) => lowerTrimmedValue(value))
password	string	Yes	No explicit default	ApiProperty({ example: 'Test@123', description: 'User password' }); IsString(); IsNotEmpty({ message: 'Password is required' }); Matches(PASSWORD_SAFE_REGEX, { message: 'Password contains unsupported control characters', })

DTO source: backend/src/modules/auth/DTO/mobile-logout.dto.ts

MobileLogoutDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
refreshToken	string	Yes	No explicit default	ApiProperty(); IsString(); IsNotEmpty()

DTO source: backend/src/modules/auth/DTO/mobile-refresh.dto.ts**MobileRefreshDto**

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
refreshToken	string	Yes	No explicit default	ApiProperty(); IsString(); IsNotEmpty()

DTO source: backend/src/modules/auth/DTO/reset-password.dto.ts**ResetPasswordDto**

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
email	string	Yes	No explicit default	ApiProperty({ example: 'student@school.edu' }); IsEmail({}, { message: 'Must be a valid email address' }); Transform(({ value }) => value?.toLowerCase().trim())
code	string	Yes	No explicit default	ApiProperty({ example: '123456', description: 'OTP code sent to email' }); IsString(); IsNotEmpty({ message: 'Code is required' })
newPassword	string	Yes	No explicit default	IsStrongPassword('NewP@ss1!', 'New password (min 8 chars, uppercase, lowercase, digit, special character)',)

DTO source: backend/src/modules/auth/DTO/set-activation-password.dto.ts**SetActivationPasswordDto**

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
email	string	Yes	No explicit default	ApiProperty({ example: 'student@school.edu' }); IsEmail({}, { message: 'Must be a valid email address' }); Transform(({ value }) => value?.toLowerCase().trim())
newPassword	string	Yes	No explicit default	IsStrongPassword('MyP@ss1!', 'New password (min 8 chars, uppercase, lowercase, digit, special character)',)

DTO source: backend/src/modules/auth/DTO/set-initial-password.dto.ts**SetInitialPasswordDto**

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
email	string	Yes	No explicit default	ApiProperty({ example: 'student@school.edu' }); IsEmail({}, { message: 'Must be a valid email address' });

Property	Type	Required	Default	Validation and transform decorators
				Transform(({ value }) => value?.toLowerCase().trim())
code	string	Yes	No explicit default	ApiProperty({ example: '123456', description: 'OTP activation code' }); IsString(); IsNotEmpty({ message: 'Code is required' })
newPassword	string	Yes	No explicit default	IsStrongPassword('MyP@ss1!', 'Initial password (min 8 chars, uppercase, lowercase, digit, special character)',)

DTO source: backend/src/modules/auth/DTO/update-profile.dto.ts

UpdateProfileDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
firstName	string	No	No explicit default	ApiProperty({ example: 'Juan', required: false }); IsOptional(); IsString(); MaxLength(30); Matches(PERSON_NAME_REGEX, { message: 'First name may only contain letters, spaces, hyphens, and apostrophes' }); Transform(({ value }: { value: string }) => value?.trim())
middleName	string	No	No explicit default	ApiProperty({ example: 'Santos', required: false }); IsOptional(); IsString(); MaxLength(30); Matches(PERSON_NAME_REGEX, { message: 'Middle name may only contain letters, spaces, hyphens, and apostrophes' }); Transform(({ value }: { value: string }) => trimValue(value))
lastName	string	No	No explicit default	ApiProperty({ example: 'Dela Cruz', required: false }); IsOptional(); IsString(); MaxLength(30); Matches(PERSON_NAME_REGEX, { message: 'Last name may only contain letters, spaces, hyphens, and apostrophes' }); Transform(({ value }: { value: string }) => value?.trim())
dob	string	No	No explicit default	ApiProperty({ example: '2005-08-15', required: false }); IsOptional(); IsDateString({}, { message: 'Date of birth must be a valid ISO date' })
dateOfBirth	string	No	No explicit default	ApiProperty({ example: '2005-08-15', required: false }); IsOptional(); IsDateString({}, { message: 'Date of birth must be a valid ISO date' })
gender	'Male' 'Female'	No	No explicit default	ApiProperty({ example: 'Male', required: false }); IsOptional(); IsIn(['Male', 'Female'], { message: 'Gender must be either Male or Female' })
lrm	string	No	No explicit default	ApiProperty({ example: '202401230001', required: false }); Is-

Property	Type	Required	Default	Validation and transform decorators
				String(); Matches(/^[0-9]{12}\$/, { message: 'LRN must be exactly 12 digits (e.g., 202401230001)', })
phone	string	No	No explicit default	ApiProperty({ example: '+639171234567', required: false }); IsOptional(); IsString(); Matches(PH_MOBILE_REGEX, { message: 'Student contact number must be a valid PH mobile format (e.g., 09171234567 or +639171234567)', })
address	string	No	No explicit default	ApiProperty({ example: '123 Main St, Manila', required: false }); IsOptional(); IsString(); MaxLength(180); Matches(ADDRESS_REGEX, { message: 'Address may only contain letters, numbers, spaces, commas, periods, number signs, apostrophes, hyphens, and slashes', }); Transform(({ value }: { value?: string }) => trimValue(value))
familyName	string	No	No explicit default	ApiProperty({ example: 'Dela Cruz', required: false }); IsOptional(); IsString(); MaxLength(80); Matches(PERSON_NAME_REGEX, { message: 'Guardian name may only contain letters, spaces, hyphens, and apostrophes', }); Transform(({ value }: { value?: string }) => trimValue(value))
familyRelationship	string	No	No explicit default	ApiProperty({ example: 'Father', enum: ['Father', 'Mother', 'Guardian', 'Sibling', 'Other'], required: false }); IsOptional(); IsIn(['Father', 'Mother', 'Guardian', 'Sibling', 'Other'], { message: 'Relationship must be one of: Father, Mother, Guardian, Sibling, Other', })
familyContact	string	No	No explicit default	ApiProperty({ example: '+639179876543', required: false }); IsOptional(); IsString(); Matches(PH_MOBILE_REGEX, { message: 'Guardian contact number must be a valid PH mobile format (e.g., 09171234567 or +639171234567)', })
profilePicture	string	No	No explicit default	ApiProperty({ example: '/api/profiles/images/student-avatar.png', required: false }); IsOptional(); IsString()

DTO source: backend/src/modules/auth/DTO/validate-credentials.dto.ts

ValidateCredentialsDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
email	string	Yes	No explicit default	ApiProperty({ example: 'student@school.edu' }); IsEmail({}, { message: 'Must be a valid email address' });

Property	Type	Required	Default	Validation and transform decorators
				Transform(({ value }) => value.toLowerCase().trim())
password	string	Yes	No explicit default	ApiProperty({ example: 'P@ssw0rd!' }); IsString(); IsNotEmpty({ message: 'Password is required' });

DTO source: backend/src/modules/class-record/DTO/bulk-record-scores.dto.ts

BulkRecordScoresDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
scores	RecordScoreDto[]	Yes	No explicit default	IsArray({ message: 'scores must be an array' }); ArrayMinSize(1, { message: 'scores must contain at least one entry' }); ValidateNested({ each: true }); Type(() => RecordScoreDto)

DTO source: backend/src/modules/class-record/DTO/create-class-record.dto.ts

CreateClassRecordDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
classId	string	Yes	No explicit default	IsUUID('4', { message: 'classId must be a valid UUID' });
gradingPeriod	'Q1' 'Q2' 'Q3' 'Q4'	Yes	No explicit default	IsIn(['Q1', 'Q2', 'Q3', 'Q4'], { message: 'gradingPeriod must be Q1, Q2, Q3, or Q4' });

DTO source: backend/src/modules/class-record/DTO/record-score.dto.ts

RecordScoreDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
studentId	string	Yes	No explicit default	IsUUID('4', { message: 'studentId must be a valid UUID' });
score	number	Yes	No explicit default	Type(() => Number); IsNumber({}, { message: 'score must be a number' }); Min(0, { message: 'score must be at least 0' });

DTO source: backend/src/modules/class-record/DTO/update-class-record-item.dto.ts

UpdateClassRecordItemDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
maxScore	number	Yes	No explicit default	Type(() => Number); IsNumber({}, { message: 'maxScore must be a number' }); Min(0, { message: 'maxScore must be at least 0' })

DTO source: backend/src/modules/class-templates/dto/class-template.dto.ts

CreateClassTemplateDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
name	string	Yes	No explicit default	IsString(); IsNotEmpty(); MaxLength(180)
subjectCode	string	Yes	No explicit default	IsString(); IsNotEmpty(); MaxLength(64)
subjectGradeLevel	string	Yes	No explicit default	IsString(); IsIn([spread operator GRADE_LEVELS])

UpdateClassTemplateDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
name	string	No	No explicit default	IsOptional(); IsString(); IsNotEmpty(); MaxLength(180)

TemplateQuestionOptionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
id	string	No	No explicit default	IsOptional(); IsString()
text	string	Yes	No explicit default	IsString()
isCorrect	boolean	No	No explicit default	IsOptional(); IsBoolean()
order	number	No	No explicit default	IsOptional(); IsInt()
imageUrl	string	No	No explicit default	IsOptional(); IsString()
imageDisplayMode	'default' 'expanded'	No	No explicit default	IsOptional(); IsString()
imageZoom	number	No	No explicit default	IsOptional(); IsInt(); Min(50); Max(200)
imagePositionX	number	No	No explicit default	IsOptional(); IsInt(); Min(0); Max(100)
imagePositionY	number	No	No explicit default	IsOptional(); IsInt(); Min(0); Max(100)

TemplateQuestionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
id	string	No	No explicit default	IsOptional(); IsString()
type	string	Yes	No explicit default	IsString()
content	string	Yes	No explicit default	IsString()
points	number	No	No explicit default	IsOptional(); IsInt(); Min(0)

Property	Type	Required	Default	Validation and transform decorators
order	number	No	No explicit default	IsOptional(); IsInt()
isRequired	boolean	No	No explicit default	IsOptional(); IsBoolean()
explanation	string	No	No explicit default	IsOptional(); IsString()
imageUrl	string	No	No explicit default	IsOptional(); IsString()
imageDisplayMode	'default' 'expanded'	No	No explicit default	IsOptional(); IsString()
imageZoom	number	No	No explicit default	IsOptional(); IsInt(); Min(50); Max(200)
imagePositionX	number	No	No explicit default	IsOptional(); IsInt(); Min(0); Max(100)
imagePositionY	number	No	No explicit default	IsOptional(); IsInt(); Min(0); Max(100)
options	TemplateQuestionOptionDto[]	No	No explicit default	IsOptional(); IsArray(); ValidateNested({ each: true }); Type() => TemplateQuestionOptionDto)

TemplateAssessmentSettingsDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
dueDateOffsetDays	number	No	No explicit default	IsOptional(); IsInt()
maxAttempts	number	No	No explicit default	IsOptional(); IsInt()
passingScore	number	No	No explicit default	IsOptional(); IsInt()
randomizeQuestions	boolean	No	No explicit default	IsOptional(); IsBoolean()
closeWhenDue	boolean	No	No explicit default	IsOptional(); IsBoolean()

ClassTemplateAssessmentDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
id	string	No	No explicit default	IsOptional(); IsUUID()
title	string	Yes	No explicit default	IsString(); IsNotEmpty()
description	string	No	No explicit default	IsOptional(); IsString()
type	string	No	No explicit default	IsOptional(); IsString()
settings	TemplateAssessmentSettingsDto	No	No explicit default	IsOptional(); ValidateNested(); Type() => TemplateAssessmentSettingsDto)
questions	TemplateQuestionDto[]	No	No explicit default	IsOptional(); IsArray(); ValidateNested({ each: true }); Type() => TemplateQuestionDto)
totalPoints	number	No	No explicit default	IsOptional(); IsInt(); Min(0)
order	number	No	No explicit default	IsOptional(); IsInt()

ClassTemplateModuleItemDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
id	string	No	No explicit default	IsOptional(); IsUUID()
itemType	ClassTemplateItemType	Yes	No explicit default	IsEnum(ClassTemplateItemType)
templateAssessmentId	string	No	No explicit default	IsOptional(); IsUUID()

Property	Type	Required	Default	Validation and transform decorators
templateLessonId	string	No	No explicit default	IsOptional(); IsUUID()
order	number	No	No explicit default	IsOptional(); IsInt()
isRequired	boolean	No	No explicit default	IsOptional(); IsBoolean()
metadata	Record	No	No explicit default	IsOptional(); IsObject()
points	number	No	No explicit default	IsOptional(); IsInt(); Min(0)

ClassTemplateLessonBlockDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
id	string	No	No explicit default	IsOptional(); IsUUID()
blockType	string	Yes	No explicit default	IsString(); IsNotEmpty()
blockVersion	number	No	No explicit default	IsOptional(); IsInt(); Min(1)
payload	Record	Yes	No explicit default	IsObject()
order	number	No	No explicit default	IsOptional(); IsInt()

ClassTemplateLessonDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
id	string	No	No explicit default	IsOptional(); IsUUID()
title	string	Yes	No explicit default	IsString(); IsNotEmpty()
summary	string	No	No explicit default	IsOptional(); IsString()
order	number	No	No explicit default	IsOptional(); IsInt()
blocks	ClassTemplateLessonBlockDto[]	No	No explicit default	IsOptional(); IsArray(); ValidateNested({ each: true }); Type(() => ClassTemplateLessonBlockDto)

ClassTemplateModuleSectionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
id	string	No	No explicit default	IsOptional(); IsUUID()
title	string	Yes	No explicit default	IsString(); IsNotEmpty()
description	string	No	No explicit default	IsOptional(); IsString()
order	number	No	No explicit default	IsOptional(); IsInt()
items	ClassTemplateModuleItemDto[]	No	No explicit default	IsOptional(); IsArray(); ValidateNested({ each: true }); Type(() => ClassTemplateModuleItemDto)

ClassTemplateModuleDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
id	string	No	No explicit default	IsOptional(); IsUUID()
title	string	Yes	No explicit default	IsString(); IsNotEmpty()
description	string	No	No explicit default	IsOptional(); IsString()

Property	Type	Required	Default	Validation and transform decorators
order	number	No	No explicit default	IsOptional(); IsInt()
themeKind	string	No	No explicit default	IsOptional(); IsString()
gradientId	string	No	No explicit default	IsOptional(); IsString()
coverImageUrl	string null	No	No explicit default	IsOptional(); IsString()
imagePositionX	number	No	No explicit default	IsOptional(); IsInt()
imagePositionY	number	No	No explicit default	IsOptional(); IsInt()
imageScale	number	No	No explicit default	IsOptional(); IsInt()
isVisible	boolean	No	No explicit default	IsOptional(); IsBoolean()
isLocked	boolean	No	No explicit default	IsOptional(); IsBoolean()
teacherNotes	string	No	No explicit default	IsOptional(); IsString()
sections	ClassTemplateModuleSectionDto[]	No	No explicit default	IsOptional(); IsArray(); ValidateNested({ each: true }); Type() => ClassTemplateModuleSectionDto)

ClassTemplateAnnouncementDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
id	string	No	No explicit default	IsOptional(); IsUUID()
title	string	Yes	No explicit default	IsString(); IsNotEmpty()
content	string	Yes	No explicit default	IsString(); IsNotEmpty()
isPinned	boolean	No	No explicit default	IsOptional(); IsBoolean()
order	number	No	No explicit default	IsOptional(); IsInt()

UpdateClassTemplateContentDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
modules	ClassTemplateModuleDto[]	No	No explicit default	IsOptional(); IsArray(); ValidateNested({ each: true }); Type() => ClassTemplateModuleDto)
assessments	ClassTemplateAssessmentDto[]	No	No explicit default	IsOptional(); IsArray(); ValidateNested({ each: true }); Type() => ClassTemplateAssessmentDto)
announcements	ClassTemplateAnnouncementDto[]	No	No explicit default	IsOptional(); IsArray(); ValidateNested({ each: true }); Type() => ClassTemplateAnnouncementDto)
lessons	ClassTemplateLessonDto[]	No	No explicit default	IsOptional(); IsArray(); ValidateNested({ each: true }); Type() => ClassTemplateLessonDto)

PublishClassTemplateDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
status	ClassTemplateStatus	No	No explicit default	IsOptional(); IsEnum(ClassTemplateStatus)

EngineImportValidateDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
manifest	string	Yes	No explicit default	IsString(); IsNotEmpty()

EngineImportDto

Inheritance or composition: `extends EngineImportValidateDto`.

Property	Type	Required	Default	Validation and transform decorators
publish	boolean	No	No explicit default	IsOptional(); IsBoolean()

DTO source: backend/src/modules/classes/DTO/bulk-class-lifecycle.dto.ts

BulkClassLifecycleDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
action	BulkClassLifecycleAction	Yes	No explicit default	IsIn(BULK_CLASS_LIFECYCLE_ACTIONS)
classIds	string[]	Yes	No explicit default	IsArray(); ArrayMinSize(1); ArrayUnique(); IsString({ each: true })

DTO source: backend/src/modules/classes/DTO/create-class.dto.ts

GradingProfileSumConstraint

Inheritance or composition: `implements ValidatorConstraintInterface`.

Property	Type	Required	Default	Validation and transform decorators

ClassGradingProfileDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
writtenWork	number	Yes	No explicit default	IsDefined({ message: 'writtenWork is required' }); IsInt({ message: 'writtenWork must be a whole number' }); Min(1, { message: 'writtenWork must be greater than 0' }); Max(99, { message: 'writtenWork must be 99 or less' }); Type(() => Number)
performanceTask	number	Yes	No explicit default	IsDefined({ message: 'performanceTask is required' }); IsInt({ message: 'performanceTask must be a whole number' }); Min(1, { message: 'performanceTask must be greater than 0' }); Max(99, { message: 'performanceTask must be 99 or less' }); Type(() => Number)
quarterlyAssessment	number	Yes	No explicit default	IsDefined({ message: 'quarterlyAssessment is required' }); IsInt({ message: 'quarterlyAssessment must be a whole number' }); Min(1, { message: 'quarterlyAssessment must be greater than 0' }); Max(99, { message: 'quarterlyAssessment must be 99 or less' }); Type(() => Number)

Property	Type	Required	Default	Validation and transform decorators
				be a whole number' }); Min(1, { message: 'quarterly-Assessment must be greater than 0' }); Max(99, { message: 'quarterlyAssessment must be 99 or less' }); Type(() => Number)

CreateClassDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
subjectName	string	Yes	No explicit default	IsString({ message: 'subject-Name must be a string' }); Transform(({ value }: { value?: string }) => trim-Value(value)); Matches(SUBJECT_NAME_REGEX, { message: 'subjectName may only contain letters, numbers, spaces, hyphens, and apostrophes', })
subjectCode	string	Yes	No explicit default	IsString({ message: 'subject-Code must be a string' }); Transform(({ value }: { value?: string }) => upper-TrimmedValue(value)); Matches(SUBJECT_CODE_REGEX, { message: 'subjectCode may only contain uppercase letters, numbers, and hyphens', })
subjectGradeLevel	string	No	No explicit default	IsOptional(); IsIn([spread operator GRADE_LEVELS], { message: 'subjectGrade-Level must be 7, 8, 9 or 10', })
sectionId	string	Yes	No explicit default	IsUUID('4', { message: 'sectionId must be a valid UUID' })
teacherId	string	Yes	No explicit default	IsUUID('4', { message: 'teacherId must be a valid UUID' })
templateId	string	No	No explicit default	IsOptional(); IsUUID('4', { message: 'templateId must be a valid UUID' })
schoolYear	string	Yes	No explicit default	IsString({ message: 'schoolYear must be a string' }); Validate(IsValidSchoolYearConstraint)
schedules	ScheduleSlotDto[]	Yes	No explicit default	IsArray({ message: 'schedules must be an array of schedule slots' }); ArrayMinSize(1, { message: 'At least one schedule slot is required' }); ValidateNested({ each: true }); Type(() => ScheduleSlotDto)
room	string	Yes	No explicit default	IsString({ message: 'room must be a string' }); Transform(({ value }: { value?: string }) => trim-Value(value)); MinLength(1, { message: 'room is required' }); IsIn([spread operator ALLOWED_ROOM_NUMBERS], { message: 'ALLOWED_ROOM_NUMBERS_MESSAGE', })
cardPreset	string	No	No explicit default	IsOptional(); IsString({ message: 'cardPreset must be a string' })

Property	Type	Required	Default	Validation and transform decorators
cardBannerUrl	string	No	No explicit default	IsOptional(); IsString({ message: 'cardBannerUrl must be a string' })
gradingProfile	ClassGradingProfileDto	No	No explicit default	IsOptional(); Validate(GradingProfileSumConstraint, { message: 'gradingProfile must sum to exactly 100', }); ValidateNested(); Type(() => ClassGradingProfileDto)

DTO source: backend/src/modules/classes/DTO/enroll-student.dto.ts

EnrollStudentDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
studentId	string	Yes	No explicit default	IsUUID('4', { message: 'studentId must be a valid UUID' })

DTO source: backend/src/modules/classes/DTO/schedule-slot.dto.ts

ScheduleSlotDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
days	string[]	Yes	No explicit default	IsArray({ message: 'days must be an array' }); ArrayMinSize(1, { message: 'days must contain at least one day' }); IsIn(VAILD_DAYS, { each: true, message: 'Each day must be one of: M, T, W, Th, F, Sa, Su', })
startTime	string	Yes	No explicit default	IsString(); Matches(/^\\d{2}:\\d{2}\$/, { message: 'start-Time must be in HH:MM format (24-hour), e.g. "09:00", })
endTime	string	Yes	No explicit default	IsString(); Matches(/^\\d{2}:\\d{2}\$/, { message: 'end-Time must be in HH:MM format (24-hour), e.g. "10:00", })

DTO source: backend/src/modules/classes/DTO/update-class-presentation.dto.ts

UpdateClassPresentationDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
cardPreset	string	No	No explicit default	IsOptional(); IsString({ message: 'cardPreset must be a string' })
cardBannerUrl	string null	No	No explicit default	IsOptional(); IsString({ message: 'cardBannerUrl must be a string' })

DTO source: backend/src/modules/classes/DTO/update-class.dto.ts**UpdateClassDto**

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
subjectName	string	No	No explicit default	IsOptional(); IsString({ message: 'subject-Name must be a string' }); Transform(({ value }: { value?: string }) => trim-Value(value)); Matches(SUBJECT_NAME_REGEX, { message: 'subjectName may only contain letters, numbers, spaces, hyphens, and apostrophes', })
subjectCode	string	No	No explicit default	IsOptional(); IsString({ message: 'subject-Code must be a string' }); Transform(({ value }: { value?: string }) => upper-TrimmedValue(value)); Matches(SUBJECT_CODE_REGEX, { message: 'subjectCode may only contain uppercase letters, numbers, and hyphens', })
subjectGradeLevel	string	No	No explicit default	IsOptional(); IsIn([spread operator GRADE_LEVELS], { message: 'subjectGrade-Level must be 7, 8, 9 or 10', })
sectionId	string	No	No explicit default	IsOptional(); IsUUID('4', { message: 'sectionId must be a valid UUID' })
teacherId	string	No	No explicit default	IsOptional(); IsUUID('4', { message: 'teacherId must be a valid UUID' })
schoolYear	string	No	No explicit default	IsOptional(); IsString({ message: 'schoolYear must be a string' }); Validate(IsValidSchoolYearConstraint)
schedules	ScheduleSlotDto[]	No	No explicit default	IsOptional(); IsArray({ message: 'schedules must be an array of schedule slots' }); ArrayMinSize(1, { message: 'At least one schedule slot is required' }); ValidateNested({ each: true }); Type(() => ScheduleSlotDto)
room	string	No	No explicit default	IsOptional(); IsString({ message: 'room must be a string' }); Transform(({ value }: { value?: string }) => trim-Value(value)); MinLength(1, { message: 'room cannot be empty' }); IsIn([spread operator ALLOWED_ROOM_NUMBERS], { message: ALLOWED_ROOM_NUMBERS_MESSAGE, })
isActive	boolean	No	No explicit default	IsOptional(); IsBoolean({ message: 'isActive must be a boolean' })
cardPreset	string	No	No explicit default	IsOptional(); IsString({ message: 'cardPreset must be a string' })
cardBannerUrl	string null	No	No explicit default	IsOptional(); IsString({ message: 'cardBannerUrl must be a string' })

DTO source: backend/src/modules/classes/DTO/update-student-class-presentation.dto.ts

UpdateStudentClassPresentationDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
styleMode	StudentPresentationMode	Yes	No explicit default	IsIn(STUDENT_PRESENTATION_MODES, { message: 'styleMode must be one of: \${STUDENT_PRESENTATION_MODES.join(', ')}', })
styleToken	string	Yes	No explicit default	IsString({ message: 'styleToken must be a string' })

DTO source: backend/src/modules/classes/DTO/update-student-course-view.dto.ts

UpdateStudentCourseViewDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
viewMode	StudentCourseViewMode	Yes	No explicit default	IsIn(STUDENT_COURSE_VIEW_MODES, { message: 'viewMode must be one of: \${STUDENT_COURSE_VIEW_MODES.join(', ')}', })

DTO source: backend/src/modules/classes/DTO/validators.ts

IsValidSchoolYearConstraint

Inheritance or composition: `implements ValidatorConstraintInterface`.

Property	Type	Required	Default	Validation and transform decorators

DTO source: backend/src/modules/content-modules/DTO/module.dto.ts

ReorderEntryDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
id	string	Yes	No explicit default	IsUUID()
order	number	Yes	No explicit default	IsInt(); Min(1)

CreateModuleDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
classId	string	Yes	No explicit default	IsUUID()

Property	Type	Required	Default	Validation and transform decorators
title	string	Yes	No explicit default	IsString(); IsNotEmpty()
description	string	No	No explicit default	IsOptional(); IsString()
order	number	No	No explicit default	IsOptional(); IsInt(); Min(1)
isVisible	boolean	No	No explicit default	IsOptional(); IsBoolean()
isLocked	boolean	No	No explicit default	IsOptional(); IsBoolean()

UpdateModuleDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
title	string	No	No explicit default	IsOptional(); IsString(); IsNotEmpty()
description	string	No	No explicit default	IsOptional(); IsString()
isVisible	boolean	No	No explicit default	IsOptional(); IsBoolean()
isLocked	boolean	No	No explicit default	IsOptional(); IsBoolean()
teacherNotes	string	No	No explicit default	IsOptional(); IsString()
themeKind	ModuleThemeKind	No	No explicit default	IsOptional(); IsEnum(ModuleThemeKind)
gradientId	string	No	No explicit default	IsOptional(); IsString(); Length(1, 64)
coverImageUrl	string null	No	No explicit default	IsOptional(); IsString()
imagePositionX	number	No	No explicit default	IsOptional(); IsInt(); Min(0); Max(100)
imagePositionY	number	No	No explicit default	IsOptional(); IsInt(); Min(0); Max(100)
imageScale	number	No	No explicit default	IsOptional(); IsInt(); Min(100); Max(220)

ReorderModulesDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
modules	ReorderEntryDto[]	Yes	No explicit default	IsArray(); ArrayMaxSize(500); ValidateNested({ each: true }); Type() => ReorderEntryDto)

CreateModuleSectionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
title	string	Yes	No explicit default	IsString(); IsNotEmpty()
description	string	No	No explicit default	IsOptional(); IsString()
order	number	No	No explicit default	IsOptional(); IsInt(); Min(1)

UpdateModuleSectionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
title	string	No	No explicit default	IsOptional(); IsString(); IsNotEmpty()

Property	Type	Required	Default	Validation and transform decorators
description	string	No	No explicit default	IsOptional(); IsString()

ReorderModuleSectionsDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
sections	ReorderEntryDto[]	Yes	No explicit default	IsArray(); ArrayMax-Size(1000); ValidateNested({ each: true }); Type(() => ReorderEntryDto)

AttachModuleItemDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
itemType	ModuleItemType	Yes	No explicit default	IsEnum(ModuleItemType)
lessonId	string	No	No explicit default	IsOptional(); IsUUID()
assessmentId	string	No	No explicit default	IsOptional(); IsUUID()
fileId	string	No	No explicit default	IsOptional(); IsUUID()
order	number	No	No explicit default	IsOptional(); IsInt(); Min(1)
isVisible	boolean	No	No explicit default	IsOptional(); IsBoolean()
isRequired	boolean	No	No explicit default	IsOptional(); IsBoolean()
isGiven	boolean	No	No explicit default	IsOptional(); IsBoolean()
metadata	Record	No	No explicit default	IsOptional(); IsObject()
points	number	No	No explicit default	IsOptional(); IsInt(); Min(0); Max(10000)

UpdateModuleItemDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
order	number	No	No explicit default	IsOptional(); IsInt(); Min(1)
isVisible	boolean	No	No explicit default	IsOptional(); IsBoolean()
isRequired	boolean	No	No explicit default	IsOptional(); IsBoolean()
isGiven	boolean	No	No explicit default	IsOptional(); IsBoolean()
metadata	Record	No	No explicit default	IsOptional(); IsObject()
points	number	No	No explicit default	IsOptional(); IsInt(); Min(0); Max(10000)

ReorderModuleItemsDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
items	ReorderEntryDto[]	Yes	No explicit default	IsArray(); ArrayMax-Size(2000); ValidateNested({ each: true }); Type(() => ReorderEntryDto)

ReleaseCoreModuleDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
isVisible	boolean	No	No explicit default	IsOptional(); IsBoolean()
isLocked	boolean	No	No explicit default	IsOptional(); IsBoolean()

ReleaseCoreModuleItemDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
isVisible	boolean	No	No explicit default	IsOptional(); IsBoolean()
isGiven	boolean	No	No explicit default	IsOptional(); IsBoolean()

ModuleGradingScaleEntryDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
letter	string	Yes	No explicit default	IsString(); IsNotEmpty()
label	string	Yes	No explicit default	IsString(); IsNotEmpty()
minScore	number	Yes	No explicit default	IsInt(); Min(0); Max(100)
maxScore	number	Yes	No explicit default	IsInt(); Min(0); Max(100)
description	string	No	No explicit default	IsOptional(); IsString()
order	number	No	No explicit default	IsOptional(); IsInt(); Min(1)

ReplaceModuleGradingScaleDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
entries	ModuleGradingScaleEntry-Dto[]	Yes	No explicit default	IsArray(); ArrayMaxSize(20); ValidateNested({ each: true }); Type() => Module-GradingScaleEntryDto)

DTO source: backend/src/modules/discussion-board/DTO/discussion-comment.dto.ts

CreateDiscussionCommentDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
bodyHtml	string	No	No explicit default	ApiPropertyOptional({ example: 'I agree with this explanation.', description: 'Optional comment body in rich text HTML.', }); IsOptional(); IsString(); MinLength(1)
attachmentFileIds	string[]	No	No explicit default	ApiPropertyOptional({ type: [String], description: 'Uploaded image file IDs for comment attachments.', }); IsOptional(); IsArray(); ArrayMaxSize(4); IsUUID('4', { each: true })

SetDiscussionReactionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
reactionType	'like' 'heart' 'wow'	Yes	No explicit default	ApiPropertyOptional({ enum: ['like', 'heart', 'wow'], example: 'heart', }); IsString(); IsIn(['like', 'heart', 'wow'])

ReportDiscussionCommentDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
reasonCode	'inappropriate' 'spam' 'off_topic' 'harassment' 'academic_dishonesty'	Yes	No explicit default	ApiPropertyOptional({ enum: ['inappropriate', 'spam', 'off_topic', 'harassment', 'academic_dishonesty',], example: 'inappropriate', }); IsString(); IsIn(['inappropriate', 'spam', 'off_topic', 'harassment', 'academic_dishonesty',]);
notes	string	No	No explicit default	ApiPropertyOptional({ example: 'Contains personal attacks toward another learner.', description: 'Optional moderator notes for audit trail follow-up.', }); IsOptional(); IsString(); MaxLength(500)

DTO source: backend/src/modules/discussion-board/DTO/discussion-thread.dto.ts

DiscussionLinkAttachmentDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
url	string	Yes	No explicit default	ApiProperty({ example: 'https://example.com/reference', description: 'External link attachment URL', }); IsUrl({ require_protocol: true })
label	string	No	No explicit default	ApiPropertyOptional({ example: 'Reference material', }); IsOptional(); IsString(); MinLength(1); MaxLength(255)

CreateDiscussionThreadDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
title	string	Yes	No explicit default	ApiProperty({ example: 'Week 2 Open Forum' }); IsString(); MinLength(1); MaxLength(255)
bodyHtml	string	Yes	No explicit default	ApiProperty({ example: 'Share your insights and questions for this week.', }); IsString(); MinLength(1)
themeld	DiscussionThemeld	No	No explicit default	ApiPropertyOptional({ enum: DISCUSSION_THEME_IDS, de-

Property	Type	Required	Default	Validation and transform decorators
				fault: 'classic' }); IsOptional(); IsString(); MaxLength(64)
commentLimitPerStudent	number	No	No explicit default	ApiPropertyOptional({ description: 'Maximum active comments per student for this thread.', example: 1, }); IsOptional(); IsInt(); Min(1); Max(20)
allowComments	boolean	No	No explicit default	ApiPropertyOptional({ default: true }); IsOptional(); IsBoolean()
isPinned	boolean	No	No explicit default	ApiPropertyOptional({ default: false }); IsOptional(); IsBoolean()
fileAttachmentIds	string[]	No	No explicit default	ApiPropertyOptional({ description: 'Uploaded file IDs for image/pdf thread attachments.', type: [String], }); IsOptional(); IsArray(); ArrayMaxSize(10); IsUUID('4', { each: true })
linkAttachments	DiscussionLinkAttachment-Dto[]	No	No explicit default	ApiPropertyOptional({ type: [DiscussionLinkAttachment-Dto], }); IsOptional(); IsArray(); ArrayMaxSize(10); ValidateNested({ each: true }); Type() => DiscussionLinkAttachment-Dto

UpdateDiscussionThreadDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
title	string	No	No explicit default	ApiPropertyOptional({ example: 'Updated Thread Title' }); IsOptional(); IsString(); MinLength(1); MaxLength(255)
bodyHtml	string	No	No explicit default	ApiPropertyOptional({ example: 'Updated thread body' }); IsOptional(); IsString(); MinLength(1)
themeld	DiscussionThemeld	No	No explicit default	ApiPropertyOptional({ enum: DISCUSSION_THEME_IDS }); IsOptional(); IsString(); MaxLength(64)
commentLimitPerStudent	number null	No	No explicit default	ApiPropertyOptional({ description: 'Maximum active comments per student for this thread.', example: 3, }); IsOptional(); IsInt(); Min(1); Max(20)
allowComments	boolean	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsBoolean()
isPinned	boolean	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsBoolean()
fileAttachmentIds	string[]	No	No explicit default	ApiPropertyOptional({ description: 'Replace existing file attachments with this file list.', type: [String], }); IsOptional(); IsArray(); ArrayMaxSize(10); IsUUID('4', { each: true })
linkAttachments	DiscussionLinkAttachment-Dto[]	No	No explicit default	ApiPropertyOptional({ description: 'Replace existing link attachments with this list.', type: [DiscussionLinkAttachment-Dto], }); IsOptional(); IsArray(); ArrayMaxSize(10); ValidateNested({ each: true }); Type()

Property	Type	Required	Default	Validation and transform decorators
				=> DiscussionLinkAttachmentDto)

QueryDiscussionThreadsDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
page	number	No	No explicit default	ApiPropertyOptional({ default: 1 }); IsOptional(); Type() => Number; IsInt(); Min(1)
limit	number	No	No explicit default	ApiPropertyOptional({ default: 20 }); IsOptional(); Type() => Number; IsInt(); Min(1); Max(100)

DTO source: backend/src/modules/file-upload/dto/file-upload.dto.ts

UploadFileDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
classId	string	No	No explicit default	IsUUID('4', { message: 'classId must be a valid UUID' }); IsOptional()
folderId	string	No	No explicit default	IsUUID('4', { message: 'folderId must be a valid UUID' }); IsOptional()
scope	FileScopeDto	No	No explicit default	IsEnum(FileScopeDto, { message: 'scope must be private or general' }); IsOptional()
subjectKey	LibrarySubjectKeyDto	No	No explicit default	IsEnum(LibrarySubjectKeyDto, { message: 'subjectKey must be one of math, science, english, filipino, ap, tle, mapeh, esp', }); IsOptional()
gradeLevel	GradeLevelDto	No	No explicit default	IsEnum(GradeLevelDto, { message: 'gradeLevel must be one of 7, 8, 9, 10', }); IsOptional()
teacherVisible	boolean	No	No explicit default	Transform(({ value }) => toBoolean(value)); IsBoolean(); IsOptional()
aiEnabled	boolean	No	No explicit default	Transform(({ value }) => toBoolean(value)); IsBoolean(); IsOptional()

FileQueryDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
classId	string	No	No explicit default	IsUUID('4', { message: 'classId must be a valid UUID' }); IsOptional()
folderId	string	No	No explicit default	IsUUID('4', { message: 'folderId must be a valid UUID' }); IsOptional()
ownerId	string	No	No explicit default	IsUUID('4', { message: 'ownerId must be a valid UUID' }); IsOptional()
scope	FileScopeDto	No	No explicit default	

Property	Type	Required	Default	Validation and transform decorators
				IsEnum(FileScopeDto, { message: 'scope must be private or general' }); IsOptional()
subjectKey	LibrarySubjectKeyDto	No	No explicit default	IsEnum(LibrarySubjectKeyDto, { message: 'subjectKey must be one of math, science, english, filipino, ap, tle, mapeh, esp' }); IsOptional()
gradeLevel	GradeLevelDto	No	No explicit default	IsEnum(GradeLevelDto, { message: 'gradeLevel must be one of 7, 8, 9, 10' }); IsOptional()
indexStatus	LibraryIndexStatusDto	No	No explicit default	IsEnum(LibraryIndexStatusDto); IsOptional()
teacherVisible	boolean	No	No explicit default	Transform(({ value }) => toBoolean(value)); IsBoolean(); IsOptional()
search	string	No	No explicit default	IsString(); IsOptional()
page	number	No	1	IsOptional(); Type(() => Number); IsInt(); Min(1)
limit	number	No	20	IsOptional(); Type(() => Number); IsInt(); Min(1); Max(100)

CreateLibraryFolderDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
name	string	Yes	No explicit default	IsString(); Length(1, 255)
parentId	string	No	No explicit default	IsUUID('4', { message: 'parentId must be a valid UUID' }); IsOptional()
scope	FileScopeDto	No	No explicit default	IsEnum(FileScopeDto, { message: 'scope must be private or general' }); IsOptional()

UpdateLibraryFolderDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
name	string	No	No explicit default	IsString(); Length(1, 255); IsOptional()
parentId	string null	No	No explicit default	IsUUID('4', { message: 'parentId must be a valid UUID' }); IsOptional()
scope	FileScopeDto	No	No explicit default	IsEnum(FileScopeDto, { message: 'scope must be private or general' }); IsOptional()

UpdateFileMetadataDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
originalName	string	No	No explicit default	IsString(); Length(1, 255); IsOptional()
folderId	string null	No	No explicit default	IsUUID('4', { message: 'folderId must be a valid UUID' }); IsOptional()
classId	string null	No	No explicit default	IsUUID('4', { message: 'classId must be a valid UUID' }); IsOptional()

Property	Type	Required	Default	Validation and transform decorators
scope	FileScopeDto	No	No explicit default	IsEnum(FileScopeDto, { message: 'scope must be private or general' }); IsOptional()
subjectKey	LibrarySubjectKeyDto	No	No explicit default	IsEnum(LibrarySubjectKeyDto, { message: 'subjectKey must be one of math, science, english, filipino, ap, tle, mapeh, esp', }); IsOptional()
gradeLevel	GradeLevelDto	No	No explicit default	IsEnum(GradeLevelDto, { message: 'gradeLevel must be one of 7, 8, 9, 10', }); IsOptional()
aiEnabled	boolean	No	No explicit default	Transform(({ value }) => to-Boolean(value)); IsBoolean(); IsOptional()
teacherVisible	boolean	No	No explicit default	IsBoolean(); IsOptional()

DTO source: backend/src/modules/ja/dto/ja-practice.dto.ts

JaPracticeRecommendationDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
id	string	Yes	No explicit default	ApiProperty(); IsString(); IsNotEmpty()
title	string	Yes	No explicit default	ApiProperty(); IsString(); IsNotEmpty()
reason	string	Yes	No explicit default	ApiProperty(); IsString(); IsNotEmpty()
focusText	string	Yes	No explicit default	ApiProperty(); IsString(); IsNotEmpty()
lessonId	string null	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsUUID()
assessmentId	string null	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsUUID()
questionId	string null	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsUUID()
sourceChunkId	string null	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsString()

JaPracticeBootstrapQueryDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
classId	string	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsUUID()

JaHubQueryDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
classId	string	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsUUID()

CreateJaPracticeSessionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
classId	string	Yes	No explicit default	ApiProperty(); IsUUID()
recommendation	JaPracticeRecommendationDto	No	No explicit default	ApiPropertyOptional({ type: JaPracticeRecommendationDto }); IsOptional(); ValidateNested(); Type() => JaPracticeRecommendationDto)

SubmitJaPracticeResponseDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
itemId	string	Yes	No explicit default	ApiProperty(); IsUUID()
answer	Record	Yes	No explicit default	ApiProperty(); IsObject()

LogJaPracticeEventDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
eventType	'focus_lost' 'focus_restored' 'focus_strike' 'resumed' 'completed' 'deleted'	Yes	No explicit default	ApiProperty({ enum: ['focus_lost', 'focus_restored', 'focus_strike', 'resumed', 'completed', 'deleted',], }); IsString(); IsIn(['focus_lost', 'focus_restored', 'focus_strike', 'resumed', 'completed', 'deleted',])
payload	Record	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsObject()

CompleteJaPracticeSessionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
metadata	Record	No	No explicit default	ApiPropertyOptional({ description: 'Reserved for replay summary metadata.' }); IsOptional(); IsObject()

JaAskBootstrapQueryDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
classId	string	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsUUID()

CreateJaAskThreadDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
classId	string	Yes	No explicit default	ApiProperty(); IsUUID()
lessonId	string	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsUUID()
title	string	No	No explicit default	

Property	Type	Required	Default	Validation and transform decorators
				ApiPropertyOptional(); IsOptional(); IsString(); IsNotEmpty()

SendJaAskMessageDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
message	string	Yes	No explicit default	ApiProperty(); IsString(); IsNotEmpty()
lessonId	string	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsUUID()
quickAction	string	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsString()

JaReviewBootstrapQueryDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
classId	string	No	No explicit default	ApiPropertyOptional(); IsOptional(); IsUUID()

CreateJaReviewSessionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
classId	string	Yes	No explicit default	ApiProperty(); IsUUID()
attemptId	string	Yes	No explicit default	ApiProperty(); IsUUID()
questionCount	number	No	10	ApiPropertyOptional({ default: 10 }); IsOptional(); IsInt(); Min(5); Max(20)

JaReviewEventDto

Inheritance or composition: `extends LogJaPracticeEventDto`.

Property	Type	Required	Default	Validation and transform decorators

JaReviewSubmitResponseDto

Inheritance or composition: `extends SubmitJaPracticeResponseDto`.

Property	Type	Required	Default	Validation and transform decorators

DTO source: backend/src/modules/lessons/DTO/lesson.dto.ts

BlockOrderItem

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
id	string	Yes	No explicit default	IsUUID()

Property	Type	Required	Default	Validation and transform decorators
order	number	Yes	No explicit default	IsInt()

LessonOrderItem

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
id	string	Yes	No explicit default	IsUUID()
order	number	Yes	No explicit default	IsInt()

CreateLessonDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
title	string	Yes	No explicit default	IsString(); IsNotEmpty()
description	string	No	No explicit default	IsOptional(); IsString()
classId	string	Yes	No explicit default	IsUUID()
order	number	No	No explicit default	IsOptional(); IsInt()

UpdateLessonDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
title	string	No	No explicit default	IsOptional(); IsString(); IsNotEmpty()
description	string	No	No explicit default	IsOptional(); IsString()
order	number	No	No explicit default	IsOptional(); IsInt()
isDraft	boolean	No	No explicit default	IsOptional(); IsBoolean()

CreateContentBlockDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
lessonId	string	No	No explicit default	IsOptional(); IsUUID()
type	ContentBlockType	Yes	No explicit default	IsString(); IsIn(CONTENT_BLOCK_TYPES)
order	number	Yes	No explicit default	IsInt()
content	any	Yes	No explicit default	IsNotEmpty()
metadata	Record	No	No explicit default	IsOptional(); IsObject()

UpdateContentBlockDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
type	ContentBlockType	No	No explicit default	IsOptional(); IsString(); IsIn(CONTENT_BLOCK_TYPES)
order	number	No	No explicit default	IsOptional(); IsInt()
content	any	No	No explicit default	IsOptional()

Property	Type	Required	Default	Validation and transform decorators
metadata	Record	No	No explicit default	IsOptional(); IsObject()

ReorderBlocksDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
blocks	BlockOrderItem[]	Yes	No explicit default	IsArray(); ArrayNotEmpty(); ValidateNested({ each: true }); Type() => BlockOrderItem)

ReorderLessonsDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
lessons	LessonOrderItem[]	Yes	No explicit default	IsArray(); ArrayNotEmpty(); ValidateNested({ each: true }); Type() => LessonOrderItem)

BulkLessonIdsDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
lessonIds	string[]	Yes	No explicit default	IsArray(); ArrayNotEmpty(); IsUUID('4', { each: true })

BulkLessonDraftStateDto

Inheritance or composition: `extends BulkLessonIdsDto`.

Property	Type	Required	Default	Validation and transform decorators
isDraft	boolean	Yes	No explicit default	IsBoolean()

CreateLessonVersionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
label	string	No	No explicit default	IsOptional(); IsString(); MaxLength(80)

DTO source: backend/src/modules/lxp/dto/lxp.dto.ts

LessonAssignmentDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
lessonId	string	Yes	No explicit default	IsUUID('4')
label	string	No	No explicit default	IsOptional(); IsString()
xpAwarded	number	Yes	No explicit default	IsInt(); Min(0); Max(10000)

AssessmentAssignmentDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
assessmentId	string	Yes	No explicit default	IsUUID('4')
label	string	No	No explicit default	IsOptional(); IsString()
xpAwarded	number	Yes	No explicit default	IsInt(); Min(0); Max(10000)

AssignInterventionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
lessonIds	string[]	No	No explicit default	IsOptional(); IsArray(); IsUUID('4', { each: true })
assessmentIds	string[]	No	No explicit default	IsOptional(); IsArray(); IsUUID('4', { each: true })
lessonAssignments	LessonAssignmentDto[]	No	No explicit default	IsOptional(); IsArray(); ValidateNested({ each: true }); Type() => LessonAssignmentDto
assessmentAssignments	AssessmentAssignmentDto[]	No	No explicit default	IsOptional(); IsArray(); ValidateNested({ each: true }); Type() => AssessmentAssignmentDto
note	string	No	No explicit default	IsOptional(); IsString()

ResolveInterventionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
note	string	No	No explicit default	IsOptional(); IsString()

GeneratedLessonDraftDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
title	string	Yes	No explicit default	IsString()
summary	string null	No	No explicit default	IsOptional(); IsString()
lessonBody	string	Yes	No explicit default	IsString()
weakConcepts	string[]	Yes	No explicit default	IsArray(); IsString({ each: true })
sourceLessonIds	string[]	Yes	No explicit default	IsArray(); IsUUID('4', { each: true })
sourceReferences	Record[]	Yes	No explicit default	IsArray()

GuidedQuestionOptionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
id	string	Yes	No explicit default	IsString()
text	string	Yes	No explicit default	IsString()
isCorrect	boolean	Yes	No explicit default	IsBoolean()

GeneratedGuidedQuestionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
id	string	Yes	No explicit default	IsString()
type	string	Yes	No explicit default	IsIn(['multiple_choice', 'multiple_select', 'true_false', 'dropdown'])
stem	string	Yes	No explicit default	IsString()
explanation	string	Yes	No explicit default	IsString()
hint	string null	No	No explicit default	IsOptional(); IsString()
reviewHint	string null	No	No explicit default	IsOptional(); IsString()
weakConceptTag	string null	No	No explicit default	IsOptional(); IsString()
sourceQuestionId	string null	No	No explicit default	IsOptional(); IsString()
options	GuidedQuestionOptionDto[]	Yes	No explicit default	IsArray(); ValidateNested({ each: true }); Type() => GuidedQuestionOptionDto)

GeneratedGuidedAssessmentDraftDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
sourceAssessmentId	string null	No	No explicit default	IsOptional(); IsUUID('4')
title	string	Yes	No explicit default	IsString()
description	string null	No	No explicit default	IsOptional(); IsString()
weakConcepts	string[]	Yes	No explicit default	IsArray(); IsString({ each: true })
formativeSummary	string null	No	No explicit default	IsOptional(); IsString()
sourceReferences	Record[]	Yes	No explicit default	IsArray()
questions	GeneratedGuidedQuestionDto[]	Yes	No explicit default	IsArray(); ValidateNested({ each: true }); Type() => GeneratedGuidedQuestionDto)

ApproveGeneratedArtifactsDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
generatedLessonDraft	GeneratedLessonDraftDto null	No	No explicit default	IsOptional(); ValidateNested(); Type() => GeneratedLessonDraftDto)
generatedGuidedAssessmentDraft	GeneratedGuidedAssessmentDraftDto null	No	No explicit default	IsOptional(); ValidateNested(); Type() => GeneratedGuidedAssessmentDraftDto)

GuidedAssessmentProgressResponseDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
questionId	string	Yes	No explicit default	IsString()
answer	string string[]	No	No explicit default	IsOptional()
isCorrect	boolean	No	No explicit default	IsOptional()

Property	Type	Required	Default	Validation and transform decorators
explanationShown	boolean	No	No explicit default	IsOptional()

UpdateGuidedAssessmentProgressDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
currentQuestionIndex	number	No	No explicit default	IsOptional(); IsInt(); Min(0)
responses	GuidedAssessmentProgress-ResponseDto[]	No	No explicit default	IsOptional(); IsArray(); ValidateNested({ each: true }); Type() => GuidedAssessmentProgressResponseDto)
hintedQuestionIds	string[]	No	No explicit default	IsOptional(); IsArray(); IsString({ each: true })

SubmitGuidedAssessmentDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
responses	GuidedAssessmentProgress-ResponseDto[]	Yes	No explicit default	IsArray(); ValidateNested({ each: true }); Type() => GuidedAssessmentProgressResponseDto)
hintedQuestionIds	string[]	Yes	No explicit default	IsArray(); IsString({ each: true })

AIEvaluationContextDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
sessionType	AISessionType	No	No explicit default	IsOptional(); IsIn(aiSessionTypes)
attemptId	string	No	No explicit default	IsOptional(); IsUUID('4')
questionId	string	No	No explicit default	IsOptional(); IsUUID('4')
classId	string	No	No explicit default	IsOptional(); IsUUID('4')
sourceFlow	string	No	No explicit default	IsOptional(); IsString(); MaxLength(64)

SubmitSystemEvaluationDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
targetModule	SystemEvaluationTarget	Yes	No explicit default	IsIn(systemTargets)
usabilityScore	number	Yes	No explicit default	IsInt(); Min(0); Max(5)
functionalityScore	number	Yes	No explicit default	IsInt(); Min(0); Max(5)
performanceScore	number	Yes	No explicit default	IsInt(); Min(0); Max(5)
satisfactionScore	number	Yes	No explicit default	IsInt(); Min(0); Max(5)
feedback	string	No	No explicit default	IsOptional(); IsString(); MaxLength(2000)
aiContextMetadata	AIEvaluationContextDto	No	No explicit default	IsOptional(); ValidateNested(); Type() => AIEvaluationContextDto)

ListSystemEvaluationsQueryDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
targetModule	SystemEvaluationTarget	No	No explicit default	IsOptional(); IsIn(systemTargets)
aiClassId	string	No	No explicit default	IsOptional(); IsUUID('4')
aiSessionType	AISessionType	No	No explicit default	IsOptional(); IsIn(aiSessionTypes)
aiSourceFlow	string	No	No explicit default	IsOptional(); IsString(); MaxLength(64)
campaignId	string	No	No explicit default	IsOptional(); IsUUID('4')
audienceRole	SystemEvaluationAudienceRole	No	No explicit default	IsOptional(); IsIn(systemEvaluationAudienceRoles)
from	string	No	No explicit default	IsOptional(); IsISO8601()
to	string	No	No explicit default	IsOptional(); IsISO8601()

SubmitAssignedSystemEvaluationDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
questionRatings	Record	Yes	No explicit default	IsObject()
feedback	string	No	No explicit default	IsOptional(); IsString(); MaxLength(2000)

CreateSystemEvaluationCampaignDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
formType	SystemEvaluationFormType	Yes	No explicit default	IsIn(systemEvaluationFormTypes)
audienceRole	SystemEvaluationAudienceRole	Yes	No explicit default	IsIn(systemEvaluationAudienceRoles)
classId	string	No	No explicit default	IsOptional(); IsUUID('4')
title	string	Yes	No explicit default	IsString(); MaxLength(160)
startsAt	string	Yes	No explicit default	IsISO8601()
endsAt	string	Yes	No explicit default	IsISO8601()
status	SystemEvaluationCampaignStatus	No	No explicit default	IsOptional(); IsIn(systemEvaluationCampaignStatuses)

ListSystemEvaluationCampaignsQueryDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
formType	SystemEvaluationFormType	No	No explicit default	IsOptional(); IsIn(systemEvaluationFormTypes)
audienceRole	SystemEvaluationAudienceRole	No	No explicit default	IsOptional(); IsIn(systemEvaluationAudienceRoles)
status	SystemEvaluationCampaignStatus	No	No explicit default	IsOptional(); IsIn(systemEvaluationCampaignStatuses)
classId	string	No	No explicit default	IsOptional(); IsUUID('4')

UpdateSystemEvaluationCampaignStatusDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
status	SystemEvaluationCampaignStatus	Yes	No explicit default	IsIn(systemEvaluationCampaignStatuses)

SubmitTeacherEvaluationDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
classId	string	Yes	No explicit default	IsUUID('4')
gradingPeriod	TeacherEvaluationGradingPeriod	Yes	No explicit default	IsIn(gradingPeriods)
evaluationType	TeacherEvaluationType	Yes	No explicit default	IsIn(teacherEvaluationTypes)
ratings	Record	Yes	No explicit default	IsObject()
comment	string	No	No explicit default	IsOptional(); IsString(); MaxLength(2000)

ListTeacherEvaluationSummaryQueryDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
evaluationType	TeacherEvaluationType	Yes	No explicit default	IsIn(teacherEvaluationTypes)
classId	string	No	No explicit default	IsOptional(); IsUUID('4')
gradingPeriod	TeacherEvaluationGradingPeriod	No	No explicit default	IsOptional(); IsIn(gradingPeriods)

DTO source: backend/src/modules/notifications/DTO/query-notifications.dto.ts

QueryNotificationsDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
page	number	No	1	ApiPropertyOptional({ default: 1 }); IsOptional(); Type() => Number; IsInt(); Min(1)
limit	number	No	20	ApiPropertyOptional({ default: 20 }); IsOptional(); Type() => Number; IsInt(); Min(1); Max(100)
isRead	boolean	No	No explicit default	ApiPropertyOptional({ description: 'Filter notifications by read status', type: Boolean, }); IsOptional(); Transform(({ value }) => { if (value === undefined value === null value === '') return undefined; if (typeof value === 'boolean') return value; if (value === 'true') return true; if (value === 'false') return false; return value; }); IsBoolean()

DTO source: backend/src/modules/otp/dto/resend-otp.dto.ts**ResendOtpDto**

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
email	string	Yes	No explicit default	IsEmail({}, { message: 'Invalid email address' }); IsNotEmpty()

DTO source: backend/src/modules/otp/dto/verify-otp.dto.ts**VerifyOtpDto**

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
email	string	Yes	No explicit default	IsEmail(); Transform(({ value }) => value.toLowerCase().trim())
code	string	Yes	No explicit default	IsNumberString({}, { message: 'OTP must contain only numbers' }); Length(6, 6, { message: 'OTP must be exactly 6 digits' })

DTO source: backend/src/modules/performance/DTO/create-performance-analysis-job.dto.ts**CreatePerformanceAnalysisJobDto**

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
studentId	string	No	No explicit default	IsOptional(); IsUUID('4')
note	string	No	No explicit default	IsOptional(); IsString()

DTO source: backend/src/modules/performance/DTO/query-performance-logs.dto.ts**QueryPerformanceLogsDto**

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
studentId	string	No	No explicit default	IsOptional(); IsUUID('4', { message: 'studentId must be a valid UUID' })
limit	number	No	50	IsOptional(); Type(() => Number); IsInt({ message: 'limit must be an integer' }); Min(1, { message: 'limit must be at least 1' }); Max(200, { message: 'limit cannot exceed 200' })

DTO source: backend/src/modules/profiles/DTO/update-profile.dto.ts**UpdateProfileDto**

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
firstName	string	No	No explicit default	IsOptional(); IsString(); MaxLength(30); Matches(PERSON_NAME_REGEX, { message: 'First name may only contain letters, spaces, hyphens, and apostrophes', }); Transform(({ value }: { value: string }) => value?.trim())
middleName	string	No	No explicit default	IsOptional(); IsString(); MaxLength(30); Matches(PERSON_NAME_REGEX, { message: 'Middle name may only contain letters, spaces, hyphens, and apostrophes', }); Transform(({ value }: { value: string }) => trimValue(value))
lastName	string	No	No explicit default	IsOptional(); IsString(); MaxLength(30); Matches(PERSON_NAME_REGEX, { message: 'Last name may only contain letters, spaces, hyphens, and apostrophes', }); Transform(({ value }: { value: string }) => value?.trim())
lrn	string	No	No explicit default	IsOptional(); IsString(); Matches(/^[0-9]{12}\$/, { message: 'LRN must be exactly 12 digits (e.g., 202401230001)', })
dob	string	No	No explicit default	IsOptional(); IsDateString({}, { message: 'Date of birth must be a valid ISO date' })
dateOfBirth	string	No	No explicit default	IsOptional(); IsDateString({}, { message: 'Date of birth must be a valid ISO date' })
gender	'Male' 'Female'	No	No explicit default	IsOptional(); IsIn(['Male', 'Female'], { message: 'Gender must be either Male or Female', })
phone	string	No	No explicit default	IsOptional(); IsString(); Matches(PH_MOBILE_REGEX, { message: 'Student contact number must be a valid PH mobile format (e.g., 09171234567 or +639171234567)', })
address	string	No	No explicit default	IsOptional(); IsString(); MaxLength(180); Matches(ADDRESS_REGEX, { message: 'Address may only contain letters, numbers, spaces, commas, periods, number signs, apostrophes, hyphens, and slashes', }); Transform(({ value }: { value?: string }) => trimValue(value))
familyName	string	No	No explicit default	IsOptional(); IsString(); MaxLength(80); Matches(PERSON_NAME_REGEX, { message: 'Guardian name may only contain letters, spaces, hyphens, and apostrophes', }); Transform(({ value }: { value?: string }) => trimValue(value))

Property	Type	Required	Default	Validation and transform decorators
familyRelationship	string	No	No explicit default	IsOptional(); IsIn(['Father', 'Mother', 'Guardian', 'Sibling', 'Other'], { message: 'Relationship must be one of: Father, Mother, Guardian, Sibling, Other', })
familyContact	string	No	No explicit default	IsOptional(); IsString(); Matches(PH_MOBILE_REGEX, { message: 'Guardian contact number must be a valid PH mobile format (e.g., 09171234567 or +639171234567)', })
gradeLevel	'7' '8' '9' '10'	No	No explicit default	IsOptional(); IsIn(['7', '8', '9', '10'], { message: 'Grade level must be one of: 7, 8, 9, 10', })
profilePicture	string	No	No explicit default	IsOptional(); IsString()

DTO source: backend/src/modules/roster-import/dto/roster-import.dto.ts

ParsedNameDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
lastName	string	Yes	No explicit default	IsString(); IsNotEmpty()
firstName	string	Yes	No explicit default	IsString(); IsNotEmpty()
middleName	string	No	No explicit default	IsOptional(); IsString()

SectionMatchDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
fileHeader	string	Yes	No explicit default	No property decorator extracted
foundSection	{ id: string; name: string; gradeLevel: string; }	Yes	No explicit default	No property decorator extracted

PreviewStudentRowDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
rowNumber	number	Yes	No explicit default	No property decorator extracted
name	ParsedNameDto	Yes	No explicit default	No property decorator extracted
lrn	string	Yes	No explicit default	No property decorator extracted
email	string	Yes	No explicit default	No property decorator extracted
userId	string	Yes	No explicit default	No property decorator extracted
alreadyEnrolled	boolean	Yes	No explicit default	No property decorator extracted

PreviewPendingRowDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
rowNumber	number	Yes	No explicit default	No property decorator extracted
name	ParsedNameDto	Yes	No explicit default	No property decorator extracted
lrm	string	Yes	No explicit default	No property decorator extracted
email	string	Yes	No explicit default	No property decorator extracted

RowErrorDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
rowNumber	number	Yes	No explicit default	No property decorator extracted
rawData	string[]	Yes	No explicit default	No property decorator extracted
issues	string[]	Yes	No explicit default	No property decorator extracted

PreviewSummaryDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
totalDataRows	number	Yes	No explicit default	No property decorator extracted
validRows	number	Yes	No explicit default	No property decorator extracted
registeredCount	number	Yes	No explicit default	No property decorator extracted
alreadyEnrolledCount	number	Yes	No explicit default	No property decorator extracted
pendingCount	number	Yes	No explicit default	No property decorator extracted
errorCount	number	Yes	No explicit default	No property decorator extracted

RosterImportPreviewResponseDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
sectionMatch	SectionMatchDto	Yes	No explicit default	No property decorator extracted
registered	PreviewStudentRowDto[]	Yes	No explicit default	No property decorator extracted
pending	PreviewPendingRowDto[]	Yes	No explicit default	No property decorator extracted
errors	RowErrorDto[]	Yes	No explicit default	No property decorator extracted
summary	PreviewSummaryDto	Yes	No explicit default	No property decorator extracted

CommitStudentRowDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
userId	string	Yes	No explicit default	IsUUID('4')
name	ParsedNameDto	Yes	No explicit default	ValidateNested(); Type(() => ParsedNameDto)
lrm	string	Yes	No explicit default	IsString(); Matches(/^\d{12}\$/, { message: 'LRN must be exactly 12 numeric digits' })
email	string	Yes	No explicit default	IsEmail()

CommitPendingRowDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
name	ParsedNameDto	Yes	No explicit default	ValidateNested(); Type(() => ParsedNameDto)
lrm	string	Yes	No explicit default	IsString(); Matches(/^\d{12}\$/, { message: 'LRN must be exactly 12 numeric digits' })
email	string	Yes	No explicit default	IsEmail()

RosterImportCommitDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
sectionId	string	Yes	No explicit default	IsUUID('4')
enrolledRows	CommitStudentRowDto[]	Yes	No explicit default	IsArray(); ValidateNested({ each: true }); Type(() => CommitStudentRowDto)
pendingRows	CommitPendingRowDto[]	Yes	No explicit default	IsArray(); ValidateNested({ each: true }); Type(() => CommitPendingRowDto)

RosterImportCommitResponseDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
enrolledUserIds	string[]	Yes	No explicit default	No property decorator extracted
pendingRosterIds	string[]	Yes	No explicit default	No property decorator extracted
alreadyEnrolledSkipped	number	Yes	No explicit default	No property decorator extracted
summary	{ enrolled: number; pending: number; total: number; }	Yes	No explicit default	No property decorator extracted

PendingRosterRowDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
id	string	Yes	No explicit default	

Property	Type	Required	Default	Validation and transform decorators
				No property decorator extracted
sectionId	string	Yes	No explicit default	No property decorator extracted
lastName	string	Yes	No explicit default	No property decorator extracted
firstName	string	Yes	No explicit default	No property decorator extracted
middleInitial	string null	Yes	No explicit default	No property decorator extracted
lrm	string	Yes	No explicit default	No property decorator extracted
rosterEmail	string	Yes	No explicit default	No property decorator extracted
resolvedAt	Date null	Yes	No explicit default	No property decorator extracted
resolvedUserId	string null	Yes	No explicit default	No property decorator extracted
importedAt	Date	Yes	No explicit default	No property decorator extracted

ResolvePendingRowDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
resolvedUserId	string	No	No explicit default	IsOptional(); IsUUID('4')

DTO source: backend/src/modules/school-events/DTO/create-school-event.dto.ts

CreateSchoolEventDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
eventType	'school_event' 'holiday_break'	Yes	No explicit default	ApiProperty({ enum: ['school_event', 'holiday_break'] }); IsIn(['school_event', 'holiday_break'], { message: 'eventType must be school_event or holiday_break', })
schoolYear	string	Yes	No explicit default	ApiProperty({ example: '2026-2027' }); IsString(); Validate(IsValidSchoolYearConstraint)
title	string	Yes	No explicit default	ApiProperty({ example: 'Foundation Day Program' }); IsString(); MinLength(1); MaxLength(180)
description	string	No	No explicit default	ApiPropertyOptional({ example: 'School-wide activity at the quadrangle.' }); IsOptional(); IsString(); MaxLength(2000)
location	string	No	No explicit default	ApiPropertyOptional({ example: 'Main Campus Quadrangle' }); Validate((payload: CreateSchoolEventDto) => payload.eventType === 'school_event' pay-

Property	Type	Required	Default	Validation and transform decorators
				load.location !== undefined,); IsString(); Transform(({ value }: { value?: string }) => value?.trim()); MinLength(1, { message: 'Location is required for school events' }); MaxLength(180)
startsAt	string	Yes	No explicit default	ApiProperty({ example: '2026-10-15T00:00:00.000Z' }); IsISO8601()
endsAt	string	Yes	No explicit default	ApiProperty({ example: '2026-10-15T23:59:59.999Z' }); IsISO8601()
allDay	boolean	No	No explicit default	ApiPropertyOptional({ default: true }); IsOptional(); IsBoolean()

DTO source: backend/src/modules/school-events/DTO/query-school-events.dto.ts

QuerySchoolEventsDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
schoolYear	string	No	No explicit default	ApiPropertyOptional({ example: '2026-2027' }); IsOptional(); IsString(); Validate(IsValidSchoolYearConstraint)
from	string	No	No explicit default	ApiPropertyOptional({ description: 'Inclusive lower bound by event range', example: '2026-06-01T00:00:00.000Z' }); IsOptional(); IsISO8601()
to	string	No	No explicit default	ApiPropertyOptional({ description: 'Inclusive upper bound by event range', example: '2027-03-31T23:59:59.999Z' }); IsOptional(); IsISO8601()

DTO source: backend/src/modules/school-events/DTO/update-school-event.dto.ts

UpdateSchoolEventDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
eventType	'school_event' 'holiday_break'	No	No explicit default	ApiPropertyOptional({ enum: ['school_event', 'holiday_break'] }); IsOptional(); IsIn(['school_event', 'holiday_break'], { message: 'eventType must be school_event or holiday_break', })
schoolYear	string	No	No explicit default	ApiPropertyOptional({ example: '2026-2027' }); IsOptional(); IsString(); Validate(IsValidSchoolYearConstraint)
title	string	No	No explicit default	ApiPropertyOptional({ example: 'Foundation Day Program' }); IsOptional();

Property	Type	Required	Default	Validation and transform decorators
				IsString(); MinLength(1); MaxLength(180)
description	string	No	No explicit default	ApiPropertyOptional({ example: 'School-wide activity at the quadrangle.' }); IsOptional(); IsString(); MaxLength(2000)
location	string	No	No explicit default	ApiPropertyOptional({ example: 'Main Campus Quadrangle' }); ValidateSelf((payload: UpdateSchoolEventDto) => payload.eventType === 'school_event' payload.location !== undefined,); IsString(); Transform(({ value }: { value?: string }) => value?.trim()); MinLength(1, { message: 'Location is required for school events' }); MaxLength(180)
startsAt	string	No	No explicit default	ApiPropertyOptional({ example: '2026-10-15T00:00:00.000Z' }); IsOptional(); IsISO8601()
endsAt	string	No	No explicit default	ApiPropertyOptional({ example: '2026-10-15T23:59:59.999Z' }); IsOptional(); IsISO8601()
allDay	boolean	No	No explicit default	ApiPropertyOptional({ default: true }); IsOptional(); IsBoolean()

DTO source: backend/src/modules/sections/DTO/access-students-overview.dto.ts

AccessStudentsOverviewQueryDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
schoolYear	string	No	No explicit default	IsOptional(); IsString(); Matches(SCHOOL_YEAR_PATTERN, { message: 'schoolYear must be in YYYY-YYYY format', });
gradeLevel	string	No	No explicit default	IsOptional(); IsString(); IsIn(['7', '8', '9', '10'], { message: 'gradeLevel must be one of: 7, 8, 9, 10', });
sectionId	string	No	No explicit default	IsOptional(); IsUUID('4', { message: 'sectionId must be a valid UUID' });
search	string	No	No explicit default	IsOptional(); IsString()

DTO source: backend/src/modules/sections/DTO/access-students-target-sections.dto.ts

AccessStudentsTargetSectionsQueryDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
fromSectionId	string	Yes	No explicit default	

Property	Type	Required	Default	Validation and transform decorators
				IsUUID('4', { message: 'from-SectionId must be a valid UUID' })
mode	'promote' 'retain'	Yes	No explicit default	IsString(); IsIn(['promote', 'retain'], { message: 'mode must be either promote or retain', })
schoolYear	string	No	No explicit default	IsOptional(); IsString(); Matches(SCHOOL_YEAR_PATTERN, { message: 'schoolYear must be in YYYY-YYYY format', })

DTO source: backend/src/modules/sections/DTO/bulk-section-lifecycle.dto.ts

BulkSectionLifecycleDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
action	BulkSectionLifecycleAction	Yes	No explicit default	IsIn(BULK_SECTION_LIFECYCLE_ACTIONS)
sectionIds	string[]	Yes	No explicit default	IsArray(); ArrayMinSize(1); ArrayUnique(); IsString({ each: true })

DTO source: backend/src/modules/sections/DTO/bulk-students.dto.ts

BulkStudentsDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
studentIds	string[]	Yes	No explicit default	IsArray({ message: 'studentIds must be an array' }); ArrayMinSize(1, { message: 'At least one student ID is required' }); ArrayMaxSize(50, { message: 'Cannot add more than 50 students at once' }); IsUUID('4', { each: true, message: 'Each student ID must be a valid UUID' })

DTO source: backend/src/modules/sections/DTO/create-section.dto.ts

CreateSectionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
name	string	Yes	No explicit default	IsString(); IsNotEmpty({ message: 'Section name is required' }); MinLength(1, { message: 'Section name must be at least 1 character' }); MaxLength(100, { message: 'Section name must not exceed 100 characters' }); Transform(({ value }) => trimValue(value)); Matches(SECTION_NAME_REGEX, { message: 'Section name may only contain let-'

Property	Type	Required	Default	Validation and transform decorators
				ters, numbers, spaces, hyphens, and apostrophes', })
gradeLevel	string	Yes	No explicit default	IsString(); IsNotEmpty({ message: 'Grade level is required' }); IsIn([VALID_GRADE_LEVELS, { message: 'gradeLevel must be one of: 7, 8, 9, 10', }]); Transform(({ value }) => value?.trim())
schoolYear	string	Yes	No explicit default	IsString(); IsNotEmpty({ message: 'School year is required' }); Validate(IsValidSchoolYearConstraint); Transform(({ value }) => value?.trim())
capacity	number	Yes	No explicit default	IsInt({ message: 'Capacity must be an integer' }); Min(1, { message: 'Capacity must be at least 1' })
roomNumber	string	No	No explicit default	IsString(); IsOptional(); Transform(({ value }) => trimValue(value)); IsIn([spread operatorALLOWED_ROOM_NUMBERS], { message: ALLOWED_ROOM_NUMBERS_MESSAGE, })
adviserId	string	No	No explicit default	IsUUID('4', { message: 'Adviser ID must be a valid UUID' }); IsOptional()

DTO source: backend/src/modules/sections/DTO/fail-students.dto.ts

FailStudentsDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
fromSectionId	string	Yes	No explicit default	IsUUID('4', { message: 'fromSectionId must be a valid UUID' })
targetSectionId	string	Yes	No explicit default	IsUUID('4', { message: 'targetSectionId must be a valid UUID' })
studentIds	string[]	Yes	No explicit default	IsArray({ message: 'studentIds must be an array' }); ArrayMinSize(1, { message: 'At least one student ID is required' }); ArrayMaxSize(200, { message: 'Cannot process more than 200 students at once', }); ArrayUnique({ message: 'studentIds must not contain duplicates' }); IsUUID('4', { each: true, message: 'Each student ID must be a valid UUID' })

DTO source: backend/src/modules/sections/DTO/finalize-access-student-grades.dto.ts

FinalizeAccessStudentGradesDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
sectionId	string	Yes	No explicit default	IsUUID('4', { message: 'sectionId must be a valid UUID' })
studentIds	string[]	No	No explicit default	IsOptional(); IsArray(); ArrayNotEmpty({ message: 'studentIds must contain at least one student when provided', }); IsUUID('4', { each: true, message: 'Each studentId must be a valid UUID' })

DTO source: backend/src/modules/sections/DTO/move-up-students.dto.ts

MoveUpStudentsDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
fromSectionId	string	Yes	No explicit default	IsUUID('4', { message: 'fromSectionId must be a valid UUID' })
targetSectionId	string	Yes	No explicit default	IsUUID('4', { message: 'targetSectionId must be a valid UUID' })
studentIds	string[]	Yes	No explicit default	IsArray({ message: 'studentIds must be an array' }); ArrayMinSize(1, { message: 'At least one student ID is required' }); ArrayMaxSize(200, { message: 'Cannot move more than 200 students at once' }); ArrayUnique({ message: 'studentIds must not contain duplicates' }); IsUUID('4', { each: true, message: 'Each student ID must be a valid UUID' })
allowFailingPromotion	boolean	No	No explicit default	IsOptional(); IsBoolean()

DTO source: backend/src/modules/sections/DTO/update-section-presentation.dto.ts

UpdateSectionPresentationDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
cardBannerUrl	string null	No	No explicit default	IsOptional(); IsString({ message: 'cardBannerUrl must be a string' })

DTO source: backend/src/modules/sections/DTO/update-section.dto.ts

UpdateSectionDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
name	string	No	No explicit default	IsString(); IsOptional(); MinLength(1, { message: 'Section name must be at least 1 character' }); MaxLength(100, { message:

Property	Type	Required	Default	Validation and transform decorators
				'Section name must not exceed 100 characters' }); Transform(({ value }) => trimValue(value)); Matches(SECTION_NAME_REGEX, { message: 'Section name may only contain letters, numbers, spaces, hyphens, and apostrophes', })
gradeLevel	string	No	No explicit default	IsString(); IsOptional(); IsIn([VALID_GRADE_LEVELS, { message: 'gradeLevel must be one of: 7, 8, 9, 10', }]); Transform(({ value }) => value?.trim())
schoolYear	string	No	No explicit default	IsString(); IsOptional(); Validate(IsValidSchoolYearConstraint); Transform(({ value }) => value?.trim())
capacity	number	No	No explicit default	IsInt({ message: 'Capacity must be an integer' }); IsOptional(); Min(1, { message: 'Capacity must be at least 1' })
roomNumber	string	No	No explicit default	IsString(); IsOptional(); Transform(({ value }) => trimValue(value)); IsIn([spread operatorALLOWED_ROOM_NUMBERS], { message: ALLOWED_ROOM_NUMBERS_MESSAGE, })
adviserId	string null	No	No explicit default	IsOptional(); ValidateIf((o) => o.adviserId !== null); IsUUID('4', { message: 'Adviser ID must be a valid UUID' })
isActive	boolean	No	No explicit default	IsBoolean(); IsOptional()

DTO source: backend/src/modules/teacher-profiles/DTO/update-teacher-profile.dto.ts

UpdateTeacherProfileDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
dateOfBirth	string	No	No explicit default	IsOptional(); IsDateString({ }, { message: 'Date of birth must be a valid ISO date' })
dob	string	No	No explicit default	IsOptional(); IsDateString({ }, { message: 'Date of birth must be a valid ISO date' })
gender	'Male' 'Female'	No	No explicit default	IsOptional(); IsIn(['Male', 'Female'], { message: 'Gender must be either Male or Female', })
address	string	No	No explicit default	IsOptional(); IsString(); MaxLength(180); Matches(ADDRESS_REGEX, { message: 'Address may only contain letters, numbers, spaces, commas, periods, number signs, apostrophes, hyphens, and slashes', }); Transform(({ value }: { value?: string }) => trimValue(value))
department	string	No	No explicit default	IsOptional(); IsString(); MaxLength(80); Matches(LABEL_TEXT_REGEX,

Property	Type	Required	Default	Validation and transform decorators
				{ message: 'Department may only contain letters, numbers, spaces, periods, apostrophes, hyphens, and slashes', }); Transform(({ value }: { value?: string }) => trim-Value(value))
specialization	string	No	No explicit default	IsOptional(); IsString(); MaxLength(80); Matches(LABEL_TEXT_REGEX, { message: 'Specialization may only contain letters, numbers, spaces, periods, apostrophes, hyphens, and slashes', }); Transform(({ value }: { value?: string }) => trim-Value(value))
profilePicture	string	No	No explicit default	IsOptional(); IsString(); Transform(({ value }: { value?: string }) => value?.trim())
contactNumber	string	No	No explicit default	IsOptional(); IsString(); Transform(({ value }: { value?: string }) => trim-Value(value)); Matches(PH_MOBILE_REGEX, { message: 'Contact number must be a valid PH mobile format (e.g., 09171234567 or +639171234567)', })
phone	string	No	No explicit default	IsOptional(); IsString(); Transform(({ value }: { value?: string }) => trim-Value(value)); Matches(PH_MOBILE_REGEX, { message: 'Phone number must be a valid PH mobile format (e.g., 09171234567 or +639171234567)', })
employeeId	string	No	No explicit default	IsOptional(); IsString(); Transform(({ value }: { value?: string }) => upperTrimmed-Value(value)); Matches(EMPLOYEE_ID_REGEX, { message: 'Employee ID must be 1-20 characters using letters, numbers, or hyphens', })

DTO source: backend/src/modules/users/DTO/bulk-user-lifecycle.dto.ts

BulkUserLifecycleDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
action	BulkUserLifecycleAction	Yes	No explicit default	IsIn(BULK_USER_LIFECYCLE_ACTIONS)
userIds	string[]	Yes	No explicit default	IsArray(); ArrayMinSize(1); ArrayUnique(); IsString({ each: true })

DTO source: backend/src/modules/users/DTO/create-user.dto.ts

CreateUserDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
email	string	Yes	No explicit default	IsEmail({}, { message: 'Must be a valid email address' }); IsPopularEmailProvider(); Transform(({ value }: { value: string }) => lowerTrimmedValue(value))
password	string	No	No explicit default	IsOptional(); IsString({ message: 'Password must be a string' }); MinLength(8, { message: 'Password must be at least 8 characters' }); Matches(/[A-Z]/, { message: 'Password must contain at least one uppercase letter' }); Matches(/[a-z]/, { message: 'Password must contain at least one lowercase letter' }); Matches(/\d/, { message: 'Password must contain at least one number' }); Matches(/[@\$!%*?&#]/, { message: 'Password must contain at least one special character' }); Matches(PASSWORD_SAFE_REGEX, { message: 'Password contains unsupported control characters' });
firstName	string	Yes	No explicit default	IsString(); Matches(PERSON_NAME_REGEX, { message: 'First name may only contain letters, spaces, hyphens, and apostrophes' }); Transform(({ value }: { value: string }) => trimValue(value))
middleName	string	No	No explicit default	IsOptional(); IsString(); Matches(PERSON_NAME_REGEX, { message: 'Middle name may only contain letters, spaces, hyphens, and apostrophes' }); Transform(({ value }: { value?: string }) => trimValue(value))
lastName	string	Yes	No explicit default	IsString(); Matches(PERSON_NAME_REGEX, { message: 'Last name may only contain letters, spaces, hyphens, and apostrophes' }); Transform(({ value }: { value: string }) => trimValue(value))
role	string	Yes	No explicit default	IsIn(['student', 'teacher', 'admin'], { message: 'Role must be student, teacher, or admin' });
employeeId	string	No	No explicit default	ValidateIf((o: { role: string }) => o.role === 'teacher'); IsString({ message: 'Employee ID is required for teacher accounts' }); Transform(({ value }: { value?: string }) => upperTrimmedValue(value)); Matches(EMPLOYEE_ID_REGEX, { message: 'Employee ID must be 1-20 characters using letters, numbers, or hyphens' });
contactNumber	string	No	No explicit default	ValidateIf((o: { role: string }) => o.role === 'teacher'); IsString({ message: 'Contact number is required for teacher accounts' }); Transform(({ value }: { value?: string }) => trimValue(value)); Matches(PH_MOBILE_REGEX, { message: 'Contact number' });

Property	Type	Required	Default	Validation and transform decorators
				must be a valid PH mobile format (e.g., 09171234567 or +639171234567)', })
lrm	string	No	No explicit default	ValidateIf((o: { role: string }) => o.role === 'student'); IsString({ message: 'LRN must be a string' }); Matches(/^[0-9]{12}\$/, { message: 'LRN must be exactly 12 digits (e.g., 202401230001)', })

DTO source: backend/src/modules/users/DTO/update-user.dto.ts

UpdateUserDto

No DTO inheritance clause is declared.

Property	Type	Required	Default	Validation and transform decorators
email	string	No	No explicit default	IsOptional(); IsEmail({}, { message: 'Must be a valid email address' }); Transform(({ value }: { value: string }) => lowerTrimmedValue(value))
password	string	No	No explicit default	IsOptional(); IsString({ message: 'Password must be a string' }); MinLength(8, { message: 'Password must be at least 8 characters' }); Matches(/[A-Z]/, { message: 'Password must contain at least one uppercase letter' }); Matches(/[a-z]/, { message: 'Password must contain at least one lowercase letter' }); Matches(/\d/, { message: 'Password must contain at least one number' }); Matches(/[!@#\$%&*?&#]/, { message: 'Password must contain at least one special character' }); Matches(PASSWORD_SAFE_REGEX, { message: 'Password contains unsupported control characters' }, })
firstName	string	No	No explicit default	IsOptional(); IsString(); MaxLength(30); Matches(PERSON_NAME_REGEX, { message: 'First name may only contain letters, spaces, hyphens, and apostrophes' }); Transform(({ value }: { value: string }) => value?.trim())
middleName	string	No	No explicit default	IsOptional(); IsString(); MaxLength(30); Matches(PERSON_NAME_REGEX, { message: 'Middle name may only contain letters, spaces, hyphens, and apostrophes' }); Transform(({ value }: { value: string }) => value?.trim())
lastName	string	No	No explicit default	IsOptional(); IsString(); MaxLength(30); Matches(PERSON_NAME_REGEX, { message: 'Last name may only contain letters, spaces, hyphens, and apostrophes' }); Transform(({ value }: { value: string }) => value?.trim())
role	string	No	No explicit default	IsOptional(); IsIn(['student', 'teacher', 'admin'], { mes-

Property	Type	Required	Default	Validation and transform decorators
				sage: 'Role must be student, teacher, or admin', })
employeeId	string	No	No explicit default	IsOptional(); IsString({ message: 'Employee ID must be a string' }); Transform(({ value }: { value?: string }) => upperTrimmedValue(value)); Matches(EMPLOYEE_ID_REGEX, { message: 'Employee ID must be 1-20 characters using letters, numbers, or hyphens', })
contactNumber	string	No	No explicit default	IsOptional(); IsString({ message: 'Contact number must be a string' }); Transform(({ value }: { value?: string }) => trimmedValue(value)); Matches(PH_MOBILE_REGEX, { message: 'Contact number must be a valid PH mobile format (e.g., 09171234567 or +639171234567)', })
lrm	string	No	No explicit default	IsOptional(); IsString({ message: 'LRN must be a string' }); Transform(({ value }: { value?: string }) => trimmedValue(value)); Matches(/^[0-9]{12}\$/, { message: 'LRN must be exactly 12 digits (e.g., 202401230001)', })
dob	string	No	No explicit default	IsOptional(); IsDateString({}, { message: 'Date of birth must be a valid ISO date' })
dateOfBirth	string	No	No explicit default	IsOptional(); IsDateString({}, { message: 'Date of birth must be a valid ISO date' })
gender	'Male' 'Female'	No	No explicit default	IsOptional(); IsIn(['Male', 'Female'], { message: 'Gender must be either Male or Female', })
phone	string	No	No explicit default	IsOptional(); IsString(); Matches(PH_MOBILE_REGEX, { message: 'Student contact number must be a valid PH mobile format (e.g., 09171234567 or +639171234567)', })
address	string	No	No explicit default	IsOptional(); IsString(); MaxLength(180); Matches(ADDRESS_REGEX, { message: 'Address may only contain letters, numbers, spaces, commas, periods, number signs, apostrophes, hyphens, and slashes', }); Transform(({ value }: { value?: string }) => trimmedValue(value))
familyName	string	No	No explicit default	IsOptional(); IsString(); MaxLength(80); Matches(PERSON_NAME_REGEX, { message: 'Guardian name may only contain letters, spaces, hyphens, and apostrophes', }); Transform(({ value }: { value?: string }) => trimmedValue(value))
familyRelationship	string	No	No explicit default	IsOptional(); IsIn(['Father', 'Mother', 'Guardian', 'Sibling', 'Other'], { message: 'Relationship must be one of: Father, Mother, Guardian, Sibling, Other', })
familyContact	string	No	No explicit default	

Property	Type	Required	Default	Validation and transform decorators
				IsOptional(); IsString(); Matches(PH_MOBILE_REGEX, { message: 'Guardian contact number must be a valid PH mobile format (e.g., 09171234567 or +639171234567)', });
gradeLevel	'7' '8' '9' '10'	No	No explicit default	IsOptional(); IsIn(['7', '8', '9', '10'], { message: 'Grade level must be one of: 7, 8, 9, 10', });
profilePicture	string	No	No explicit default	IsOptional(); IsString()

Interface contracts

Interface	Definition	Source
BulkClassLifecycleFailure	export interface BulkClassLifecycleFailure	backend/src/modules/classes/DTO/bulk-class-lifecycle.dto.ts
BulkClassLifecycleResult	export interface BulkClassLifecycleResult { message: string; data: { action: BulkClassLifecycleAction; requested: number; succeeded: string[]; failed: BulkClassLifecycleFailure[]; }; }	backend/src/modules/classes/DTO/bulk-class-lifecycle.dto.ts
ReportQuery	export interface ReportQuery	backend/src/modules/reports/dto/report-query.dto.ts
BulkSectionLifecycleFailure	export interface BulkSectionLifecycleFailure	backend/src/modules/sections/DTO/bulk-section-lifecycle.dto.ts
BulkSectionLifecycleResult	export interface BulkSectionLifecycleResult { message: string; data: { action: BulkSectionLifecycleAction; requested: number; succeeded: string[]; failed: BulkSectionLifecycleFailure[]; }; }	backend/src/modules/sections/DTO/bulk-section-lifecycle.dto.ts
BulkLifecycleFailure	export interface BulkLifecycleFailure	backend/src/modules/users/DTO/bulk-user-lifecycle.dto.ts
BulkUserLifecycleResult	export interface BulkUserLifecycleResult { message: string; data: { action: BulkUserLifecycleAction; requested: number; succeeded: string[]; failed: BulkLifecycleFailure[]; }; }	backend/src/modules/users/DTO/bulk-user-lifecycle.dto.ts

Type-alias contracts

Type alias	Definition	Source
AiPolicySourceScope	(typeof aiPolicySourceScopeEnum.enumValues)[number]	backend/src/modules/ai-mentor/DTO/class-ai-policy.dto.ts
BulkClassLifecycleAction	(typeof BULK_CLASS_LIFECYCLE_ACTIONS)[number]	backend/src/modules/classes/DTO/bulk-class-lifecycle.dto.ts
DayAbbreviation	(typeof VALID_DAYS)[number]	backend/src/modules/classes/DTO/schedule-slot.dto.ts
StudentPresentationMode	(typeof STUDENT_PRESENTATION_MODES)[number]	backend/src/modules/classes/DTO/update-student-class-presentation.dto.ts
StudentCourseViewMode	(typeof STUDENT_COURSE_VIEW_MODES)[number]	backend/src/modules/classes/DTO/update-student-course-view.dto.ts
DiscussionThemeld	(typeof DISCUSSION_THEME_IDS)[number]	backend/src/modules/discussion-board/DTO/discussion-thread.dto.ts
ContentBlockType	(typeof CONTENT_BLOCK_TYPES)[number]	backend/src/modules/lessons/DTO/lesson.dto.ts
SystemEvaluationTarget	(typeof systemTargets)[number]	backend/src/modules/lxp/dto/lxp.dto.ts
SystemEvaluationFormType	(typeof systemEvaluationFormTypes)[number]	backend/src/modules/lxp/dto/lxp.dto.ts
SystemEvaluationAudienceRole	(typeof systemEvaluationAudienceRoles)[number]	backend/src/modules/lxp/dto/lxp.dto.ts
SystemEvaluationCampaignStatus	(typeof systemEvaluationCampaignStatuses)[number]	backend/src/modules/lxp/dto/lxp.dto.ts
TeacherEvaluationType	(typeof teacherEvaluationTypes)[number]	backend/src/modules/lxp/dto/lxp.dto.ts
TeacherEvaluationGradingPeriod	(typeof gradingPeriods)[number]	backend/src/modules/lxp/dto/lxp.dto.ts

Type alias	Definition	Source
aiSessionType	(typeof aiSessionTypes)[number]	backend/src/modules/lxp/dto/lxp.dto.ts
ReportExportFormat	'csv'	backend/src/modules/reports/dto/report-query.dto.ts
BulkSectionLifecycleAction	(typeof BULK_SECTION_LIFECYCLE_ACTIONS)[number]	backend/src/modules/sections/DTO/bulk-section-lifecycle.dto.ts
BulkUserLifecycleAction	(typeof BULK_USER_LIFECYCLE_ACTIONS)[number]	backend/src/modules/users/DTO/bulk-user-lifecycle.dto.ts

Backend extension procedure

1. Select the owning module and confirm whether the change crosses a public API, database, queue, AI, web, or mobile contract.
2. Define or update the DTO first, including validation, transformation, Swagger metadata, and negative tests.
3. Add the controller route with explicit access metadata and delegate business decisions to the service.
4. Enforce object ownership, class membership, publication, lifecycle, and audit rules in the service.
5. Use a transaction when multiple authoritative writes must succeed or fail together.
6. If work is long-running, persist intent before enqueueing and make the processor idempotent.
7. Update all typed consumers and this catalog in the same change.
8. Run the focused unit and integration tests, migration checks when applicable, backend typecheck or build, and the smallest cross-client smoke that proves the contract.

Chapter 06 — FastAPI AI Service and Vector Engine

Snapshot authority. This chapter describes commit `3d0c93e5270d44b9912deee0218e95c9a311dd5` on branch `development`. Source paths named below are the authority if the implementation changes after 2026-07-13.

This chapter documents the assistive AI execution process: every FastAPI route and Pydantic contract, runtime configuration, model selection, prompts, retrieval and indexing, vector storage, background-execution limits, guardrails, error behavior, and the backend trust boundary.

Source map

- `ai-service/app/`
- `ai-service/tests/`
- `ai-service/requirements.txt`
- `backend/src/modules/ai-mentor/`
- `backend/src/modules/rag/`
- `backend/src/modules/file-upload/`
- `backend/src/drizzle/schema/rag.schema.ts`

AI authority boundary

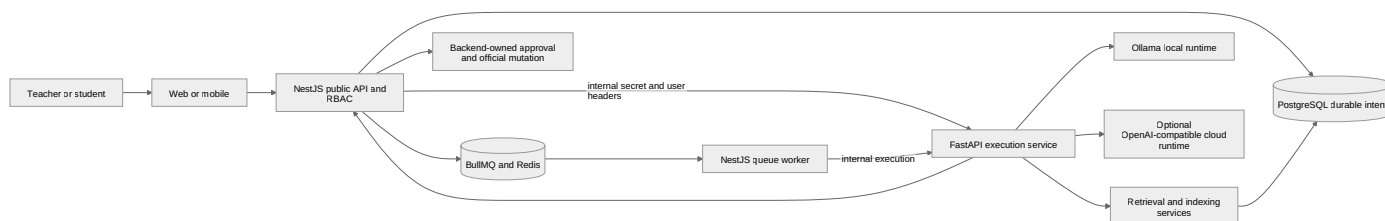


Diagram 14

- FastAPI is an assistive compute and retrieval boundary, not the public auth authority and not the owner of official grades or role state.
- Durable teacher generation, extraction, content chunks, embeddings, and outputs live in PostgreSQL; BullMQ owns restart-safe orchestration and retries.
- The process also holds bounded runtime-only maps and concurrency controls. Those improve coordination within one process but are not durable job authority.
- Direct browser or mobile access is outside the supported architecture.

Runtime configuration dictionary

Exhaustive inventory rule. The 40 AI settings below were extracted from `ai-service/app/config.py` at commit `3d0c93e`. A later source change requires regenerating or manually reconciling this chapter.

Setting	Type	Default and environment alias
database_url	str	'postgresql+asyncpg://postgres:CHANGE_ME_DB_PASSWORD@localhost:5432/capstone'
ollama_base_url	str	'http://localhost:11434'
ollama_text_model	str	Field(default='qwen2.5:3b', validation_alias=AliasChoices('OLLAMA_TEXT_MODEL', 'OLLAMA_MODEL'))
ollama_vision_model	str	Field(default='gemma3:4b', validation_alias='OLLAMA_VISION_MODEL')
ollama_embed_model	str	'nomic-embed-text'
embedding_dimensions	int	768
ollama_timeout_chat_s	int	Field(default=60, validation_alias=AliasChoices('OLLAMA_TIMEOUT_CHAT_S', 'OLLAMA_TIMEOUT'))
ollama_timeout_extraction_s	int	Field(default=240, validation_alias=AliasChoices('OLLAMA_TIMEOUT_EXTRACTION_S', 'OLLAMA_TIMEOUT'))
ollama_timeout_quiz_generation_s	int	Field(default=150, validation_alias=AliasChoices('OLLAMA_TIMEOUT_QUIZ_GENERATION_S', 'OLLAMA_TIMEOUT_QUIZ_S', 'OLLAMA_TIMEOUT'))
ollama_keep_alive	str	'15m'
upload_dir	str	'../backend/uploads'
backend_internal_url	str	''
backend_upload_fetch_timeout_s	int	60
max_raw_text	int	50000
db_pool_size	int	Field(default=10, validation_alias='DB_POOL_SIZE')
db_max_overflow	int	Field(default=15, validation_alias='DB_MAX_OVERFLOW')
db_pool_timeout_s	int	Field(default=30, validation_alias='DB_POOL_TIMEOUT_S')
db_pool_recycle_s	int	Field(default=1800, validation_alias='DB_POOL_RECYCLE_S')
db_pool_pre_ping	bool	Field(default=True, validation_alias='DB_POOL_PRE_PING')
log_level	str	'INFO'
ai_tutor_max_inflight	int	Field(default=8, validation_alias='AI_TUTOR_MAX_INFLIGHT')
ai_tutor_reject_status	int	Field(default=429, validation_alias='AI_TUTOR_REJECT_STATUS')
ai_tutor_retry_after_s	int	Field(default=5, validation_alias='AI_TUTOR_RETRY_AFTER_S')
ai_teacher_bg_max_concurrency	int	Field(default=2, validation_alias='AI_TEACHER_BG_MAX_CONCURRENCY')
ai_extraction_bg_max_concurrency	int	Field(default=1, validation_alias='AI_EXTRACTION_BG_MAX_CONCURRENCY')
ai_service_shared_secret	str	''
ai_degraded_allowed	bool	False
retrieval_min_final_score	float	Field(default=2.2, validation_alias='RETRIEVAL_MIN_FINAL_SCORE')
retrieval_min_semantic_score	float	Field(default=0.2, validation_alias='RETRIEVAL_MIN_SEMANTIC_SCORE')
retrieval_min_distinct_sources	int	Field(default=1, validation_alias='RETRIEVAL_MIN_DISTINCT_SOURCES')
ai_cloud_fallback_enabled	bool	Field(default=False, validation_alias='AI_CLOUD_FALLBACK_ENABLED')

Setting	Type	Default and environment alias
ai_runtime_mode	str	Field(default='auto', validation_alias=AliasChoices('AI_RUNTIME_MODE', 'ai_runtime_mode'))
ai_cloud_fallback_provider	str	Field(default='openai', validation_alias='AI_CLOUD_FALLBACK_PROVIDER')
ai_cloud_fallback_model	str	Field(default='gpt-4o-mini', validation_alias=AliasChoices('AI_CLOUD_FALLBACK_MODEL', 'OPENROUTER_TEXT_MODEL'))
ai_cloud_fallback_vision_model	str	Field(default='', validation_alias='OPENROUTER_VISION_MODEL')
ai_cloud_fallback_embedding_model	str	Field(default='google/gemini-embedding-2-preview', validation_alias='OPENROUTER_EMBEDDING_MODEL')
ai_cloud_fallback_api_key	str	Field(default='', validation_alias=AliasChoices('AI_CLOUD_FALLBACK_API_KEY', 'OPENROUTER_API_KEY'))
ai_cloud_fallback_base_url	str	Field(default='https://api.openai.com/v1', validation_alias=AliasChoices('AI_CLOUD_FALLBACK_BASE_URL', 'OPENROUTER_BASE_URL'))
ai_cloud_fallback_referer	str	Field(default='', validation_alias='OPENROUTER_HTTP_REFERER')
ai_cloud_fallback_title	str	Field(default='', validation_alias='OPENROUTER_X_TITLE')

Runtime and model selection

Task profile	Default local model or provider	Timeout or limit	Fallback behavior
General chat and grounded tutor text	qwen2.5:3b through Ollama	60-second chat timeout; tutor maximum eight in-flight requests	Cloud fallback is disabled by default and requires explicit compatible-runtime configuration.
Vision input	gemma3:4b through Ollama	Uses the chat execution boundary	Attachment validation and media pre-processing occur before model invocation.
Embeddings	nomic-embed-text through Ollama	Exactly 768 dimensions	Configured cloud embedding model is google/gemini-embedding-2-preview when cloud embedding is deliberately enabled.
Module extraction	Configured local text or vision runtime	240-second extraction timeout; extraction background concurrency one	Rule-based and structured repair paths supplement model extraction.
Quiz generation	Configured local text runtime	150-second quiz-generation timeout	Schema validation and repair prompts run before accepting the draft.
Teacher generation	Configured local text runtime	Teacher background concurrency two	Optional cloud runtime fallback model defaults to gpt-4o-mini.

- Runtime mode defaults to auto. The adapter chooses the configured local or OpenAI-compatible provider according to settings and task support.
- Cloud fallback must be explicitly enabled. A configured API key alone must not silently broaden data egress.
- Keep embedding model, dimensions, and stored embedding_model identity aligned during any provider migration.

Complete FastAPI route catalog

Exhaustive inventory rule. The 60 FastAPI routes below were extracted from `ai-service/app/**/*.py` at commit `3d0c93e`. A later source change requires regenerating or manually reconciling this chapter.

At this snapshot, 13 routes explicitly depend on `require_internal_service` and 43 depend on forwarded current-user context. Operational routes without an auth dependency still require network-level exposure controls.

Route source: ai-service/app/main.py

Method	Path	Access	Handler	Inputs and dependencies	Return annotation	Status	Line
POST	/chat	Forwarded authenticated user; secret enforced when configured	chat	body: ChatRequest; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	1901
POST	/admin/chat	Forwarded authenticated user; secret enforced when configured	admin_chat	body: AdminChatRequest; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	2025
POST	/mentor/explain	Forwarded authenticated user; secret enforced when configured	mentor_explain	body: MentorExplainRequest; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	2141
GET	/student/ja/practice/bootstrap	Forwarded authenticated user; secret enforced when configured	ja_practice_bootstrap	class_id: str None = Query(None, alias='classId'); user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	2167
POST	/student/ja/practice/sessions/generate	Forwarded authenticated user; secret enforced when configured	ja_practice_generate	body: JaPracticeGenerateRequest; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	2181
GET	/student/ja/ask/bootstrap	Forwarded authenticated user; secret enforced when configured	ja_ask_bootstrap	class_id: str None = Query(None, alias='classId'); user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	2203
POST	/student/ja/ask/respond	Forwarded authenticated user; secret enforced when configured	ja_ask_respond	body: JaAskResponseRequest; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	2217
GET	/student/ja/review/bootstrap	Forwarded authenticated user; secret enforced when configured	ja_review_bootstrap	class_id: str None = Query(None, alias='classId'); user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	2246
POST	/student/ja/review/sessions/generate	Forwarded authenticated user;	ja_review_generate	body: JaReviewGenerateRequest; user:	inferred	Framework default	2260

Method	Path	Access	Handler	Inputs and dependencies	Return annotation	Status	Line
		secret enforced when configured		RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)			
GET	/student/tutor/bootstrap	Forwarded authenticated user; secret enforced when configured	student_tutor_bootstrap	class_id: str None = Query(None, alias='classId'); user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	2287
POST	/student/tutor/session	Forwarded authenticated user; secret enforced when configured	student_tutor_start	body: StudentTutorStartRequest; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	2301
GET	/student/tutor/session/{session_id}	Forwarded authenticated user; secret enforced when configured	student_tutor_get_session	session_id: str; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	2321
POST	/student/tutor/session/{session_id}/message	Forwarded authenticated user; secret enforced when configured	student_tutor_message	session_id: str; body: StudentTutorMessageRequest; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	2335
POST	/student/tutor/session/{session_id}/answers	Forwarded authenticated user; secret enforced when configured	student_tutor_answers	session_id: str; body: StudentTutorAnswerRequest; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	2359
GET	/health	Operational endpoint; network exposure must be controlled	health	No handler arguments	inferred	Framework default	2388
GET	/metrics	Operational endpoint; network exposure must be controlled	metrics	No handler arguments	inferred	Framework default	2422
GET	/live	No authentication dependency extracted; deploy only behind the intended private boundary	live	No handler arguments	inferred	Framework default	2427
GET	/ready	Operational endpoint; network exposure must be controlled	ready	db: AsyncSession = Depends(get_db)	inferred	Framework default	2441
POST	/demo/intervention-plan	Internal service secret required	generate_demo_intervention_plan	body: DemoInterventionPlanRequest; _auth: None = Depends(require_internal_service)	inferred	Framework default	2458
GET	/internal/retrieval/preview	Internal service secret required	internal_retrieval_preview	class_id: str = Query([required], alias='classId'); query_text: str = Query([required], alias='query');	inferred	Framework default	2608

Method	Path	Access	Handler	Inputs and dependencies	Return annotation	Status	Line
				<pre> policy: str = Query('general'); top_k: int = Query(8, ali- as='topK'); sub- ject_key: str None = Query(None, ali- as='subjectKey'); grade_level: str None = Query(None, ali- as='gradeLevel'); include_library: bool = Query(True, ali- as='includeLib- rary'); db: Async- Session = De- pends(get_db); _auth: None = De- pends(require_int ernal_service) </pre>			
GET	/internal/ extractions/{ex- traction_id}/audit	Internal service secret required	intern- al_extrac- tion_audit	<pre> extraction_id: str; db: AsyncSession = De- pends(get_db); _auth: None = De- pends(require_int ernal_service) </pre>	inferred	Framework de- fault	2644
GET	/index/classes/ {class_id}/status	Forwarded au- thenticated user; secret enforced when configured	get_in- dex_class_status	<pre> class_id: str; user: RequestUser = De- pends(get_curren t_user); db: AsyncSession = Depends(get_db) </pre>	inferred	Framework de- fault	2689
POST	/index/classes/ {class_id}	Forwarded au- thenticated user; secret enforced when configured	index_class	<pre> class_id: str; user: RequestUser = De- pends(get_curren t_user); db: AsyncSession = Depends(get_db) </pre>	inferred	Framework de- fault	2720
POST	/internal/index/ classes/ {class_id}	Internal service secret required	intern- al_index_class	<pre> class_id: str; pay- load: dict[str, Any] None = Body(de- fault=None); _au- thorized: None = De- pends(require_int ernal_service); db: AsyncSession = De- pends(get_db) </pre>	inferred	Framework de- fault	2746
POST	/internal/index/ library-files/ {file_id}	Internal service secret required	intern- al_in- dex_library_file	<pre> file_id: str; pay- load: dict[str, Any] None = Body(de- fault=None); _au- thorized: None = De- pends(require_int ernal_service); db: AsyncSession = De- pends(get_db) </pre>	inferred	Framework de- fault	2764
DELETE	/internal/index/ library-files/ {file_id}/chunks	Internal service secret required	intern- al_de- letе_lib- rary_file_chunks	<pre> file_id: str; _au- thorized: None = De- pends(require_int ernal_service); db: AsyncSession = De- pends(get_db) </pre>	inferred	Framework de- fault	2782
POST	/internal/index/ library/backfill	Internal service secret required	intern- al_back- fill_library_index	<pre> payload: dict[str, Any] None = Body(de- fault=None); _au- thorized: None = </pre>	inferred	Framework de- fault	2796

Method	Path	Access	Handler	Inputs and dependencies	Return annotation	Status	Line
				Depends(require_internal_service); db: AsyncSession = Depends(get_db)			
POST	/internal/index/backfill	Internal service secret required	internal_backfill_index	payload: dict[str, Any] None = Body(default=None); _authorized: None = Depends(require_internal_service); db: AsyncSession = Depends(get_db)	inferred	Framework default	2813
GET	/extractions/{extraction_id}/status	Forwarded authenticated user; secret enforced when configured	get_extraction_status	extraction_id: str; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	3409
GET	/extractions	Forwarded authenticated user; secret enforced when configured	list_extractions	class_id: str = Query([required], alias='classId'); user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	3462
GET	/extractions/{extraction_id}	Forwarded authenticated user; secret enforced when configured	get_extraction	extraction_id: str; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	3523
PATCH	/extractions/{extraction_id}	Forwarded authenticated user; secret enforced when configured	update_extraction	extraction_id: str; body: UpdateExtractionRequest; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	3568
POST	/extractions/{extraction_id}/apply/preview	Forwarded authenticated user; secret enforced when configured	preview_apply_extraction	extraction_id: str; body: ApplyExtractionRequest; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	3773
POST	/extractions/{extraction_id}/apply	Forwarded authenticated user; secret enforced when configured	apply_extraction	extraction_id: str; body: ApplyExtractionRequest; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	201	3813
POST	/extractions/{extraction_id}/cancel	Forwarded authenticated user; secret enforced when configured	cancel_extraction	extraction_id: str; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	4266
POST	/extractions/{extraction_id}/retry	Forwarded authenticated user; secret enforced when configured	retry_extraction	extraction_id: str; body: RetryExtractionRequest; user: RequestUser = Depends(get_current_user); db:	inferred	202	4297

Method	Path	Access	Handler	Inputs and dependencies	Return annotation	Status	Line
				AsyncSession = Depends(get_db)			
DELETE	/extractions/{extraction_id}	Forwarded authenticated user; secret enforced when configured	delete_extraction	extraction_id: str; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	4348
GET	/history	Forwarded authenticated user; secret enforced when configured	interaction_history	user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	4389
GET	/admin/history	Forwarded authenticated user; secret enforced when configured	admin_chat_history	user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	4410
GET	/admin/sessions/{session_id}	Forwarded authenticated user; secret enforced when configured	admin_chat_session	session_id: str; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	4438
POST	/teacher/interventions/{case_id}/jobs	Forwarded authenticated user; secret enforced when configured	queue_intervention_recommendation_job	case_id: str; body: InterventionRecommendationRequest; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	202	4476
POST	/teacher/quizzes/jobs	Forwarded authenticated user; secret enforced when configured	queue_teacher_quiz_draft_job	body: GenerateQuizDraftRequest; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	202	4534
POST	/teacher/lesson-plans/jobs	Forwarded authenticated user; secret enforced when configured	queue_teacher_lesson_plan_job	body: GenerateLessonPlanRequest; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	202	4604
POST	/internal/teacher/lesson-plans/jobs/{job_id}/run	Internal service secret required	run_teacher_lesson_plan_job	job_id: str; meta: dict[str, Any] None = Body(None); _auth: None = Depends(require_internal_service); db: AsyncSession = Depends(get_db)	inferred	Framework default	4664
POST	/internal/teacher/quizzes/jobs/{job_id}/run	Internal service secret required	run_teacher_quiz_job	job_id: str; meta: dict[str, Any] None = Body(None); _auth: None = Depends(require_internal_service); db: AsyncSession = Depends(get_db)	inferred	Framework default	4761
POST	/internal/teacher/interventions/jobs/{job_id}/run	Internal service secret required	run_teacher_intervention_job	job_id: str; meta: dict[str, Any] None = Body(None); _auth: None = Depends(require_int	inferred	Framework default	4850

Method	Path	Access	Handler	Inputs and dependencies	Return annotation	Status	Line
				ernal_service); db: AsyncSession = Depends(get_db)			
PATCH	/teacher/lesson-plans/jobs/{job_id}/draft	Forwarded authenticated user; secret enforced when configured	update_teacher_lesson_plan_draft	job_id: str; body: UpdateLessonPlanDraftRequest; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	4924
PATCH	/teacher/quizzes/jobs/{job_id}/draft	Forwarded authenticated user; secret enforced when configured	update_teacher_quiz_draft	job_id: str; body: UpdateQuizDraftRequest; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	4966
POST	/teacher/quizzes/jobs/{job_id}/apply/preview	Forwarded authenticated user; secret enforced when configured	preview_teacher_quiz_draft_apply	job_id: str; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	5008
POST	/teacher/quizzes/jobs/{job_id}/apply	Forwarded authenticated user; secret enforced when configured	apply_teacher_quiz_draft	job_id: str; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	5027
POST	/teacher/quizzes/jobs/{job_id}/retry	Forwarded authenticated user; secret enforced when configured	retry_teacher_quiz_draft_job	job_id: str; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	202	5057
POST	/teacher/quizzes/jobs/{job_id}/cancel	Forwarded authenticated user; secret enforced when configured	cancel_teacher_quiz_draft_job	job_id: str; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	5092
GET	/teacher/jobs/{job_id}	Forwarded authenticated user; secret enforced when configured	get_teacher_ai_job_status	job_id: str; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	5106
GET	/teacher/jobs/{job_id}/result	Forwarded authenticated user; secret enforced when configured	get_teacher_ai_job_result	job_id: str; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	5136
DELETE	/teacher/jobs/{job_id}	Forwarded authenticated user; secret enforced when configured	delete_teacher_ai_job	job_id: str; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	5185
POST	/teacher/interventions/{case_id}/recommend	Forwarded authenticated user; secret enforced when configured	recommend_intervention	case_id: str; body: InterventionRecommendationRequest; user: RequestUser = Depends(get_current_user); db:	inferred	Framework default	5245

Method	Path	Access	Handler	Inputs and dependencies	Return annotation	Status	Line
				AsyncSession = Depends(get_db)			
POST	/teacher/quizzes/generate-draft	Forwarded authenticated user; secret enforced when configured	teacher_generate_quiz_draft	body: GenerateQuizDraftRequest; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	Framework default	5270

Route source: ai-service/app/routers/extractions.py

Method	Path	Access	Handler	Inputs and dependencies	Return annotation	Status	Line
POST	/extract	Forwarded authenticated user; secret enforced when configured	extract_module	body: ExtractRequest; user: RequestUser = Depends(get_current_user); db: AsyncSession = Depends(get_db)	inferred	202	50
POST	/internal/extractions/{extraction_id}/run	Internal service secret required	run_internal_extraction	extraction_id: str; meta: dict[str, Any] None = Body(None); _auth: None = Depends(require_internal_service); db: AsyncSession = Depends(get_db)	inferred	Framework default	83
POST	/internal/extractions/{extraction_id}/fail	Internal service secret required	fail_pending_internal_extraction	extraction_id: str; payload: dict[str, Any] None = Body(None); _auth: None = Depends(require_internal_service); db: AsyncSession = Depends(get_db)	inferred	Framework default	160

Complete Pydantic contract dictionary

Exhaustive inventory rule. The 37 Pydantic model classes below were extracted from `ai-service/app/**/*.*py` at commit `3d0c93e`. A later source change requires regenerating or manually reconciling this chapter.

ImageAttachment

Base classes: `BaseModel`. Source: `ai-service/app/schemas.py`.

Field	Type	Default, alias, or constraint
file_path	str None	Field(default=None, alias='filePath')
base64_data	str None	Field(default=None, alias='base64Data')
mime_type	str None	Field(default=None, alias='mimeType')

ChatRequest

Base classes: `BaseModel`. Source: `ai-service/app/schemas.py`.

Field	Type	Default, alias, or constraint
message	str	Field([required], max_length=2000)
session_id	str None	Field(None, alias='sessionId')
attachments	list[ImageAttachment] None	None

ChatData

Base classes: `BaseModel` . Source: `ai-service/app/schemas.py` .

Field	Type	Default, alias, or constraint
reply	str	Required
session_id	str	Field(alias='sessionId')
model_used	str	Field(alias='modelUsed')

AdminAnalyticsSource

Base classes: `BaseModel` . Source: `ai-service/app/schemas.py` .

Field	Type	Default, alias, or constraint
source	str	Required
filters	dict[str, Any]	Field(default_factory=dict)
window	str None	None

AdminAnalyticsChartSeries

Base classes: `BaseModel` . Source: `ai-service/app/schemas.py` .

Field	Type	Default, alias, or constraint
name	str	Required
data	list[float int]	Required

AdminAnalyticsChart

Base classes: `BaseModel` . Source: `ai-service/app/schemas.py` .

Field	Type	Default, alias, or constraint
type	str	Required
title	str	Required
labels	list[str]	Required
series	list[AdminAnalyticsChartSeries]	Required
y_axis_label	str None	Field(default=None, alias='yAxisLabel')
x_axis_label	str None	Field(default=None, alias='xAxisLabel')

AdminChatRequest

Base classes: `BaseModel` . Source: `ai-service/app/schemas.py` .

Field	Type	Default, alias, or constraint
message	str	Field([required], max_length=2000)
session_id	str None	Field(None, alias='sessionId')
context	dict[str, Any]	Required

ExtractRequest

Base classes: `BaseModel` . Source: `ai-service/app/schemas.py` .

Field	Type	Default, alias, or constraint
file_id	str	Field([required], alias='fileId')
target_section_count	Literal[3, 4, 5]	Field([required], alias='targetSectionCount')
extraction_style	Literal['faithful', 'clean', 'student_friendly']	Field(default='clean', alias='extractionStyle')

RetryExtractionRequest

Base classes: `BaseModel` . Source: `ai-service/app/schemas.py` .

Field	Type	Default, alias, or constraint
target_section_count	Literal[3, 4, 5] None	Field(default=None, alias='targetSectionCount')
extraction_style	Literal['faithful', 'clean', 'student_friendly'] None	Field(default=None, alias='extractionStyle')

ApplyExtractionRequest

Base classes: `BaseModel` . Source: `ai-service/app/schemas.py` .

Field	Type	Default, alias, or constraint
section_indices	list[int] None	Field(None, alias='sectionIndices')
lesson_indices	list[int] None	Field(None, alias='lessonIndices')

ExtractionBlockDto

Base classes: `BaseModel` . Source: `ai-service/app/schemas.py` .

Field	Type	Default, alias, or constraint
type	str	Required
order	int	Required
content	dict[str, Any] str	Required
metadata	dict[str, Any] None	None

ExtractionLessonDto

Base classes: `BaseModel` . Source: `ai-service/app/schemas.py` .

Field	Type	Default, alias, or constraint
title	str	Required
description	str None	None
blocks	list[ExtractionBlockDto]	Required

ExtractionAssessmentOptionDto

Base classes: `BaseModel` . Source: `ai-service/app/schemas.py` .

Field	Type	Default, alias, or constraint
text	str	Required
is_correct	bool	Field(default=False, alias='isCorrect')
order	int None	None

ExtractionAssessmentQuestionDto

Base classes: `BaseModel` . Source: `ai-service/app/schemas.py` .

Field	Type	Default, alias, or constraint
content	str	Required
type	str	'multiple_choice'
points	int	1
order	int None	None
explanation	str None	None
image_url	str None	Field(default=None, alias='imageUrl')
concept_tags	list[str] None	Field(default=None, alias='conceptTags')
options	list[ExtractionAssessmentOptionDto] None	None

ExtractionAssessmentDraftDto

Base classes: `BaseModel` . Source: `ai-service/app/schemas.py` .

Field	Type	Default, alias, or constraint
title	str	Required
description	str None	None
type	str	'quiz'
passing_score	int	Field(default=60, alias='passingScore')
feedback_level	str	Field(default='standard', alias='feedbackLevel')
question_type	str	Field(default='multiple_choice', alias='questionType')
questions	list[ExtractionAssessmentQuestionDto]	Field(default_factory=list)

ExtractionMediaCandidateDto

Base classes: `BaseModel` . Source: `ai-service/app/schemas.py` .

Field	Type	Default, alias, or constraint
section_index	int	Field(alias='sectionIndex')
score	float	Required
explicit_match	bool None	Field(default=None, alias='explicitMatch')
score_breakdown	dict[str, float] None	Field(default=None, alias='scoreBreakdown')

ExtractionMediaAssetDto

Base classes: `BaseModel` . Source: `ai-service/app/schemas.py` .

Field	Type	Default, alias, or constraint
id	str	Required
url	str	Required
page_number	int None	Field(default=None, alias='pageNumber')
caption	str None	None
anchor_text	str None	Field(default=None, alias='anchorText')
keywords	list[str]	Field(default_factory=list)
figure_references	list[str]	Field(default_factory=list, alias='figureReferences')
selected_section_index	int None	Field(default=None, alias='selectedSectionIndex')
assignment_confidence	float None	Field(default=None, alias='assignmentConfidence')

Field	Type	Default, alias, or constraint
assignment_breakdown	dict[str, float] None	Field(default=None, alias='assignmentBreakdown')
candidate_sections	list[ExtractionMediaCandidateDto]	Field(default_factory=list, alias='candidateSections')
teacher_reviewed	bool	Field(default=False, alias='teacherReviewed')
review_state	str None	Field(default=None, alias='reviewState')

ExtractionSectionDto

Base classes: `BaseModel`. Source: `ai-service/app/schemas.py`.

Field	Type	Default, alias, or constraint
title	str	Required
description	str None	None
order	int None	None
lesson_blocks	list[ExtractionBlockDto]	Field(default_factory=list, alias='lessonBlocks')
assessment_draft	ExtractionAssessmentDraftDto None	Field(default=None, alias='assessmentDraft')
confidence	float None	None
graph_keywords	list[str]	Field(default_factory=list, alias='graphKeywords')
figure_references	list[str]	Field(default_factory=list, alias='figureReferences')

UpdateExtractionRequest

Base classes: `BaseModel`. Source: `ai-service/app/schemas.py`.

Field	Type	Default, alias, or constraint
title	str None	None
description	str None	None
sections	list[ExtractionSectionDto] None	None
lessons	list[ExtractionLessonDto] None	None
review_issues	list[dict[str, Any]] None	Field(default=None, alias='reviewIssues')
review_state	str None	Field(default=None, alias='reviewState')
media_assets	list[ExtractionMediaAssetDto] None	Field(default=None, alias='mediaAssets')

MentorExplainRequest

Base classes: `BaseModel`. Source: `ai-service/app/schemas.py`.

Field	Type	Default, alias, or constraint
attempt_id	str	Field([required], alias='attemptId')
question_id	str	Field([required], alias='questionId')
message	str None	None
attachments	list[ImageAttachment] None	None

InterventionRecommendationRequest

Base classes: `BaseModel`. Source: `ai-service/app/schemas.py`.

Field	Type	Default, alias, or constraint
note	str None	None

DemoInterventionPlanRequest

Base classes: BaseModel . Source: ai-service/app/schemas.py .

Field	Type	Default, alias, or constraint
subject_id	str	Field([required], alias='subjectId')
quarter_exam_score	int	Field([required], alias='quarterExamScore')
weak_concepts	list[str]	Field(default_factory=list, alias='weakConcepts')
module_scores	list[int] None	Field(default=None, alias='moduleScores')

GenerateQuizDraftRequest

Base classes: BaseModel . Source: ai-service/app/schemas.py .

Field	Type	Default, alias, or constraint
class_id	str	Field([required], alias='classId')
lesson_ids	list[str] None	Field(default=None, alias='lessonIds')
extraction_ids	list[str] None	Field(default=None, alias='extractionIds')
title	str None	None
question_count	int	Field(default=5, alias='questionCount')
question_type	str	Field(default='multiple_choice', alias='questionType')
assessment_type	str	Field(default='quiz', alias='assessmentType')
passing_score	int	Field(default=60, alias='passingScore')
teacher_note	str None	Field(default=None, alias='teacherNote')
feedback_level	str	Field(default='standard', alias='feedbackLevel')
class_record_category	str None	Field(default=None, alias='classRecordCategory')
quarter	str None	None
source_policy	str	Field(default='published_default', alias='sourcePolicy')
allow_draft_sources	bool	Field(default=False, alias='allowDraftSources')
retry_of_job_id	str None	Field(default=None, alias='retryOfJobId')

LessonPlanHeaderRequest

Base classes: BaseModel . Source: ai-service/app/schemas.py .

Field	Type	Default, alias, or constraint
instructional_format	str None	Field(default=None, alias='instructionalFormat')
school_name	str None	Field(default=None, alias='schoolName')
quarter	str None	None
date	str None	None
start_time	str None	Field(default=None, alias='startTime')
end_time	str None	Field(default=None, alias='endTime')

GenerateLessonPlanRequest

Base classes: BaseModel . Source: ai-service/app/schemas.py .

Field	Type	Default, alias, or constraint
class_id	str	Field([required], alias='classId')
anchor_type	str	Field([required], alias='anchorType')
anchor_id	str	Field([required], alias='anchorId')

Field	Type	Default, alias, or constraint
teacher_note	str None	Field(default=None, alias='teacherNote')
header	LessonPlanHeaderRequest None	None

UpdateLessonPlanDraftRequest

Base classes: `BaseModel`. Source: `ai-service/app/schemas.py`.

Field	Type	Default, alias, or constraint
structured_output	dict[str, Any]	Field([required], alias='structuredOutput')

UpdateQuizDraftRequest

Base classes: `BaseModel`. Source: `ai-service/app/schemas.py`.

Field	Type	Default, alias, or constraint
structured_output	dict[str, Any]	Field([required], alias='structuredOutput')

StudentTutorBootstrapRequest

Base classes: `BaseModel`. Source: `ai-service/app/schemas.py`.

Field	Type	Default, alias, or constraint
class_id	str None	Field(default=None, alias='classId')

TutorRecommendationDto

Base classes: `BaseModel`. Source: `ai-service/app/schemas.py`.

Field	Type	Default, alias, or constraint
id	str	Required
title	str	Required
reason	str	Required
focus_text	str	Field(alias='focusText')
lesson_id	str None	Field(default=None, alias='lessonId')
assessment_id	str None	Field(default=None, alias='assessmentId')
question_id	str None	Field(default=None, alias='questionId')
source_chunk_id	str None	Field(default=None, alias='sourceChunkId')

StudentTutorStartRequest

Base classes: `BaseModel`. Source: `ai-service/app/schemas.py`.

Field	Type	Default, alias, or constraint
class_id	str	Field([required], alias='classId')
recommendation	TutorRecommendationDto	Required

StudentTutorMessageRequest

Base classes: `BaseModel`. Source: `ai-service/app/schemas.py`.

Field	Type	Default, alias, or constraint
session_id	str	Field([required], alias='sessionId')
message	str	Field([required], max_length=2000)

Field	Type	Default, alias, or constraint
attachments	list[ImageAttachment] None	None

StudentTutorAnswerRequest

Base classes: `BaseModel`. Source: `ai-service/app/schemas.py`.

Field	Type	Default, alias, or constraint
session_id	str	Field([required], alias='sessionId')
answers	list[str]	Field([required], min_length=1, max_length=3)
attachments	list[ImageAttachment] None	None

JaPracticeGenerateRequest

Base classes: `BaseModel`. Source: `ai-service/app/schemas.py`.

Field	Type	Default, alias, or constraint
class_id	str	Field([required], alias='classId')
question_count	int	Field(default=10, alias='questionCount')
recommendation	TutorRecommendationDto None	None
allowed_lesson_ids	list[str] None	Field(default=None, alias='allowedLessonIds')
allowed_assessment_ids	list[str] None	Field(default=None, alias='allowedAssessmentIds')

JaAskHistoryMessageDto

Base classes: `BaseModel`. Source: `ai-service/app/schemas.py`.

Field	Type	Default, alias, or constraint
role	str	Required
content	str	Required

JaAskResponseRequest

Base classes: `BaseModel`. Source: `ai-service/app/schemas.py`.

Field	Type	Default, alias, or constraint
class_id	str	Field([required], alias='classId')
thread_id	str	Field([required], alias='threadId')
message	str	Field([required], min_length=1, max_length=2000)
quick_action	str None	Field(default=None, alias='quickAction')
lesson_id	str None	Field(default=None, alias='lessonId')
lesson_title	str None	Field(default=None, alias='lessonTitle')
history	list[JaAskHistoryMessageDto] None	None
allowed_lesson_ids	list[str] None	Field(default=None, alias='allowedLessonIds')
allowed_assessment_ids	list[str] None	Field(default=None, alias='allowedAssessmentIds')

JaReviewGenerateRequest

Base classes: `BaseModel`. Source: `ai-service/app/schemas.py`.

Field	Type	Default, alias, or constraint
class_id	str	Field([required], alias='classId')
attempt_id	str	Field([required], alias='attemptId')

Field	Type	Default, alias, or constraint
question_count	int	Field(default=10, alias='questionCount')
allowed_lesson_ids	list[str] None	Field(default=None, alias='allowedLessonIds')
allowed_assessment_ids	list[str] None	Field(default=None, alias='allowedAssessmentIds')

RequestUser

Base classes: `BaseModel` . Source: `ai-service/app/schemas.py` .

Field	Type	Default, alias, or constraint
id	str	Required
email	str	Required
roles	list[str]	Required

Prompt ownership register

Prompt or builder	Owner	Purpose and guardrail
JAKIPIR_SYSTEM_PROMPT	ai-service/app/main.py	General Ja assistant identity, school context, grounding, and safe-answer behavior.
ADMIN_ANALYTICS_SYSTEM_PROMPT	ai-service/app/main.py	Administrative analytics explanation grounded in backend-provided aggregates.
DEMO_INTERVENTION_PLAN_SYSTEM_PROMPT	ai-service/app/main.py	Concise practical intervention-plan output for the explicit demonstration workflow.
MENTOR_SYSTEM_PROMPT	ai-service/app/mentor_service.py	Grounded AI mentor behavior and source discipline.
TUTOR_SYSTEM_PROMPT	ai-service/app/student_tutor_service.py	Student-facing instructional guidance with class and source boundaries.
LESSON_PLAN_SYSTEM_PROMPT	ai-service/app/lesson_plan_service.py	Teacher-facing daily lesson-plan draft generation from retrieved evidence.
QUIZ_GENERATION_SYSTEM_PROMPT	ai-service/app/quiz_generation_service.py	Structured high-school assessment drafts with validity and grounding rules.
INTERVENTION_RECOMMENDATION_SYSTEM_PROMPT	ai-service/app/remedial_service.py	Teacher-facing intervention recommendations based on provided performance evidence.
EXTRACTION_SYSTEM_PROMPT	ai-service/app/extraction_pipeline.py	Source extraction into controlled content structures.
STRUCTURE_SYSTEM_PROMPT	ai-service/app/extraction_pipeline.py	Repairs or derives valid module structure from extraction evidence.
Ja practice builder	ai-service/app/ja_practice_service.py	Constructs practice guidance from current session, questions, and learner evidence.
Content sanitizer prompt	ai-service/app/content_sanitizer.py	Repairs unsafe or malformed generated content after deterministic sanitization checks.

- Prompt text is source code and must be reviewed, tested, versioned, and treated as behavior.
- Never interpolate a raw retrieved document into a privileged instruction channel. Source material is untrusted context.
- Structured outputs must pass schema and domain validation; a syntactically valid model response is not automatically a publishable academic artifact.

Retrieval and indexing pipeline

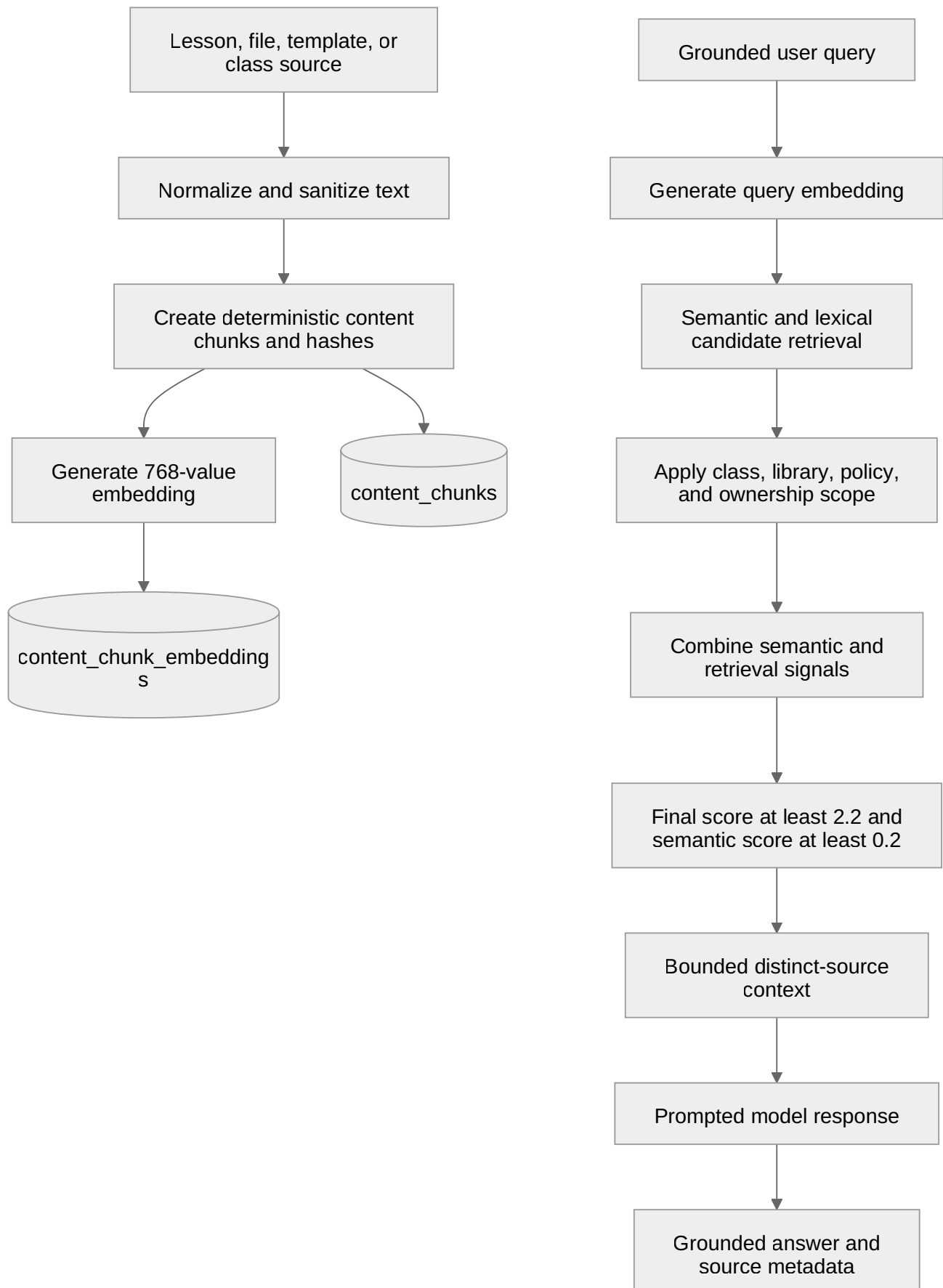


Diagram 15

Storage item	Contract
content_chunks	Normalized text, source identity, class or library scope, checksum, ordinal, and metadata.
content_chunk_embeddings	One required vector(768) value per chunk, embedding model identity, and embedded timestamp.
Vector extension	Enabled by active migration 0003 before vector-dependent use.
Vector index	No HNSW or IVFFlat index is declared in the active schema or migrations.
Model index	A B-tree index exists for embedding_model filtering only.
Thresholds	Default final retrieval score 2.2, semantic score 0.2, and at least one distinct source.

Indexing ownership

- Class reindexing is serialized by the rag-indexing queue and the internal class-index route.
- Library-file indexing uses a separate queue with concurrency two and a file-specific internal route.
- Extraction can create structured source material, but applying an extraction remains a backend-governed lifecycle step.
- Reindex operations must be idempotent by source identity and checksum so a retry replaces or reconciles stale chunks rather than multiplying them.

Concurrency, timeouts, and overload behavior

- Tutor requests are bounded by an in-process maximum of eight. Rejection status defaults to 429 with Retry-After of five seconds.
- Teacher background execution is bounded to two and extraction to one, matching backend queue concurrency for those workloads.
- Database pool defaults are size ten, overflow fifteen, checkout timeout thirty seconds, recycle 1,800 seconds, with pre-ping enabled.
- Model requests have task-specific timeouts. A timeout is a failure signal, not proof that a remote runtime performed no work.
- Backend queue workers must record durable attempt state and rethrow retryable failures so BullMQ can apply backoff.

Health and readiness behavior

- Liveness proves the FastAPI process can answer. Readiness also considers required database and runtime dependencies.
- ai_degraded_allowed defaults false. A deployment must explicitly opt into degraded readiness behavior.
- Ollama availability, configured model presence, database connectivity, and cloud runtime configuration must be interpreted together with the selected runtime mode.
- Metrics are exposed through the FastAPI metrics route and scraped by Prometheus as documented in Chapter 09.

AI safety and correctness gates

Risk	Required gate
Hallucinated academic fact	Ground retrieval to allowed sources, preserve citations, and reject weak-evidence output.
Cross-class data leakage	Apply backend-validated user context, class ownership or enrollment, class AI policy, and retrieval scope.
Prompt injection in uploaded content	

Risk	Required gate
	Treat source content as data, sanitize it, keep system instructions separate, and validate structured output.
Invalid structured draft	Pydantic validation, deterministic domain validation, and repair with a bounded retry.
Official-state mutation	Return a draft to a backend-owned approval or apply workflow; never finalize grades directly from model output.
Cloud data egress	Require explicit cloud enablement, approved provider configuration, and deployment-level secret control.
Duplicate background execution	Use durable job identity, idempotent writes, and current-state checks before applying output.
Oversized attachment or text	Enforce file, media, raw-text, and request-body bounds before model invocation.

AI feature maintenance checklist

1. Define the backend-owned public contract and authorization before adding a FastAPI route.
2. Decide whether the task is synchronous or a durable BullMQ job; long-running extraction or generation must remain restart-safe.
3. Add a typed Pydantic request and response contract with negative tests.
4. Select the task profile, timeout, concurrency budget, and local or cloud data policy explicitly.
5. Ground the prompt to allowed source scope and validate the response deterministically.
6. Persist durable intent, attempts, output, failure reason, and approval state in PostgreSQL.
7. Test runtime unavailable, database unavailable, malformed model output, weak retrieval, duplicate execution, timeout, and cancellation.
8. Update backend consumers, both clients when exposed, metrics, runbooks, and this chapter.

Chapter 07 — Next.js 16 Web Frontend

Snapshot authority. This chapter describes commit `3d0c93e5270d44b9912deee0218e95c9a311dd5` on branch `development`. Source paths named below are the authority if the implementation changes after 2026-07-13.

This chapter is the web-client service manual. It records the App Router tree, role-scoped navigation, every page, API service, provider, query-cache policy, session bootstrap, notification delivery, reusable UI boundary, and design-token declaration.

Source map

- `next-frontend/app/`
- `next-frontend/src/components/`
- `next-frontend/src/lib/`
- `next-frontend/src/providers/`
- `next-frontend/src/services/`
- `next-frontend/src/types/`
- `next-frontend/proxy.ts`
- `next-frontend/next.config.ts`

Web-client boundary



Diagram 16

- The web frontend consumes backend `/api` contracts. It does not call FastAPI directly.
- The production build uses Next standalone output. The rewrite destination is selected from the explicit internal API origin, Railway backend origin, or local backend fallback.
- Browser requests stay relative to `/api`, keeping deployment origin and cookie behavior under the Next.js boundary.

Provider composition and responsibilities

Provider	Responsibility	Key behavior	Source
ThemeProvider	Application theme state	Supplies consistent theme selection to App Router content.	<code>next-frontend/src/providers/ThemeProvider.tsx</code>
QueryProvider	Server-state cache	30-second stale time, two retries, exponential jitter, no refetch on window focus, development devtools.	<code>next-frontend/src/providers/QueryProvider.tsx</code>
AuthProvider	Session bootstrap and role helpers	Refreshes the httpOnly-cookie session, stores access token in memory, fetches current user, and exposes bootstrapping, authenticated, or unauthenticated state.	<code>next-frontend/src/providers/AuthProvider.tsx</code>
NotificationProvider	Inbox, sockets, polling, extraction tracking, and student reminders	Authenticates Socket.IO with current access token and combines durable notification fetches with bounded polling and route-aware toasts.	<code>next-frontend/src/providers/NotificationProvider.tsx</code>

Authentication and request lifecycle

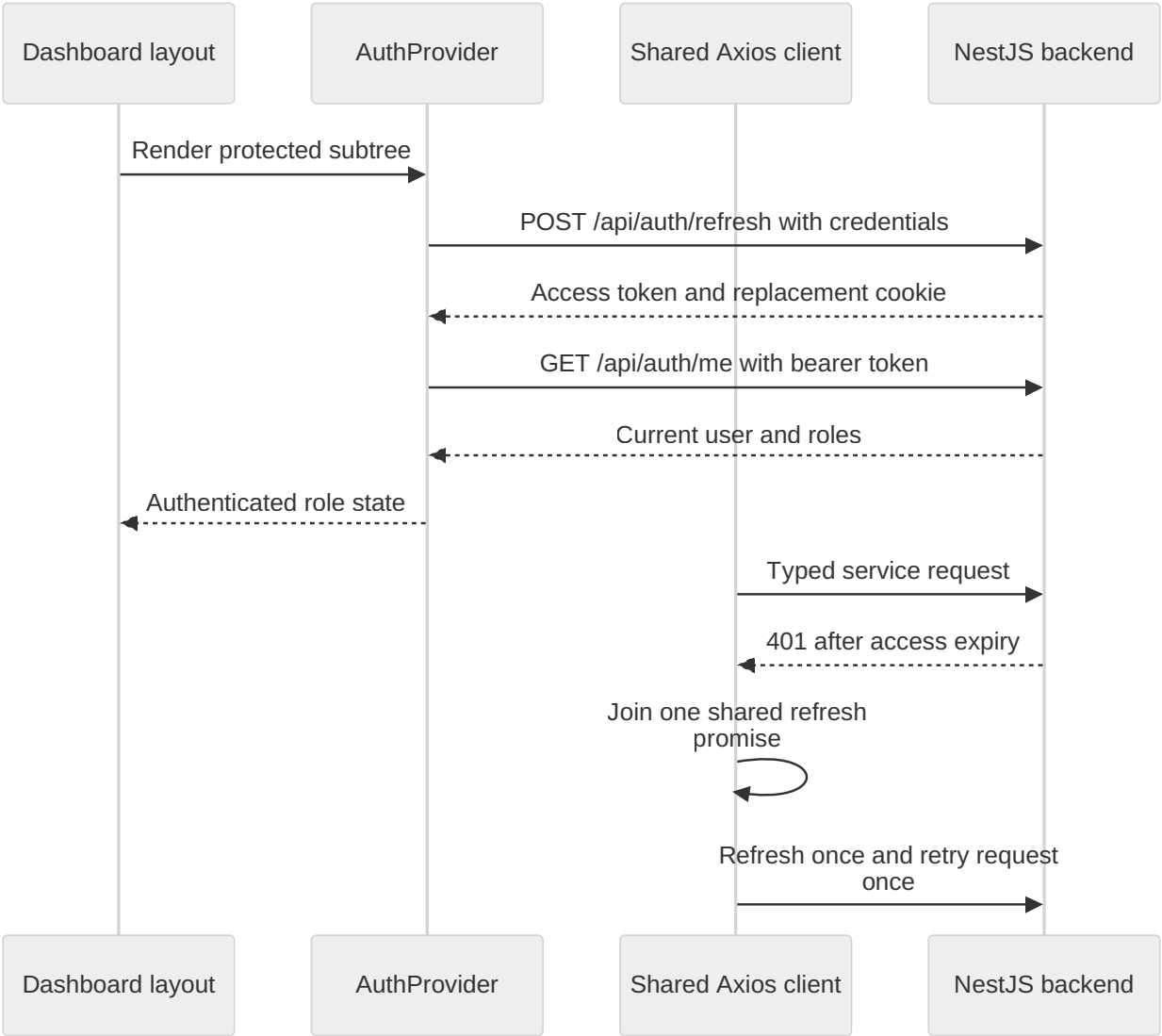


Diagram 17

Client control	Current setting
Base URL	Relative /api
Credentials	withCredentials true for refresh-cookie transport
Request timeout	30 seconds
Access token	Process-memory variable in the browser bundle; not localStorage
401 recovery	One shared refresh promise and one retry marker per failed request
Auth bootstrap refresh timeout	5 seconds
Current-user lookup	Timeout-protected with one delayed retry

Route access and navigation

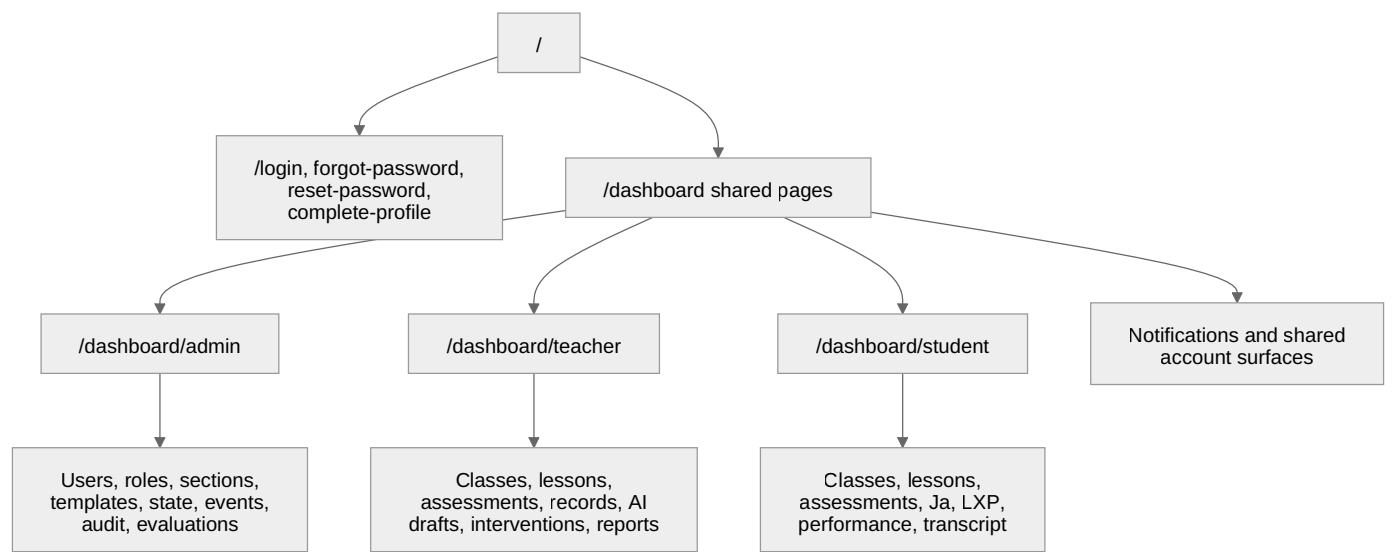


Diagram 18

Role	Scoped prefix	Default post-login route
Admin	/dashboard/admin	/dashboard/admin
Teacher	/dashboard/teacher	/dashboard/teacher/classes
Student	/dashboard/student	/dashboard/student

- Next proxy checks for the refresh cookie before dashboard navigation. It is an early session hint, not a role verifier.
- Dashboard layouts use AuthProvider and dashboard-route-access to prevent a user from navigating into another role prefix.
- Requested post-login paths must be local single-slash paths. External-origin and protocol-relative paths are rejected.
- Backend JWT, role, ownership, and enrollment checks remain authoritative even when a page is hidden.

Complete App Router page catalog

Exhaustive inventory rule. The 103 App Router pages below were extracted from `next-frontend/app/**/page.tsx` at commit `3d0c93e`. A later source change requires regenerating or manually reconciling this chapter.

Route	Role slice	Exports	Local API-path evidence	Source
/complete-profile	PUBLIC/SHARED	metadata, CompleteProfile-Page	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(auth)/complete-profile/page.tsx
/forgot-password	PUBLIC/SHARED	metadata, ForgotPassword-Page	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(auth)/forgot-password/page.tsx
/login	PUBLIC/SHARED	metadata, LoginPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(auth)/login/page.tsx
/reset-password	PUBLIC/SHARED	metadata, ResetPassword-Page	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(auth)/reset-password/page.tsx
/set-initial-password	PUBLIC/SHARED			

Route	Role slice	Exports	Local API-path evidence	Source
		metadata, SetInitialPasswordPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(auth)/set-initial-password/page.tsx
/verify-email	PUBLIC/SHARED	metadata, VerifyEmailPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(auth)/verify-email/page.tsx
/dashboard/admin/access-students	ADMIN	AdminAccessStudentsPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/admin/access-students/page.tsx
/dashboard/admin/announcements	ADMIN	AdminAnnouncementsPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/admin/announcements/page.tsx
/dashboard/admin/audit	ADMIN	AuditTrailPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/admin/audit/page.tsx
/dashboard/admin/calendar	ADMIN	AdminCalendarPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/admin/calendar/page.tsx
/dashboard/admin/chatbot	ADMIN	AdminChatbotPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/admin/chatbot/page.tsx
/dashboard/admin/class-templates/:id/announcements/:announcementKey/edit	ADMIN	AdminTemplateAnnouncementEditPage	/dashboard/admin/class-templates/:param	next-frontend/app/(dashboard)/dashboard/admin/class-templates/[id]/announcements/[announcementKey]/edit/page.tsx
/dashboard/admin/class-templates/:id/announcements/new	ADMIN	AdminTemplateAnnouncementCreatePage	/dashboard/admin/class-templates/:param	next-frontend/app/(dashboard)/dashboard/admin/class-templates/[id]/announcements/new/page.tsx
/dashboard/admin/class-templates/:id/assessments/:assessmentKey/edit	ADMIN	AdminTemplateAssessmentEditorPage	/dashboard/admin/class-templates/:param	next-frontend/app/(dashboard)/dashboard/admin/class-templates/[id]/assessments/[assessmentKey]/edit/page.tsx
/dashboard/admin/class-templates/:id/lessons/:lessonKey/edit	ADMIN	LessonEditorPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/admin/class-templates/[id]/lessons/[lessonKey]/edit/page.tsx
/dashboard/admin/class-templates/:id/modules/:moduleKey	ADMIN	AdminTemplateModuleWorkspacePage	/dashboard/admin/class-templates/:param, /dashboard/admin/class-templates/:param/assessments/:param/edit, /dashboard/admin/class-templates/:param/lessons/:param/edit	next-frontend/app/(dashboard)/dashboard/admin/class-templates/[id]/modules/[moduleKey]/page.tsx
/dashboard/admin/class-templates/:id	ADMIN	ClassTemplateEditorPage	/dashboard/admin/class-templates, /dashboard/admin/class-templates/:param/announcements/:param/edit, /dashboard/admin/class-templates/:param/announcements/new, /dashboard/admin/class-templates/:param/assessments/:param/edit, /dashboard/admin/class-templates/:param/modules/:param	next-frontend/app/(dashboard)/dashboard/admin/class-templates/[id]/page.tsx
/dashboard/admin/class-templates	ADMIN	ClassTemplatesPage	/dashboard/admin/class-templates/:param, /dashboard/admin/classes, /dashboard/admin/classes/new?:param	next-frontend/app/(dashboard)/dashboard/admin/class-templates/page.tsx
/dashboard/admin/classes/:id/edit	ADMIN	EditClassPage	/dashboard/admin/classes	next-frontend/app/(dashboard)/dashboard/admin/classes/[id]/edit/page.tsx
/dashboard/admin/classes/:id	ADMIN	AdminClassDetailPage	/dashboard/admin/calendar, /dashboard/admin/classes, /dashboard/admin/classes/:param/edit, /dashboard/admin/classes/:param/students/add, /dashboard/admin/classes/:param?view=:param, /dashboard/admin/sections/:param/roster, /	next-frontend/app/(dashboard)/dashboard/admin/classes/[id]/page.tsx

Route	Role slice	Exports	Local API-path evidence	Source
			dashboard/admin/users/:param	
/dashboard/admin/classes/:id/students/add	ADMIN	AdminAddStudentsPage	/dashboard/admin/classes/:param, /dashboard/admin/classes/:param/students/add, /dashboard/admin/classes/:param/students/add?:param, /dashboard/admin/users/:param	next-frontend/app/(dashboard)/dashboard/admin/classes/[id]/students/add/page.tsx
/dashboard/admin/classes/new	ADMIN	CreateClassPage	/dashboard/admin/classes	next-frontend/app/(dashboard)/dashboard/admin/classes/new/page.tsx
/dashboard/admin/classes	ADMIN	ClassManagementPage	/, /dashboard/admin/class-templates, /dashboard/admin/classes/:param, /dashboard/admin/classes/:param/edit, /dashboard/admin/classes/new	next-frontend/app/(dashboard)/dashboard/admin/classes/page.tsx
/dashboard/admin/diagnostics	ADMIN	AdminDiagnosticsPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/admin/diagnostics/page.tsx
/dashboard/admin/evaluations	ADMIN	AdminEvaluationsPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/admin/evaluations/page.tsx
/dashboard/admin/library	ADMIN	DashboardAdminLibraryPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/admin/library/page.tsx
/dashboard/admin	ADMIN	AdminDashboardPage	/dashboard/admin/classes, /dashboard/admin/diagnostics, /dashboard/admin/sections, /dashboard/admin/users	next-frontend/app/(dashboard)/dashboard/admin/page.tsx
/dashboard/admin/profile	ADMIN	AdminProfileRoute	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/admin/profile/page.tsx
/dashboard/admin/reports	ADMIN	AdminReportsPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/admin/reports/page.tsx
/dashboard/admin/roster-import	ADMIN	RosterImportPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/admin/roster-import/page.tsx
/dashboard/admin/sections/:id/edit	ADMIN	EditSectionPage	/dashboard/admin/sections, /dashboard/admin/sections/:param/students/add, /dashboard/admin/users/:param	next-frontend/app/(dashboard)/dashboard/admin/sections/[id]/edit/page.tsx
/dashboard/admin/sections/:id/roster	ADMIN	AdminSectionRosterPage	/dashboard/admin/sections, /dashboard/admin/sections/:param/edit, /dashboard/admin/sections/:param/students/add, /dashboard/admin/users/:param	next-frontend/app/(dashboard)/dashboard/admin/sections/[id]/roster/page.tsx
/dashboard/admin/sections/:id/students/add	ADMIN	AddSectionStudentsPage	/dashboard/admin/sections, /dashboard/admin/sections/:param/roster, /dashboard/admin/sections/:param/students/add, /dashboard/admin/sections/:param/students/add?:param, /dashboard/admin/users/:param	next-frontend/app/(dashboard)/dashboard/admin/sections/[id]/students/add/page.tsx
/dashboard/admin/sections/:id/students	ADMIN	StudentsLandingPage	/dashboard/admin/sections/:param/students/add	next-frontend/app/(dashboard)/dashboard/admin/sections/[id]/students/page.tsx
/dashboard/admin/sections/new	ADMIN	CreateSectionPage	/dashboard/admin/sections	next-frontend/app/(dashboard)/dashboard/admin/sections/new/page.tsx
/dashboard/admin/sections	ADMIN	SectionManagementPage	/dashboard/admin/sections/:param/edit, /dashboard/admin/sections/:param/roster, /dashboard/admin/sections/new	next-frontend/app/(dashboard)/dashboard/admin/sections/page.tsx
/dashboard/admin/system-settings	ADMIN	AdminSystemSettingsPage		

Route	Role slice	Exports	Local API-path evidence	Source
			Uses components, hooks, or services; no local path literal extracted	next-frontent/app/(dashboard)/dashboard/admin/system-settings/page.tsx
/dashboard/admin/user-reports	ADMIN	AdminUserReportsPage	Uses components, hooks, or services; no local path literal extracted	next-frontent/app/(dashboard)/dashboard/admin/user-reports/page.tsx
/dashboard/admin/users/:id	ADMIN	AdminUserDetailPage	/dashboard/admin/users	next-frontent/app/(dashboard)/dashboard/admin/users/[id]/page.tsx
/dashboard/admin/users/create	ADMIN	AdminCreateUserPage	/dashboard/admin/users	next-frontent/app/(dashboard)/dashboard/admin/users/create/page.tsx
/dashboard/admin/users	ADMIN	UserManagementPage	/dashboard/admin/users/:param, /dashboard/admin/users/create	next-frontent/app/(dashboard)/dashboard/admin/users/page.tsx
/dashboard/library	PUBLIC/SHARED	DashboardLibraryAliasPage	/dashboard/admin/library, /dashboard/teacher/library	next-frontent/app/(dashboard)/dashboard/library/page.tsx
/dashboard/notifications	PUBLIC/SHARED	NotificationsPage	Uses components, hooks, or services; no local path literal extracted	next-frontent/app/(dashboard)/dashboard/notifications/page.tsx
/dashboard	PUBLIC/SHARED	DashboardPage	Uses components, hooks, or services; no local path literal extracted	next-frontent/app/(dashboard)/dashboard/page.tsx
/dashboard/profile	PUBLIC/SHARED	ProfileRedirectPage	Uses components, hooks, or services; no local path literal extracted	next-frontent/app/(dashboard)/dashboard/profile/page.tsx
/dashboard/student/announcements	STUDENT	StudentAnnouncementsPage	Uses components, hooks, or services; no local path literal extracted	next-frontent/app/(dashboard)/dashboard/student/announcements/page.tsx
/dashboard/student/assessment-history	STUDENT	StudentAssessmentHistoryPage	/dashboard/student/profile	next-frontent/app/(dashboard)/dashboard/student/assessment-history/page.tsx
/dashboard/student/assessments/:id	STUDENT	StudentAssessmentPage	/dashboard/student, /dashboard/student/assessments/:param/results/:param, /dashboard/student/assessments/:param/take, /dashboard/student/assessments/:param/take?attemptId=:param, /dashboard/student/classes/:param?view=assignments	next-frontent/app/(dashboard)/dashboard/student/assessments/[id]/page.tsx
/dashboard/student/assessments/:id/results/:attemptId	STUDENT	StudentAssessmentResultsPage	/dashboard/student/assessments/:param, /dashboard/student/classes	next-frontent/app/(dashboard)/dashboard/student/assessments/[id]/results/[attemptId]/page.tsx
/dashboard/student/assessments/:id/take	STUDENT	StudentAssessmentTakePage	/dashboard/student/assessments/:param, /dashboard/student/assessments/:param/results/:param, /dashboard/student/assessments/:param?view=submitted	next-frontent/app/(dashboard)/dashboard/student/assessments/[id]/take/page.tsx
/dashboard/student/assessments	STUDENT	StudentAssessmentsIndexPage	/dashboard/student/assessment-history	next-frontent/app/(dashboard)/dashboard/student/assessments/page.tsx
/dashboard/student/calendar	STUDENT	StudentCalendarPage	/dashboard/student/classes/:param, /dashboard/student/classes/:param?view=announcements, /dashboard/student/classes/:param?view=calendar, /dashboard/student/courses	next-frontent/app/(dashboard)/dashboard/student/calendar/page.tsx
/dashboard/student/chatbot	STUDENT	StudentChatbotRedirectPage	/dashboard/student/ja?mode=ask	next-frontent/app/(dashboard)/dashboard/student/chatbot/page.tsx
/dashboard/student/classes/:id/modules/:moduleId	STUDENT	StudentModuleDetailPage	/dashboard/student/assessments/:param, /dashboard/student/assessments/:param/results/:param, /dashboard/student/assessments/:param/take?attemptId=:param, /	next-frontent/app/(dashboard)/dashboard/student/classes/[id]/modules/[moduleId]/page.tsx

Route	Role slice	Exports	Local API-path evidence	Source
			dashboard/student/ classes/:param/ modules/:param, /dashboard/ student/classes/:param/mod- ules/:param?assessment- Id=:param, /dashboard/stu- dent/classes/:param/mod- ules/:param? fileItemId=:param, /dash- board/student/ classes/:param/mod- ules/:param?les- sonId=:param, /dashboard/ student/classes/:param? view=modules	
/dashboard/student/ classes/:id	STUDENT	StudentClassDetailPage	/, /dashboard/student/assess- ments/:param? classId=:param, /dashboard/ student/classes/:param/mod- ules/:param, /dashboard/stu- dent/classes/:param? view=:param, /dashboard/ student/courses, /images/JA/ ja_cheer.png, /images/JA/ ja_thinking.png, /images/JA/ ja_wave.png, /images/mod- ules/module-stock- board.svg, /images/modules/ module-stock-library.svg, /im- ages/modules/module-stock- science.svg	next-frontend/app/(dash- board)/dashboard/student/ classes/[id]/page.tsx
/dashboard/student/classes	STUDENT	StudentClassesIndexPath	/dashboard/student/courses	next-frontend/app/(dash- board)/dashboard/student/ classes/page.tsx
/dashboard/student/courses	STUDENT	StudentCoursesPage	/dashboard/student/ classes/:param, /dashboard/ student/classes/:param? view=announcements, /dash- board/student/ classes/:param?view=assign- ments, /dashboard/student/ classes/:param?view=calen- dar, /images/JA/ ja_cheer.png, /images/JA/ ja_thinking.png, /images/JA/ ja_wave.png	next-frontend/app/(dash- board)/dashboard/student/ courses/page.tsx
/dashboard/student/evalu- ations	STUDENT	StudentEvaluationsPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dash- board)/dashboard/student/ evaluations/page.tsx
/dashboard/student/ja	STUDENT	StudentJaPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dash- board)/dashboard/student/ja/ page.tsx
/dashboard/student/les- sons/:id	STUDENT	StudentLessonViewPage	/dashboard/student/, /dash- board/student/ classes/:param/ modules/:param, /dashboard/ student/classes/:param/mod- ules/:param?les- sonId=:param, /dashboard/ student/classes/:param? view=modules	next-frontend/app/(dash- board)/dashboard/student/ lessons/[id]/page.tsx
/dashboard/student/lessons	STUDENT	StudentLessonsIndexPath	/dashboard/student/lxp	next-frontend/app/(dash- board)/dashboard/student/ lessons/page.tsx
/dashboard/student/ lxp/:classId/generated-les- sons/:assignmentId	STUDENT	StudentGeneratedLesson- Page	/dashboard/student/ lxp/:param	next-frontend/app/(dash- board)/dashboard/student/ lxp/[classId]/generated-les- sons/[assignmentId]/page.tsx
/dashboard/student/ lxp/:classId/guided-assess- ment/:assignmentId	STUDENT	StudentGuidedAssessment- Page	/dashboard/student/ lxp/:param?tab=replays, /im- ages/JA/ja_cheer.png, /im- ages/JA/ja_wave.png	next-frontend/app/(dash- board)/dashboard/student/ lxp/[classId]/guided-assess- ment/[assignmentId]/ page.tsx
/dashboard/student/ lxp/:classId	STUDENT	StudentLxpDetailPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dash- board)/dashboard/student/ lxp/[classId]/page.tsx
/dashboard/student/lxp	STUDENT	StudentLxpPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dash- board)/dashboard/student/ lxp/page.tsx

Route	Role slice	Exports	Local API-path evidence	Source
/dashboard/student	STUDENT	StudentDashboardPage	/dashboard/student/announcements, /dashboard/student/assessments/:param, /dashboard/student/classes/:param, /dashboard/student/classes/:param?view=announcements, /dashboard/student/courses, /dashboard/student/lessons/:param, /images/JA/ja_cheer.png, /images/JA/ja_thinking.png, /images/JA/ja_wave.png	next-frontend/app/(dashboard)/dashboard/student/page.tsx
/dashboard/student/performance	STUDENT	StudentPerformancePage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/student/performance/page.tsx
/dashboard/student/profile	STUDENT	StudentProfileRoute	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/student/profile/page.tsx
/dashboard/student/transcript	STUDENT	StudentTranscriptPage	/dashboard/student/profile	next-frontend/app/(dashboard)/dashboard/student/transcript/page.tsx
/dashboard/teacher/announcements	TEACHER	TeacherAnnouncementsPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/teacher/announcements/page.tsx
/dashboard/teacher/assessments/:id/edit	TEACHER	AssessmentEditorPage	/dashboard/teacher/assessments, /dashboard/teacher/classes/:param?view=assignments	next-frontend/app/(dashboard)/dashboard/teacher/assessments/[id]/edit/page.tsx
/dashboard/teacher/assessments/:id	TEACHER	TeacherAssessmentDetailPage	/dashboard/teacher/assessments, /dashboard/teacher/assessments/:param/edit, /dashboard/teacher/classes/:param?view=assignments	next-frontend/app/(dashboard)/dashboard/teacher/assessments/[id]/page.tsx
/dashboard/teacher/assessments/:id/results:attemptId	TEACHER	TeacherAttemptResultsPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/teacher/assessments/[id]/results/[attemptId]/page.tsx
/dashboard/teacher/assessments	TEACHER	TeacherAssessmentsPage	/dashboard/teacher, /dashboard/teacher/assessments/:param, /dashboard/teacher/assessments/:param/edit	next-frontend/app/(dashboard)/dashboard/teacher/assessments/page.tsx
/dashboard/teacher/calendar	TEACHER	TeacherCalendarPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/teacher/calendar/page.tsx
/dashboard/teacher/class-record	TEACHER	ClassRecordPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/teacher/class-record/page.tsx
/dashboard/teacher/classes/:id/ai-draft	TEACHER	TeacherAiDraftQuizPage	/dashboard/teacher/assessments/:param/edit, /dashboard/teacher/classes/:param?view=assignments	next-frontend/app/(dashboard)/dashboard/teacher/classes/[id]/ai-draft/page.tsx
/dashboard/teacher/classes/:id/modules/:moduleId/files/:fileId	TEACHER	TeacherModuleFileEditorPage	/dashboard/teacher/classes/:param/modules/:param, /dashboard/teacher/classes/:param/modules/:param/files/:param	next-frontend/app/(dashboard)/dashboard/teacher/classes/[id]/modules/[moduleId]/files/[fileId]/page.tsx
/dashboard/teacher/classes/:id/modules/:moduleId	TEACHER	TeacherModuleDetailPage	/dashboard/teacher/assessments, /dashboard/teacher/assessments/:param/edit, /dashboard/teacher/assessments/:param/edit?classId=:param&moduleId=:param, /dashboard/teacher/classes/:param, /dashboard/teacher/classes/:param/modules/:param/files/:param, /dashboard/teacher/lessons/:param/edit, /dashboard/teacher/lessons/:param/view?classId=:param&moduleId=:param	next-frontend/app/(dashboard)/dashboard/teacher/classes/[id]/modules/[moduleId]/page.tsx
	TEACHER	TeacherClassDetailPage		

Route	Role slice	Exports	Local API-path evidence	Source
/dashboard/teacher/classes/:id			/, /dashboard/teacher/assessments/:param, /dashboard/teacher/assessments/:param/edit, /dashboard/teacher/calendar?classId=:param, /dashboard/teacher/classes, /dashboard/teacher/classes/:param/ai-draft, /dashboard/teacher/classes/:param/modules/:param, /dashboard/teacher/classes/:param/students/:param, /dashboard/teacher/classes/:param/students/add, /dashboard/teacher/classes/:param?view=:param, /dashboard/teacher/classes/:param?view=assignments, /dashboard/teacher/classes/:param?view=calendar, /dashboard/teacher/extractions/:param	next-frontend/app/(dashboard)/dashboard/teacher/classes/[id]/page.tsx
/dashboard/teacher/classes/:id/students/:studentId	TEACHER	TeacherStudentProfilePage	/dashboard/teacher/classes/:param?view=students	next-frontend/app/(dashboard)/dashboard/teacher/classes/[id]/students/[studentId]/page.tsx
/dashboard/teacher/classes/:id/students/add	TEACHER	TeacherAddStudentsPage	/dashboard/teacher/classes/:param, /dashboard/teacher/classes/:param/students/:param, /dashboard/teacher/classes/:param/students/add, /dashboard/teacher/classes/:param/students/add?:param	next-frontend/app/(dashboard)/dashboard/teacher/classes/[id]/students/add/page.tsx
/dashboard/teacher/classes	TEACHER	TeacherClassesPage	/dashboard/teacher/announcements, /dashboard/teacher/classes/:param, /dashboard/teacher/classes/:param?view=announcements, /dashboard/teacher/classes/:param?view=assignments, /dashboard/teacher/classes/:param?view=modules	next-frontend/app/(dashboard)/dashboard/teacher/classes/page.tsx
/dashboard/teacher/evaluations	TEACHER	TeacherEvaluationsPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/teacher/evaluations/page.tsx
/dashboard/teacher/extractions/:id	TEACHER	ExtractionReviewPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/teacher/extractions/[id]/page.tsx
/dashboard/teacher/interventions/:caselId	TEACHER	TeacherInterventionWorkspacePage	/dashboard/teacher/interventions, /dashboard/teacher/interventions?classId=:param	next-frontend/app/(dashboard)/dashboard/teacher/interventions/[caselId]/page.tsx
/dashboard/teacher/interventions	TEACHER	TeacherInterventionsPage	/dashboard/teacher/interventions/:param, /dashboard/teacher/interventions/:param?classId=:param	next-frontend/app/(dashboard)/dashboard/teacher/interventions/page.tsx
/dashboard/teacher/lessons/:id/edit	TEACHER	LessonEditorPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/teacher/lessons/[id]/edit/page.tsx
/dashboard/teacher/lessons/:id/view	TEACHER	TeacherLessonViewPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/teacher/lessons/[id]/view/page.tsx
/dashboard/teacher/lessons	TEACHER	TeacherLessonsPage	/dashboard/teacher, /dashboard/teacher/lessons/:param/edit	next-frontend/app/(dashboard)/dashboard/teacher/lessons/page.tsx
/dashboard/teacher/library	TEACHER	DashboardTeacherLibraryPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/teacher/library/page.tsx
/dashboard/teacher/modules	TEACHER	TeacherModulesIndexPage	/dashboard/teacher/library	next-frontend/app/(dashboard)/dashboard/teacher/modules/page.tsx
/dashboard/teacher	TEACHER	TeacherDashboardRedirectPage	/dashboard/teacher/classes	next-frontend/app/(dashboard)/dashboard/teacher/page.tsx

Route	Role slice	Exports	Local API-path evidence	Source
/dashboard/teacher/performance	TEACHER	TeacherPerformancePage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/teacher/performance/page.tsx
/dashboard/teacher/profile	TEACHER	TeacherProfileRoute	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/teacher/profile/page.tsx
/dashboard/teacher/reports	TEACHER	TeacherReportsPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/teacher/reports/page.tsx
/dashboard/teacher/sections/:id/roster	TEACHER	SectionRosterPage	/dashboard/teacher/sections, /dashboard/teacher/sections/:param, /dashboard/teacher/sections/:param/students/add	next-frontend/app/(dashboard)/dashboard/teacher/sections/[id]/roster/page.tsx
/dashboard/teacher/sections/:id/students/:studentId	TEACHER	TeacherSectionStudentProfilePage	/dashboard/teacher/sections/:param/roster	next-frontend/app/(dashboard)/dashboard/teacher/sections/[id]/students/[studentId]/page.tsx
/dashboard/teacher/sections/:id/students/add	TEACHER	TeacherAddSectionStudentsPage	/dashboard/teacher/sections/:param/roster, /dashboard/teacher/sections/:param/students/:param, /dashboard/teacher/sections/:param/students/add, /dashboard/teacher/sections/:param/students/add?:param	next-frontend/app/(dashboard)/dashboard/teacher/sections/[id]/students/add/page.tsx
/dashboard/teacher/sections	TEACHER	TeacherSectionsPage	/dashboard/teacher/announcements, /dashboard/teacher/announcements?classId=:param, /dashboard/teacher/sections/:param/roster	next-frontend/app/(dashboard)/dashboard/teacher/sections/page.tsx
/dashboard/theme-test	PUBLIC/SHARED	ThemeTestPage	Uses components, hooks, or services; no local path literal extracted	next-frontend/app/(dashboard)/dashboard/theme-test/page.tsx
/demo	PUBLIC/SHARED	DemoPage	/, /images/JA/ja_cheer.png, /images/JA/ja_sad.png, /images/JA/ja_thinking.png, /images/JA/ja_wave.png	next-frontend/app/demo/page.tsx
/	PUBLIC/SHARED	LandingPage	/, /Gatbg.png, /NexoraHome.png, /dashboard, /demo, /downloads/nexora-student-mobile-release.apk, /taguigpic.png	next-frontend/app/page.tsx

Complete frontend API service catalog

Exhaustive inventory rule. The 30 frontend service modules below were extracted from `next-frontend/src/services/*.ts` at commit `3d0c93e`. A later source change requires regenerating or manually reconciling this chapter.

Service source	Exports	Backend path literals
next-frontend/src/services/academic-state-service.ts	academicStateService	/academic-state/current, /academic-state/impact-preview, /academic-state/transition
next-frontend/src/services/admin-chatbot-service.ts	adminChatbotService	/ai/admin/chat, /ai/admin/history, /ai/admin/sessions/:param, /ai/health, /api/ai/admin/chat
next-frontend/src/services/admin-service.ts	adminService	/admin/audit-logs, /admin/overview, /admin/usage-summary, /api/admin/activity-export:param, /health/live, /health/ready
next-frontend/src/services/ai-service.ts	aiService	/ai/index/classes/:param, /ai/index/classes/:param/status, /ai/mentor/explain, /ai/student/tutor/bootstrap, /ai/student/tutor/session, /ai/student/tutor/session/:param, /ai/student/tutor/session/:param/answers, /ai/student/tutor/session/:param/message, /ai/teacher/classes/:param/policy, /ai/teacher/interventions/:param/jobs, /ai/teacher/interven-

Service source	Exports	Backend path literals
		tions/:param/recommend, /ai/teacher/jobs/:param, /ai/teacher/jobs/:param/result, /ai/teacher/lesson-plans/jobs, /ai/teacher/lesson-plans/jobs/:param/draft, /ai/teacher/quizzes/generate-draft, /ai/teacher/quizzes/jobs, /ai/teacher/quizzes/jobs/:param/apply, /ai/teacher/quizzes/jobs/:param/apply/preview, /ai/teacher/quizzes/jobs/:param/cancel, /ai/teacher/quizzes/jobs/:param/draft, /ai/teacher/quizzes/jobs/:param/retry
next-frontend/src/services/analytics-service.ts	analyticsService	/analytics/admin/overview, /analytics/classes/:param/intervention-outcomes, /analytics/classes/:param/trends, /analytics/teachers/:param/workload
next-frontend/src/services/announcement-service.ts	announcementService	/classes/:param/announcements, /classes/:param/announcements/:param, /classes/:param/announcements/:param/core-release
next-frontend/src/services/assessment-service.ts	assessmentService	/assessments, /assessments/:param, /assessments/:param/all-attempts, /assessments/:param/core-release, /assessments/:param/ongoing-attempt, /assessments/:param/question-analytics, /assessments/:param/return-all, /assessments/:param/rubric-review, /assessments/:param/rubric-source, /assessments/:param/start, /assessments/:param/stats, /assessments/:param/student-attempts, /assessments/:param/submission-file, /assessments/:param/submission-files/:param, /assessments/:param/submissions, /assessments/:param/teacher-attachment, /assessments/:param/teacher-attachment/download, /assessments/:param/unsubmit-file-upload, /assessments/attempts/:param/progress, /assessments/attempts/:param/results, /assessments/attempts/:param/return, /assessments/attempts/:param/submission-file/download, /assessments/attempts/:param/submission-files/:param/download, /assessments/attempts/:param/unreturn, /assessments/attempts/bulk-return, /assessments/attempts/ongoing, /assessments/class/:param, /assessments/options/:param/image, /assessments/questions, /assessments/questions/:param, /assessments/questions/:param/image, /assessments/submit
next-frontend/src/services/class-record-service.ts	classRecordService	/class-record, /class-record/:param, /class-record/:param/final-grades, /class-record/:param/final-grades/:param, /class-record/:param/finalize, /class-record/:param/preview-grades, /class-record/:param/reopen, /class-record/:param/reports/class-average, /class-record/:param/reports/distribution, /class-record/:param/reports/intervention, /class-record/:param/spreadsheet, /class-record/adviser/section/:param, /class-record/by-class/:param, /class-record/by-class/:param/slot-overview, /class-record/items/:param, /class-record/items/:param/scores, /class-record/items/:param/scores/bulk, /class-record/items/:param/sync-scores
next-frontend/src/services/class-service.ts	classService	/classes, /classes/:param, /classes/:param/banner, /classes/:param/candidates, /classes/:param/enrollments, /classes/:param/enrollments/:param, /classes/:param/hide, /classes/:param/presentation, /classes/:param/purge, /classes/:param/student-presentation, /classes/:param/students/:param/overview, /classes/:param/students/:param/profile, /classes/:param/students/masterlist, /classes/:param/toggle-status, /classes/:param/unhide, /classes/all, /classes/bulk/lifecycle, /classes/section/:param, /classes/student/:param, /classes/student/:param/preferences/presentation, /classes/student/:param/preferences/view, /classes/subject/:param, /classes/teacher/:param
next-frontend/src/services/class-template-service.ts	classTemplateService	/class-templates, /class-templates/:param, /class-templates/:param/assessment-images, /class-templates/:param/content, /class-templates/:param/engine-export, /class-templates/:param/publish, /class-templates/compatible, /class-templates/engine-import, /class-templates/engine-import/validate
next-frontend/src/services/dashboard-service.ts	dashboardService	/admin/dashboard/stats, /teacher/assessments, /teacher/classes, /teacher/lessons
next-frontend/src/services/demo-ai-plan-service.ts	generateDemoAiPlan	/api/ai/demo/intervention-plan
	discussionBoardService	

Service source	Exports	Backend path literals
next-frontend/src/services/discussion-board-service.ts		/classes/:param/discussion-threads, /classes/:param/discussion-threads/:param, /classes/:param/discussion-threads/:param/close, /classes/:param/discussion-threads/:param/comments, /classes/:param/discussion-threads/:param/comments/:param, /classes/:param/discussion-threads/:param/comments/:param/reaction, /classes/:param/discussion-threads/:param/comments/:param/report, /classes/:param/discussion-threads/:param/comments/uploads, /classes/:param/discussion-threads/:param/publish, /classes/:param/discussion-threads/:param/reopen, /classes/:param/discussion-threads/uploads
next-frontend/src/services/extraction-service.ts	extractionService	/ai/extract-module, /ai/extractions, /ai/extractions/:param, /ai/extractions/:param/apply, /ai/extractions/:param/apply/preview, /ai/extractions/:param/cancel, /ai/extractions/:param/retry, /ai/extractions/:param/status
next-frontend/src/services/file-service.ts	fileService	/files, /files/:param, /files/:param/download, /files/:param/index/retry, /files/folders, /files/folders/:param, /files/storage-summary, /files/upload
next-frontend/src/services/health-service.ts	FRONTEND_APP_VERSION, healthService	/ai/health, /health/live, /health/ready
next-frontend/src/services/index.ts	No named export extracted	No static API path literal extracted
next-frontend/src/services/ja-service.ts	jaService	/ai/student/ja/ask/bootstrap, /ai/student/ja/ask/threads, /ai/student/ja/ask/threads/:param, /ai/student/ja/ask/threads/:param/messages, /ai/student/ja/hub, /ai/student/ja/practice/bootstrap, /ai/student/ja/practice/sessions, /ai/student/ja/practice/sessions/:param, /ai/student/ja/practice/sessions/:param/complete, /ai/student/ja/practice/sessions/:param/events, /ai/student/ja/practice/sessions/:param/responses, /ai/student/ja/review/bootstrap, /ai/student/ja/review/sessions, /ai/student/ja/review/sessions/:param, /ai/student/ja/review/sessions/:param/complete, /ai/student/ja/review/sessions/:param/events, /ai/student/ja/review/sessions/:param/responses
next-frontend/src/services/lesson-service.ts	lessonService	/lessons, /lessons/:param, /lessons/:param/blocks, /lessons/:param/complete, /lessons/:param/completion-status, /lessons/:param/publish, /lessons/:param/reorder-blocks, /lessons/:param/versions, /lessons/:param/versions/:param/restore, /lessons/blocks/:param, /lessons/class/:param, /lessons/class/:param/bulk-delete, /lessons/class/:param/bulk-status, /lessons/class/:param/completed, /lessons/class/:param/reorder
next-frontend/src/services/lxp-service.ts	lxpService	/lxp/evaluations, /lxp/me/eligibility, /lxp/me/overview/:param, /lxp/me/playlist/:param, /lxp/me/playlist/:param/checkpoints/:param/complete, /lxp/me/playlist/:param/generated-lessons/:param, /lxp/me/playlist/:param/guided-assessments/:param/progress, /lxp/me/playlist/:param/guided-assessments/:param/result, /lxp/me/playlist/:param/guided-assessments/:param/start, /lxp/me/playlist/:param/guided-assessments/:param/submit, /lxp/me/system-evaluations, /lxp/me/system-evaluations/:param/submit, /lxp/me/teacher-evaluations, /lxp/system-evaluation-campaigns, /lxp/system-evaluation-campaigns/:param/status, /lxp/teacher/classes/:param/interventions, /lxp/teacher/classes/:param/interventions/history, /lxp/teacher/classes/:param/reports/summary, /lxp/teacher/evaluations/summary, /lxp/teacher/interventions/:param, /lxp/teacher/interventions/:param/activate, /lxp/teacher/interventions/:param/assign, /lxp/teacher/interventions/:param/detail, /lxp/teacher/interventions/:param/generated-content/approve, /lxp/teacher/interventions/:param/generated-content/reject, /lxp/teacher/interventions/:param/regenerate, /lxp/teacher/interventions/:param/resolve, /lxp/teacher/interventions/pending-count
next-frontend/src/services/module-service.ts	moduleService	/modules, /modules/:param, /modules/:param/core-release, /modules/:param/cover, /modules/:param/grading-scale, /modules/:param/sections, /modules/:param/sections/reorder, /modules/class/:param, /modules/class/:param/:param, /modules/class/:param/reorder, /modules/items/:param, /modules/items/:param/core-release, /modules/items/:param/file/download, /modules/sections/:param, /modules/sections/:param/items, /modules/sections/:param/items/reorder

Service source	Exports	Backend path literals
next-frontend/src/services/notification-service.ts	normalizeNotification, notificationService	/notifications, /notifications/:param/read, /notifications/read-all, /notifications/unread-count
next-frontend/src/services/performance-service.ts	performanceService	/performance/admin/analytics, /performance/analysis/jobs/:param, /performance/analysis/jobs/:param/result, /performance/classes/:param/analysis/jobs, /performance/classes/:param/at-risk, /performance/classes/:param/diagnostics, /performance/classes/:param/intervention-quiz-comparison, /performance/classes/:param/logs, /performance/classes/:param/recompute, /performance/classes/:param/summary, /performance/students/me/summary
next-frontend/src/services/profile-service.ts	profileService	/profiles/me, /profiles/me/academic-summary, /profiles/me/assessment-history, /profiles/me/avatar, /profiles/me/transcript, /profiles/update/:param
next-frontend/src/services/report-service.ts	reportService	/reports/assessment-summary, /reports/class-enrollment, /reports/intervention-participation, /reports/student-master-list, /reports/student-performance, /reports/system-usage
next-frontend/src/services/roster-import-service.ts	rosterImportService	/roster-import/:param/commit, /roster-import/:param/pending, /roster-import/:param/preview, /roster-import/pending/:param/resolve
next-frontend/src/services/school-event-service.ts	schoolEventService	/school-events, /school-events/:param
next-frontend/src/services/section-service.ts	sectionService	/sections/:param, /sections/:param/banner, /sections/:param/candidates, /sections/:param/hide, /sections/:param/presentation, /sections/:param/restore, /sections/:param/roster, /sections/:param/roster/:param, /sections/:param/schedule, /sections/:param/students/:param/profile, /sections/:param/unhide, /sections/access-students/fail, /sections/access-students/finalize-grades, /sections/access-students/move-up, /sections/access-students/overview, /sections/access-students/target-sections, /sections/all, /sections/bulk/lifecycle, /sections/create, /sections/delete/:param, /sections/my, /sections/permanent/:param, /sections/update/:param
next-frontend/src/services/teacher-profile-service.ts	teacherProfileService	/teacher-profiles/:param, /teacher-profiles/me, /teacher-profiles/me/avatar
next-frontend/src/services/user-service.ts	userService	/users/:param, /users/:param/export, /users/:param/purge, /users/:param/reactivate, /users/:param/reset-password, /users/:param/soft-delete, /users/:param/suspend, /users/all, /users/bulk/lifecycle, /users/create, /users/delete/:param, /users/reports/monitoring, /users/update/:param

Service-layer rules

- Components and pages should use the owning service or query hook instead of constructing a second Axios client.
- Service methods normalize the backend response contract and keep route parameters, query parameters, and payload types in one place.
- Dynamic `:param` entries in the table represent a template expression extracted from source, not a literal backend route segment.
- A path already beginning with `/api` must be reviewed carefully because the shared client base is also `/api`; intentional direct fetches and Axios calls have different joining behavior.

React Query policy and invalidation

Concern	Current policy	Maintenance rule
Staleness	30 seconds by default	Override only when the domain has a clearer freshness requirement.
Retries	Two retries with exponential delay, 30-second cap, and random jitter	Do not retry a known validation or authorization failure through a query wrapper.
Window focus	No automatic refetch	Mutations and notification events must invalidate affected keys deliberately.

Concern	Current policy	Maintenance rule
QueryClient lifetime	One client created with useState per provider instance	Never create a new client during each render.
Mutation reconciliation	Service or hook-specific invalidateQueries or cache update	Invalidate list, detail, counts, and role dashboard summaries affected by the mutation.
Devtools	Mounted closed by default	Exclude sensitive values from query keys and cached debug content.

Real-time notification behavior

- NotificationProvider fetches the durable inbox and connects Socket.IO after authentication.
- Disconnected polling runs every 15 seconds; connected fallback polling runs every 30 seconds.
- Extraction status polling runs every 10 seconds for tracked nonterminal extraction work.
- Student pending-task and intervention reminders are synthesized on a 90-second cadence from backend data and use route-aware display logic.
- Socket delivery accelerates user feedback; the durable notifications table and fetch endpoint remain the recovery source after reconnect.

Reusable UI component register

Component source	Ownership
next-frontend/src/components/ui/avatar.tsx	Reusable primitive under src/components/ui; feature-specific composition belongs outside this directory.
next-frontend/src/components/ui/badge.tsx	Reusable primitive under src/components/ui; feature-specific composition belongs outside this directory.
next-frontend/src/components/ui/button.tsx	Reusable primitive under src/components/ui; feature-specific composition belongs outside this directory.
next-frontend/src/components/ui/card.tsx	Reusable primitive under src/components/ui; feature-specific composition belongs outside this directory.
next-frontend/src/components/ui/dialog.tsx	Reusable primitive under src/components/ui; feature-specific composition belongs outside this directory.
next-frontend/src/components/ui/dropdown-menu.tsx	Reusable primitive under src/components/ui; feature-specific composition belongs outside this directory.
next-frontend/src/components/ui/input.tsx	Reusable primitive under src/components/ui; feature-specific composition belongs outside this directory.
next-frontend/src/components/ui/label.tsx	Reusable primitive under src/components/ui; feature-specific composition belongs outside this directory.
next-frontend/src/components/ui/popover.tsx	Reusable primitive under src/components/ui; feature-specific composition belongs outside this directory.
next-frontend/src/components/ui/progress.tsx	Reusable primitive under src/components/ui; feature-specific composition belongs outside this directory.
next-frontend/src/components/ui/select.tsx	Reusable primitive under src/components/ui; feature-specific composition belongs outside this directory.
next-frontend/src/components/ui/separator.tsx	Reusable primitive under src/components/ui; feature-specific composition belongs outside this directory.
next-frontend/src/components/ui/sheet.tsx	Reusable primitive under src/components/ui; feature-specific composition belongs outside this directory.
next-frontend/src/components/ui/skeleton.tsx	Reusable primitive under src/components/ui; feature-specific composition belongs outside this directory.
next-frontend/src/components/ui/sonner.tsx	Reusable primitive under src/components/ui; feature-specific composition belongs outside this directory.
next-frontend/src/components/ui/table.tsx	Reusable primitive under src/components/ui; feature-specific composition belongs outside this directory.
next-frontend/src/components/ui/tabs.tsx	Reusable primitive under src/components/ui; feature-specific composition belongs outside this directory.
next-frontend/src/components/ui/textarea.tsx	Reusable primitive under src/components/ui; feature-specific composition belongs outside this directory.

Component source	Ownership
next-frontend/src/components/ui/tooltip.tsx	Reusable primitive under src/components/ui; feature-specific composition belongs outside this directory.

Design token dictionary

The style inventory contains 459 CSS custom-property declarations and 408 distinct name/value pairs across base and theme selectors. Repeated declarations are theme or selector overrides; all distinct pairs are listed.

Custom property	Declared value
--color-background	#ffffff
--color-foreground	#000000
--color-primary	#dc2626
--color-primary-foreground	#ffffff
--color-secondary	#374151
--color-secondary-foreground	#ffffff
--color-muted	#f3f4f6
--color-muted-foreground	#6b7280
--color-accent	#f59e0b
--color-accent-foreground	#000000
--color-destructive	#dc2626
--color-destructive-foreground	#ffffff
--color-border	#e5e7eb
--color-input	#ffffff
--color-ring	#dc2626
--color-card	#ffffff
--color-card-foreground	#000000
--color-popover	#ffffff
--color-popover-foreground	#000000
--color-sidebar	#ffffff
--color-sidebar-foreground	#000000
--color-sidebar-border	#e5e7eb
--color-sidebar-accent	#f3f4f6
--color-sidebar-accent-foreground	#000000
--color-sidebar-ring	#dc2626
--radius-sm	0.25rem
--radius-md	0.375rem
--radius-lg	0.5rem
--radius-xl	0.75rem
--color-foreground	#111827
--color-accent	#f3f4f6
--color-accent-foreground	#111827
--color-card-foreground	#111827
--color-popover-foreground	#111827
--color-sidebar-foreground	#111827
--color-sidebar-accent-foreground	#111827

Custom property	Declared value
--student-shell	#fffaf9
--student-page-gradient	radial-gradient(circle at 12% 0%, rgba(220, 38, 38, 0.14), transparent 34%), linear-gradient(180deg, #fffaf9 0%, #fff3f1 52%, #ffdfc 100%)
--student-topbar-gradient	linear-gradient(135deg, rgba(255, 255, 255, 0.96), rgba(255, 240, 240, 0.92))
--student-sidebar-gradient	linear-gradient(180deg, rgba(255, 255, 255, 0.98), rgba(255, 244, 244, 0.96))
--student-surface	#fff8f7
--student-surface-soft	#fff0ee
--student-elevated	#ffffff
--student-glass	rgba(255, 255, 255, 0.72)
--student-outline	rgba(248, 113, 113, 0.22)
--student-outline-strong	rgba(239, 68, 68, 0.34)
--student-text-strong	#0f172a
--student-text-muted	#64748b
--student-accent	#ef4444
--student-accent-soft	rgba(239, 68, 68, 0.12)
--student-accent-soft-strong	rgba(239, 68, 68, 0.28)
--student-accent-contrast	#ffffff
--student-hero-stripe	rgba(239, 68, 68, 0.08)
--student-progress-track	rgba(226, 232, 240, 0.9)
--student-progress-fill	linear-gradient(90deg, #ef4444 0%, #fb7185 100%)
--student-success-bg	#ecfdf5
--student-success-border	#a7f3d0
--student-success-text	#047857
--student-danger-bg	#fff1f2
--student-danger-border	#fecdd3
--student-danger-text	#be123c
--student-shadow	0 18px 44px -28px rgba(127, 29, 29, 0.38)
--student-shadow-hover	0 24px 60px -28px rgba(127, 29, 29, 0.46)
--student-radius	18px
--student-motion-fast	0.18s
--student-motion-normal	0.28s
--teacher-page-gradient	radial-gradient(circle at 10% 0%, rgba(220, 38, 38, 0.14), transparent 34%), linear-gradient(180deg, #fffaf9 0%, #fff4f3 54%, #ffdfc 100%)
--teacher-surface	rgba(255, 255, 255, 0.92)
--teacher-surface-soft	rgba(255, 244, 244, 0.92)
--teacher-surface-strong	#fff1f2
--teacher-outline	rgba(248, 113, 113, 0.2)
--teacher-outline-strong	rgba(239, 68, 68, 0.3)
--teacher-text-strong	#0f172a
--teacher-text-muted	#64748b
--teacher-accent	#dc2626
--teacher-accent-strong	#ef4444
--teacher-accent-warm	#fb7185
--teacher-glow	0 22px 52px -34px rgba(127, 29, 29, 0.28)
--teacher-glow-hover	0 28px 64px -34px rgba(127, 29, 29, 0.34)
--teacher-shadow	0 22px 52px -34px rgba(127, 29, 29, 0.28)
--teacher-radius	24px
--teacher-motion-fast	0.18s

Custom property	Declared value
--teacher-motion-normal	0.28s
--student-shell	#09111f
--student-page-gradient	radial-gradient(circle at 14% 0%, rgba(56, 189, 248, 0.18), transparent 34%), linear-gradient(180deg, #09111f 0%, #0f172a 52%, #060b16 100%)
--student-topbar-gradient	linear-gradient(135deg, rgba(8, 15, 28, 0.9), rgba(17, 24, 39, 0.88))
--student-sidebar-gradient	linear-gradient(180deg, rgba(8, 15, 28, 0.98), rgba(15, 23, 42, 0.94))
--student-surface	#0f172a
--student-surface-soft	#162033
--student-elevated	#111c31
--student-glass	rgba(9, 17, 31, 0.72)
--student-outline	rgba(96, 165, 250, 0.2)
--student-outline-strong	rgba(56, 189, 248, 0.32)
--student-text-strong	#e5eefc
--student-text-muted	#94a3b8
--student-accent	#38bdf8
--student-accent-soft	rgba(56, 189, 248, 0.14)
--student-accent-soft-strong	rgba(56, 189, 248, 0.28)
--student-accent-contrast	#08111f
--student-hero-stripe	rgba(56, 189, 248, 0.1)
--student-progress-track	rgba(30, 41, 59, 0.92)
--student-progress-fill	linear-gradient(90deg, #38bdf8 0%, #2dd4bf 100%)
--student-success-bg	rgba(6, 95, 70, 0.24)
--student-success-border	rgba(16, 185, 129, 0.34)
--student-success-text	#86efac
--student-danger-bg	rgba(127, 29, 29, 0.2)
--student-danger-border	rgba(251, 113, 133, 0.24)
--student-danger-text	#fecdd3
--student-shadow	0 24px 55px -30px rgba(2, 6, 23, 0.74)
--student-shadow-hover	0 30px 70px -30px rgba(2, 6, 23, 0.82)
--teacher-page-gradient	radial-gradient(circle at 12% 0%, rgba(34, 211, 238, 0.16), transparent 28%), radial-gradient(circle at 100% 12%, rgba(251, 191, 36, 0.14), transparent 28%), linear-gradient(180deg, #08111d 0%, #0d1725 48%, #111827 100%)
--teacher-surface	rgba(8, 18, 33, 0.9)
--teacher-surface-soft	rgba(13, 31, 54, 0.88)
--teacher-surface-strong	#102034
--teacher-outline	rgba(34, 211, 238, 0.16)
--teacher-outline-strong	rgba(56, 189, 248, 0.28)
--teacher-text-strong	#e6f1ff
--teacher-text-muted	rgba(255, 255, 255, 0.88)
--teacher-accent	#38bdf8
--teacher-accent-strong	#2dd4bf
--teacher-accent-warm	#fbbf24
--teacher-glow	0 26px 66px -34px rgba(2, 6, 23, 0.76)
--teacher-glow-hover	0 32px 82px -34px rgba(34, 211, 238, 0.18)
--student-shell	#f2fbff
--student-page-gradient	radial-gradient(circle at 12% 0%, rgba(14, 165, 233, 0.16), transparent 32%), linear-gradient(180deg, #f2fbff 0%, #ecfeff 54%, #f7fffd 100%)
--student-topbar-gradient	linear-gradient(135deg, rgba(255, 255, 255, 0.9), rgba(237, 253, 255, 0.92))
--student-sidebar-gradient	linear-gradient(180deg, rgba(248, 254, 255, 0.98), rgba(237, 252, 255, 0.94))

Custom property	Declared value
--student-surface	#f8feff
--student-surface-soft	#e8f9ff
--student-outline	rgba(14, 165, 233, 0.18)
--student-outline-strong	rgba(13, 148, 136, 0.26)
--student-text-muted	#4b6477
--student-accent	#0f766e
--student-accent-soft	rgba(15, 118, 110, 0.12)
--student-accent-soft-strong	rgba(14, 165, 233, 0.26)
--student-accent-contrast	#f8feff
--student-hero-stripe	rgba(14, 165, 233, 0.1)
--student-progress-track	rgba(207, 250, 254, 0.9)
--student-progress-fill	linear-gradient(90deg, #0ea5e9 0%, #14b8a6 100%)
--student-success-bg	#ecfeff
--student-success-border	#99f6e4
--student-success-text	#0f766e
--student-danger-bg	#fff7ed
--student-danger-border	#fdb74
--student-danger-text	#c2410c
--student-shadow	0 22px 48px -30px rgba(8, 47, 73, 0.3)
--student-shadow-hover	0 28px 66px -28px rgba(8, 47, 73, 0.36)
--student-shell	#0a0a0a
--student-page-gradient	radial-gradient(circle at 16% 0%, rgba(168, 85, 247, 0.22), transparent 36%), linear-gradient(180deg, #0a0a0a 0%, #111111 56%, #080808 100%)
--student-topbar-gradient	linear-gradient(135deg, rgba(10, 10, 10, 0.95), rgba(20, 20, 20, 0.92))
--student-sidebar-gradient	linear-gradient(180deg, rgba(10, 10, 10, 0.98), rgba(17, 17, 17, 0.96))
--student-surface	#121212
--student-surface-soft	#1a1a1a
--student-elevated	#1e1e1e
--student-glass	rgba(20, 20, 20, 0.75)
--student-outline	rgba(168, 85, 247, 0.22)
--student-outline-strong	rgba(147, 51, 234, 0.36)
--student-text-strong	#f1f5f9
--student-accent	#a855f7
--student-accent-soft	rgba(168, 85, 247, 0.16)
--student-accent-soft-strong	rgba(147, 51, 234, 0.30)
--student-accent-contrast	#0a0a0a
--student-hero-stripe	rgba(168, 85, 247, 0.12)
--student-progress-track	rgba(30, 30, 30, 0.94)
--student-progress-fill	linear-gradient(90deg, #a855f7 0%, #6366f1 100%)
--student-success-bg	rgba(34, 197, 94, 0.18)
--student-success-border	rgba(34, 197, 94, 0.32)
--student-danger-bg	rgba(239, 68, 68, 0.18)
--student-danger-border	rgba(239, 68, 68, 0.32)
--student-danger-text	#fecaca
--student-shadow	0 26px 60px -32px rgba(0, 0, 0, 0.85)
--student-shadow-hover	0 32px 75px -32px rgba(0, 0, 0, 0.90)
--student-shell	#fff8f0
--student-page-gradient	

Custom property	Declared value
	radial-gradient(circle at 10% 0%, rgba(255, 107, 107, 0.18), transparent 30%), linear-gradient(180deg, #fff8f0 0%, #fff0e6 50%, #fff7f0 100%)
--student-topbar-gradient	linear-gradient(135deg, rgba(255, 248, 240, 0.95), rgba(255, 240, 230, 0.92))
--student-sidebar-gradient	linear-gradient(180deg, rgba(255, 250, 245, 0.98), rgba(255, 245, 238, 0.96))
--student-surface	#fffaf3
--student-surface-soft	#fff3e6
--student-glass	rgba(255, 255, 255, 0.80)
--student-outline	rgba(255, 107, 107, 0.20)
--student-outline-strong	rgba(255, 107, 107, 0.34)
--student-text-strong	#2d3748
--student-text-muted	#718096
--student-accent	#ff6b6b
--student-accent-soft	rgba(255, 107, 107, 0.14)
--student-accent-soft-strong	rgba(255, 107, 107, 0.28)
--student-accent-contrast	#fffaf3
--student-hero-stripe	rgba(255, 107, 107, 0.12)
--student-progress-track	rgba(255, 229, 229, 0.92)
--student-progress-fill	linear-gradient(90deg, #ff6b6b 0%, #f56565 100%)
--student-success-bg	#e6fffa
--student-success-border	#9ae6b4
--student-success-text	#2f855a
--student-danger-bg	#fed7d7
--student-danger-border	#feb2b2
--student-danger-text	#c53030
--student-shadow	0 20px 40px -24px rgba(255, 107, 107, 0.25)
--student-shadow-hover	0 24px 50px -24px rgba(255, 107, 107, 0.35)
--student-shell	#fdf6ff
--student-page-gradient	radial-gradient(circle at 12% 0%, rgba(79, 70, 229, 0.16), transparent 32%), linear-gradient(180deg, #fdf6ff 0%, #faf5ff 52%, #f7f0ff 100%)
--student-topbar-gradient	linear-gradient(135deg, rgba(253, 246, 255, 0.95), rgba(250, 245, 255, 0.92))
--student-sidebar-gradient	linear-gradient(180deg, rgba(253, 246, 255, 0.98), rgba(250, 245, 255, 0.96))
--student-surface	#faf5ff
--student-surface-soft	#f7f0ff
--student-glass	rgba(255, 255, 255, 0.78)
--student-outline	rgba(79, 70, 229, 0.18)
--student-outline-strong	rgba(120, 113, 255, 0.30)
--student-text-strong	#1e293b
--student-accent	#f59e0b
--student-accent-soft	rgba(245, 158, 11, 0.12)
--student-accent-soft-strong	rgba(245, 158, 11, 0.26)
--student-accent-contrast	#faf5ff
--student-hero-stripe	rgba(79, 70, 229, 0.10)
--student-progress-track	rgba(238, 236, 255, 0.90)
--student-progress-fill	linear-gradient(90deg, #f59e0b 0%, #eab308 100%)
--student-success-border	#bbf7d0
--student-success-text	#166534
--student-danger-border	#fecaca
--student-danger-text	#b91c1c

Custom property	Declared value
--student-shadow	0 22px 45px -28px rgba(79, 70, 229, 0.22)
--student-shadow-hover	0 26px 55px -28px rgba(79, 70, 229, 0.32)
--student-shell	#fff7ed
--student-page-gradient	radial-gradient(circle at 14% 0%, rgba(249, 115, 22, 0.18), transparent 34%), linear-gradient(180deg, #fff7ed 0%, #fff3e0 54%, #fff0db 100%)
--student-topbar-gradient	linear-gradient(135deg, rgba(255, 247, 237, 0.95), rgba(255, 243, 224, 0.92))
--student-sidebar-gradient	linear-gradient(180deg, rgba(255, 247, 237, 0.98), rgba(255, 243, 224, 0.96))
--student-surface	#fff3e0
--student-surface-soft	#ffeaa7
--student-glass	rgba(255, 255, 255, 0.76)
--student-outline	rgba(249, 115, 22, 0.20)
--student-outline-strong	rgba(234, 88, 12, 0.32)
--student-text-strong	#1f2937
--student-text-muted	#6b7280
--student-accent	#f97316
--student-accent-soft	rgba(249, 115, 22, 0.14)
--student-accent-soft-strong	rgba(234, 88, 12, 0.28)
--student-accent-contrast	#fff3e0
--student-hero-stripe	rgba(249, 115, 22, 0.12)
--student-progress-track	rgba(254, 215, 170, 0.92)
--student-progress-fill	linear-gradient(90deg, #f97316 0%, #f59e0b 100%)
--student-shadow	0 24px 50px -30px rgba(249, 115, 22, 0.28)
--student-shadow-hover	0 30px 65px -30px rgba(249, 115, 22, 0.38)
--student-shell	#0a1628
--student-page-gradient	radial-gradient(circle at 16% 0%, rgba(0, 217, 255, 0.2), transparent 36%), linear-gradient(180deg, #0a1628 0%, #0d1f36 52%, #081422 100%)
--student-topbar-gradient	linear-gradient(135deg, rgba(10, 22, 40, 0.95), rgba(13, 31, 54, 0.92))
--student-sidebar-gradient	linear-gradient(180deg, rgba(10, 22, 40, 0.98), rgba(13, 31, 54, 0.96))
--student-surface	#0f2438
--student-surface-soft	#132d45
--student-elevated	#113456
--student-glass	rgba(10, 22, 40, 0.72)
--student-outline	rgba(0, 217, 255, 0.18)
--student-outline-strong	rgba(0, 255, 136, 0.26)
--student-text-strong	#e0f7ff
--student-text-muted	#7aa3b8
--student-accent	#00d9ff
--student-accent-soft	rgba(0, 217, 255, 0.14)
--student-accent-soft-strong	rgba(0, 255, 136, 0.28)
--student-accent-contrast	#0a1628
--student-hero-stripe	rgba(0, 217, 255, 0.12)
--student-progress-track	rgba(15, 36, 56, 0.92)
--student-progress-fill	linear-gradient(90deg, #00d9ff 0%, #00ff88 100%)
--student-success-bg	rgba(0, 255, 136, 0.18)
--student-success-border	rgba(0, 255, 136, 0.32)
--student-success-text	#7effcc
--student-danger-bg	rgba(255, 71, 87, 0.18)
--student-danger-border	rgba(255, 71, 87, 0.32)

Custom property	Declared value
--student-danger-text	#ff9aa2
--student-shadow-hover	0 32px 75px -32px rgba(0, 217, 255, 0.25)
--student-shell	#f5ede0
--student-page-gradient	radial-gradient(circle at 12% 0%, rgba(139, 115, 85, 0.16), transparent 32%), linear-gradient(180deg, #f5ede0 0%, #ede5d8 52%, #f9f3eb 100%)
--student-topbar-gradient	linear-gradient(135deg, rgba(245, 237, 224, 0.95), rgba(237, 229, 216, 0.92))
--student-sidebar-gradient	linear-gradient(180deg, rgba(245, 237, 224, 0.98), rgba(237, 229, 216, 0.96))
--student-surface	#f1e8dc
--student-surface-soft	#e8dcd0
--student-elevated	#faf6f0
--student-glass	rgba(245, 237, 224, 0.76)
--student-outline	rgba(139, 115, 85, 0.18)
--student-outline-strong	rgba(120, 90, 60, 0.28)
--student-text-strong	#3d3d3d
--student-text-muted	#6b6055
--student-accent	#8b7355
--student-accent-soft	rgba(139, 115, 85, 0.12)
--student-accent-soft-strong	rgba(139, 115, 85, 0.26)
--student-accent-contrast	#f5ede0
--student-hero-stripe	rgba(139, 115, 85, 0.10)
--student-progress-track	rgba(224, 214, 202, 0.90)
--student-progress-fill	linear-gradient(90deg, #8b7355 0%, #a0826d 100%)
--student-success-bg	#e8f5e9
--student-success-border	#c8e6c9
--student-success-text	#2e7d32
--student-danger-bg	#ffebee
--student-danger-border	#ffcdd2
--student-danger-text	#c62828
--student-shadow	0 22px 48px -30px rgba(61, 61, 61, 0.3)
--student-shadow-hover	0 28px 66px -28px rgba(139, 115, 85, 0.36)
--student-surface	#fff7f7
--student-surface-soft	#fef2f2
--student-border-strong	#fecaca
--student-text-soft	#6b7280
--student-shadow	0 10px 24px -16px rgba(127, 29, 29, 0.35)
--student-shadow-hover	0 16px 36px -18px rgba(127, 29, 29, 0.42)
--student-radius	16px
--admin-accent	#f70a10
--admin-accent-strong	#d9070e
--admin-accent-warm	#4f86ff
--admin-accent-soft	#fff0f0
--admin-text-strong	#13223a
--admin-text-muted	#8da0bf
--admin-outline	#e6edf7
--admin-outline-strong	#d5dfec
--admin-surface	#ffffff
--admin-surface-soft	#f6f8fc
--admin-shell-bg	#f4f6fb

Custom property	Declared value
--admin-sidebar-bg	#151e33
--admin-sidebar-border	rgba(255, 255, 255, 0.08)
--admin-sidebar-hover	rgba(255, 255, 255, 0.06)
--admin-sidebar-active	#f70a10
--admin-sidebar-text	#8ea0bc
--admin-sidebar-text-strong	#ffffff
--admin-shadow	0 20px 36px -30px rgba(19, 34, 58, 0.22)
--teacher-page-gradient	#f4f6fb
--teacher-surface	#ffffff
--teacher-surface-soft	#f6f8fc
--teacher-surface-strong	#eff4fb
--teacher-outline	#e6edf7
--teacher-outline-strong	#d5dfec
--teacher-text-strong	#13223a
--teacher-text-muted	#8da0bf
--teacher-accent	var(--admin-accent)
--teacher-accent-strong	var(--admin-accent-strong)
--teacher-accent-warm	var(--admin-accent-warm)
--teacher-glow	var(--admin-shadow)
--teacher-glow-hover	var(--admin-shadow)
--teacher-shadow	var(--admin-shadow)
--active	:before { background: #16a34a
--archived	:before, .admin-status-pill--suspended::before { background: #7b90b3
--pending	:before { background: #d97706
--landing-ink	var(--student-text-strong)
--landing-ink-soft	#334155
--landing-surface	rgba(255, 255, 255, 0.8)
--landing-surface-strong	rgba(255, 244, 244, 0.92)
--landing-border	rgba(239, 68, 68, 0.14)
--landing-border-strong	rgba(239, 68, 68, 0.24)
--landing-text-muted	#5f6b7c
--teacher-page-gradient	linear-gradient(180deg, #f7f8fc 0%, #f3f6fb 52%, #f7f9fd 100%)
--teacher-surface-soft	#f7f9fd
--teacher-surface-strong	#eef3fb
--teacher-outline	#e4eaf4
--teacher-outline-strong	#cad6e8
--teacher-text-muted	#5d6b82
--teacher-accent	#0f172a
--teacher-accent-strong	#1e293b
--teacher-accent-warm	#3b82f6
--teacher-glow	0 16px 36px -28px rgba(15, 23, 42, 0.24)
--teacher-glow-hover	0 20px 42px -28px rgba(15, 23, 42, 0.3)
--teacher-shadow	0 14px 28px -22px rgba(15, 23, 42, 0.2)
--teacher-radius	14px
--teacher-motion-fast	0.16s
--teacher-motion-normal	0.24s
--teacher-page-gradient	linear-gradient(180deg, #0b1322 0%, #101a2d 56%, #101827 100%)

Custom property	Declared value
--teacher-surface	#111d32
--teacher-surface-soft	#142238
--teacher-surface-strong	#1a2b44
--teacher-outline	rgba(148, 163, 184, 0.22)
--teacher-outline-strong	rgba(148, 163, 184, 0.36)
--teacher-text-strong	#e6edf8
--teacher-text-muted	#b6c3d7
--teacher-accent	#e6edf8
--teacher-accent-strong	#f8fafc
--teacher-accent-warm	#60a5fa
--teacher-glow	0 18px 34px -28px rgba(2, 6, 23, 0.86)
--teacher-glow-hover	0 24px 42px -28px rgba(2, 6, 23, 0.9)
--teacher-shadow	0 16px 30px -24px rgba(2, 6, 23, 0.75)
--danger	hover { border-color: #b91c1c
--interactive	hover { border-color: #c6d7ec
--interactive	focus-visible { outline: 2px solid #315fde
--library-bg	#f3f4f7
--library-surface	#ffffff
--library-border	#d8dde7
--library-text-strong	#122033
--library-text-muted	#5f6c80
--library-accent	#cf2027
--library-chip-bg	#eef1f6
--library-chip-text	#495769
--library-focus	rgba(207, 32, 39, 0.18)
--library-bg	#f4f6fb
--library-border	#d5dce8
--library-text-strong	#132338
--library-text-muted	#617187
--library-chip-bg	#eef2f7
--library-chip-text	#4d5a6d
--library-focus	rgba(220, 38, 38, 0.18)
--library-bg	#f3f5f8
--library-border	#d4dae4
--library-text-muted	#607085
--library-chip-bg	#edf1f6
--library-chip-text	#506073
--library-focus	rgba(242, 39, 39, 0.2)
--solid	hover { background: #b91c23
--intervention-border	#dbe2ec
--intervention-border-soft	#edf1f6
--intervention-surface	#ffffff
--intervention-muted	#637083
--intervention-strong	#111827
--intervention-red	#a32d2d
--intervention-red-soft	#fcebeb
--intervention-amber	#ba7517

Custom property	Declared value
--intervention-green	#3b6d11
--progress	0%

UI maintenance rules

- Reuse semantic tokens for background, foreground, border, status, and focus state; avoid cloning page-specific color literals into shared components.
- Preserve keyboard focus, screen-reader names, contrast, reduced-motion behavior, and mobile-width layout when changing an interactive primitive.
- Keep loading, empty, error, forbidden, and stale-data states explicit on data-heavy pages.
- A button that starts durable background work must expose queued, running, retryable, failed, cancelled, and completed states where the backend contract supports them.

Web feature extension procedure

1. Confirm the backend route, DTO, response, role, ownership, and lifecycle contract.
2. Add or update shared frontend types and the owning service method.
3. Define stable query keys and mutation invalidation before wiring the page.
4. Place the page under the correct App Router role prefix and add navigation only for authorized roles.
5. Build explicit loading, empty, failure, forbidden, and success states with reusable UI primitives.
6. Test access bootstrap, direct deep-link navigation, refresh after access expiry, mutation invalidation, and narrow viewport behavior.
7. Run targeted component tests, service tests, Next lint or typecheck, production build, and a browser smoke of the critical path.

Chapter 08 — Expo 54 Mobile Application

Snapshot authority. This chapter describes commit `3d0c93e5270d44b9912deee0218e95c9a311dd5` on branch `developement`. Source paths named below are the authority if the implementation changes after 2026-07-13.

This chapter is the mobile-client service manual. It records provider boot order, secure session storage, API refresh behavior, role resolution, student and teacher navigation, admin workspace behavior, every registered navigation entry, every screen file, every mobile API service, notifications, update policy, and offline boundaries.

Source map

- `mobile/App.tsx`
- `mobile/src/navigation/`
- `mobile/src/providers/`
- `mobile/src/api/`
- `mobile/src/screens/`
- `mobile/src/types/`
- `mobile/src/theme/`
- `mobile/app.json`

Application boot topology

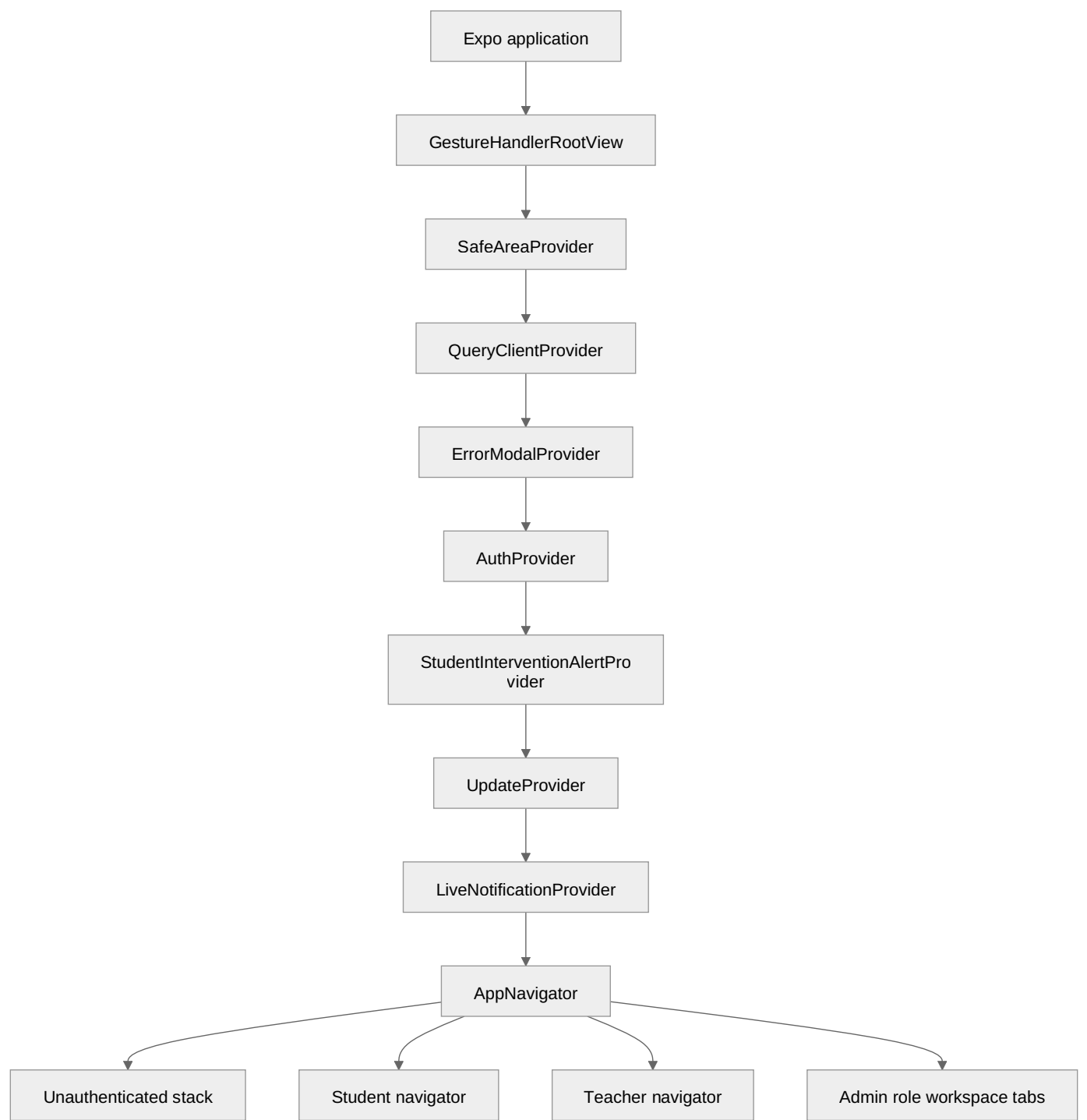


Diagram 19

Session and API transport

Concern	Current behavior
API base URL	Resolved by mobile API configuration; every app request targets the NestJS / api contract.
Timeout	30 seconds for public and authenticated Axios clients.
Access token	Held in memory and persisted by the storage layer; attached as a bearer credential.
Refresh token	Returned by mobile auth endpoints and persisted by the storage layer.

Concern	Current behavior
Native primary storage	Expo SecureStore.
Compatibility copy	AsyncStorage is written alongside SecureStore; web uses AsyncStorage.
Hydration	Tokens are loaded before authenticated requests when memory is empty.
401 recovery	One shared refresh promise rotates tokens and retries each failed request at most once.
Transient refresh failure	Does not clear the session unless the backend returns 401 or 403.
Definitive rejection	Clears access token, refresh token, and session snapshot.

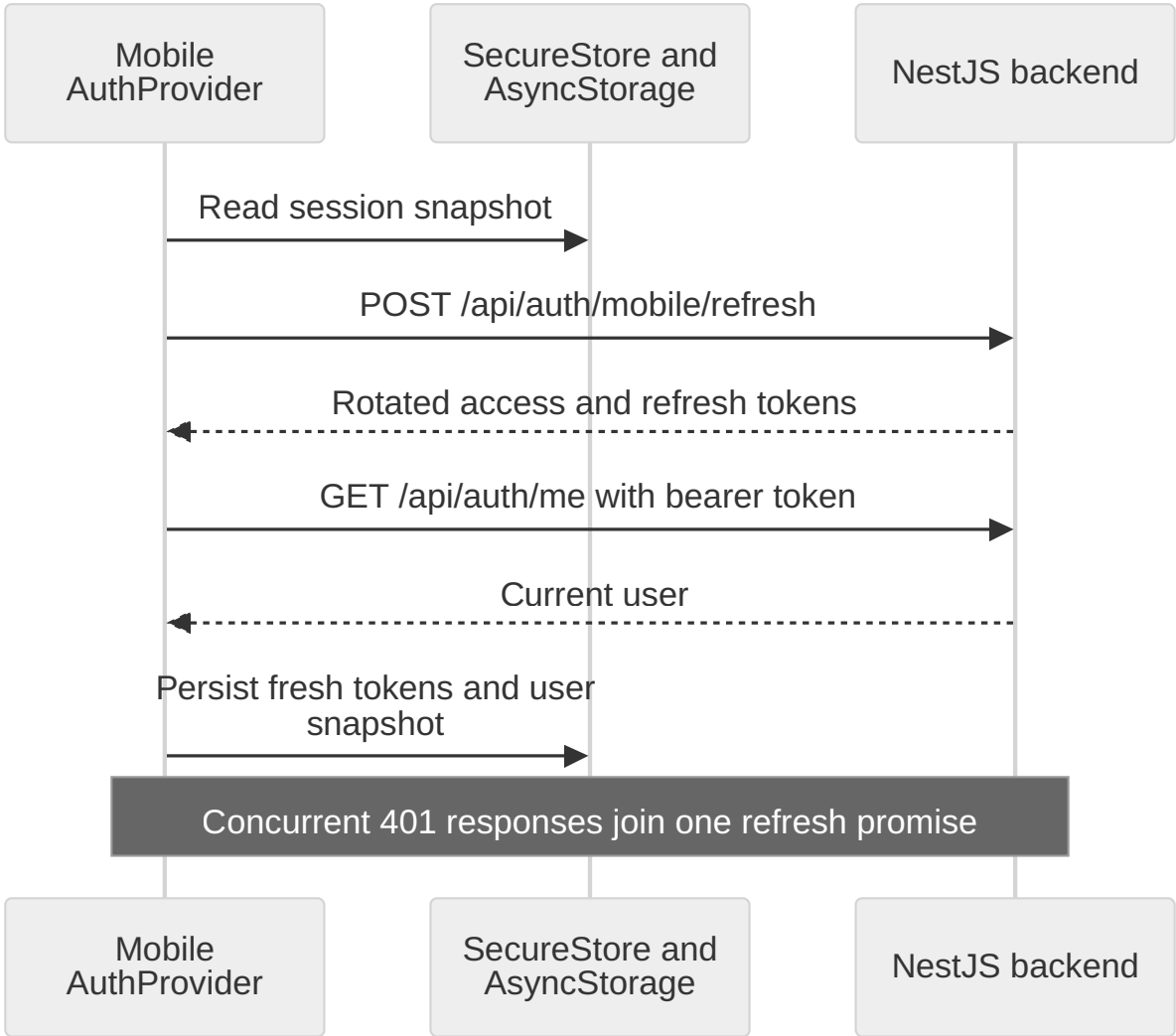


Diagram 20

The session snapshot improves startup presentation but is not proof of a valid current session. Bootstrap refresh and /auth/me establish current authority.

Query-client policy

Option	Value
Stale time	30 seconds
Retry count	2
Retry delay	Exponential with 30-second cap and random jitter

Option	Value
Refetch on window focus	False

Role resolution and navigator selection

- Roles may be strings or objects containing name. Matching is case-insensitive.
- Admin takes precedence, teacher is second, and all other authenticated role sets fall back to student.
- That fallback makes an unrecognized role appear as student navigation; backend authorization still blocks unauthorized calls, but role additions must update the resolver deliberately.

Resolved role	Root experience	Primary tabs
Student	Student stack with tabs and deep learning routes	Dashboard, Classes, Assessments, JA, Announcements, Profile
Teacher	Teacher stack with full parity routes	Home, Assessments, Classes, Announcements, Sections, Profile
Admin	RoleWorkspace tab composition	Home, Classes, Assessments, Announcements, Profile
Unauthenticated	Authentication stack	Login, VerifyEmail, ForgotPassword, ResetPassword, SetInitialPassword

Navigation map

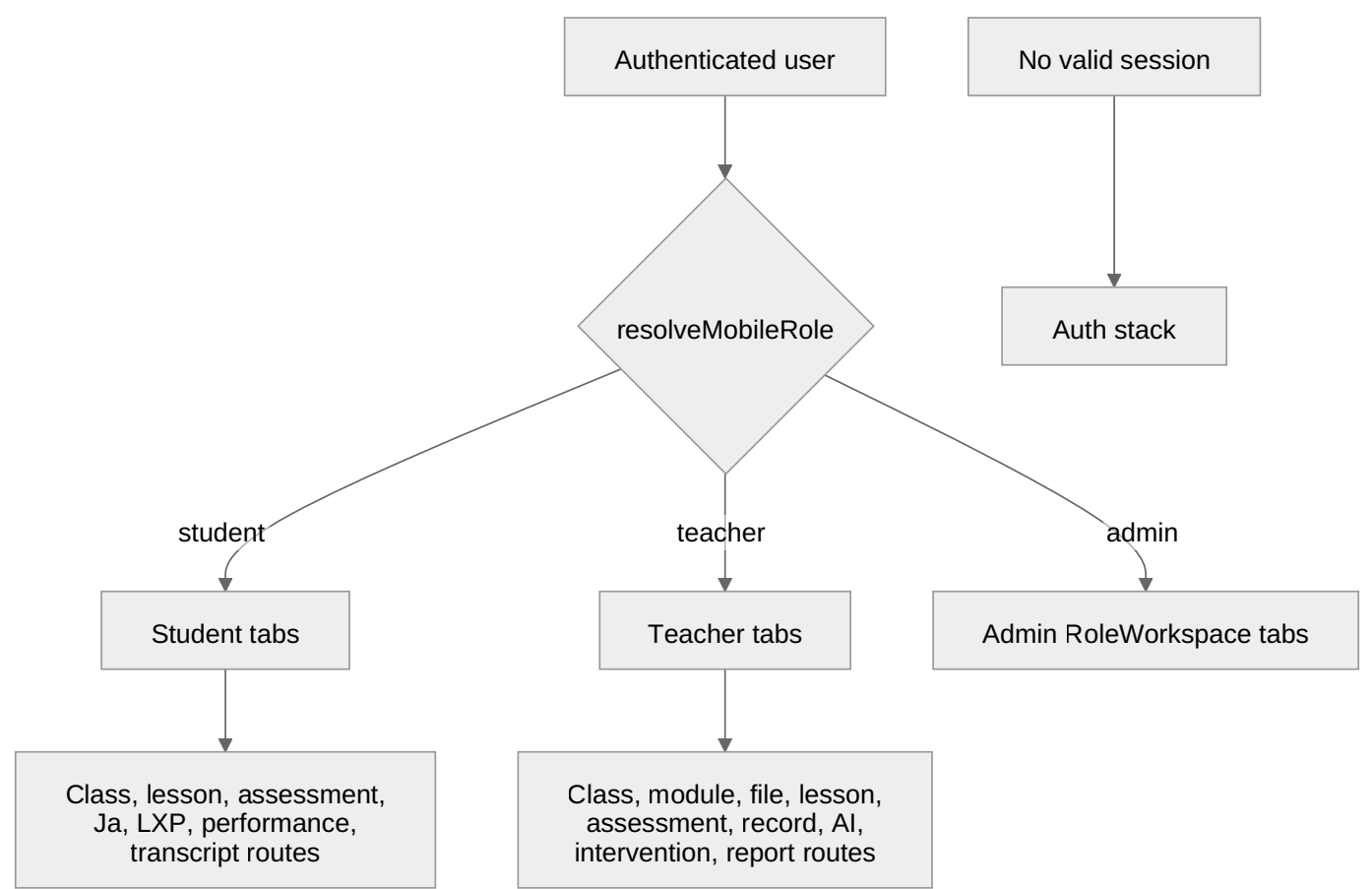


Diagram 21

Complete registered navigation catalog

Exhaustive inventory rule. The 70 JSX screen registrations below were extracted from `mobile/src/navigation/**/*.*tsx` at commit `3d0c93e`. A later source change requires regenerating or manually reconciling this chapter.

Navigator	Resolved route name	Component expression
Tab	Dashboard	studentTabScreens.Dashboard
Tab	Classes	studentTabScreens.Classes
Tab	Assessments	studentTabScreens.Assessments
Tab	JA	studentTabScreens.JA
Tab	Announcements	studentTabScreens.Announcements
Tab	Profile	studentTabScreens.Profile
RootStack	ClassDetail	studentStackScreens.ClassDetail
RootStack	ModuleDetail	studentStackScreens.ModuleDetail
RootStack	Calendar	studentStackScreens.Calendar
RootStack	Courses	studentStackScreens.Courses
RootStack	Lessons	studentStackScreens.Lessons
RootStack	LessonDetail	studentStackScreens.LessonDetail
RootStack	AssessmentDetail	studentStackScreens.AssessmentDetail
RootStack	AssessmentTake	studentStackScreens.AssessmentTake
RootStack	AssessmentResults	studentStackScreens.AssessmentResults
RootStack	AssessmentHistory	studentStackScreens.AssessmentHistory
RootStack	Chatbot	studentStackScreens.Chatbot
RootStack	Performance	studentStackScreens.Performance
RootStack	Transcript	studentStackScreens.Transcript
RootStack	LXP	studentStackScreens.LXP
RootStack	ClassWorkspace	studentSupportScreens.ClassWorkspace
RootStack	AiTutor	studentSupportScreens.AiTutor
AuthStack	Login	LoginScreen
AuthStack	VerifyEmail	VerifyEmailScreen
AuthStack	ForgotPassword	ForgotPasswordScreen
AuthStack	ResetPassword	ResetPasswordScreen
AuthStack	SetInitialPassword	SetInitialPasswordScreen
RootStack	MainTabs	StudentTabs
RootStack	StudentGuidedAssessment	StudentGuidedAssessmentScreen
RootStack	StudentJaReviewAssessment	StudentJaReviewAssessmentScreen
Tab	Home	TeacherHomeScreen
Tab	Assessments	TeacherAssessmentsScreen
Tab	Classes	TeacherClassesScreen
Tab	Announcements	NotificationsInboxScreen
Tab	Sections	TeacherSectionsScreen
Tab	Profile	TeacherProfileScreen
RootStack	MainTabs	TeacherTabs
RootStack	TeacherClassDetail	TeacherClassDetailScreen
RootStack	TeacherModuleDetail	TeacherModuleDetailScreen
RootStack	TeacherModuleFileDetail	TeacherModuleFileDetailScreen

Navigator	Resolved route name	Component expression
RootStack	TeacherLessonDetail	TeacherLessonDetailScreen
RootStack	TeacherLessonEditor	TeacherLessonEditorScreen
RootStack	TeacherAssessmentDetail	TeacherAssessmentDetailScreen
RootStack	TeacherAssessmentEditor	TeacherAssessmentEditorScreen
RootStack	TeacherAssessmentReview	TeacherAssessmentReviewScreen
RootStack	TeacherAssessmentAttemptResult	TeacherAssessmentAttemptResultScreen
RootStack	TeacherCalendar	TeacherCalendarScreen
RootStack	TeacherCreateModule	TeacherCreateModuleScreen
RootStack	TeacherCreateAssessment	TeacherCreateAssessmentScreen
RootStack	TeacherClassAddStudents	TeacherClassAddStudentsScreen
RootStack	TeacherClassStudentOverview	TeacherClassStudentOverviewScreen
RootStack	TeacherSectionDetail	TeacherSectionDetailScreen
RootStack	TeacherSectionAddStudents	TeacherSectionAddStudentsScreen
RootStack	TeacherSectionStudentProfile	TeacherSectionStudentProfileScreen
RootStack	TeacherExtractionDetail	TeacherExtractionDetailScreen
RootStack	TeacherAiDraft	TeacherAiDraftScreen
RootStack	TeacherInterventionDetail	TeacherInterventionDetailScreen
RootStack	TeacherLibrary	TeacherLibraryScreen
RootStack	TeacherClassRecord	TeacherClassRecordScreen
RootStack	TeacherReports	TeacherReportsScreen
RootStack	TeacherInterventions	TeacherInterventionsScreen
RootStack	TeacherPerformance	TeacherPerformanceScreen
RootStack	TeacherEvaluations	TeacherEvaluationsScreen
RootStack	TeacherAnnouncements	TeacherAnnouncementsScreen
RootStack	TeacherMore	TeacherMoreScreen
Tab	Home	None
Tab	Classes	None
Tab	Assessments	None
Tab	Announcements	NotificationsInboxScreen
Tab	Profile	None

Student route contract

Layer	Routes
Tabs	Dashboard, Classes, Assessments, JA, Announcements, Profile
Stack	ClassDetail, ModuleDetail, Calendar, Courses, Lessons, LessonDetail, AssessmentDetail, AssessmentTake, AssessmentResults, AssessmentHistory, Chatbot, Performance, Transcript, LXP
Support	ClassWorkspace, AiTutor
Additional mounted assessment routes	StudentGuidedAssessment, StudentJaReviewAssessment

Teacher route contract

Layer	Routes
Tabs	Home, Assessments, Classes, Announcements, Sections, Profile
Class and content	TeacherClassDetail, TeacherModuleDetail, TeacherModuleFileDetail, TeacherLessonDetail, TeacherLessonEditor, TeacherCreateModule
Assessment	TeacherAssessmentDetail, TeacherAssessmentEditor, TeacherAssessmentReview, TeacherAssessmentAttemptResult, TeacherCreateAssessment
Roster and sections	

Layer	Routes
	TeacherClassAddStudents, TeacherClassStudentOverview, TeacherSectionDetail, TeacherSectionAddStudents, TeacherSectionStudentProfile
AI and extraction	TeacherExtractionDetail, TeacherAiDraft, TeacherInterventionDetail, TeacherInterventions
Operations	TeacherLibrary, TeacherClassRecord, TeacherReports, TeacherPerformance, TeacherEvaluations, TeacherAnnouncements, TeacherCalendar, TeacherMore

Complete screen-file catalog

Exhaustive inventory rule. The 57 non-test screen implementation files below were extracted from `mobile/src/screens/**/*.tsx`, excluding test files and barrel exports, at commit `3d0c93e`. A later source change requires regenerating or manually reconciling this chapter.

Screen source	Exports	Local API-path literals
mobile/src/screens/AiTutorScreen.tsx	AiTutorScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/AnnouncementsScreen.tsx	AnnouncementsScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/AssessmentDetailScreen.tsx	AssessmentDetailScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/AssessmentHistoryScreen.tsx	AssessmentHistoryScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/AssessmentResultsScreen.tsx	AssessmentResultsScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/AssessmentTakeScreen.tsx	AssessmentTakeScreen	/, /:param
mobile/src/screens/AssessmentsScreen.tsx	AssessmentsScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/CalendarScreen.tsx	CalendarScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/ClassDetailScreen.tsx	StudentClassDetailContent, ClassDetailScreen	/
mobile/src/screens/CoursesScreen.tsx	CoursesScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/DashboardScreen.tsx	DashboardScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/ForgotPasswordScreen.tsx	ForgotPasswordScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/JaScreen.tsx	JaScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/LessonDetailScreen.tsx	LessonDetailScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/LessonsScreen.tsx	LessonsScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/LoginScreen.tsx	LoginScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/LxpScreen.tsx	LxpScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/ModuleDetailScreen.tsx	ModuleDetailScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/NotificationsInboxScreen.tsx	NotificationsInboxScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/PerformanceScreen.tsx	PerformanceScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/ProfileScreen.tsx	ProfileScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/ProgressScreen.tsx	ProgressScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/ResetPasswordScreen.tsx	ResetPasswordScreen	

Screen source	Exports	Local API-path literals
		Uses hooks or API services; no local path literal extracted
mobile/src/screens/RoleWorkspaceScreen.tsx	RoleWorkspaceScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/SetInitialPasswordScreen.tsx	SetInitialPasswordScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/StudentGuidedAssessmentScreen.tsx	StudentGuidedAssessmentScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/StudentJaReviewAssessmentScreen.tsx	StudentJaReviewAssessmentScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/SubjectLessonsScreen.tsx	SubjectLessonsScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherAiDraftScreen.tsx	TeacherAiDraftScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherAnnouncementsScreen.tsx	TeacherAnnouncementsScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherAssessmentDetailScreen.tsx	TeacherAssessmentDetailScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherAssessmentEditorScreen.tsx	TeacherAssessmentEditorScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherAssessmentReviewScreen.tsx	TeacherAssessmentReviewScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherAssessmentsScreen.tsx	TeacherAssessmentsScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherCalendarScreen.tsx	TeacherCalendarScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherClassDetailScreen.tsx	TeacherClassDetailScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherClassRecordScreen.tsx	TeacherClassRecordScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherClassesScreen.tsx	TeacherClassesScreen	/
mobile/src/screens/TeacherCreateAssessmentScreen.tsx	TeacherCreateAssessmentScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherCreateModuleScreen.tsx	TeacherCreateModuleScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherDeepParityScreens.tsx	TeacherAssessmentAttemptResultScreen, TeacherClassStudentOverviewScreen, TeacherClassAddStudentsScreen, TeacherSectionAddStudentsScreen, TeacherSectionStudentProfileScreen, TeacherModuleFileDetailScreen, TeacherLessonEditorScreen, TeacherInterventionWorkspaceContent, TeacherInterventionDetailScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherEvaluationsScreen.tsx	TeacherEvaluationsScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherExtractionDetailScreen.tsx	TeacherExtractionDetailScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherHomeScreen.tsx	TeacherHomeScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherInterventionsScreen.tsx	TeacherInterventionsScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherLessonDetailScreen.tsx	TeacherLessonDetailScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherLibraryScreen.tsx	TeacherLibraryScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherModuleDetailScreen.tsx	TeacherModuleDetailScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherMoreScreen.tsx	TeacherMoreScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherPerformanceScreen.tsx	TeacherPerformanceScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherProfileScreen.tsx	TeacherProfileScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherReportsScreen.tsx	TeacherReportsScreen	Uses hooks or API services; no local path literal extracted

Screen source	Exports	Local API-path literals
mobile/src/screens/TeacherSectionDetailScreen.tsx	TeacherSectionDetailScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherSectionsScreen.tsx	TeacherSectionsScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TeacherUnsupportedScreen.tsx	TeacherUnsupportedScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/TranscriptScreen.tsx	TranscriptScreen	Uses hooks or API services; no local path literal extracted
mobile/src/screens/VerifyEmailScreen.tsx	VerifyEmailScreen	Uses hooks or API services; no local path literal extracted

Complete mobile API service catalog

Exhaustive inventory rule. The 22 mobile API service modules below were extracted from `mobile/src/api/services/*.ts` at commit `3d0c93e`. A later source change requires regenerating or manually reconciling this chapter.

Service source	Exports	Backend path literals
mobile/src/api/services/ai.ts	aiApi	/ai/index/classes/:param, /ai/index/classes/:param/status, /ai/student/tutor/bootstrap:param, /ai/student/tutor/session, /ai/student/tutor/session/:param, /ai/student/tutor/session/:param/answers, /ai/student/tutor/session/:param/message, /ai/teacher/classes/:param/policy, /ai/teacher/interventions/:param/jobs, /ai/teacher/jobs/:param, /ai/teacher/jobs/:param/result, /ai/teacher/quizzes/jobs
mobile/src/api/services/announcements.ts	announcementsApi	/classes/:param/announcements, /classes/:param/announcements/:param
mobile/src/api/services/assessments.ts	assessmentsApi	/assessments, /assessments/:param, /assessments/:param/ongoing-attempt, /assessments/:param/start, /assessments/:param/student-attempts, /assessments/:param/submission-file, /assessments/:param/submission-files/:param, /assessments/:param/submissions, /assessments/:param/teacher-attachment/download, /assessments/:param/unsubmit-file-upload, /assessments/attempts/:param/progress, /assessments/attempts/:param/results, /assessments/attempts/:param/return, /assessments/attempts/:param/submission-file/download, /assessments/attempts/:param/submission-files/:param/download, /assessments/attempts/:param/unreturn, /assessments/class/:param, /assessments/questions, /assessments/questions/:param, /assessments/submit, /profiles/me/assessment-history
mobile/src/api/services/auth.ts	authApi	/auth/forgot-password, /auth/me, /auth/mobile/login, /auth/mobile/logout, /auth/profile, /auth/reset-password, /auth/set-activation-password, /auth/validate-credentials, /otp/resend, /otp/verify
mobile/src/api/services/class-record.ts	classRecordApi	/class-record, /class-record/:param, /class-record/:param/final-grades, /class-record/:param/finalize, /class-record/:param/preview-grades, /class-record/:param/reopen, /class-record/:param/reports/class-average, /class-record/:param/reports/distribution, /class-record/:param/reports/intervention, /class-record/:param/spreadsheet, /class-record/by-class/:param, /class-record/items/:param, /class-record/items/:param/scores, /class-record/items/:param/scores/bulk, /class-record/items/:param/sync-scores
mobile/src/api/services/classes.ts	classesApi	/classes/:param, /classes/:param/enrollments, /classes/:param/enrollments/:param, /classes/:param/students/:param/overview, /classes/:param/students/:param/profile, /classes/:param/students/masterlist, /classes/all, /classes/student/:param, /classes/teacher/:param
mobile/src/api/services/dashboard.ts	dashboardApi	No static API path literal extracted
mobile/src/api/services/discussion-board.ts	discussionBoardApi	/classes/:param/discussion-threads, /classes/:param/discussion-threads/:param, /

Service source	Exports	Backend path literals
		classes/:param/discussion-threads/:param/close, /classes/:param/discussion-threads/:param/comments, /classes/:param/discussion-threads/:param/comments/:param, /classes/:param/discussion-threads/:param/comments/:param/reaction, /classes/:param/discussion-threads/:param/comments/:param/report, /classes/:param/discussion-threads/:param/comments/uploads, /classes/:param/discussion-threads/:param/publish, /classes/:param/discussion-threads/:param/reopen, /classes/:param/discussion-threads/uploads
mobile/src/api/services/extractions.ts	extractionsApi	/ai/extract-module, /ai/extractions, /ai/extractions/:param, /ai/extractions/:param/apply
mobile/src/api/services/file-upload.ts	fileUploadApi	/files/:param, /files/:param/download, /files/upload
mobile/src/api/services/ja.ts	jaApi	/ai/student/ja/ask/threads, /ai/student/ja/ask/threads/:param, /ai/student/ja/ask/threads/:param/messages, /ai/student/ja/hub/:param, /ai/student/ja/practice/sessions, /ai/student/ja/practice/sessions/:param, /ai/student/ja/practice/sessions/:param/complete, /ai/student/ja/practice/sessions/:param/events, /ai/student/ja/practice/sessions/:param/responses, /ai/student/ja/review/sessions, /ai/student/ja/review/sessions/:param, /ai/student/ja/review/sessions/:param/complete, /ai/student/ja/review/sessions/:param/events, /ai/student/ja/review/sessions/:param/responses
mobile/src/api/services/lessons.ts	lessonsApi	/lessons/:param, /lessons/:param/blocks, /lessons/:param/complete, /lessons/:param/completion-status, /lessons/:param/publish, /lessons/:param/reorder-blocks, /lessons/blocks/:param, /lessons/class/:param, /lessons/class/:param/bulk-status, /lessons/class/:param/completed
mobile/src/api/services/lxp.ts	lxpApi	/lxp/me/eligibility, /lxp/me/intervention-alerts, /lxp/me/overview/:param, /lxp/me/playlist/:param, /lxp/me/playlist/:param/checkpoints/:param/complete, /lxp/me/playlist/:param/guided-assessments/:param/progress, /lxp/me/playlist/:param/guided-assessments/:param/result, /lxp/me/playlist/:param/guided-assessments/:param/start, /lxp/me/playlist/:param/guided-assessments/:param/submit, /lxp/teacher/classes/:param/interventions, /lxp/teacher/classes/:param/interventions/history, /lxp/teacher/classes/:param/reports/summary, /lxp/teacher/evaluations/summary, /lxp/teacher/interventions/:param, /lxp/teacher/interventions/:param/activate, /lxp/teacher/interventions/:param/detail, /lxp/teacher/interventions/:param/generated-content/approve, /lxp/teacher/interventions/:param/generated-content/reject, /lxp/teacher/interventions/:param/regenerate, /lxp/teacher/interventions/:param/resolve, /lxp/teacher/interventions/pending-count
mobile/src/api/services/modules.ts	modulesApi	/modules, /modules/:param, /modules/class/:param, /modules/class/:param/:param, /modules/items/:param, /modules/items/:param/file/download
mobile/src/api/services/notifications.ts	notificationsApi	/notifications, /notifications/:param/read, /notifications/unread-count
mobile/src/api/services/performance.ts	performanceApi	/performance/classes/:param/at-risk, /performance/classes/:param/intervention-quiz-comparison, /performance/classes/:param/summary, /performance/students/me/summary
mobile/src/api/services/profile.ts	profileApi	/profiles/me, /profiles/me/avatar, /profiles/update/:param
mobile/src/api/services/protected-files.ts	buildProtectedUrl, openLocalFile, downloadProtectedFile, buildProtectedImageSource	/, /:param, /api/
mobile/src/api/services/reports.ts	reportsApi	/profiles/me/assessment-history, /profiles/me/transcript, /reports/assessment-summary, /reports/class-enrollment, /reports/intervention-participation, /reports/student-performance, /reports/system-usage
mobile/src/api/services/school-events.ts	schoolEventsApi	/school-events
mobile/src/api/services/sections.ts	sectionsApi	/sections/:param, /sections/:param/candidates, /sections/:param/roster, /sections/:param/roster/:param, /sections/:param/schedule, /sec-

Service source	Exports	Backend path literals
		tions/:param/students/:param/profile, /sections/all, /sections/my
mobile/src/api/services/teacher-profile.ts	teacherProfileApi	/teacher-profiles/:param, /teacher-profiles/me, /teacher-profiles/me/avatar

Notifications, intervention alerts, and updates

- LiveNotificationProvider owns real-time notification reception and mobile presentation after authentication.
- StudentInterventionAlertProvider evaluates current learner intervention state and surfaces actionable alerts without changing official records.
- UpdateProvider consumes backend app-version policy so a deployment can recommend or require a supported mobile version.
- Notification taps must resolve through the typed navigation contract and revalidate backend access before showing protected detail.

Offline and degraded-network truth

- Persisted tokens and the user snapshot permit startup continuity, but protected data still requires backend validation.
- React Query caches in memory under the current provider. No durable offline mutation queue or conflict-resolution engine was found.
- A transient refresh-network failure does not immediately erase valid local state; a definitive 401 or 403 does.
- Mutations must present pending and retry states and must not imply success until the backend confirms the authoritative write.
- Socket disconnection must fall back to refetch or polling where the provider implements it; push delivery alone is not durable evidence.

Mobile security and accessibility

- Never place credentials, OTPs, internal secrets, or full response bodies in production console logs.
- SecureStore should remain available on native production builds. AsyncStorage compatibility does not provide equivalent secret protection.
- Deep links and notification payloads are untrusted input; validate route name, required parameters, current role, and backend object access.
- Preserve safe-area handling, touch target size, text scaling, loading announcements, keyboard navigation on web target, and error recovery.

Mobile feature extension procedure

1. Confirm the backend contract and add or update mobile types and the owning API service.
2. Add the route name and exact params to the appropriate typed manifest and navigator.
3. Build the screen against service or query hooks with explicit loading, empty, offline, failure, forbidden, and success states.
4. Persist only the minimum state necessary; never create a second authority for official academic records.
5. Reconcile mutations by invalidating or updating every affected query key.
6. Test cold boot, token rotation, transient network loss, definitive session rejection, role selection, deep link, back navigation, and small-screen accessibility.

7. Run mobile unit tests, typecheck, Expo configuration validation, and focused device or simulator smoke tests.

Chapter 09 — Observability, Telemetry, and Diagnostics

Snapshot authority. This chapter describes commit `3d0c93e5270d44b9912deee0218e95c9a311dd5` on branch `development`. Source paths named below are the authority if the implementation changes after 2026-07-13.

This chapter is the operator service sheet for health, metrics, dashboards, alerts, logs, traces, container probes, failure isolation, and cross-process diagnostics. It distinguishes currently instrumented evidence from correlation gaps that must not be guessed around.

Source map

- `monitoring/`
- `docker-compose.yml`
- `docker-compose.debug.yml`
- `backend/src/monitoring/`
- `backend/src/modules/health/`
- `backend/src/tracing.ts`
- `backend/src/common/logger/`
- `ai-service/app/main.py`

Telemetry topology

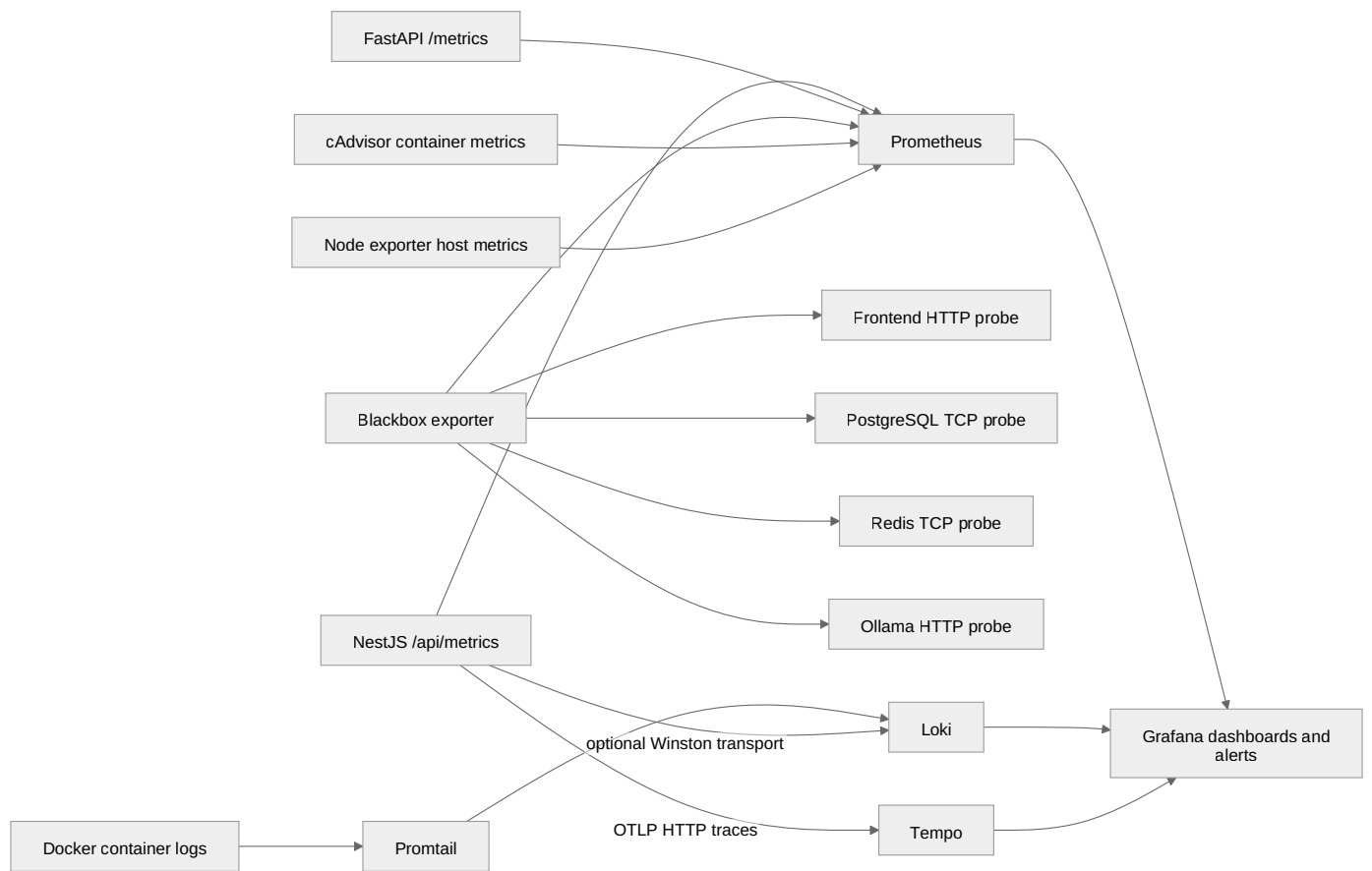


Diagram 22

- The entire observability suite is opt-in through the Compose `observability` profile. Core application startup does not require it.
- Prometheus stores metrics, Loki stores logs, Tempo stores traces, and Grafana presents provisioned data sources, dashboards, and alert rules.
- Blackbox exporter probes HTTP or TCP reachability where an application-native metric endpoint is unavailable.
- cAdvisor observes containers and node-exporter observes the host.

Prometheus scrape register

Exhaustive inventory rule. The 10 scrape jobs below were extracted from `monitoring/prometheus.yml` at commit `3d0c93e`. A later source change requires regenerating or manually reconciling this chapter.

Job	Metrics path	Targets	Evidence type
prometheus	/metrics	prometheus:9090	Native metrics scrape
backend	/api/metrics	backend:3000	Native metrics scrape
ai-service	/metrics	ai-service:8000	Native metrics scrape
frontend	/probe	http://frontend:3001	Blackbox reachability probe
loki	/probe	http://loki:3100/ready	Blackbox reachability probe
postgres	/probe	postgres:5432	Blackbox reachability probe
redis	/probe	redis:6379	Blackbox reachability probe

Job	Metrics path	Targets	Evidence type
ollama	/probe	http://ollama:11434/	Blackbox reachability probe
cadvisor	/metrics	cadvisor:8080	Native metrics scrape
node-exporter	/metrics	node-exporter:9100	Native metrics scrape

Application metric dictionary

Metric	Type	Labels or value	Owner
http_request_duration_seconds	Histogram	method, route, status_code	NestJS metrics interceptor
http_requests_total	Counter	method, route, status_code	NestJS metrics interceptor
http_request_errors_total	Counter	method, route, status_code	NestJS metrics interceptor
storage_cleanup_failures_total	Counter	Storage cleanup failure count	NestJS storage maintenance
db_pool_total_connections	Gauge	Current pg Pool totalCount	NestJS MetricsController
db_pool_idle_connections	Gauge	Current pg Pool idleCount	NestJS MetricsController
db_pool_waiting_requests	Gauge	Current pg Pool waitingCount	NestJS MetricsController
nexora_ai_http_requests_total	Counter	method, route, status	FastAPI middleware
nex-ora_ai_http_request_duration_seconds	Histogram	method and route latency	FastAPI middleware
nexora_ai_ready	Gauge	1 ready, 0 degraded or not ready	FastAPI readiness
nexora_ai_ollama_available	Gauge	1 reachable, 0 unavailable	FastAPI runtime health

No active backend metric declaration for `bullmq_waiting_total` was found. The provisioned queue-backlog alert therefore enters NoData unless another deployment component exports that exact metric. Instrument queue state before relying on the alert.

Health and readiness endpoint register

Endpoint	Meaning	Expected use
GET /api/health	Compatibility alias of backend liveness	Simple process check.
GET /api/health/live	Backend process and service metadata	Container liveness and first diagnostic.
GET /api/health/ready	Backend database, Redis, and AI dependency readiness	Traffic admission; returns 503 when a required dependency is unavailable.
GET /api/ai/health	Backend proxy view of AI or Ollama availability	Public compatibility check through the backend boundary.
GET ai-service:8000/health	FastAPI process and runtime status	Private network diagnostics.
GET ai-service:8000/ready	FastAPI database and configured runtime readiness	AI-service container readiness.
GET ai-service:8000/metrics	FastAPI Prometheus exposition	Prometheus scrape only.
GET backend:3000/api/metrics	NestJS Prometheus exposition and pool gauges	Prometheus scrape only.

Alert rule register

Exhaustive inventory rule. The 15 Grafana alert rules below were extracted from `monitoring/grafana/provisioning/alerting/` at commit `3d0c93e`. A later source change requires regenerating or manually reconciling this chapter.

Group	Alert	Expression	Pending duration	No-data state	Execution-error state	UID
nexora-platform-availability	Backend Down	<code>up{job="backend"} == 0</code>	2m	Alerting	Alerting	backend-down
nexora-platform-availability	AI Service Down	<code>up{job="ai-service"} == 0</code>	2m	Alerting	Alerting	ai-service-down
nexora-platform-availability	Frontend Down	<code>probe_success{job="frontend"} == 0</code>	2m	Alerting	Alerting	frontend-down
nexora-platform-availability	Postgres Down	<code>probe_success{job="postgres"} == 0</code>	2m	Alerting	Alerting	postgres-down
nexora-platform-availability	Redis Down	<code>probe_success{job="redis"} == 0</code>	2m	Alerting	Alerting	redis-down
nexora-platform-availability	Ollama Unavailable	<code>probe_success{job="ollama"} == 0</code>	2m	Alerting	Alerting	ollama-unavailable
nexora-platform-availability	AI Readiness Degraded	<code>nexora_ai_ready == 0</code>	2m	Alerting	Alerting	ai-ready-degraded
nexora-platform-pressure	Host Memory Pressure	<code>(1 - (node_memory_MemAvailable_bytes / node_memory_MemTotal_bytes)) > 0.90</code>	5m	NoData	Alerting	host-memory-pressure
nexora-platform-pressure	Host Disk Pressure	<code>(1 - (node_filesystem_avail_bytes{fstype!~"tmpfs squashfs"} / node_filesystem_size_bytes{fstype!~"tmpfs squashfs"})) > 0.90</code>	5m	NoData	Alerting	host-disk-pressure
nexora-platform-pressure	Backend 5xx Rate High	<code>sum(rate(http_requests_total{job="backend",status_code=~"5.."}[5m])) > 0.05</code>	5m	NoData	Alerting	backend-5xx-rate
nexora-platform-pressure	AI Service Error Rate High	<code>sum(rate(nexora_ai_http_requests_total{status=~"5.."}[5m])) > 0</code>	5m	NoData	Alerting	ai-service-error-rate
nexora-platform-pressure	Container Lifecycle Churn High	<code>sum(changes(container_start_time_seconds[15m])) > 3</code>	10m	NoData	Alerting	container-lifecycle-churn
nexora-platform-pressure	BullMQ Queue Backlog High	<code>bullmq_waiting_total > 10</code>	5m	NoData	Alerting	bullmq-waiting-jobs
nexora-platform-pressure	AI Job Failure Rate High	<code>sum(rate(nexora_ai_http_requests_total{status=~"5.."}[10m])) > 0.1</code>	10m	NoData	Alerting	ai-job-failure-rate
nexora-platform-pressure	DB Pool Connection Starvation	<code>db_pool_waiting_requests > 5</code>	2m	NoData	Alerting	db-pool-starvation

Alert interpretation

- Availability alerts deliberately treat no data as alerting because the monitoring path itself may be down.
- Pressure alerts generally use NoData so a missing optional metric does not masquerade as measured pressure; operators must still investigate absent expected telemetry.
- A firing probe proves the probe failed, not the root cause. Confirm DNS, network, process, dependency, and application response in that order.
- The AI failure alerts currently derive from AI HTTP 5xx metrics. They do not count every durable BullMQ terminal failure unless that failure also produced the measured HTTP response.

Grafana dashboard register

Dashboard	Panels	Provisioned source
Nexora AI Service	AI Ready, Ollama Available, AI Requests, AI Latency	monitoring/grafana/dashboards/nexora-ai-service.json
Nexora Backend & DB Pool	Backend Up, Backend Request Rate, Backend P95 / P99 Latency (s), Backend 5xx Error Rate, Database Connection Pool Diagnostics, HTTP Error Spike Breakdown by Route	monitoring/grafana/dashboards/nexora-backend.json
Nexora Containers	Container CPU, Container Memory, Container Lifecycle Churn	monitoring/grafana/dashboards/nexora-containers.json
Nexora Infrastructure	Host CPU, Host Memory, Host Disk Utilization	monitoring/grafana/dashboards/nexora-infrastructure.json
Nexora Logs	Docker Logs	monitoring/grafana/dashboards/nexora-logs.json
Nexora Overview	Backend Up, AI Service Up, Prometheus Up, Loki Ready, Container CPU, Trace Volume	monitoring/grafana/dashboards/nexora-overview.json
Nexora Traces	Backend Trace Search, Trace Volume	monitoring/grafana/dashboards/nexora-traces.json

Logging pipeline

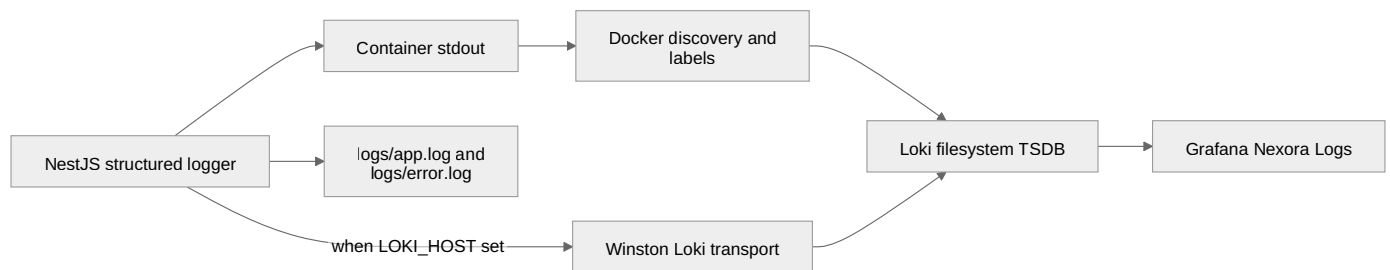


Diagram 23

- Console logs always run. JSON file transports write all events to `logs/app.log` and errors to `logs/error.log`.
- When `LOKI_HOST` is set, the backend also sends logs directly with app, service_name, and environment labels.
- Promtail discovers Docker containers through the Docker socket and labels container ID, name, Compose project, service, container number, and image.
- If both direct Loki and Promtail paths are active, equivalent backend events can appear twice unless queries or deployment policy select one ingestion path.
- Loki is configured as a single local filesystem instance with authentication disabled inside the deployment network. Do not expose it publicly.

Trace pipeline

- Backend startup imports `backend/src/tracing.ts` before Nest bootstrap.
- OpenTelemetry Node auto-instrumentations export OTLP HTTP spans to the configured endpoint, defaulting to `http://tempo:4318/v1/traces`.
- Resource attributes identify service name, namespace `nexora`, container or process instance, and application version.
- Tempo stores traces locally and compacts blocks with 24-hour retention.
- Trace initialization can be disabled by setting the OTLP endpoint to an empty value.

End-to-end request correlation: current truth

No application-wide X-Request-ID generator, response header, structured request-ID log field, BullMQ correlation field, or explicit FastAPI request-ID propagation contract was found. A guaranteed Next.js → NestJS → BullMQ → FastAPI request-ID trace is not currently available.

Use the strongest identifiers that do exist:

1. Record client time, authenticated user ID, HTTP method, backend path, target entity ID, and response status.
2. Search the backend trace dashboard around that time for the HTTP span and inspect auto-instrumented downstream HTTP and PostgreSQL spans.
3. For queued work, obtain the durable AI job UUID, extraction UUID, file ID, class ID, or deterministic BullMQ job ID returned or logged by the workflow.
4. Search backend logs by the durable ID and queue job name. Compare attemptsMade and durable database status.
5. Search FastAPI logs and metrics in the same interval by route and durable ID. Confirm the internal call result and database transition.
6. Treat time-only matching as inference. Do not claim one event caused another without a shared durable ID or trace context.

Recommended correlation improvement

Introduce one validated X-Request-ID at the backend ingress, return it to clients, add it to structured logs and audit-safe metadata, copy it into every BullMQ payload, forward it to FastAPI, and include it as a trace attribute. Keep durable job IDs separate because one request may create multiple jobs.

Diagnostic command sheet

```
# Render the effective Compose configuration without starting services
docker compose config

# Inspect core and observability container state
docker compose --profile observability ps

# Read recent service logs without following forever
docker compose logs --since=15m backend ai-service redis postgres ollama

# Check public backend liveness and readiness
curl --fail --silent --show-error http://localhost:3000/api/health/live
curl --fail --silent --show-error http://localhost:3000/api/health/ready

# Check private dependencies from their containers
docker compose exec redis redis-cli ping
docker compose exec postgres pg_isready -U postgres -d capstone
docker compose exec ollama ollama list

# Check Prometheus target health and Grafana health when the profile is running
curl --fail --silent --show-error http://localhost:9090/-/ready
curl --fail --silent --show-error http://localhost:3002/api/health

# Check Loki and Tempo readiness when their default host ports are published
curl --fail --silent --show-error http://localhost:3100/ready
curl --fail --silent --show-error http://localhost:3200/ready
```


Incident isolation decision tree

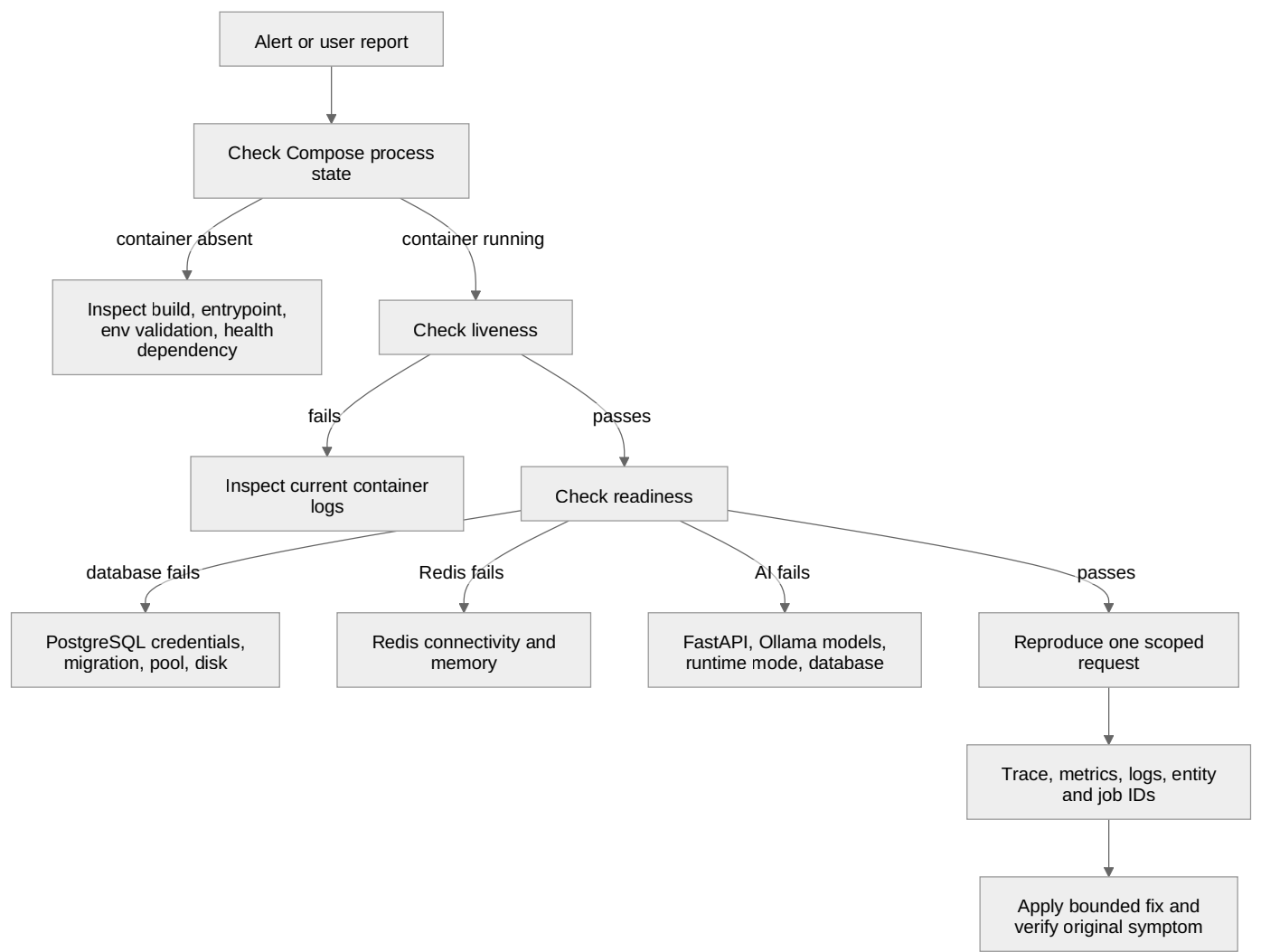


Diagram 24

Common symptom runbooks

Symptom	First evidence	Likely branches	Safe next action
Backend readiness 503	/api/health/ready response and backend logs	Database, Redis, or AI dependency failure	Test each dependency from the backend network and correct only the failed branch.
AI readiness degraded	/ready response, nexora_ai_ready, Ollama probe	Missing model, runtime unavailable, database failure, cloud config mismatch	Confirm selected runtime mode and exact configured model names.
Requests slow but healthy	Backend and AI latency histograms plus DB pool gauges	Pool waiting, model saturation, slow query, container pressure	Match the slow route to pool, trace, and resource evidence before tuning.
Queue work not completing	Durable job row, BullMQ job state, worker logs	Worker absent, retained failure, dependency timeout, invalid payload	Reconcile durable and Redis state before retrying.
Notifications delayed	notifications queue, socket connection, inbox fetch	Worker failure, socket disconnect, polling delay	Verify durable notification rows before focusing on real-time transport.
Frontend down alert	Blackbox probe and frontend logs	Next process, rewrite target, backend health, DNS	Probe the frontend and backend separately.
Disk pressure	node filesystem metric and Docker disk usage	Logs, image layers, volumes, model data	Identify the consuming path before any deletion; preserve database and upload backups.

Observability maintenance checklist

1. Add a metric only with a stable name, type, labels, cardinality budget, dashboard consumer, alert rationale, and test.
2. Never place user content, tokens, cookies, secrets, or full request bodies in metric labels.
3. Verify every alert expression against the currently scraped metric names before enabling notification delivery.
4. Keep core Compose independent of the observability profile.
5. Test no-data and monitoring-stack failure behavior, not only the nominal alert threshold.
6. Confirm retention and disk capacity for Prometheus, Loki, Tempo, and Grafana volumes.
7. After a change, reproduce a request and show its metric, log, and trace evidence where instrumentation supports it.

Chapter 10 — Developer Cookbook and Modification Recipes

Snapshot authority. This chapter describes commit `3d0c93e5270d44b9912deee0218e95c9a311dd5` on branch `development`. Source paths named below are the authority if the implementation changes after 2026-07-13.

This chapter turns the architecture into repeatable maintenance procedures. Each recipe names the current owning paths, provides concrete code patterns, identifies every contract consumer, and ends with evidence gates. Adapt names only after tracing the current source at the edition commit.

Source map

- `backend/src/`
- `backend/drizzle/`
- `ai-service/app/`
- `next-frontend/app/`
- `next-frontend/src/`
- `mobile/src/`
- `docker-compose.yml`
- `.github/workflows/`

Universal change workflow

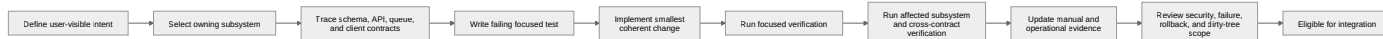


Diagram 25

Pre-change record

```
git status --short
git branch --show-current
git rev-parse HEAD
docker compose config --quiet
```

- Preserve unrelated dirty worktree changes.
- Read the root and owning subsystem `AGENTS.md` files.
- Use current source paths. The active database schema is `backend/src/drizzle/schema/`, not the legacy path named in older prose.
- State whether the change modifies official state, derived state, AI assistive output, a public contract, or only presentation.
- Do not broaden an implementation because a nearby subsystem could also be improved.

Recipe 10.1 — Add a database entity and CRUD API

This concrete example adds admin-maintained operational notes. It demonstrates the complete schema → migration → DTO → service → controller → module → test chain without granting AI or clients direct database authority.

Step 1 — Add the Drizzle schema

Create `backend/src/drizzle/schema/maintenance-notes.schema.ts`:

```
import { index, pgTable, text, timestamp, uuid, varchar } from 'drizzle-orm/pg-core';
import { users } from '../base.schema';

export const maintenanceNotes = pgTable(
  'maintenance_notes',
  {
    id: uuid('id').defaultRandom().primaryKey(),
    targetType: varchar('target_type', { length: 80 }).notNull(),
    targetId: uuid('target_id'),
    body: text('body').notNull(),
    createdBy: uuid('created_by')
      .notNull()
      .references(() => users.id, { onDelete: 'restrict' }),
    createdAt: timestamp('created_at', { withTimezone: true }).defaultNow().notNull(),
    updatedAt: timestamp('updated_at', { withTimezone: true }).defaultNow().notNull(),
  },
  (table) => ({
    targetIdx: index('maintenance_notes_target_idx').on(table.targetType, table.targetId),
    createdByIdx: index('maintenance_notes_created_by_idx').on(table.createdBy),
  }),
);
```

Add this exact export to `backend/src/drizzle/schema/index.ts`:

```
export * from './maintenance-notes.schema';
```

Design checks before migration:

- The table has an explicit primary key, bounded target type, required actor, timezone-aware timestamps, and indexes matching list predicates.
- Actor deletion is restricted so maintenance evidence cannot silently lose provenance.
- `targetId` is intentionally nullable because some notes can describe a subsystem rather than one row.

Step 2 — Generate, inspect, and apply the migration

```
cd backend
npx drizzle-kit generate --name add_maintenance_notes
npm run check:migrations
cd ..

# Apply through the repository migration runner in an already-running disposable stack
docker compose exec backend node run-migrations.js
```

Open the generated SQL and journal entry. Confirm table name, foreign key delete action, both indexes, and no unrelated drop or rename. Never use Drizzle push against a shared or deployed database. Exercise the migration on a disposable database before a persistent volume.

Step 3 — Add validated DTOs

Create `backend/src/modules/maintenance-notes/DTO/maintenance-note.dto.ts`:

```
import { ApiProperty, ApiPropertyOptional } from '@nestjs/swagger';
import { IsOptional, IsString, IsUUID, MaxLength, MinLength } from 'class-validator';

export class CreateMaintenanceNoteDto {
```

```

@ApiProperty({ example: 'queue-job' })
@IsString()
@MinLength(2)
@MaxLength(80)
targetType!: string;

@ApiPropertyOptional({ example: '42e8dbe5-3797-4fc7-a64a-c288747e401b' })
@IsOptional()
@IsUUID()
targetId?: string;

@ApiProperty({ example: 'Reviewed retained failure and approved one retry.' })
@IsString()
@MinLength(3)
@MaxLength(4000)
body!: string;
}

export class UpdateMaintenanceNoteDto {
  @ApiProperty({ example: 'Retry completed and durable state was reconciled.' })
  @IsString()
  @MinLength(3)
  @MaxLength(4000)
  body!: string;
}

```

Step 4 — Implement service and audit behavior

Create `backend/src/modules/maintenance-notes/maintenance-notes.service.ts`:

```

import { Injectable, NotFoundException } from '@nestjs/common';
import { desc, eq } from 'drizzle-orm';
import { DatabaseService } from '../../../database/database.service';
import { maintenanceNotes } from '../../../drizzle/schema';
import { AuditService } from '../../../audit/audit.service';
import { CreateMaintenanceNoteDto, UpdateMaintenanceNoteDto } from './DTO/maintenance-note.dto';

@Injectable()
export class MaintenanceNotesService {
  constructor(
    private readonly databaseService: DatabaseService,
    private readonly auditService: AuditService,
  ) {}

  private get db() {
    return this.databaseService.db;
  }

  findAll() {
    return this.db.query.maintenanceNotes.findMany({
      orderBy: [desc(maintenanceNotes.createdAt)],
      limit: 200,
    });
  }

  async create(dto: CreateMaintenanceNoteDto, actorId: string) {
    const [created] = await this.db.insert(maintenanceNotes).values({
      targetType: dto.targetType.trim(),
      targetId: dto.targetId ?? null,
      body: dto.body.trim(),
      createdBy: actorId,
    }).returning();
    await this.auditService.log({
      actorId,
      action: 'maintenance_note.created',
      targetType: 'maintenance_note',
      targetId: created.id,
      metadata: { targetType: created.targetType, targetId: created.targetId },
    });
    return created;
  }

  async update(id: string, dto: UpdateMaintenanceNoteDto, actorId: string) {
    const [updated] = await this.db.update(maintenanceNotes).set({
      body: dto.body.trim(),
      updatedAt: new Date(),
    }).where(eq(maintenanceNotes.id, id)).returning();
    if (!updated) throw new NotFoundException('Maintenance note not found');
    await this.auditService.log({
      actorId,
      action: 'maintenance_note.updated',
      targetType: 'maintenance_note',
    });
  }
}

```

```

        targetId: updated.id,
    });
    return updated;
}

async remove(id: string, actorId: string) {
    const [removed] = await this.db.delete(maintenanceNotes)
        .where(eq(maintenanceNotes.id, id)).returning();
    if (!removed) throw new NotFoundException('Maintenance note not found');
    await this.auditService.log({
        actorId,
        action: 'maintenance_note.deleted',
        targetType: 'maintenance_note',
        targetId: removed.id,
    });
    return { id: removed.id };
}
}

```

Step 5 — Add controller and RBAC

Create `backend/src/modules/maintenance-notes/maintenance-notes.controller.ts`:

```

import { Body, Controller, Delete, Get, Param, ParseUUIDPipe, Patch, Post, UseGuards } from '@nestjs/common';
import { ApiBearerAuth, ApiTags } from '@nestjs/swagger';
import { RoleName } from '.../common/constants/role.constants';
import { CurrentUser } from '.../auth/decorators/current-user.decorator';
import { Roles } from '.../auth/decorators/roles.decorator';
import { JwtAuthGuard } from '.../auth/guards/jwt-auth.guard';
import { RolesGuard } from '.../auth/guards/roles.guard';
import { CreateMaintenanceNoteDto, UpdateMaintenanceNoteDto } from '.../DTO/maintenance-note.dto';
import { MaintenanceNotesService } from '.../maintenance-notes.service';

@ApiTags('Maintenance Notes')
@ApiBearerAuth('token')
@UseGuards(JwtAuthGuard, RolesGuard)
@Roles(RoleName.Admin)
@Controller('maintenance-notes')
export class MaintenanceNotesController {
    constructor(private readonly service: MaintenanceNotesService) {}

    @Get()
    async findAll() {
        return { success: true, data: await this.service.findAll() };
    }

    @Post()
    async create(
        @Body() dto: CreateMaintenanceNoteDto,
        @CurrentUser() user: { userId: string },
    ) {
        return { success: true, data: await this.service.create(dto, user.userId) };
    }

    @Patch('/:id')
    async update(
        @Param('id', ParseUUIDPipe) id: string,
        @Body() dto: UpdateMaintenanceNoteDto,
        @CurrentUser() user: { userId: string },
    ) {
        return { success: true, data: await this.service.update(id, dto, user.userId) };
    }

    @Delete('/:id')
    async remove(
        @Param('id', ParseUUIDPipe) id: string,
        @CurrentUser() user: { userId: string },
    ) {
        return { success: true, data: await this.service.remove(id, user.userId) };
    }
}

```

Create `backend/src/modules/maintenance-notes/maintenance-notes.module.ts`:

```

import { Module } from '@nestjs/common';
import { AuditModule } from '.../audit/audit.module';
import { MaintenanceNotesController } from '.../maintenance-notes.controller';
import { MaintenanceNotesService } from '.../maintenance-notes.service';

```

```
@Module({
  imports: [AuditModule],
  controllers: [MaintenanceNotesController],
  providers: [MaintenanceNotesService],
  exports: [MaintenanceNotesService],
})
export class MaintenanceNotesModule {}
```

Import `MaintenanceNotesModule` in `backend/src/app.module.ts` and add it once to the `AppModule` imports array.

Step 6 — Lock the route contract with tests

Create focused DTO, service, and controller specs. This controller example verifies delegation and response shape:

```
import { MaintenanceNotesController } from './maintenance-notes.controller';

describe('MaintenanceNotesController', () => {
  const service = {
    findAll: jest.fn(),
    create: jest.fn(),
    update: jest.fn(),
    remove: jest.fn(),
  };
  const controller = new MaintenanceNotesController(service as never);

  afterEach(() => jest.clearAllMocks());

  it('creates an admin maintenance note through the service', async () => {
    const dto = {
      targetType: 'queue-job',
      targetId: '42e8dbe5-3797-4fc7-a64a-c288747e401b',
      body: 'Approved one retry after durable-state reconciliation.',
    };
    const created = { id: 'note-1', targetType: dto.targetType, targetId: dto.targetId, body: dto.body };
    service.create.mockResolvedValue(created);
    await expect(controller.create(dto, { userId: 'admin-1' })).resolves.toEqual({
      success: true,
      data: created,
    });
    expect(service.create).toHaveBeenCalledWith(dto, 'admin-1');
  });
});
```

Also test invalid UUID, unknown properties, missing body, overlong body, unauthenticated access, teacher and student rejection, not-found update, delete audit behavior, and migration rollback in a disposable database.

Verification gate

```
npm --prefix backend run check:migrations
npm --prefix backend test -- maintenance-notes
npm --prefix backend run lint
npm --prefix backend run build
```

Recipe 10.2 — Add a BullMQ queue and worker

The current repository registers queues in the owning module; there is no central `QueueModule`. This example adds a durable manual-export queue whose worker writes a bounded audit export through the existing storage abstraction.

Step 1 — Define constants and contracts

Create `backend/src/modules/manual-export/manual-export.contract.ts`:

```
export const MANUAL_EXPORT_QUEUE = 'manual-export';
export const BUILD_MANUAL_EXPORT_JOB = 'build-manual-export';
```

```
export interface BuildManualExportJobData {
  exportId: string;
  requestedByUserId: string;
  queuedAt: string;
}

export interface BuildManualExportResult {
  exportId: string;
  storageKey: string;
  rowCount: number;
}
```

Step 2 — Implement bounded domain work

Create `backend/src/modules/manual-export/manual-export.service.ts`:

```
import { Injectable } from '@nestjs/common';
import { desc } from 'drizzle-orm';
import { DatabaseService } from '../../../database/database.service';
import { auditLogs } from '../../../drizzle/schema';
import { StorageService } from '../../file-upload/storage/storage.service';
import { BuildManualExportResult } from './manual-export.contract';

@Injectable()
export class ManualExportService {
  constructor(
    private readonly databaseService: DatabaseService,
    private readonly storageService: StorageService,
  ) {}

  async build(exportId: string): Promise<BuildManualExportResult> {
    const rows = await this.databaseService.db.query.auditLogs.findMany({
      orderBy: [desc(auditLogs.createdAt)],
      limit: 10000,
    });
    const storageKey = "maintenance-exports/" + exportId + ".json";
    await this.storageService.putObject({
      key: storageKey,
      body: Buffer.from(JSON.stringify(rows), "utf8"),
      contentType: 'application/json',
    });
    return { exportId, storageKey, rowCount: rows.length };
  }
}
```

Step 3 — Implement producer and processor

Create `backend/src/modules/manual-export/manual-export-queue.service.ts`:

```
import { InjectQueue } from '@nestjs/bullmq';
import { Injectable } from '@nestjs/common';
import type { Queue } from 'bullmq';
import { BUILD_MANUAL_EXPORT_JOB, BuildManualExportJobData, MANUAL_EXPORT_QUEUE } from './manual-export.contract';

@Injectable()
export class ManualExportQueueService {
  constructor(@InjectQueue(MANUAL_EXPORT_QUEUE) private readonly queue: Queue) {}

  async enqueue(exportId: string, requestedByUserId: string): Promise<void> {
    const data: BuildManualExportJobData = {
      exportId,
      requestedByUserId,
      queuedAt: new Date().toISOString(),
    };
    await this.queue.add(BUILD_MANUAL_EXPORT_JOB, data, {
      jobId: "manual-export:" + exportId,
      attempts: 3,
      backoff: { type: 'exponential', delay: 5000 },
      removeOnComplete: 100,
      removeOnFail: 200,
    });
  }
}
```

Create `backend/src/modules/manual-export/manual-export.processor.ts`:


```
import { Processor, WorkerHost } from '@nestjs/bullmq';
import { Injectable } from '@nestjs/common';
import type { Job } from 'bullmq';
import { BUILD_MANUAL_EXPORT_JOB, BuildManualExportJobData, BuildManualExportResult, MANUAL_EXPORT_QUEUE } from './manual-export.contract';
import { ManualExportService } from './manual-export.service';

@Injectable()
@Processor(MANUAL_EXPORT_QUEUE, { concurrency: 1 })
export class ManualExportProcessor extends WorkerHost {
  constructor(private readonly manualExportService: ManualExportService) {
    super();
  }

  async process(job: Job<BuildManualExportJobData>): Promise<BuildManualExportResult> {
    if (job.name !== BUILD_MANUAL_EXPORT_JOB) {
      throw new Error("Unsupported manual-export job: " + job.name);
    }
    return this.manualExportService.build(job.data.exportId);
  }
}
```

The processor throws unknown-job and domain failures so BullMQ owns retry behavior. Storage keys are deterministic, making a retry replace the same export object rather than create an unbounded duplicate.

Step 4 — Register the queue in its owner

Create `backend/src/modules/manual-export/manual-export.module.ts`:

```
import { BullModule } from '@nestjs/bullmq';
import { Module } from '@nestjs/common';
import { DatabaseModule } from '../../database/database.module';
import { StorageModule } from '../../file-upload/storage/storage.module';
import { MANUAL_EXPORT_QUEUE } from './manual-export.contract';
import { ManualExportProcessor } from './manual-export.processor';
import { ManualExportQueueService } from './manual-export-queue.service';
import { ManualExportService } from './manual-export.service';

@Module({
  imports: [
    DatabaseModule,
    StorageModule,
    BullModule.registerQueue({ name: MANUAL_EXPORT_QUEUE }),
  ],
  providers: [ManualExportService, ManualExportQueueService, ManualExportProcessor],
  exports: [ManualExportQueueService],
})
export class ManualExportModule {}
```

Import `ManualExportModule` in `AppModule`. Persist an export-request row before enqueueing if users must poll status after restarts; do not rely on Redis as the only business record.

Step 5 — Verify job semantics

- Unit-test exact queue name, job name, payload, deterministic ID, attempts, backoff, completion retention, and failure retention.
- Processor-test success, unknown job, storage failure, database failure, retry idempotency, and the 10,000-row bound.
- Integration-test Redis unavailable at enqueue, worker restart during execution, duplicate enqueue, and retained terminal failure.
- Add queue waiting, active, completed, failed, and duration metrics before creating a backlog alert.

```
npm --prefix backend test -- manual-export
npm --prefix backend run lint
npm --prefix backend run build
```

Recipe 10.3 — Add a grounded AI feature

This example adds an internal study-summary execution route. A public NestJS controller must authorize the teacher and either queue durable work or call the proxy with a bounded timeout. The example performs class-scoped retrieval before generation.

Step 1 — Add Pydantic contracts

Add to `ai-service/app/schemas.py`:

```
class StudySummaryRequest(BaseModel):
    class_id: str = Field(alias='classId')
    topic: str = Field(min_length=3, max_length=300)
    top_k: int = Field(default=6, ge=2, le=10, alias="topK")

    model_config = ConfigDict(populate_by_name=True)

class StudySummaryResponse(BaseModel):
    summary: str
    source_ids: list[str] = Field(alias="sourceIds")

    model_config = ConfigDict(populate_by_name=True)
```

Step 2 — Add the internal router

Create `ai-service/app/routers/study_summary.py`:

```
from collections.abc import Callable

from fastapi import APIRouter, Depends, HTTPException
from sqlalchemy.ext.asyncio import AsyncSession

from ..database import get_db
from ..ollama_client import generate
from ..retrieval_service import similarity_search
from ..schemas import StudySummaryRequest, StudySummaryResponse

STUDY_SUMMARY_SYSTEM_PROMPT = (
    "You create concise teacher-facing study summaries. "
    "Use only the supplied evidence. State when evidence is insufficient. "
    "Do not invent grades, student facts, citations, or official decisions."
)

def build_study_summary_router(require_internal_service: Callable):
    router = APIRouter()

    @router.post(
        "/internal/study-summary",
        response_model=StudySummaryResponse,
    )
    async def create_study_summary(
        body: StudySummaryRequest,
        _auth: None = Depends(require_internal_service),
        db: AsyncSession = Depends(get_db),
    ):
        chunks = await similarity_search(
            db,
            query_text=body.topic,
            class_id=body.class_id,
            top_k=body.top_k,
            only_published=True,
            policy_name="general",
        )
        if len(chunks) < 2:
            raise HTTPException(422, "Insufficient grounded evidence for a study summary")
        evidence = "\n\n".join(
            "SOURCE " + str(index + 1) + ": " + str(chunk["chunkText"])
            for index, chunk in enumerate(chunks)
        )
        prompt = "TOPIC: " + body.topic + "\n\nEVIDENCE:\n" + evidence
        summary = await generate(
            prompt=prompt,
            system=STUDY_SUMMARY_SYSTEM_PROMPT,
            task="chat",
        )
```

```

        temperature=0.1,
        num_predict=700,
    )
    return StudySummaryResponse(
        summary=summary.strip(),
        sourceIds=[str(chunk["sourceId"]) for chunk in chunks],
    )

    return router

```

In `ai-service/app/main.py`, import the builder after dependency definitions, build it with `require_internal_service`, and call `app.include_router` once. Keep the route internal.

Step 3 — Add the backend execution seam

Add a bounded method to `backend/src/modules/ai-mentor/ai-proxy.service.ts` or a dedicated client. The shared proxy already attaches user headers and the internal secret:

```

async createInternalStudySummary(
  user: { id?: string; userId?: string; email?: string; roles?: string[] },
  body: { classId: string; topic: string; topK: number },
): Promise<unknown> {
  return this.forward('POST', '/internal/study-summary', user, body);
}

```

Before calling it, the NestJS controller and service must enforce teacher or admin role and class ownership. If the operation can exceed the request budget, create a durable `ai_generation_jobs` row, enqueue a typed BullMQ job, call this method from the worker, and persist output or failure.

Step 4 — Validate grounding and failure behavior

- Unit-test the Pydantic aliases, topic length, topK bounds, internal secret, insufficient evidence, and source ID response.
- Mock `similarity_search` and generate separately so retrieval failures are distinguishable from model failures.
- Test cross-class attempts through the public backend route, not only the internal FastAPI function.
- Test Ollama timeout, malformed runtime response, database outage, duplicate queued execution, and cancellation where supported.
- Add latency, outcome, and evidence-count metrics without student content in labels.

```

python -m pytest ai-service/tests -q
npm --prefix backend test -- ai-mentor
npm --prefix backend run build

```

Recipe 10.4 — Add a role-restricted web and mobile screen

This example adds a teacher-only study-summary surface backed by the Chapter 10.3 backend contract.

Step 1 — Add the web service and query hook

Add the method to `next-frontend/src/services/ai-service.ts` using the existing shared API client:

```

export type StudySummaryRequest = {
  classId: string;
  topic: string;
  topK: number;
};

export type StudySummaryResult = {
  summary: string;
  sourceIds: string[];
};

```

```
export async function createStudySummary(
  payload: StudySummaryRequest,
): Promise<StudySummaryResult> {
  const response = await apiClient.post('/ai/teacher/study-summary', payload);
  return response.data.data as StudySummaryResult;
}
```

Create `next-frontend/src/hooks/use-study-summary.ts`:

```
'use client';

import { useMutation } from '@tanstack/react-query';
import { createStudySummary } from '@services/ai-service';

export function useStudySummary() {
  return useMutation({ mutationFn: createStudySummary });
}
```

Step 2 — Add the teacher App Router page

Create `next-frontend/app/(dashboard)/dashboard/teacher/study-summary/page.tsx`. The page must use `AuthProvider` role state or the enclosing teacher layout, expose topic and class selection, disable duplicate submission while pending, render backend failure text safely, and show source references returned by the contract. Add the route to teacher navigation and dashboard-route-access tests.

Step 3 — Add mobile API and route types

Add this method to the owning mobile AI service:

```
export type MobileStudySummaryRequest = {
  classId: string;
  topic: string;
  topK: number;
};

export async function createStudySummary(payload: MobileStudySummaryRequest) {
  const response = await apiClient.post('/ai/teacher/study-summary', payload);
  return response.data.data as { summary: string; sourceIds: string[] };
}
```

Add `TeacherStudySummary: { classId: string }` to `RootStackParamList` in `mobile/src/navigation/types.ts`, add `TeacherStudySummary` to `teacherRouteManifest.stack`, create `mobile/src/screens/TeacherStudySummaryScreen.tsx`, import it in `AppNavigator`, and register:

```
<RootStack.Screen name="TeacherStudySummary" component={TeacherStudySummaryScreen} />
```

Add a teacher web-to-mobile mapping entry in the teacher route manifest so parity tests cover the new surface.

Step 4 — Verify authorization and client behavior

- Direct web and mobile navigation as teacher succeeds only for an owned class.
- Student access is hidden in navigation and rejected by the backend.
- Admin behavior matches the backend policy rather than being inferred by the clients.
- Access-token expiry performs one refresh and one request retry.
- Offline submission does not display success; retry remains explicit.
- Loading, empty evidence, validation, 403, 422, 429, 503, and timeout states are accessible and actionable.

```
npm --prefix next-frontend test -- study-summary dashboard-route-access
npm --prefix next-frontend run lint
npm --prefix next-frontend run build
```

```
npm --prefix mobile test -- study-summary app-navigator-role-resolution
npm --prefix mobile run typecheck
```

Recipe 10.5 — Safely upgrade dependencies and run tests

Step 1 — Establish the before-state

```
git status --short
node --version
npm --version
npm --prefix backend outdated
npm --prefix next-frontend outdated
npm --prefix mobile outdated
python --version
python -m pip check
```

Read release notes and migration guides for every direct major upgrade. Keep backend, web, mobile, and AI upgrades in separate commits unless one contract requires coordinated versions.

Step 2 — Change one dependency family

- Use the package manager and lockfile already owned by the subsystem.
- Keep Nest peer packages on compatible major versions and Next, React, React DOM, and test renderer versions aligned.
- Keep Expo SDK packages on versions accepted by the selected Expo SDK.
- Regenerate Python requirements from the repository input workflow rather than manually editing transitive pins.
- Do not combine an ORM upgrade with an unrelated schema change.

After editing the declared dependency, run the deterministic install:

```
npm --prefix backend install
npm --prefix next-frontend install
npm --prefix mobile install
python -m pip check
```

Step 3 — Run subsystem gates

```
# Backend
npm --prefix backend run check:src-clean
npm --prefix backend run check:migrations
npm --prefix backend run lint
npm --prefix backend test -- --runInBand
npm --prefix backend run build

# Web
npm --prefix next-frontend run lint
npm --prefix next-frontend test -- --runInBand
npm --prefix next-frontend run build

# Mobile
npm --prefix mobile run typecheck
npm --prefix mobile test

# AI service
python -m pytest ai-service/tests -q
```

Step 4 — Run integration gates

```
docker compose config --quiet
docker compose build backend ai-service frontend
docker compose up -d postgres redis ollama backend ai-service frontend
docker compose ps
curl --fail --silent --show-error http://localhost:3000/api/health/ready
curl --fail --silent --show-error http://localhost:3001/
npm --prefix next-frontend run test:e2e
```

Use a disposable Compose project when migration, database bootstrap, Redis state, model bootstrap, or startup ordering is under test. Do not delete persistent volumes as a routine upgrade step.

Step 5 — Review and rollback readiness

- Inspect package manifests and lockfiles for only intended dependency movement.
- Compare application image size, startup time, health timing, route latency, bundle warnings, and advisory output with the recorded before-state.
- Confirm migrations remain forward-only and the prior application version can coexist during a rolling deployment when the deployment model requires it.
- Document the exact rollback unit: application image, lockfile, configuration, and migration compatibility.

Recipe 10.6 — Add or change a public API contract safely

1. Identify every consumer in web services, mobile services, FastAPI SQL or schemas, queue payloads, tests, seed data, and reports.
2. Prefer additive fields and routes. Keep old fields until all deployed consumers have migrated.
3. Update DTO validation and response types before client wiring.
4. Add contract tests for allowed roles, ownership, validation failure, success body, and backward compatibility.
5. Deploy backend compatibility first, then clients, then remove deprecated fields in a later gated release.

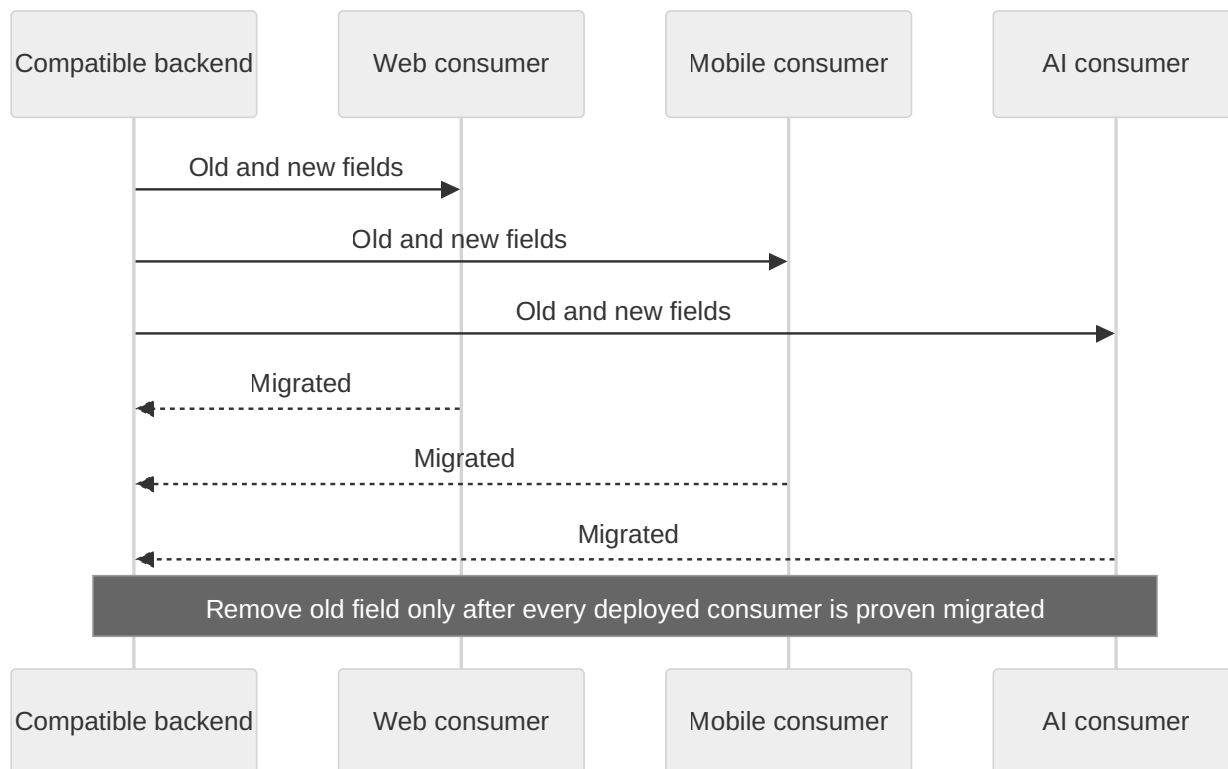


Diagram 26

Recipe 10.7 — Release evidence packet

For a release-affecting change, retain:

- Commit SHA and dirty-tree status.
- Exact install and verification commands with exit codes.
- Migration integrity result and disposable migration evidence when schema changed.
- Health and readiness responses from the candidate stack.
- Targeted browser, device, queue, or AI scenario evidence.
- Security review covering auth, ownership, secrets, logging, upload handling, and data egress.
- Known limitations, rollback trigger, and rollback unit.

Final maintenance rule

A change is complete only when the owning source, every affected contract consumer, tests, migration or queue behavior, operational telemetry, and this manual agree. Passing one subsystem build does not prove a cross-system workflow.